

Solutions

NCERT Class XII Chemistry · Chapter 1 · Solutions

15 marks on the blueprint — the heaviest chapter in the paper. Theory you know; the marks live in the numericals. Every formula below is followed by the method that actually gets used in the exam.

Types of solution — *nine combinations, one table*

A solution is a homogeneous mixture of two or more components. Composition and properties are uniform throughout. Two components only = **binary solution**.

The component in the largest quantity is the **solvent** — and the solvent fixes the physical state of the solution. Everything else is **solute**.

Solvent	Solute	Example
Gas	Gas	Air (O ₂ in N ₂)
Liquid	Chloroform in nitrogen	
Solid	Camphor in nitrogen	
Liquid	Gas	O ₂ in water; CO ₂ in soft drinks
Liquid	Ethanol in water	
Solid	Glucose in water	
Solid	Gas	H ₂ in palladium
Liquid	Amalgam (Na in Hg)	
Solid	Alloys — Cu in Au	

Examiner asks: one-markers naming the solute/solvent type for a given example. Camphor-in-nitrogen and hydrogen-in-palladium are the two people forget.

Concentration terms — *seven of them — and which one the question wants*

The whole game is reading which term the question gives you and which it wants. Get that right and the arithmetic is trivial.

Mass percentage (w/w)

$$\text{mass \%} = \frac{\text{mass of component}}{\text{mass of solution}} \times 100$$

"10% glucose by mass" means **10 g glucose in 100 g solution** — so 90 g water. Used for industrial chemicals.

Volume percentage (v/v)

$$\text{volume \%} = \frac{\text{volume of component}}{\text{volume of solution}} \times 100$$

Used when both components are liquids. 35% v/v ethylene glycol is car antifreeze — it drops water's freezing point to 255.4 K.

Mass by volume (w/v)

Mass of solute in 100 mL of solution. Medicine and pharmacy. **0.9% w/v NaCl = normal saline** — remember this one, it reappears in the osmosis questions.

Parts per million

$$\text{ppm} = \frac{\text{parts of component}}{\text{total parts}} \times 10^6$$

For trace quantities — pollutants, dissolved oxygen in seawater.

Mole fraction (x)

$$x_A = \frac{n_A}{n_A + n_B} \quad x_A + x_B = 1$$

No units. This is the one Raoult's and Henry's laws are written in.

Molarity (M)

$$M = \frac{\text{moles of solute}}{\text{volume of solution in L}}$$

Depends on temperature — volume expands on heating, so molarity falls.

Molality (m)

$$m = \frac{\text{moles of solute}}{\text{mass of solvent in kg}}$$

Independent of temperature — mass doesn't change with T. This is why every colligative-property formula uses molality, not molarity.

Trap: molarity uses *volume of solution*; molality uses *mass of solvent*. Not solution. Losing the solute mass from the denominator is the single most common arithmetic slip in this chapter.

Worked · 2014 — molality and molarity from mass %

10% by mass glucose ($M = 180 \text{ g/mol}$) in water, density 1.2 g/mL . Find molality and molarity.

Set up: 10% by mass $\rightarrow$ 10 g glucose in 100 g solution $\rightarrow$ water = 90 g.

Molality:

$$m = \frac{10/180}{90/1000} = 0.62 \text{ mol kg}^{-1}$$

Molarity: need volume. From $d = m/V$: $V = 100/1.2 = 83.3 \text{ mL} = 0.0833 \text{ L}$.

$$M = \frac{10/180}{0.0833} = 0.67 \text{ mol L}^{-1}$$

Examiner asks: "given mass % and density, find molality and molarity" is a standing 3-marker. The density is only ever there to get you the volume.

Solubility — *what dissolves in what, and what temperature does to it*

Solubility = maximum amount of solute that dissolves in a specified amount of solvent at a specified temperature.

Solid in liquid

Like dissolves like. Polar solutes (NaCl, sugar) dissolve in polar solvents (water); non-polar solutes (naphthalene, anthracene) dissolve in non-polar solvents (benzene).

Add solute and two opposing processes run: **dissolution** and **crystallisation**. When their rates become equal, dynamic equilibrium is reached, concentration goes constant, and the solution is **saturated**.

- **Temperature:** if dissolution is endothermic ($\Delta H > 0$), solubility rises with T. If exothermic ($\Delta H < 0$), solubility falls with T. (Le Chatelier.)
- **Pressure:** no significant effect — solids and liquids are incompressible.

Gas in liquid

Solubility **increases with pressure**: more gas particles per unit volume above the solution → more strike the surface → more dissolve. Solubility **decreases with temperature** (dissolution of a gas is exothermic).

Examiner asks: "aquatic species are more comfortable in cold water" (2019) — because oxygen's solubility falls as water warms. Straight 1–2 marker.

Henry's law — *the gas-solubility law and its three applications*

At constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of that gas above the liquid. Equivalently:

$$p = K_H x$$

where p is partial pressure of the gas, x its mole fraction in solution, and K_H the Henry's law constant.

Read K_H backwards: higher K_H means *lower* solubility. K_H rises with temperature — which is exactly why gases get less soluble in warm water.

Worked · 2020, repeated 2026 — solubility from Henry's law

Solubility of CO₂ in water at 298 K under 760 mm Hg. $K_H = 1.25 \times 10^6$ mm Hg.

Solubility here means mole fraction. From $p = K_H x$:

$$x = \frac{p}{K_H} = \frac{760}{1.25 \times 10^6} = 6.08 \times 10^{-4}$$

Mole fraction has no unit.

The three applications

- **Carbonated drinks** — bottles sealed under high pressure to force CO₂ into solution.
- **Deep-sea diving / bends** — high pressure underwater dissolves N₂ into blood; on ascent it comes out as bubbles that block capillaries. Divers' tanks use helium-diluted air (11.7% He, 56.2% N₂, 32.1% O₂) because helium's high K_H means it barely dissolves.

- **Anoxia at altitude** — low partial pressure of $O_2 \rightarrow$ low oxygen in blood and tissue $\rightarrow$ climbers feel weak and can't think clearly.

Examiner asks: "state Henry's law + calculate mole fraction" as one 2–3 marker, or the bends/anoxia reasoning as a standalone. Both are near-certain appearances.

Raoult's law and vapour pressure — *volatile pairs, then non-volatile solutes*

Two volatile liquids

Partial vapour pressure of each component is proportional to its mole fraction *in the liquid*:

$$p_1 = p_1^\circ x_1 \quad p_2 = p_2^\circ x_2$$

$$p_{\text{total}} = p_1^\circ x_1 + p_2^\circ x_2$$

For the mole fraction in the **vapour phase**, use pressures instead of moles:

$$y_1 = \frac{p_1}{p_{\text{total}}}$$

Worked · 2023 — total vapour pressure of an ideal solution

$p_X^\circ = 120$ mm Hg, $p_Y^\circ = 160$ mm Hg, equal moles of X and Y mixed, ideal solution. Find p_{total} .

Equal moles $\rightarrow x_X = x_Y = 0.5$.

$$p_{\text{total}} = (120)(0.5) + (160)(0.5) = 60 + 80 = 140 \text{ mm Hg}$$

Non-volatile solute in a volatile solvent

Only the solvent contributes to vapour pressure, so $p_{\text{solution}} = p_1^\circ x_1$. Adding a non-volatile solute always *lowers* vapour pressure — solute particles occupy surface area that solvent molecules would otherwise escape from.

Raoult's law as a special case of Henry's law

Henry's law is $p = K_H x$; Raoult's is $p = p^\circ x$. When the solute–solvent interaction happens to equal the solute–solute interaction, K_H becomes equal to p° and the two laws coincide.

Ideal and non-ideal solutions — *two deviations, with the examples that prove them*

Ideal solution

Obeys Raoult's law over the entire concentration range, and:

$$\Delta H_{\text{mix}} = 0 \quad \Delta V_{\text{mix}} = 0$$

Happens when A–A, B–B and A–B interactions are all nearly equal. Examples: n-hexane + n-heptane, bromoethane + chloroethane, benzene + toluene.

Positive deviation

A–B interaction **weaker** than A–A and B–B. Molecules escape more easily $\rightarrow$ vapour pressure *higher* than Raoult predicts.

$$p_A > p_A^\circ x_A, \quad \Delta H_{\text{mix}} > 0, \quad \Delta V_{\text{mix}} > 0$$

Examples: ethanol + acetone (ethanol's own H-bonding is broken up), carbon disulphide + acetone.

Negative deviation

A–B interaction **stronger** than A–A and B–B. Molecules held tighter → vapour pressure *lower* than predicted.

$$p_A < p_A^\circ x_A, \quad \Delta H_{\text{mix}} < 0, \quad \Delta V_{\text{mix}} < 0$$

Examples: phenol + aniline, chloroform + acetone (they form an H-bond between them), nitric acid + water.

Trap: chloroform + acetone is negative deviation because a *new* hydrogen bond forms *between* the two. Ethanol + acetone is positive because an *existing* hydrogen bond is broken. Both are asked; the reasoning direction is what earns the mark.

Azeotropes

Binary mixtures with the **same composition in liquid and vapour phase**, boiling at constant temperature — so they cannot be separated by fractional distillation.

- **Minimum boiling azeotrope** ← large *positive* deviation. Ethanol + water, 95% ethanol by volume.
- **Maximum boiling azeotrope** ← large *negative* deviation. Nitric acid + water, 68% HNO₃ by mass, boils at 393.5 K.

Examiner asks: "define azeotrope + which type from negative deviation, with example" — a 2–3 marker that has run repeatedly.

Colligative properties — *four of them, four formulas, one idea*

Properties that depend on the **number** of solute particles, not their identity. That one sentence is the reason every question in this section works.

1 · Relative lowering of vapour pressure

$$\frac{p_1^\circ - p_1}{p_1^\circ} = x_2 = \frac{n_2}{n_1 + n_2}$$

For a dilute solution $n_1 \gg n_2$, so this simplifies to the working form:

$$\frac{p_1^\circ - p_1}{p_1^\circ} = \frac{w_2/M_2}{w_1/M_1}$$

2 · Elevation of boiling point

$$\Delta T_b = K_b m = \frac{K_b w_2 \times 1000}{M_2 w_1}$$

K_b = molal elevation constant (ebullioscopic constant), units K kg mol^{−1}. For water, $K_b = 0.52$.

3 · Depression of freezing point

$$\Delta T_f = K_f m = \frac{K_f w_2 \times 1000}{M_2 w_1}$$

K_f = molal depression constant (cryoscopic constant). For water, $K_f = 1.86$, freezing point 273 K.

Both constants depend only on the solvent, and can themselves be calculated:

$$K_f = \frac{R M_1 T_f^2}{1000 \Delta H_{\text{fus}}} \quad K_b = \frac{R M_1 T_b^2}{1000 \Delta H_{\text{vap}}}$$

4 · Osmotic pressure

Osmosis: solvent flows through a semi-permeable membrane from pure solvent into solution. **Osmotic pressure** (π) is the extra pressure applied on the solution side that just stops that flow.

$$\pi = CRT = \frac{n_2 RT}{V} = \frac{w_2 RT}{M_2 V}$$

Why it's preferred for macromolecules (proteins, polymers): works at room temperature, so heat-unstable biomolecules survive; uses molarity, which is easy to measure; and its magnitude is large even for dilute solutions, which is all you get from a poorly soluble polymer.

Trap: π uses *molarity* and volume in litres. ΔT_b and ΔT_f use *molality* and solvent mass in kg. Mixing these up is the standard way to lose an otherwise-correct 3-marker.

Isotonic, hypertonic, hypotonic

- **Isotonic** — same osmotic pressure, no net osmosis. Blood cells are isotonic with 0.9% w/v NaCl.
- **Hypertonic** — higher concentration outside; water leaves the cell; cell **shrinks**.
- **Hypotonic** — lower concentration outside; water enters; cell **swells** and may burst.

Reverse osmosis: apply pressure *greater than* π on the solution side and solvent flows backwards, out of the solution. Used to desalinate seawater, with a cellulose acetate membrane that passes water but not ions.

Abnormal molar mass and van't Hoff factor — when the measured mass is wrong, and by how much

Colligative properties give a molar mass. When the solute associates or dissociates, that measured value differs from the true one — an **abnormal molar mass**.

Association

Molecules combine → fewer particles → colligative property falls → measured molar mass comes out **too high**. Ethanoic acid in benzene dimerises through hydrogen bonds: measured 120 g/mol against a true 60.

Dissociation

Electrolyte splits into ions → more particles → colligative property rises → measured molar mass comes out **too low**. KCl gives 37.25 g/mol against a true 74.5.

van't Hoff factor

$$i = \frac{\text{normal molar mass}}{\text{abnormal molar mass}} = \frac{\text{observed colligative property}}{\text{calculated colligative property}}$$

Equivalently, particles after dissociation/association $\div$ particles before.

- **Dissociation** $\rightarrow i > 1$. KCl: 2. MgSO_4 : 2. K_2SO_4 : 3. CaCl_2 : 3.
- **Association** $\rightarrow i < 1$. Complete dimerisation gives $i = 0.5$.
- **Neither** $\rightarrow i = 1$. Glucose, urea, sucrose.

Every colligative formula then carries i :

$$\frac{p_1^\circ - p_1}{p_1^\circ} = i x_2 \quad \Delta T_b = i K_b m \quad \Delta T_f = i K_f m \quad \pi = \frac{i n_2 R T}{V}$$

Partial dissociation or association

$$\text{dissociation: } i = 1 + (n - 1)\alpha \quad \text{association: } i = 1 + \left(\frac{1}{n} - 1\right)\beta$$

α = degree of dissociation, β = degree of association, n = particles produced or combined.

Trap: a degree of dissociation cannot exceed 1. If your α comes out at 1.056, the intended answer is "essentially complete dissociation" — say so rather than writing an impossible number.

Numerical patterns, collected — every calculation type in this chapter, one model each

Six patterns cover essentially every numerical this chapter sets. Method first, then one worked model.

A · Concentration conversion — 3 marks

Recognise it: gives a mass %, asks for molality/molarity, and hands you a density you'd otherwise have no use for.

1. Take 100 g of solution. Solute mass = the percentage; solvent mass = 100 – that.
2. Moles of solute = w_2/M_2 .
3. Molality: divide by solvent mass in kg.
4. Molarity: get volume from $V = \text{mass}/d$, convert to litres, divide.

B · Henry's law — 2 marks

Recognise it: a K_H value appears.

1. $x = p/K_H$. Answer is a mole fraction, unitless. That's the whole question.

C · Total vapour pressure — 3 marks

Recognise it: two p° values for two volatile liquids.

1. Find mole fractions in the liquid (equal moles $\rightarrow$ 0.5 each).
2. $p_{\text{total}} = p_1^\circ x_1 + p_2^\circ x_2$.
3. If it asks for vapour-phase composition: $y_1 = p_1/p_{\text{total}}$.

D · $\Delta T_b / \Delta T_f \rightarrow$ molar mass or temperature — 3 marks

Recognise it: a K_b or K_f is given. This is the most-set numerical in the chapter.

1. Decide whether i is needed — is the solute an electrolyte, or does it dimerise?
2. $\Delta T = iK \frac{w_2 \times 1000}{M_2 w_1}$. Rearrange for whatever's missing.
3. If asked for the actual boiling/freezing point, *add* ΔT_b to T_b° or *subtract* ΔT_f from T_f° . Don't stop at ΔT .

Worked · 2018 — freezing point of a glucose solution

60 g glucose ($M = 180$) in 250 g water, $K_f = 1.86$. Find the freezing point.

Glucose doesn't dissociate, so $i = 1$.

$$\Delta T_f = \frac{1.86 \times 60 \times 1000}{180 \times 250} = 2.48 \text{ K}$$

$$T_f = 273 - 2.48 = 270.52 \text{ K}$$

E · Osmotic pressure — 3 marks

Recognise it: R is given, or the word isotonic appears.

1. $\pi = \frac{w_2 RT}{M_2 V}$, volume in **litres**.
2. For isotonic pairs, set the two concentrations equal: $\frac{w_1}{M_1} = \frac{w_2}{M_2}$ (volumes cancel) and solve for the unknown M .

F · van't Hoff factor $\rightarrow \alpha$ or β — 3–5 marks

Recognise it: asks for degree of dissociation/association, or "predict the nature of the solute".

1. Get i from the colligative-property equation with i left in as the unknown.
2. $i > 1 \rightarrow$ dissociation, use $i = 1 + (n - 1)\alpha$. $i < 1 \rightarrow$ association, use the association form.
3. Read n off the actual dissociation equation — write it out, don't guess.

Worked · 2026 — degree of dissociation of fluoroacetic acid

19.5 g FCH_2COOH ($M = 78$) in 500 g water gives $\Delta T_f = 1 \text{ K}$. $K_f = 1.86$.

$$1 = i \times \frac{1.86 \times 19.5 \times 1000}{78 \times 500} \Rightarrow i = 1.075$$

$\text{FCH}_2\text{COOH} \rightarrow \text{FCH}_2\text{COO}^- + \text{H}^+$, so $n = 2$:

$$1.075 = 1 + (2 - 1)\alpha \Rightarrow \alpha = 0.075 = 7.5\%$$

Past year questions · question types — ranked by how often they turn up

From the last 10 years of boards. 22 questions are captured below with their years; the source video states it covers 25, so treat this as the bulk of the set rather than all of it.

1 · Colligative-property numerical — 3 marks

Recognise it: K_b , K_f or R is supplied.

1. Pick the right formula from the constant given.
2. Decide on i .
3. Convert to the final temperature if asked.

Trap: stopping at ΔT when the question asked for the boiling or freezing point.

2 · Deviation from Raoult's law, with reason — 2–3 marks

Recognise it: names a specific liquid pair and asks "what type of deviation, why".

1. Compare A–B interaction against A–A and B–B.
2. Stronger → negative, vapour pressure falls. Weaker → positive, vapour pressure rises.
3. Name the actual interaction — which H-bond forms or breaks.

Trap: answering "negative deviation" without saying *which* bond forms between the two. The reason carries the mark.

3 · van't Hoff factor / degree of dissociation — 3–5 marks

Recognise it: asks for i , α , β , or "predict association or dissociation".

1. Solve the colligative equation for i .
2. Compare against 1 to decide association vs dissociation.
3. Apply the matching α/β relation.

Trap: using $n = 2$ automatically. CaCl_2 gives 3 ions, K_2SO_4 gives 3.

4 · Henry's law statement + calculation — 2–3 marks

Recognise it: K_H appears, often paired with "state Henry's law".

1. State the law in mole-fraction form.
2. $x = p/K_H$.
3. If asked for applications, give carbonated drinks and the bends.

5 · Osmosis reasoning (cells, meat, isotonic) — 2–3 marks

Recognise it: blood cells, salting meat, or a % NaCl solution.

1. Compare concentration inside vs outside.
2. Water moves toward the higher concentration.
3. State the consequence: shrink, swell, or preserved.

6 · Definitions and two-difference questions — 2 marks

Recognise it: "define...", "write two differences between..."

1. Give the definition in the book's own words.
2. For differences, answer in pairs — ideal vs non-ideal on both ΔH_{mix} and ΔV_{mix} .

Past year questions · mark slots — *what each type is worth, and the time that buys*

Question type	Slot	Time
Colligative numerical	3-marker	4–5 min
Deviation + reason	2–3 marker	3 min
van't Hoff / α	3-marker, sometimes inside a 5	5 min
Henry's law	2–3 marker	3 min
Osmosis reasoning	2–3 marker	3 min
Definitions / differences	2-marker	2 min
Concentration conversion	3-marker	4 min

Slots marked from the question shapes in the paper — the video groups them as short-answer type I (2 marks) and type II (3 marks).

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

These are the reason the PYQ video is worth watching. Each has run at least twice.

2020 Q3 · 2026 Q1

Henry's law statement plus mole fraction of CO_2 at 760 mm Hg — the identical numbers reappeared six years apart.

2019 Q11(a) · lecture PYQ

Aquatic species more comfortable in cold water — gas solubility falls as temperature rises.

2019 Q11(b) · 2019 Q12

Anoxia at high altitude, and chloroform + acetone negative deviation. Both recur as reason-type one/two markers.

2018 Q10 · 2026 Q18

Freezing-point depression with a given K_f — glucose in water, then fluoroacetic acid in water with a degree of dissociation on top.

2023 Q5 · lecture PYQ

Total vapour pressure of an ideal solution from two p° values and equal moles. Same 120/160 mm Hg numbers in both.

2016 Q7 · 2025 Q6

van't Hoff factor: definition, and why osmotic pressure is preferred for macromolecular molar masses.

2017 Q13 · 2020 Q2

Blood cells in 1.2% and 0.4% NaCl, and why 0.1 M KCl boils higher than 0.1 M glucose. Both test "colligative = number of particles".

Past year questions · cold practice — *answers only — work them before you look*

One model per pattern is above. These are the same patterns with different numbers: work each one cold, then check.

Colligative numericals

- 2026 Q14 — 8 g non-volatile solute in 100 g diethyl ether, bp rises $35.60 \rightarrow 36.86^\circ\text{C}$, $K_b = 2.02$. Find M_2 . 128.25 g/mol
- 2017 — 10 g CaCl_2 ($M = 111$) in 200 g water, $K_b = 0.52$. Find ΔT_b . 0.702 K (remember $i = 3$)
- 2026 Q8 — 1 molal trichloroacetic acid boils at 100.18°C , $K_b = 0.512$. Find i . 0.35
- 2021 Q15 — 2.56 g sulphur in 100 g CS_2 , $\Delta T_f = 1.62$ K, $K_f = 4.9$, $M(\text{S}_8) = 256$. Find % association. $i = 0.846$; 15.4%

Vapour pressure

- 2026 Q19 — 61 g benzoic acid in 500 g benzene, $p^\circ = 51.2$ mm Hg, complete dimerisation. Find p_s . 50.42 mm Hg (use $i = 0.5$)
- Lecture PYQ — pure water $p^\circ = 32$ mm Hg at 308 K falls to 31.84 with 10 g solute in 200 g water. Find M_2 . 180 g/mol

Osmotic pressure

- 2020 Q3 — 5% urea solution ($M = 60$) at 300 K, $R = 0.0821$. Find π . 20.51 atm
- 2019 Q17 — 6% glucose ($M = 180$) isotonic with 2.5% of an unknown. Find the unknown's M . 75 g/mol

Freezing point, two-stage

- 2021 Q21 — 5% cane sugar ($M = 342$) freezes at 271 K. Find the freezing point of 5% glucose ($M = 180$). 269.2 K — find K_f from the sugar first

Built from Sourabh Raina's Solutions one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 1 (Rationalised 2022–23). Constants verified: water $K_b = 0.52$, $K_f = 1.86$ K kg mol $^{-1}$.

Electrochemistry

NCERT Class XII Chemistry • Chapter 2 • Electrochemistry

14 marks — second heaviest on the blueprint. Two engines run this chapter: the Nernst equation and Faraday's laws. Nearly every numerical is one of those two wearing a different hat.

The two cell types — *the distinction every question hangs off*

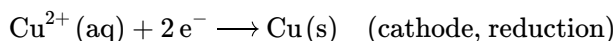
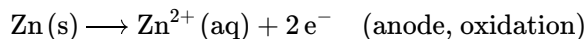
	Electrochemical (galvanic)	Electrolytic
Does	Chemical → electrical	Electrical → chemical
Reaction	Spontaneous	Non-spontaneous
ΔG	Negative	Positive
Anode	Negative	Positive
Cathode	Positive	Negative

Constant across both: **oxidation at the anode, reduction at the cathode**. Only the charge signs flip.

Examiner asks: "two points of difference" (2020) is a standing 2-marker. Give the ΔG and the anode-sign pair — they're the two that can't be waffled.

Daniell cell and cell notation — *how to write a cell down, and why the salt bridge is there*

Zinc rod in 1 M ZnSO_4 , copper rod in 1 M CuSO_4 , joined by a salt bridge and an external wire. EMF = **1.1 V** at unit concentration.



The zinc rod thins; the copper rod thickens.

Cell notation

Anode on the left, cathode on the right. Single line = phase boundary, double line = salt bridge:



What the salt bridge actually does

It's a U-tube of electrolyte (KCl , KNO_3 , NH_4Cl) set in agar or gelatin. Without it, Zn^{2+} builds up in the anode compartment and SO_4^{2-} in the cathode compartment; the charge separation stalls the cell. The bridge's ions migrate in to neutralise both halves and complete the circuit.

Examiner asks: "why is a salt bridge necessary" (2026) — answer with both jobs: maintains electrical neutrality *and* completes the circuit.

Electrode potential and SHE — *why everything is written as reduction potential*

Dip a metal in a solution of its own ions and charge separates until equilibrium. That potential difference between metal and solution is the **electrode potential**.

By IUPAC convention **every electrode potential is quoted as a reduction potential** — because you can't subtract an oxidation potential from a reduction potential meaningfully. Oxidation and reduction potentials of the same electrode are numerically equal and opposite in sign.

Standard electrode potential (E°): ion concentration 1 M, 298 K, and 1 bar for any gas.

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ$$

Also written $E_{\text{right}} - E_{\text{left}}$, since cathode sits on the right in cell notation. E_{cell} for a working cell is always positive.

Standard hydrogen electrode

A single half-cell's absolute potential can't be measured — you need a second electrode to get a reading at all. So SHE is defined as the reference with $E^\circ = 0.00$ V exactly. It's a platinum foil coated with platinum black, H_2 at 1 bar, H^+ at 1 M.

Connect zinc to SHE: electrons flow zinc $\rightarrow$ SHE, so zinc is the anode, reading 0.76 V. Then $0.76 = 0 - E_{\text{Zn}^{2+}/\text{Zn}}^\circ$, giving $E_{\text{Zn}^{2+}/\text{Zn}}^\circ = -0.76$ V. Copper against SHE reads 0.34 V with electrons flowing the other way, giving +0.34 V.

Electrochemical series — reading a table of E° values

Arrange standard electrode potentials in order and the series tells you three things at a glance.

- **High E°** $\rightarrow$ reduces easily $\rightarrow$ **strong oxidising agent**. Fluorine, +2.87 V, is the strongest.
- **Low (negative) E°** $\rightarrow$ oxidises easily $\rightarrow$ **strong reducing agent**. Lithium, -3.05 V, is the strongest.
- **Feasibility:** the species actually being reduced must have the higher E° . If a proposed reaction has it backwards, the reaction isn't feasible.

Worked • 2023 — E° of a cell

$E_{\text{Ag}^+/\text{Ag}}^\circ = 0.80$ V, $E_{\text{Fe}^{2+}/\text{Fe}}^\circ = -0.44$ V. Find E_{cell}° .

Higher E° becomes the cathode, so silver is cathode, iron is anode.

$$E_{\text{cell}}^\circ = 0.80 - (-0.44) = 1.24 \text{ V}$$

Examiner asks: "which of A and B liberates H_2 from dilute H_2SO_4 " (2026) — the one with the more negative E° , because it oxidises more readily.

Nernst equation — the workhorse — for when concentrations aren't 1 M

Standard potentials assume 1 M. Change the concentration and the potential changes. Nernst gives the relation:

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{2.303 RT}{nF} \log Q$$

At 298 K, with $R = 8.314$ and $F = 96500$, that collapses to the form you'll actually use:

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.0591}{n} \log \frac{[\text{products}]}{[\text{reactants}]}$$

n = electrons exchanged. **Solids and pure liquids are omitted** — their concentration is constant, taken as unity.

Trap: n is the electrons exchanged *after balancing*, not the charge on one ion. And stoichiometric coefficients become powers inside the log. Both are where marks vanish.

Worked · 2019 — EMF with unequal concentrations

$\text{Al} \mid \text{Al}^{3+} (0.001 \text{ M}) \parallel \text{Ni}^{2+} (0.1 \text{ M}) \mid \text{Ni}$. $E_{\text{Ni}^{2+}/\text{Ni}}^{\circ} = -0.25 \text{ V}$, $E_{\text{Al}^{3+}/\text{Al}}^{\circ} = -1.66 \text{ V}$.

Balance: Al loses 3e^{-} , Ni^{2+} gains $2\text{e}^{-} \rightarrow$ multiply by 2 and 3 $\rightarrow n = 6$.

$$E_{\text{cell}}^{\circ} = -0.25 - (-1.66) = 1.41 \text{ V}$$

$$E_{\text{cell}} = 1.41 - \frac{0.0591}{6} \log \frac{(10^{-3})^2}{(10^{-1})^3} = 1.41 - \frac{0.0591}{6} \log(10^{-3})$$

$$= 1.41 + \frac{0.0591 \times 3}{6} = 1.41 + 0.0295 = 1.44 \text{ V}$$

Nernst at equilibrium $\rightarrow$ equilibrium constant

At equilibrium $E_{\text{cell}} = 0$ and $Q = K_c$, so:

$$E_{\text{cell}}^{\circ} = \frac{0.0591}{n} \log K_c$$

Examiner asks: "why is K_c related to E_{cell}° and not E_{cell} ?" (2026) — because at equilibrium E_{cell} is zero and generates no current, while E° stays constant.

Gibbs energy and cell EMF — one equation, two exam uses

$$\Delta_r G = -nFE_{\text{cell}} \quad \Delta_r G^{\circ} = -nFE_{\text{cell}}^{\circ}$$

The electrical work a cell does equals the fall in Gibbs energy.

The intensive/extensive point the examiner likes: E_{cell} is **intensive** — multiply the equation by 2 and it doesn't change. ΔG is **extensive** — it depends on n , so doubling the equation doubles it.

Worked · 2025 — ΔG° and $\log K_c$

$2 \text{ Cr} + 3 \text{ Cd}^{2+} \rightarrow 2 \text{ Cr}^{3+} + 3 \text{ Cd}$. $E_{\text{Cd}^{2+}/\text{Cd}}^{\circ} = -0.40 \text{ V}$, $E_{\text{Cr}^{3+}/\text{Cr}}^{\circ} = -0.74 \text{ V}$, $F = 96500$.

Cd is reduced $\rightarrow$ cathode. $E_{\text{cell}}^{\circ} = -0.40 - (-0.74) = 0.34 \text{ V}$. Cr loses $3\text{e}^{-} \times 2 = 6$, so $n = 6$.

$$\Delta_r G^{\circ} = -6 \times 96500 \times 0.34 = -196860 \text{ J mol}^{-1} = -196.86 \text{ kJ mol}^{-1}$$

Conductance, conductivity, molar conductivity — *four quantities that are easy to confuse*

Quantity	Symbol	Relation	Unit
Resistance	R	$R = \rho \frac{l}{A}$	Ω
Resistivity	ρ	$\rho = R \frac{A}{l}$	$\Omega \text{ cm}$
Conductance	G	$G = 1/R$	S (siemens)
Conductivity	κ	$\kappa = 1/\rho$	S cm^{-1}
Molar conductivity	Λ_m	$\Lambda_m = \frac{\kappa \times 1000}{M}$	$\text{S cm}^2 \text{ mol}^{-1}$

Cell constant

l/A is fixed for a given conductivity cell, so it's called the **cell constant** G^* :

$$G^* = \frac{l}{A} = \kappa \times R$$

Measuring l and A directly is unreliable, so G^* is found by filling the cell with a solution of known conductivity (usually KCl) and measuring resistance. Once known, the same cell gives κ for any solution: $\kappa = G^*/R$.

Resistance is measured on a Wheatstone bridge with an AC source (DC would electrolyse the solution and change its composition) and a purpose-built conductivity cell (a solution can't be wired into the bridge directly).

Worked · 2024 — resistivity, conductivity, molar conductivity

0.05 M NaOH, cell constant 50 cm^{-1} , resistance $4.5 \times 10^3 \Omega$.

$$\kappa = \frac{G^*}{R} = \frac{50}{4.5 \times 10^3} = 0.011 \text{ S cm}^{-1}$$

$$\rho = \frac{1}{\kappa} = 90 \Omega \text{ cm}$$

$$\Lambda_m = \frac{0.011 \times 1000}{0.05} = 220 \text{ S cm}^2 \text{ mol}^{-1}$$

Variation with dilution — the two go opposite ways

- **Conductivity falls** on dilution: fewer ions per cm^3 .
- **Molar conductivity rises** on dilution: the volume holding one mole of electrolyte increases, and interionic attraction weakens.

Trap: these two moving in opposite directions is itself an exam question. Conductivity is per unit *volume*; molar conductivity is per *mole*.

Strong vs weak electrolyte on the graph

Plot Λ_m against $\sqrt{c}$:

- **Strong electrolyte** — starts high, rises gently and **linearly**. Extrapolate to zero concentration to read Λ_m° straight off the graph.
- **Weak electrolyte** — starts low, then shoots up steeply near zero concentration, running almost **parallel to the y-axis**. Cannot be extrapolated; you must use Kohlrausch's law instead.

Kohlrausch's law — *and its three applications*

Kohlrausch law of independent migration of ions: the limiting molar conductivity of an electrolyte is the sum of the individual contributions of its cation and anion.

$$\Lambda_m^\circ(\text{NaCl}) = \lambda_{\text{Na}^+}^\circ + \lambda_{\text{Cl}^-}^\circ$$

$$\Lambda_m^\circ(\text{BaCl}_2) = \lambda_{\text{Ba}^{2+}}^\circ + 2\lambda_{\text{Cl}^-}^\circ$$

Trap: the stoichiometric multiplier. $\text{Al}_2(\text{SO}_4)_3$ needs $2\lambda_{\text{Al}^{3+}}^\circ + 3\lambda_{\text{SO}_4^{2-}}^\circ$. Forgetting the 2 and 3 is the whole mistake.

1 · Λ° of a weak electrolyte

Can't be read off a graph, so build it from strong electrolytes that share ions:

$$\Lambda^\circ(\text{CH}_3\text{COOH}) = \Lambda^\circ(\text{CH}_3\text{COONa}) + \Lambda^\circ(\text{HCl}) - \Lambda^\circ(\text{NaCl})$$

The Na^+ and Cl^- terms cancel, leaving exactly $\lambda_{\text{CH}_3\text{COO}^-}^\circ + \lambda_{\text{H}^+}^\circ$.

2 · Degree of dissociation

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

3 · Dissociation constant

$$K_a = \frac{c \alpha^2}{1 - \alpha}$$

Worked · 2020 — limiting ionic conductivity from Kohlrausch

$\Lambda^\circ(\text{Al}_2(\text{SO}_4)_3) = 858$, $\lambda_{\text{SO}_4^{2-}}^\circ = 160 \text{ S cm}^2 \text{ mol}^{-1}$. Find $\lambda_{\text{Al}^{3+}}^\circ$.

$$858 = 2\lambda_{\text{Al}^{3+}}^\circ + 3(160) \Rightarrow 2\lambda_{\text{Al}^{3+}}^\circ = 858 - 480 = 378$$

$$\lambda_{\text{Al}^{3+}}^\circ = 189 \text{ S cm}^2 \text{ mol}^{-1}$$

Faraday's laws of electrolysis — *the second engine of this chapter*

First law

Mass deposited is proportional to the charge passed:

$$w = \frac{M I t}{n F}$$

M = molar mass, I = current in amperes, t = time in **seconds**, n = electrons gained per ion, $F = 96500 \text{ C mol}^{-1}$.

Second law

Same charge through different electrolytes deposits masses in the ratio of their **equivalent weights**:

$$\frac{w_1}{w_2} = \frac{E_1}{E_2}, \quad E = \frac{\text{atomic mass}}{n}$$

So Na (23/1), Mg (24/2 = 12), Al (27/3 = 9) deposit in the ratio 23 : 12 : 9 per faraday.

Trap: time in minutes. Convert to seconds first — this single slip has cost more marks in this chapter than any conceptual error.

Worked · 2017 — mass deposited

2 A through AgNO_3 for 15 min. $M(\text{Ag}) = 108$.

$t = 15 \times 60 = 900 \text{ s}$. $\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}$, so $n = 1$.

$$w = \frac{108 \times 2 \times 900}{1 \times 96500} = 2.014 \text{ g}$$

Charge in faradays — without the formula

Often faster to reason directly. To reduce 1 mol Zn^{2+} you need 2 mol of electrons = **2 F** = 193000 C. To reduce 1 mol MnO_4^- to Mn^{2+} , Mn goes +7 $\rightarrow$ +2, so 5 mol electrons = **5 F**.

Products of electrolysis — the E° comparison, and the overpotential exception

In aqueous solution, water competes with the dissolved ions at both electrodes. Two rules decide the winner:

- **At the cathode:** the species with the **higher** E° is reduced.
- **At the anode:** the species with the **lower** E° is oxidised.

Aqueous NaCl — the standard case

Cathode: Na^+ (−2.71 V) vs H^+ (0.00 V). Hydrogen wins $\rightarrow \text{H}_2$ gas.

Anode: Cl^- (1.36 V) vs water $\rightarrow \text{O}_2$ (1.23 V). By the rule oxygen should win — **but it doesn't**. Oxygen evolution is kinetically slow and needs extra voltage (**overpotential**) to proceed at a useful rate, so chlorine is released instead.

Net: $\text{NaCl}(\text{aq}) + \text{H}_2\text{O} \longrightarrow \text{NaOH} + \frac{1}{2} \text{H}_2 + \frac{1}{2} \text{Cl}_2$

Trap: the overpotential exception is the entire point of this question. Answering "oxygen, because $1.23 < 1.36$ " is the trap being set. Name overpotential explicitly.

Molten vs aqueous

Molten NaCl has no water, so there's no competition: sodium at the cathode, chlorine at the anode.

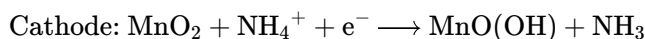
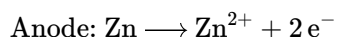
Sulphuric acid

Dilute $\text{H}_2\text{SO}_4 \rightarrow$ oxygen at the anode. Concentrated $\rightarrow$ peroxodisulphate, $2\text{SO}_4^{2-} \rightarrow \text{S}_2\text{O}_8^{2-} + 2\text{e}^-$ ($E^\circ = 1.96\text{ V}$).

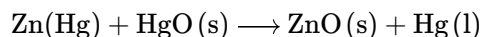
Batteries and fuel cells — *four devices, and the one fact asked about each*

Primary — cannot be recharged

Dry cell (Leclanché): zinc container = anode, graphite rod surrounded by MnO_2 and carbon = cathode, moist $\text{NH}_4\text{Cl}/\text{ZnCl}_2$ paste = electrolyte. Used in transistors and clocks.

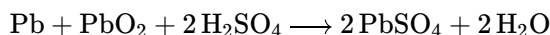


Mercury cell: used in hearing aids and watches. Its selling point — and the exam answer — is that **the potential stays constant through its life**, because no ions appear in the overall reaction, so no concentration changes:



Secondary — rechargeable

Lead storage battery (automobiles, inverters): Pb anode, PbO_2 on a lead grid as cathode, 38% H_2SO_4 electrolyte.

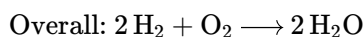
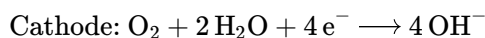


On **charging** the whole thing runs backwards: $2\text{PbSO}_4 + 2\text{H}_2\text{O} \longrightarrow \text{Pb} + \text{PbO}_2 + 2\text{H}_2\text{SO}_4$.

Nickel–cadmium: Cd anode, NiO_2 cathode, KOH electrolyte. Longer life than lead storage but more expensive to make.

Fuel cell

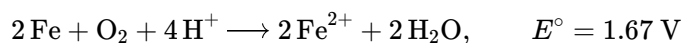
Converts a fuel's chemical energy *directly* to electricity, with reactants fed in continuously. The $\text{H}_2\text{--O}_2$ cell powered the **Apollo programme**, and its product water was condensed into the astronauts' drinking supply.



Two advantages (the standard 2-marker): ~70% efficiency against ~40% for a thermal plant, and it's pollution-free.

Corrosion — the electrochemical explanation, and two preventions

Corrosion is chemical attack by atmospheric gases and moisture on a metal surface, giving oxides, sulphides and carbonates. Rusting is the electrochemical case:



Fe^{2+} is then oxidised further by atmospheric oxygen to Fe^{3+} , and rust is **hydrated ferric oxide**, $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.

Prevention

- **Barrier protection** — paint, bisphenol, or a coat of another metal (Sn, Zn) between iron and the air.
- **Sacrificial protection** — attach a *more reactive* metal (Mg, Zn). It has the more negative E° , so it oxidises in preference to the iron and corrodes away while the iron survives.

Examiner asks: "why are magnesium blocks fixed to iron pipelines" — sacrificial anode; magnesium is more reactive, more negative E° , oxidises preferentially.

Numerical patterns, collected — five patterns, one model each

A · Nernst / EMF — 3 marks

Recognise it: concentrations given that aren't 1 M.

1. Write both half-reactions; balance electrons to get n .
2. $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$.
3. Apply $E = E^\circ - \frac{0.0591}{n} \log Q$, coefficients as powers, solids omitted.

B · ΔG° and K_c — 3 marks

Recognise it: F is given, or the question says "at equilibrium".

1. Get E°_{cell} and n .
2. $\Delta_r G^\circ = -nFE^\circ$; divide by 1000 for kJ.
3. For K_c : $E^\circ = \frac{0.0591}{n} \log K_c$.

C · Conductivity chain — 3–5 marks

Recognise it: a resistance and either a cell constant or a known- κ reference solution.

1. Cell constant: $G^* = \kappa \times R$ from the reference solution, or read it off directly.
2. $\kappa = G^*/R$ for the unknown.
3. $\Lambda_m = \kappa \times 1000/M$.
4. If asked for α : divide by Λ_m° from Kohlrausch.

D · Faraday first law — 3 marks

Recognise it: current in amperes and a time.

1. Time $\rightarrow$ seconds.
2. Find n from the ion's charge.
3. $w = MIt/nF$, rearranged for whatever's missing.

E · Faraday second law (cells in series) — 5 marks

Recognise it: two electrolytic cells "connected in series".

1. Series means identical charge through both.
2. Use the first law on the cell you have full data for, to get time.
3. Then $w_1/w_2 = E_1/E_2$ with $E = \text{atomic mass}/n$.

Worked · 2020 — two cells in series

2 A through ZnSO_4 (cell A) and CuSO_4 (cell B) in series; 2 g Cu deposited. $M(\text{Cu}) = 63.5$, $M(\text{Zn}) = 65$.

Time: $2 = \frac{63.5 \times 2 \times t}{2 \times 96500} \Rightarrow t = \frac{2 \times 96500}{63.5} = 3039 \text{ s}$

Zinc: both gain 2 electrons, so

$$\frac{w_{\text{Zn}}}{2} = \frac{65/2}{63.5/2} \Rightarrow w_{\text{Zn}} = \frac{65 \times 2}{63.5} = 2.04 \text{ g}$$

Past year questions · question types — ranked by how often they turn up

From the last 10 years of boards — 25 questions, grouped by pattern.

1 · Nernst / EMF calculation — 3 marks

Recognise it: a cell notation with concentrations attached.

1. Balance for n .
2. E°_{cell} from the two E° values.
3. Nernst, watching the powers in Q .

Trap: sign error when log of a negative power comes down. $-\frac{0.0591}{n} \log 10^{-3}$ becomes *plus*.

2 · Conductivity / molar conductivity chain — 3–5 marks

Recognise it: a resistance value in ohms.

1. Cell constant.
2. κ , then ρ if asked.
3. Λ_m , then α if asked.

Trap: forgetting the $\times 1000$ in Λ_m , or dividing by the wrong concentration.

3 · Faraday's laws numerical — 3–5 marks

Recognise it: amperes, minutes, or "how many faradays".

1. Seconds.
2. n from the ion.
3. $w = MIt/nF$, or reason directly in moles of electrons.

4 · Electrolysis products with reasoning — 2–3 marks

Recognise it: "predict the products", with E° values supplied.

1. Cathode: higher E° reduced.
2. Anode: lower E° oxidised.
3. Check for the overpotential exception before answering.

5 · Cell/battery definitions and comparisons — 2 marks

Recognise it: "define", "name the cell used in...", "two differences".

1. Name the device.
2. Give both electrode reactions if asked.
3. For differences, answer in matched pairs.

6 · Kohlrausch application — 2–3 marks

Recognise it: limiting molar conductivities of ions given.

1. Write the electrolyte as its ions with correct multipliers.
2. Sum or subtract as needed.
3. For weak electrolytes, build from three strong ones.

Past year questions · mark slots — *what each type is worth*

Question type	Slot	Time
Nernst / EMF	3-marker	5 min
Conductivity chain	3-marker, or part of a 5	5–7 min
Faraday's laws	3-marker; series cells is a 5	4–7 min
Electrolysis products	2–3 marker	3 min
Definitions / cell names	2-marker	2 min
Kohlrausch	2–3 marker	3 min
ΔG° / K_c	3-marker	4 min

The video splits these as short-answer type I (2 marks), type II (3 marks), and long answer (5 marks) — the conductivity chain and series-cell problems are the ones that appear as 5-markers.

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2019 Q6 · 2019 Q8 · lecture PYQ

Electrolysis of aqueous NaCl, and the overpotential reason chlorine appears instead of oxygen. Runs almost every year in some form.

2020 Q16 · lecture PYQ

Two electrolytic cells in series, ZnSO_4 and CuSO_4 , 2 g copper deposited — identical numbers both times.

2024 Q18 · 2026 Q22 · 2026 Q24

The conductivity → resistivity → molar conductivity chain, sometimes with degree of dissociation added. The single most reliable numerical in the chapter.

2025 Q1 · 2026 Q7(a)

Fuel cells: definition and two advantages, then the Apollo hydrogen–oxygen cell and its drinking-water by-product.

2022 Q13 · 2020 Q21 · 2026 Q22

Kohlrausch's law statement, then applied — degree of dissociation of acetic acid, limiting ionic conductivity of Al^{3+} , Λ° of NaCl.

2020 Q3 · 2026 Q23(b)

Electrochemical vs electrolytic cell differences, and primary vs secondary battery differences. Straight recall, always available.

2017 Q5 · 2020 Q21(a)

Which cell is used where — dry cell in transistors, mercury cell in hearing aids and watches, with electrode reactions.

Past year questions · cold practice — *answers only* — *work them before you look*

Nernst / EMF

- 2020 Q2 — write cell notation and the Nernst equation for $\text{Mg} + 2 \text{Ag}^+ \longrightarrow \text{Mg}^{2+} + 2 \text{Ag}$.
 $\text{Mg}|\text{Mg}^{2+}||\text{Ag}^+|\text{Ag}; E = E^\circ - \frac{0.0591}{2} \log \frac{[\text{Mg}^{2+}]}{[\text{Ag}^+]^2}$
- 2026 Q19 — EMF of $\text{Sn}|\text{Sn}^{2+}||\text{H}^+|\text{H}_2$ at 298 K with concentrations given. Apply Nernst with $n = 2$; H_2 omitted as a gas at 1 bar

ΔG° and K_c

- 2026 Q23 — E° for $\text{Cu} + 2 \text{Ag}^+ \rightleftharpoons \text{Cu}^{2+} + 2 \text{Ag}$ at equilibrium, $K_c = 10^{15}$. $E^\circ = \frac{0.0591}{2} \times 15 = 0.44 \text{ V}$

Conductivity

- Lecture PYQ — 0.05 M KCl, $l = 50 \text{ cm}$, $A = 0.625 \text{ cm}^2$, $R = 5 \times 10^3 \Omega$. Find ρ , κ , Λ_m . $62.5 \Omega \text{ cm}$; 0.016 S cm^{-1} ; $320 \text{ S cm}^2 \text{ mol}^{-1}$
- Lecture PYQ — cell gives 164Ω with 0.02 M KCl ($\kappa = 2.768 \times 10^{-3}$), then 78.5Ω with 0.05 M AgNO_3 . Find κ and Λ_m of AgNO_3 . $G^* = 0.4539$; $\kappa = 5.78 \times 10^{-3}$; $\Lambda_m = 115.6 \text{ S cm}^2 \text{ mol}^{-1}$
- 2026 Q22 — 0.1 M NaCl, $\kappa = 1.06 \times 10^{-2} \text{ S cm}^{-1}$, $\lambda^\circ \text{Na}^+ = 50.1$, $\text{Cl}^- = 76.5$. Find Λ_m and α . $106 \text{ S cm}^2 \text{ mol}^{-1}$; $\Lambda^\circ = 126.6$; $\alpha = 0.837$
- 2022 Q13(b) — α of acetic acid if $\Lambda_m = 48$ and $\Lambda_m^\circ = 400$. 0.12

Kohlrausch

- 2021–22 — Λ° of MgCl_2 given $\lambda^\circ \text{Mg}^{2+} = 106$, $\text{Cl}^- = 76.3$. $285.6 \text{ S cm}^2 \text{ mol}^{-1}$ (remember the $2 \times$)

Faraday's laws

- 2014 — 5 A through $\text{Ni}(\text{NO}_3)_2$ for 20 min, $M = 58.7$. Mass of Ni? 1.825 g
- 2015 — charge to reduce 1 mol Zn^{2+} to Zn. $2 \text{ F} = 193000 \text{ C}$
- 2026 Q25(b) — faradays to produce 40 g Al from molten Al_2O_3 , $M = 27$. 4.44 F

Built from Sourabh Raina's Electrochemistry one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 2 (Rationalised 2022–23). Verified against NCERT: "Kohlrausch law of independent migration of ions"; Faraday constant 96487 C mol^{-1} (taken as 96500 in working); Daniell cell 1.1 V at unit concentration.

Chemical Kinetics

NCERT Class XII Chemistry • Chapter 3 • Chemical Kinetics

13 marks. Almost pure numerical territory — order from data, integrated rate equations, half-life, activation energy. Four formulas do nearly all the work; the marks go to knowing which one the question is pointing at.

What kinetics answers — *and what it deliberately doesn't*

Three different questions about a reaction, three different branches:

Question	Answered by
Is it feasible? ($\Delta G < 0$)	Thermodynamics
How far does it go?	Chemical equilibrium (K)
How fast, and by what mechanism?	Chemical kinetics

The standard illustration: thermodynamics says diamond $\rightarrow$ graphite is feasible. Kinetics explains why your ring is safe — the conversion is immeasurably slow.

Rate of reaction — *average, instantaneous, and the coefficient division*

Rate = change in concentration per unit time. Reactant concentration falls, so its rate carries a minus sign to keep the rate positive:

$$\text{rate} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$$

Average rate uses Δ over a finite interval. **Instantaneous rate** is the slope of the tangent at one moment:

$$-\frac{d[R]}{dt}.$$

When coefficients differ — *divide by them*

For $aA + bB \rightarrow cC + dD$:

$$\text{rate} = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$$

Trap: "rate of reaction" and "rate of disappearance of B" are different numbers whenever B's coefficient isn't 1. Read which one is asked — this is the entire content of a 2-marker that runs most years.

Worked • 2022 — rate of reaction vs rate of disappearance

$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, rate of formation of $\text{NH}_3 = 3.6 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$. Find the rate of reaction and the rate of disappearance of H_2 .

$$\text{rate} = \frac{1}{2} \frac{\Delta[\text{NH}_3]}{\Delta t} = \frac{1}{2}(3.6 \times 10^{-4}) = 1.8 \times 10^{-4}$$

$$-\frac{\Delta[\text{H}_2]}{\Delta t} = 3 \times \text{rate} = 5.4 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

Examiner asks: this exact shape, with $\text{N}_2/\text{H}_2/\text{NH}_3$ or $A + 3B \rightarrow 2C$. Near-guaranteed 2–3 marker.

Rate law and finding the order from data — *the standard 3-marker, step by step*

The rate law is **experimental** — powers are not read off the balanced equation:

$$\text{rate} = k[A]^x[B]^y$$

x and y may or may not equal the stoichiometric coefficients. **Order** = $x + y$. Order can be zero, fractional, or negative; molecularity cannot.

Getting x and y from a data table — *the method*

1. Pick two experiments where **one** concentration is held constant and the other changes.
2. Write the rate law for each; divide one by the other. k and the constant term cancel.
3. Read off the power that makes the ratio work.
4. Repeat with a different pair for the other exponent.
5. Substitute any single experiment back to find k .

Trap: choosing two experiments where *both* concentrations change. Nothing cancels and the algebra becomes unsolvable. Scan the table for the constant column first.

Order vs molecularity — *the comparison they ask for*

Order	Molecularity
Sum of powers in the experimental rate law	Number of species colliding in one elementary step
Experimental	Theoretical
Can be 0, fractional, negative	Whole number only (1, 2, 3)
Applies to overall reaction	Only defined for an elementary step

Examiner asks: "can this be an elementary reaction?" (2026) with a fractional order like $3/2$ — **no**, because an elementary reaction's order equals its molecularity, which must be a whole number.

Units of the rate constant — *one formula, every order*

From $\text{rate} = k[\]^n$, so $k = \text{rate}/[\]^n$:

$$\text{unit of } k = \text{mol}^{1-n} \text{ L}^{n-1} \text{ s}^{-1}$$

Order	Unit of k
Zero	$\text{mol L}^{-1} \text{ s}^{-1}$
First	s^{-1}
Second	$\text{L mol}^{-1} \text{ s}^{-1}$

Examiner asks: given a rate law, state the units — or, run backwards, given the units, state the order. Both directions appear.

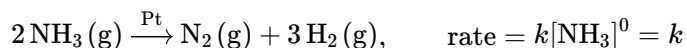
Integrated rate equations — zero and first order, plus their graphs

Zero order

$$[R] = -kt + [R]_0 \quad k = \frac{[R]_0 - [R]}{t}$$

Plot $[R]$ against t : straight line, **slope** = $-k$, intercept $[R]_0$.

Example: decomposition of gaseous ammonia on a hot platinum surface at 1130 K. At high pressure the metal surface saturates with ammonia, so adding more changes nothing — rate becomes independent of concentration:



First order

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

Plot $\log \frac{[R]_0}{[R]}$ against t : straight line through the origin, **slope** = $k/2.303$. Plot $\ln[R]$ against t instead and the slope is $-k$, intercept $\ln[R]_0$.

Examples: hydrogenation of ethene; all natural and artificial radioactive decay; decomposition of N_2O_5 and N_2O .

First order in the gas phase — the closed form worth memorising

Concentration is proportional to partial pressure, so the same equation runs on pressures. For $A(\text{g}) \rightarrow B(\text{g}) + C(\text{g})$ with all coefficients 1, let x be the pressure decomposed:

$$p_{\text{total}} = (p_i - x) + x + x = p_i + x \Rightarrow x = p_T - p_i$$

So the reactant's partial pressure is $p_A = p_i - x = 2p_i - p_T$, and:

$$k = \frac{2.303}{t} \log \frac{p_i}{2p_i - p_T}$$

Use this directly whenever the question gives you **initial and total** pressure. If it gives the reactant's partial pressure outright, the plain $\log(p_i/p_A)$ form is enough.

Worked · 2016 — rate constant from total pressure

$\text{C}_2\text{H}_5\text{Cl}(\text{g}) \rightarrow \text{C}_2\text{H}_4(\text{g}) + \text{HCl}(\text{g})$. At $t = 0$, $p_i = 0.30$ atm; at $t = 300$ s, $p_T = 0.50$ atm. ($\log 3 = 0.4771$)

$$\begin{aligned} k &= \frac{2.303}{300} \log \frac{0.30}{2(0.30) - 0.50} = \frac{2.303}{300} \log \frac{0.30}{0.10} \\ &= \frac{2.303 \times 0.4771}{300} = 3.66 \times 10^{-3} \text{ s}^{-1} \end{aligned}$$

The unit is s^{-1} , which itself confirms first order.

Trap: feeding the *total* pressure straight into the log. It must be $2p_i - p_T$, not p_T .

Half-life — *and the graph question that turns on it*

Time for half the reactant to be consumed.

$$\text{First order: } t_{1/2} = \frac{0.693}{k} \quad \text{Zero order: } t_{1/2} = \frac{[R]_0}{2k}$$

The distinction that gets examined: first-order half-life is **independent of initial concentration**; zero-order half-life is **directly proportional** to it.

Examiner asks: "predict the order from the graph" (2019) — a flat $t_{1/2}$ vs $[R]_0$ line means first order; a straight rising line means zero order.

Percentage-completion problems

Set $[R]_0 = 100$ and subtract the percentage completed. 75% done $\rightarrow [R] = 25$; 99% done $\rightarrow [R] = 1$.

Worked · 2026 — time for 3/4 decomposition

First order, $k = 2.54 \times 10^{-3} \text{ s}^{-1}$. Time for 3/4 of the reactant to decompose? ($\log 4 = 0.6$)

$[R]_0 = a$, so $[R] = a - \frac{3}{4}a = \frac{a}{4}$.

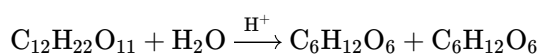
$$t = \frac{2.303}{k} \log \frac{a}{a/4} = \frac{2.303 \times 0.6}{2.54 \times 10^{-3}} = 544 \text{ s}$$

A neat consequence worth remembering: for first order, $t_{99\%} = 2 \times t_{90\%}$ — because $\log 100 = 2$ and $\log 10 = 1$.

Pseudo first order — *two examples, one idea*

A reaction that is **bimolecular but follows first-order kinetics**, because one reactant is present in such large excess that its concentration doesn't measurably change.

Hydrolysis of sucrose:



Water is the solvent, so $[\text{H}_2\text{O}]$ is effectively constant. Rate = $k[\text{C}_{12}\text{H}_{22}\text{O}_{11}]$: order 1, molecularity 2.

Hydrolysis of an ester in dilute acid behaves the same way.

Examiner asks: (2024) write the rate law, then state order *and* molecularity separately, then name the reaction type. The gap between order 1 and molecularity 2 is the whole point.

Temperature dependence and Arrhenius — *the 5-marker's favourite*

Rate roughly **doubles for every 10 K rise** near room temperature. Quantitatively:

$$k = A e^{-E_a/RT}$$

A = frequency factor, E_a = activation energy, the minimum energy colliding molecules need.

The logarithmic form and its graph

$$\ln k = -\frac{E_a}{RT} + \ln A$$

Against $y = mx + c$ with $y = \ln k$ and $x = 1/T$:

- **Slope** = $-E_a/R$ (negative, so the line falls)
- **Intercept** = $\ln A$

Two-temperature form — the one used in numericals

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Worked · 2025 — E_a when the rate doubles over 10 K

Rate doubles from 298 K to 308 K. $2.303R = 19.15$, $\log 2 = 0.3$.

Concentrations are unchanged, so a doubled rate means a doubled k : $k_2/k_1 = 2$.

$$0.3 = \frac{E_a}{19.15} \left[\frac{10}{308 \times 298} \right]$$

$$E_a = \frac{0.3 \times 19.15 \times 308 \times 298}{10} = 52729.9 \text{ J mol}^{-1} = 52.7 \text{ kJ mol}^{-1}$$

Worked · 2025 — E_a from two half-lives

First order, 50% complete in 20 min at 300 K and in 5 min at 350 K. ($\log 4 = 0.602$)

50% completion is the half-life, so $k = 0.693/t_{1/2}$:

$$\frac{k_2}{k_1} = \frac{0.693/5}{0.693/20} = 4$$

$$0.602 = \frac{E_a}{2.303 \times 8.314} \left[\frac{50}{300 \times 350} \right] \Rightarrow E_a = 24205 \text{ J mol}^{-1} = 24.2 \text{ kJ mol}^{-1}$$

Trap: "50% completed in 20 minutes" is a half-life in disguise. Spot that and the question collapses into two lines.

Collision theory and catalysis — the definitions that carry marks

Activated complex and the energy profile

Colliding reactant molecules first form a short-lived **activated complex**. The energy needed to reach it is the **activation energy**. The reaction's own energy change is separate:

$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$

On a potential-energy vs reaction-coordinate plot, the barrier height is E_a either way; what changes is where the products land. Products lower than reactants $\rightarrow \Delta H$ negative $\rightarrow$ **exothermic**. Products higher $\rightarrow \Delta H$ positive $\rightarrow$ **endothermic**.

Maxwell–Boltzmann distribution

Plot the fraction of molecules (n_E/n_T) against kinetic energy. Most molecules sit near the **most probable kinetic energy**; only a small tail — roughly 10–20% — carries energy above E_a , and only those can react.

Raise the temperature by 10 K and the curve flattens and shifts right: the fraction with energy at or above E_a roughly **doubles**. That is precisely why the rate doubles for a 10 K rise.

Collision theory

Treats molecules as hard spheres that must collide to react.

Collision frequency (Z): number of collisions per second per unit volume of reaction mixture.

$$\text{rate} = Z_{AB} e^{-E_a/RT}$$

Comparing this with Arrhenius shows the frequency factor A is essentially the **collision frequency**. The theory is accurate for atoms and simple molecules but deviates for complex ones — because not every collision works. Two conditions must both hold:

1. Colliding molecules have energy at least equal to the **threshold energy**.
2. They collide in the **proper orientation**, so old bonds can break and new ones form.

Catalysis

A catalyst increases the rate **without itself undergoing any permanent chemical change**. By **intermediate complex theory**, it forms a temporary bond with the reactant, making a transitory intermediate that decomposes into product and releases the catalyst again — which is why a small amount catalyses a large amount of reactant.

The effect is an **alternative pathway of lower activation energy**, so more molecules clear the barrier and the rate rises.

What a catalyst does not change: the Gibbs energy of the reaction, the equilibrium constant, or the position of equilibrium. It catalyses forward and backward reactions *to the same extent*, so equilibrium is reached **earlier** but at the same place. And since it can't change ΔG , it cannot make a non-spontaneous reaction happen — only speed up one that already is.

Examiner asks: (2017) "effect of a catalyst on activation energy and on ΔG " — lowers E_a , leaves ΔG untouched. Answering only the first half loses the mark.

Reaction mechanism and the rate-determining step

For a multi-step reaction, the **slowest step determines the rate** — write the rate law directly from that step's reactants.

Example (2025): $2 \text{H}_2\text{O}_2 \xrightarrow{\text{I}^-} 2 \text{H}_2\text{O} + \text{O}_2$ via a slow step $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$ and a fast step. So rate = $k[\text{H}_2\text{O}_2][\text{I}^-]$, overall order 2 — even though the balanced equation shows $2 \text{H}_2\text{O}_2$.

Numerical patterns, collected — five patterns, one model each

A · Rate from stoichiometry — 2 marks

Recognise it: one species' rate is given, another is asked for.

1. Write the full rate expression with 1/coefficient on every term.
2. Equate the known term to the wanted one.
3. Check whether "rate of reaction" or "rate of disappearance" is asked.

B · Order from a data table — 3 marks

Recognise it: a table of concentrations and initial rates.

1. Find two rows with one concentration constant.
2. Divide the two rate laws; solve for the exponent.
3. Repeat for the other exponent, then back-substitute for k .

C · Integrated first-order equation — 3 marks

Recognise it: a percentage decomposed, or concentrations at two times.

1. Set $[R]_0 = 100$ and subtract the percentage.
2. $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$, rearranged for the unknown.
3. Gas phase $\rightarrow$ convert total pressure to reactant partial pressure first.

D · Half-life $\leftrightarrow$ rate constant — 2–3 marks

Recognise it: "half-life", or "50% completed".

1. Identify the order from the rate law or the units of k .
2. First order: $t_{1/2} = 0.693/k$. Zero order: $t_{1/2} = [R]_0/2k$.

Worked · 2026 — half-life from percentage decomposition

First order, 25% decomposed in 25 min. Find $t_{1/2}$. ($\log 4 = 0.6021$, $\log 3 = 0.4771$)

$$k = \frac{2.303}{25} \log \frac{100}{75} = \frac{2.303}{25} (0.6021 - 0.4771) = 0.0115 \text{ min}^{-1}$$

$$t_{1/2} = \frac{0.693}{0.0115} = 60.26 \text{ min}$$

E · Activation energy from two temperatures — 5 marks

Recognise it: two temperatures, or "rate doubles when temperature rises by...".

1. Get k_2/k_1 — from rates directly, or from $0.693/t_{1/2}$ at each temperature.
2. $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$.
3. Answer in J mol^{-1} , then divide by 1000 for kJ.

Past year questions · question types — ranked by how often they turn up

From the last 10 years of boards — 25 questions, grouped by pattern.

1 · First-order integrated rate equation — 3 marks

Recognise it: a percentage decomposed, or a concentration/pressure at two times.

1. $[R]_0 = 100$, subtract the percentage.
2. Apply the 2.303 formula.

3. Convert total pressure to partial pressure if gaseous.

Trap: using the percentage *decomposed* as $[R]$ instead of what remains.

2 · Activation energy (Arrhenius) — 5 marks

Recognise it: two temperatures, or $2.303R = 19.15$ handed to you.

1. Get k_2/k_1 .
2. Two-temperature Arrhenius form.
3. Convert to kJ if the answer is large.

Trap: inverting $T_2 - T_1$ over $T_1 T_2$, or forgetting that the half-life ratio inverts the k ratio.

3 · Order from a data table — 3–5 marks

Recognise it: a table of initial concentrations and rates.

1. Pair rows sharing a constant concentration.
2. Divide; solve the exponent.
3. Back-substitute for k and state its units.

4 · Rate expression from stoichiometry — 2–3 marks

Recognise it: one species' rate given, another asked.

1. Full expression with 1/coefficient terms.
2. Equate and solve.

5 · Definitions and comparisons — 2 marks

Recognise it: "define", "distinguish between", "write the unit of".

1. Order vs molecularity: answer in matched pairs.
2. Half-life, collision frequency, effective collision, activation energy — book definitions.

6 · Graph interpretation — 3 marks

Recognise it: a plot of $\ln k$ vs $1/T$, or $t_{1/2}$ vs $[R]_0$, or $\log \frac{[R]_0}{[R]}$ vs t .

1. Match the plot to its equation.
2. Read slope and intercept off $y = mx + c$.
3. State what each represents in symbols, not words alone.

Past year questions · mark slots — *what each type is worth*

Question type	Slot	Time
First-order integrated equation	3-marker	4 min
Activation energy	5-marker (or part of one)	6–7 min
Order from data table	3-marker, sometimes a 5	5–6 min
Rate from stoichiometry	2–3 marker	3 min
Definitions / comparisons	2-marker	2 min
Graph interpretation	3-marker	4 min
Pseudo first order	3-marker	3 min

The video splits these as short-answer type I (2 marks), type II (3 marks) and long answer (5 marks) — activation energy and the multi-part data questions are where the 5-markers sit.

Past year questions · repeat offenders — *appeared more than once* — *highest probability in the chapter*

2018 Q15 · lecture PYQ

"Second order in A, first order in B" — write the differential rate equation, then how the rate changes when concentrations are tripled or doubled. Identical wording both times.

2025 Q16 · 2025 Q23(a)

Activation energy from two temperatures — once from two half-lives, once from a rate that doubles over 10 K. Same formula, two disguises.

2019 Q5 · 2022 Q17

Graph questions: $\ln k$ vs $1/T$ (intercept and slope), and $\log \frac{[R]_0}{[R]}$ vs t (order and slope). Both reduce to $y = mx + c$.

2022 Q8 · 2026 Q6

Order vs molecularity — as a straight comparison, and as "can this be an elementary reaction" with a fractional order.

2024 Q21 · lecture PYQ

Sucrose hydrolysis: rate law, order 1 but molecularity 2, and the name — pseudo first order.

2019 Q9 · 2019 Q10

Half-life: predicting order from a $t_{1/2}$ vs $[R]_0$ graph, and computing $t_{1/2}$ from a first-order rate law.

2026 Q1 · 2024 Q3

Definitions: collision frequency, effective collisions, half-life period, and the units of k for zero and first order.

Past year questions · cold practice — *answers only* — *work them before you look*

Integrated rate equation

- 2017 Q11 — N_2O_5 decomposition: 1.6×10^{-2} M at $t = 0$, 0.8×10^{-2} M at 300 s. Show first order and find $t_{1/2}$. $k = 2.3 \times 10^{-3} \text{ s}^{-1}$ at both times; $t_{1/2} = 301 \text{ s}$
- 2026 Q22 — $\text{C}_2\text{H}_5\text{Cl}$: $p_i = 0.30$ atm, total $p = 0.50$ atm at 30 s. Find k . 0.0368 s^{-1}
- 2026 Q4 — $k = 2.54 \times 10^{-3} \text{ s}^{-1}$, time for $3/4$ decomposition. 544 s
- 2024 — first order, $k = 1.25 \times 10^{-3} \text{ s}^{-1}$. Time for 5 g to fall to 2.5 g. 554.6 s — note this is just $t_{1/2}$ in disguise
- 2024 — show $t_{99\%} = 2 t_{90\%}$ for first order. $\log 100 = 2$ vs $\log 10 = 1$
- 2021–22 — first order, 75% decomposed in 30 min. Find $t_{1/2}$. $k = 0.046 \text{ min}^{-1}$; $t_{1/2} = 15 \text{ min}$
- Lecture PYQ — zero order, $[R]_0 = 0.1 \text{ M}$ falls to 0.064 M, $k = 4 \times 10^{-3}$. Find t . 9 s

Half-life

- 2019 Q10 — rate = $5.5 \times 10^{-14} [\text{C}_2\text{H}_4]$. Units of k and $t_{1/2}$. s^{-1} ; $1.26 \times 10^{13} \text{ s}$
- 2026 Q24(b) — first order, 25% decomposed in 25 min. Find $t_{1/2}$. $k = 0.0115 \text{ min}^{-1}$; $t_{1/2} = 60.26 \text{ min}$

Activation energy

- 2025 Q16 — 50% complete in 20 min at 300 K, 5 min at 350 K. Find E_a . 24.2 kJ mol^{-1}
- 2025 Q23(a) — rate doubles from 298 K to 308 K. Find E_a . 52.7 kJ mol^{-1}

Rate and order

- 2020 — $A + 3B \rightarrow 2C$, rate of formation of C = 2.5×10^{-4} . Find rate of reaction and rate of disappearance of B. 1.25×10^{-4} ; $3.75 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
- 2026 Q6 — rate = $k[A][B]^{3/2}$. Overall order, and can it be elementary? 2.5; no — elementary reactions have whole-number order
- 2025 Q18 — order with respect to A and B from a rate table where doubling B leaves the rate unchanged. Order in B is zero

Mechanism

- 2025 Q23(b) — $2 \text{H}_2\text{O}_2 \xrightarrow{\text{I}^-} 2 \text{H}_2\text{O} + \text{O}_2$, slow step $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$. Rate law and overall order. rate = $k[\text{H}_2\text{O}_2][\text{I}^-]$; order 2

Built from Sourabh Raina's Chemical Kinetics one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 3 (Rationalised 2022–23). NCERT confirms the Arrhenius form $k = Ae^{-E_a/RT}$ and the term "pseudo first order reaction".

The d- and f-Block Elements

NCERT Class XII Chemistry • Chapter 4 • d and f Block Elements

*11 marks, and material you haven't met before. Everything here is written from scratch — terms tagged [exposure] are ones you're seeing for the first time, and each gets a proper explanation before it's used. Almost every answer in this chapter traces back to one idea: how many unpaired d electrons are there?**

What the d block is — where these elements sit, and why they're called "transition"

[exposure] d-block elements are the elements of groups 3 to 12 — the wide middle block of the periodic table, sitting between the s block on the left and the p block on the right. What defines them is that electrons are filling a **d orbital**. Each atom has an outermost shell; the shell just inside it is the **penultimate** one, and it's that shell's d orbital being filled. General configuration:

$$(n - 1)d^{1-10} ns^{1-2}$$

So if the outermost shell is the 4th, the 3d orbital is filling. There are four such series — 3d, 4d, 5d, 6d.

[exposure] Transition element is a narrower term with a precise definition: an element with a **partially filled d orbital**, either in its atom or in any of its stable oxidation states. The word "transition" comes from their position, bridging the reactive metals of the s block and the non-metals of the p block. The definition matters because it excludes some d-block members — see zinc below. Transition elements share a family of properties: variable oxidation states, coloured ions, catalytic behaviour, complex formation, and hardness.

The four series

Series	Orbital	Runs from	Period
First	3d	Sc (21) → Zn (30)	4th
Second	4d	Y (39) → Cd (48)	5th
Third	5d	La (57), then Hf (72) → Hg (80)	6th
Fourth	6d	Ac (89), then Rf (104) → Cn (112)	7th

The gaps in the third and fourth series (58–71 and 90–103) are where the f-block elements sit — pulled out and studied separately.

3d series configurations — learn these, most questions start here

All are [Ar] $3d^n 4s^2$ except two exceptions:

Sc $3d^1 4s^2$	Ti $3d^2 4s^2$	V $3d^3 4s^2$
Cr $3d^5 4s^1$	Mn $3d^5 4s^2$	Fe $3d^6 4s^2$
Co $3d^7 4s^2$	Ni $3d^8 4s^2$	Cu $3d^{10} 4s^1$
Zn $3d^{10} 4s^2$		

Why Cr and Cu break the pattern: a **half-filled** (d^5) or **completely filled** (d^{10}) d subshell is extra stable, so an electron shifts from 4s to 3d to reach it.

Why Zn, Cd, Hg are not transition elements

Their configuration is $(n-1)d^{10}ns^2$ — the d orbital is **completely filled**, in the ground state and in their common +2 oxidation state (Zn^{2+} is $3d^{10}$). No partially filled d orbital in any stable state means they fail the definition.

Examiner asks: the flip side — "why is Cu a transition element despite a filled d orbital in the ground state?" Because its common oxidation state Cu^{2+} is $3d^9$, which is partially filled. Same for Sc: $3d^1$ in the ground state, so it qualifies even though Sc^{3+} is colourless.

Enthalpy of atomization — the unpaired-electron idea, introduced properly

[exposure] Enthalpy of atomization is the energy needed to take one mole of a solid metal and pull it apart into free gaseous atoms. It exists as a measure because it tells you directly how strong the metallic bonding is — strong bonds mean the atoms resist being separated, so the value is high. Transition metals have unusually high values: unpaired d electrons participate in metallic bonding, and more unpaired electrons means a stronger bond. Chromium, with six unpaired electrons ($3d^5 4s^1$), sits near the top; zinc, with none ($3d^{10} 4s^2$), is the lowest in the 3d series.

Two trends carry nearly all the marks:

- **Maximum in the middle of each series**, where unpaired electrons peak.
- **Second and third series > first series**. The 4d and 5d elements have poorly shielding inner electrons, so the effective nuclear charge is higher, metal–metal bonding is stronger, and atomization needs more energy.

Examiner asks: "zinc has the lowest enthalpy of atomization" (fully filled d, no unpaired electrons, weak metallic bond) and "why do 2nd/3rd series have greater enthalpy of atomization" (poor shielding → stronger metallic bond). Both recur.

Melting and boiling points follow the same logic

Highest in the middle: Cr in the 3d series, Mo in 4d, W in 5d. Mn and Tc are anomalously **low** — their $d^5 s^2$ configuration is so stable that the electrons are less free to delocalise into metallic bonding.

Zn, Cd, Hg are **soft metals with low melting points** for the same reason as their low atomization enthalpy: all d electrons paired, weak metallic bond.

Ionization enthalpy and the exchange-energy idea — why the trend is so irregular

Ionization enthalpy depends on three competing things:

1. **Nuclear attraction** — higher nuclear charge holds electrons tighter, raising it.
2. **Electron repulsion (screening)** — electrons pushing each other out lowers it.
3. **Exchange energy** — see below.

[exposure] Exchange energy is a stabilising effect that appears when several electrons with **parallel spins** occupy orbitals of equal energy. Such electrons can swap positions with each other, and every possible swap releases a little energy. More possible swaps means more energy released, and losing energy means greater stability. This is the real reason d^5 and d^{10} configurations are so stable — they maximise the number of parallel-spin exchanges. It's why chromium is $3d^5 4s^1$ rather than $3d^4 4s^2$.

The comparisons they set

- **Mn^+ has lower IE than Cr^+** . Mn^+ is $3d^5 4s^1$; losing one electron gives the stable $3d^5$ Mn^{2+} , so it goes easily. Cr^+ is already $3d^5$; removing an electron breaks that stability.

- Fe^{2+} **has lower IE than** Mn^{2+} . Fe^{2+} ($3d^6$) loses one to reach the stable $3d^5$; Mn^{2+} is already $3d^5$ and resists.
- **Second IE of Cr and Cu is high.** Cr^+ ($3d^5$) and Cu^+ ($3d^{10}$) are both stable, so the next electron is hard to remove.
- **Third IE of Mn and Zn is high** — Mn^{2+} ($3d^5$) and Zn^{2+} ($3d^{10}$) are stable.

Trap: every one of these is answered by writing the configuration *before and after* the electron leaves, and pointing at whichever side is d^5 or d^{10} . Don't reason in words alone — write both configurations.

Variable oxidation states — *the defining property, and why P-block differs*

[exposure] Variable oxidation state means one element can lose different numbers of electrons in different compounds — iron exists as both Fe^{2+} and Fe^{3+} , manganese runs from +2 all the way to +7. It happens because in transition elements the $(n-1)d$ and ns orbitals are **very close in energy**. With so small a gap, electrons from both can be pulled into bonding, and the number that participates can vary. Main-group elements have no such near-degenerate pair, so their oxidation states are far more restricted.

- **Most common state is** +2 across the 3d series (from losing the two 4s electrons) — except Sc, which is +3.
- **Maximum variety in the middle:** Mn shows +2 to +7, the most of any 3d element, because it has the most unpaired electrons available.
- **Fewest at the ends:** Sc and Ti have too few electrons to lose; Cu and Zn have too few vacant d orbitals to use.
- **Cu shows** +1 ($3d^{10}$, stable) — the only 3d element with a common +1 state.

Transition vs p-block variability — *a standing question*

Transition metal oxidation states differ by **one** unit (Fe^{2+} , Fe^{3+} ; V shows 2, 3, 4, 5) because both $(n-1)d$ and ns contribute. P-block states differ by **two** (Sn^{2+} , Sn^{4+}) because there the p orbital dominates and electrons are lost in pairs.

Zero oxidation state in carbonyls

In $\text{Ni}(\text{CO})_4$ and $\text{Fe}(\text{CO})_5$ the metal is in oxidation state **zero**. CO is neutral and forms a sigma bond to the metal while accepting d-electron density back as a pi bond — so no charge is transferred overall.

Standard electrode potentials — *reading the E° table for this series*

Across the 3d series, $E^\circ(\text{M}^{2+}/\text{M})$ becomes **less negative** — the tendency to form the +2 ion decreases — because ionization enthalpies rise as nuclear charge increases across the row.

Copper is the exception — *positive E°*

$E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$, meaning copper prefers to stay as the solid metal. Converting $\text{Cu}(\text{s})$ to Cu^{2+} needs a high enthalpy of atomization *plus* a high sum of first and second ionization enthalpies, and the energy released on hydration doesn't compensate.

Consequence asked directly: copper does not displace hydrogen from acids. It lies below hydrogen in the electrochemical series with a positive reduction potential, so it can't reduce H^+ to H_2 . Only oxidising acids (HNO_3 , hot conc. H_2SO_4) dissolve it.

Three more anomalies, each with the same explanation shape

- **Mn, Ni, Zn are more negative than the trend predicts.** Mn^{2+} is $3d^5$ (half-filled, stable); Zn^{2+} is $3d^{10}$ (filled, stable); Ni^{2+} is small so its hydration enthalpy is exceptionally large and negative.
- $E^\circ(\text{Mn}^{3+}/\text{Mn}^{2+})$ **is highly positive** — Mn^{3+} ($3d^4$) readily gains an electron to become the stable $3d^5$. So Mn^{3+} is a **good oxidising agent**.
- $E^\circ(\text{Cr}^{3+}/\text{Cr}^{2+})$ **is negative** — Cr^{2+} ($3d^4$) readily loses an electron to become $3d^3$, which is stable (t_{2g}^3 , half-filled in the lower set). So Cr^{2+} is a **strong reducing agent**.

Trap: Mn^{3+} oxidising and Cr^{2+} reducing are a matched pair, both answered by "which side is the stable configuration". Learn them together — the exam often asks for both in one question.

Cu^+ **is unstable in aqueous solution** — it disproportionates to Cu^{2+} and Cu , because the much more negative hydration enthalpy of the smaller Cu^{2+} more than compensates for the second ionization enthalpy.

Magnetic properties — and the one formula you'll actually compute

[exposure] Paramagnetic substances contain **unpaired electrons** and are weakly *attracted* into a magnetic field. **Diamagnetic** substances have all electrons paired and are weakly *repelled*. The distinction exists because an unpaired electron behaves like a tiny magnet whose field isn't cancelled by a partner. Most transition metal ions have unpaired d electrons, so most of their compounds are paramagnetic — Mn^{2+} ($3d^5$, five unpaired) is strongly so, while Zn^{2+} ($3d^{10}$) is diamagnetic.

The strength is measured as **spin-only magnetic moment**:

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

n = number of unpaired electrons; BM = Bohr magneton.

Ion	Config	n	μ (BM)
Sc^{3+}	$3d^0$	0	0
Ti^{3+}	$3d^1$	1	1.73
V^{3+}	$3d^2$	2	2.83
Cr^{3+}	$3d^3$	3	3.87
Fe^{2+}	$3d^6$	4	4.90
Mn^{2+}	$3d^5$	5	5.92

Worked · 2020 — spin-only moment of Co^{2+}

Co has $Z = 27$: $[\text{Ar}] 3d^7 4s^2$. Removing two electrons (the 4s pair) gives $\text{Co}^{2+} = [\text{Ar}] 3d^7$.

Filling five d orbitals with 7 electrons: two orbitals paired, three singly occupied $\rightarrow n = 3$.

$$\mu = \sqrt{3(3+2)} = \sqrt{15} = 3.87 \text{ BM}$$

Trap: the 4s electrons come off **first**, before any 3d electron. Co^{2+} is $3d^7$, not $3d^5 4s^2$.

Why transition compounds are coloured — *d–d transition, and the one exception to it*

[exposure] A **d–d transition** is what happens when an electron absorbs light and jumps from a lower-energy d orbital to a higher-energy one. In a complex, the five d orbitals no longer all have the same energy — they split into two sets (called t_{2g} and e_g , covered properly in Coordination Compounds). The gap between them happens to match visible light, so the compound absorbs one colour and we see the **complementary** colour. Cu^{2+} ($3d^9$) has an unpaired electron available to jump, so its salts are blue; Zn^{2+} ($3d^{10}$) has no vacancy to jump into, so its salts are white.

The rule: colour requires **partially filled** d orbitals. Both d^0 and d^{10} are colourless.

Ion	Config	Colour
Sc^{3+} , Ti^{4+}	$3d^0$	Colourless
Ti^{3+}	$3d^1$	Purple
Cr^{3+}	$3d^3$	Green
Mn^{2+}	$3d^5$	Pink
Fe^{2+}	$3d^6$	Green
Cu^{2+}	$3d^9$	Blue
Zn^{2+}	$3d^{10}$	Colourless

The exception: charge transfer

[exposure] **Ligand-to-metal charge transfer** is a second, different way a compound can be coloured — an electron jumps from the surrounding ligand onto the metal, rather than between two d orbitals. It matters because it explains the compounds that are intensely coloured despite having *no* d electrons at all. KMnO_4 is the standard case: Mn is +7, so $3d^0$, and no d–d transition is possible — yet it's deep purple, because charge transfers from oxygen's 2p orbital to manganese's 3d. The same explains $\text{K}_2\text{Cr}_2\text{O}_7$'s orange.

Examiner asks: "why is KMnO_4 coloured despite $3d^0$?" — charge transfer, not d–d. Getting this distinction right is a whole mark.

Catalysis, interstitial compounds, alloys — *three properties, three clean explanations*

Catalytic behaviour

Two reasons, and good answers give both:

1. **Variable oxidation state** lets the metal form intermediates by changing state, then revert. The classic: Fe^{3+} catalysing the iodide–persulphate reaction — Fe^{3+} oxidises I^- to I_2 and becomes Fe^{2+} , which is then re-oxidised by persulphate back to Fe^{3+} .
2. **Surface adsorption** — reactants stick to the metal surface, which raises their local concentration and weakens their bonds, making reaction easier.

V_2O_5 is asked specifically: large surface area, and it changes oxidation state easily to form an unstable intermediate, providing a lower-energy path.

Interstitial compounds

[exposure] An **interstitial compound** forms when small atoms — hydrogen, boron, carbon, nitrogen — get trapped in the gaps between metal atoms in a crystal lattice. Transition metals sit in three-dimensional lattices with spaces (**interstitial sites**) between the atoms, and small atoms slot into them without

displacing anything. The result isn't a normal compound: the ratio isn't a whole number, so these are called **non-stoichiometric**. $\text{VH}_{0.56}$ means 56 hydrogen atoms per 100 vanadium atoms; $\text{TiH}_{1.7}$ means 170 per 100.

Properties: higher melting points than the pure metal, very hard (some borides approach diamond), retain metallic conductivity, and are chemically inert.

Alloy formation

[exposure] An **alloy** is a homogeneous solid solution of two or more metals — melt them together and let them solidify. It forms readily when the metals' atomic radii are within about **15%** of each other, because then one metal's atoms can simply take the other's positions in the crystal lattice. Transition metals have very similar atomic sizes, so they alloy easily. Ferrous alloys (Fe with Cr, V, W, Mo, Mn) give steel and stainless steel; brass is Cu–Zn and bronze is Cu–Sn, both pairing a transition with a non-transition metal.

Examiner asks: "why do transition metals form alloys" — similar atomic size, so one metal replaces the other in the lattice. Two lines, two marks.

Oxides and oxoanions — the acid–base trend that gets asked every year

The **highest oxidation number in an oxide coincides with the group number** from Sc to Mn: Sc_2O_3 (+3, group 3) up to Mn_2O_7 (+7, group 7). Beyond group 7 no higher oxide than Fe_2O_3 is known.

The acid–base rule

Oxidation state of metal	Oxide is	Chromium example
Low	Basic	CrO (+2)
Intermediate	Amphoteric	Cr_2O_3 (+3)
High	Acidic	CrO_3 (+6)

Why: in a low oxidation state the metal has d electrons available to *donate*, so it behaves as a Lewis base. In a high oxidation state the large positive charge lets it *accept* an electron pair — a Lewis acid. As oxidation state rises, metallic character falls and covalent character rises.

Examiner asks: " Mn_2O_3 is basic whereas Mn_2O_7 is acidic" (2019) — a direct application. Give the oxidation numbers (+3 and +7) and the Lewis reasoning.

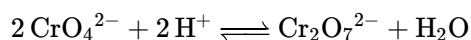
Oxygen beats fluorine at stabilising high oxidation states

Manganese's highest fluoride is MnF_4 (+4), but its highest oxide is Mn_2O_7 (+7) — even though fluorine is more electronegative. The reason: **oxygen can form multiple bonds** with the metal, fluorine only single bonds. More bonds per atom means a higher oxidation state can be supported.

Potassium dichromate — preparation, the pH equilibrium, structures, oxidising action

The pH interconversion — the most-asked single fact

Chromate and dichromate exist in equilibrium, and pH decides which:

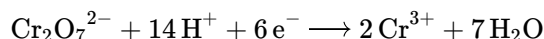


- **Acidic** (high H^+) → equilibrium shifts right → **dichromate**, $\text{Cr}_2\text{O}_7^{2-}$, **orange**.
- **Alkaline** → shifts left → **chromate**, CrO_4^{2-} , **yellow**.

Structures

Chromate CrO_4^{2-} : **tetrahedral**, yellow. **Dichromate** $\text{Cr}_2\text{O}_7^{2-}$: **two tetrahedra sharing one corner** through a Cr–O–Cr bridge, bond angle **126°**, orange.

Oxidising action in acidic medium



Combine this reduction half with any oxidation half, matching electrons:

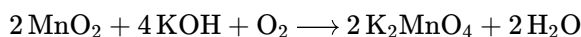
- $6\text{Fe}^{2+} \longrightarrow 6\text{Fe}^{3+} + 6\text{e}^- \rightarrow \text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \longrightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$
- $\text{I}^- \rightarrow \text{I}_2$, $\text{H}_2\text{S} \rightarrow \text{S}$, $\text{Sn}^{2+} \rightarrow \text{Sn}^{4+}$ all work the same way.

Potassium dichromate is used as a **primary standard** in volumetric analysis; sodium dichromate is more soluble and preferred in organic oxidations.

Potassium permanganate — *preparation, why it's purple, and both media*

Preparation — two steps

1. Fusion. MnO_2 fused with KOH in air (or with KNO_3 as oxidant) gives green potassium manganate:

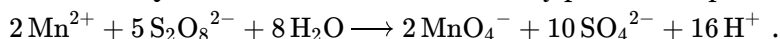


2. Oxidation of manganate to permanganate, either by disproportionation in acid or — commercially — by electrolytic oxidation in alkaline solution:



[exposure] That second reaction is a **disproportionation** — a reaction where *the same element* is simultaneously oxidised and reduced. It's a distinct category because normally one species oxidises while a different one reduces. Here manganese starts at +6 in MnO_4^{2-} and ends up both higher (+7 in MnO_4^-) and lower (+4 in MnO_2) in the same step.

Laboratory route: a Mn^{2+} salt oxidised by peroxodisulphate:



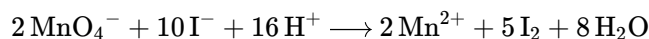
Structure and colour

Both MnO_4^- and MnO_4^{2-} are **tetrahedral**. Manganate (MnO_4^{2-} , Mn is +6, $3d^1$) is **green and paramagnetic** — one unpaired electron. Permanganate (MnO_4^- , Mn is +7, $3d^0$) is **purple and diamagnetic**. Its colour comes from **charge transfer**, not d–d, since it has no d electrons.

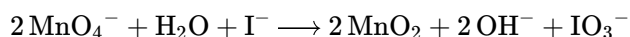
Heating above 513 K decomposes it: $2\text{KMnO}_4 \longrightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$.

Oxidising action — both media, both asked

Acidic: $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ (5 electrons)



Alkaline/neutral: $\text{MnO}_4^- \rightarrow \text{MnO}_2$ (3 electrons), and iodide goes to **iodate**, not iodine:



Trap: the products differ by medium. Acidic gives Mn^{2+} and I_2 ; alkaline gives MnO_2 and IO_3^- . Writing the acidic products for an alkaline question loses the whole answer.

Examiner asks: "why is permanganate titration not carried out in presence of HCl ?" — KMnO_4 oxidises HCl to Cl_2 , a side reaction giving an inaccurate end point. Use H_2SO_4 instead.

Lanthanoids — *the f block, and the contraction that explains everything*

[exposure] The **lanthanoids** are the 14 elements following lanthanum (Ce, $Z = 58$, to Lu, $Z = 71$) in which the **4f orbital** progressively fills. They're pulled out of group 3 and printed as a separate row because they're so chemically alike that keeping them in the main table would be unwieldy. General configuration: $4f^{1-14} 5d^{0-1} 6s^2$. Their dominant oxidation state is +3, with a few showing +2 or +4 where that gets them closer to an empty, half-filled or full f subshell.

[exposure] **Lanthanoid contraction** is the steady decrease in atomic and ionic radii across the lanthanoid series, from lanthanum to lutetium, as atomic number rises. It happens because 4f electrons **shield poorly** — an f electron is diffuse and doesn't screen the outer electrons from the nucleus effectively. So each added proton pulls the whole electron cloud in a little more, and the atoms shrink steadily instead of staying the same size. The effect is small per element but accumulates over 14 elements into something chemically decisive.

Its three consequences — the reason it's examined so heavily

1. **Lanthanoids are hard to separate.** Similar radii → similar chemical properties → separating a mixture is difficult.
2. **4d and 5d elements have nearly identical sizes** (Zr and Hf, Nb and Ta), because the contraction cancels the expected size increase.
3. **Second and third transition series resemble each other** far more than either resembles the first.

Eu^{2+} **is a strong reducing agent** — +3 is the stable lanthanoid state, so Eu^{2+} readily loses an electron to reach it, reducing something else in the process.

[exposure] **Misch metal** is a commercially useful lanthanoid alloy — roughly **95% lanthanoid, 5% iron**, with traces of S, C, Ca and Al. It exists because separating individual lanthanoids is expensive and often unnecessary; the mixture works fine. It's used in magnesium-based alloys and to make **bullets, shells and lighter flints**, since it sparks when struck.

Actinoids — *and the three-way comparison with lanthanoids*

[exposure] The **actinoids** are the 14 elements after actinium (Th, $Z = 90$, to Lr, $Z = 103$) in which the **5f orbital** fills. What sets them apart from lanthanoids is that **all of them are radioactive**, and their 5f, 6d and 7s orbitals are so close in energy that electrons from all three join in bonding — giving a much wider range of oxidation states (+3 through +6 and beyond) than the lanthanoids' near-uniform +3. That, plus the radioactivity, is why their chemistry is described as more complicated.

Actinoid contraction is greater than lanthanoid contraction, because 5f electrons shield even more poorly than 4f, so effective nuclear charge rises faster and the radii shrink more sharply.

The three differences to write

	Lanthanoids	Actinoids
Orbital filling	4f	5f
Shielding	Poor	Even poorer → greater contraction
Oxidation states	Mainly +3; some +2, +4	+3 common, but also +4, +5, +6
Radioactivity	Only promethium	All of them

Examiner asks: "three points of difference" (2020) — pick orbital, contraction magnitude, and oxidation-state range. Radioactivity is the easy fourth if you need it.

Numerical and equation patterns, collected — *the calculable parts of a mostly-descriptive chapter*

A · Spin-only magnetic moment — 2–3 marks

Recognise it: an atomic number is given, or "calculate the magnetic moment".

1. Write the neutral atom's configuration from Z .
2. Remove electrons for the ion — **4s first**, then 3d.
3. Fill the five d orbitals singly before pairing; count unpaired = n .
4. $\mu = \sqrt{n(n+2)}$ BM.

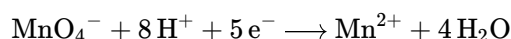
B · Balancing a redox equation — 3–5 marks

Recognise it: "complete and balance", usually with MnO_4^- or $\text{Cr}_2\text{O}_7^{2-}$ in acidic medium.

1. Write reduction and oxidation halves separately.
2. Balance the metal, then O with H_2O , then H with H^+ .
3. Balance charge with electrons.
4. Scale both halves to equal electrons, add, cancel.

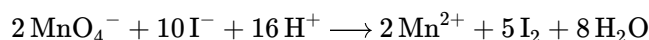
Worked · 2025 — permanganate and iodide in acid

Reduction: $\text{MnO}_4^- \longrightarrow \text{Mn}^{2+}$. Add 4 H_2O right, 8 H^+ left. Charge: left $-1 + 8 = +7$, right $+2$. Add 5 electrons left:



Oxidation: $2 \text{I}^- \longrightarrow \text{I}_2 + 2 \text{e}^-$

Multiply reduction by 2, oxidation by 5:



C · Oxidation number from a formula — 1–2 marks

Recognise it: "write the oxoanion in which the metal shows a state equal to its group number".

1. Set the metal as x ; oxygen is -2 each; sum to the overall charge.
2. For Mn (group 7): MnO_4^- gives $x - 8 = -1$, so $x = +7$. For Cr (group 6): CrO_4^{2-} or $\text{Cr}_2\text{O}_7^{2-}$, both $+6$.

D · Identify the compound (A, B, C chains) — 5 marks

Recognise it: a colour-coded sequence — "black-brown solid A fused with KOH gives dark green B, which on oxidation gives dark purple C".

1. Match colours to compounds: black-brown $\text{MnO}_2 \rightarrow$ dark green $\text{K}_2\text{MnO}_4 \rightarrow$ dark purple KMnO_4 .
2. Write each step's equation.

Past year questions · question types — ranked by how often they turn up

From the last 10 years of boards — 25 questions, grouped by pattern. This chapter is overwhelmingly reason-based: most marks come from explaining, not calculating.

1 · "Give reason" on configuration stability — 2–3 marks

Recognise it: any comparison of two ions, oxidising/reducing strength, or an E° anomaly.

1. Write both configurations, before and after electron transfer.
2. Point at whichever is d^5 or d^{10} .
3. State the consequence in the question's own language.

Trap: answering "because it's stable" without naming *which* configuration. The $3d^5$ is the mark.

2 · Balancing redox equations — 3–5 marks

Recognise it: "complete and balance the following".

1. Halves separately.
2. Metal $\rightarrow$ O with water $\rightarrow$ H with H^+ $\rightarrow$ charge with electrons.
3. Scale, add, cancel.

3 · Lanthanoid / actinoid comparison — 3 marks

Recognise it: "lanthanoid contraction", "three differences", "why is actinoid chemistry more complicated".

1. Name the orbital (4f vs 5f).
2. Poor shielding $\rightarrow$ contraction; 5f worse than 4f.
3. Add the oxidation-state range and radioactivity.

4 · Magnetic and colour properties — 2–3 marks

Recognise it: "which is colourless/paramagnetic and why", or a μ calculation.

1. Get the ion's d-configuration.
2. Count unpaired electrons.
3. None $\rightarrow$ colourless and diamagnetic. Some $\rightarrow$ coloured and paramagnetic.
4. Check for the charge-transfer exception (KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$).

5 · KMnO_4 / $\text{K}_2\text{Cr}_2\text{O}_7$ preparation and reactions — 3–5 marks

Recognise it: named reagent questions, or the A/B/C identification chain.

1. Preparation equations.
2. The pH interconversion for chromate/dichromate.
3. Oxidising action — check which medium is asked.

6 · Physical-property trends — 2–3 marks

Recognise it: enthalpy of atomization, melting point, or a graph across the series.

1. Count unpaired electrons.
2. More unpaired $\rightarrow$ stronger metallic bond $\rightarrow$ higher value.
3. For 2nd/3rd series, add the poor-shielding argument.

Past year questions · mark slots — *what each type is worth*

Question type	Slot	Time
"Give reason" (configuration)	2–3 marker	2–3 min
Balancing redox equations	3-marker; both media is a 5	5–7 min
Lanthanoid/actinoid comparison	3-marker	3 min
Magnetic moment calculation	2–3 marker	3 min
KMnO ₄ /K ₂ Cr ₂ O ₇ chemistry	3–5 marker	5–7 min
Graph/trend interpretation	3-marker	4 min
Definitions (misch metal, oxoanion)	1–2 marker	2 min

The video splits these as short-answer type I (2 marks), type II (3 marks) and long answer (5 marks) — the KMnO₄ chemistry and multi-part "answer the following" sets are where the 5-markers sit.

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2017 Q3 · 2017 Q8 · 2019 Q2

"Transition metals show variable oxidation state" and "Zn, Cd, Hg are soft metals" — asked together, repeatedly, in the same two-part format.

2024 Q5 · lecture PYQ

Define lanthanoid contraction, then why actinoid contraction is greater. The single most reliable f-block question.

2025 Q4 · 2026 Q25(b)

Balance MnO₄[−] with I[−] — once in acidic medium alone, once acidic *and* alkaline in the same question.

2022 Q15 · 2022 Q16 · lecture PYQ

Graph questions on enthalpy of atomization and E° values — lowest in 3d series (Zn), why 2nd/3rd series are higher, why Mn's E° is exceptionally negative, why Cu's is exceptionally positive.

2020 Q11 · 2026 Q22

Lanthanoids vs actinoids — three differences, then why actinoid chemistry is more complicated.

2020 Q10 · 2022 Q12 · 2020 Q6

Colour and magnetism: Sc³⁺ vs Cr³⁺, which ion is colourless, spin-only moment of Co²⁺. All answered by counting unpaired electrons.

2023 Q17 · 2017 Q13 · 2020 Q20

Mn's +7 state, transition metals as catalysts, and interstitial compound formation. Standard reason-based trio.

Past year questions · cold practice — *answers only* — *work them before you look*

Configuration reasoning

- 2026 Q7(a) — why is the third ionization enthalpy of Mn high? Mn^{2+} is $3d^5$, half-filled and stable; removing a third electron breaks it
- 2019 Q13(c) — why is $E^\circ(\text{Mn}^{3+}/\text{Mn}^{2+})$ highly positive compared with $\text{Cr}^{3+}/\text{Cr}^{2+}$? Mn^{3+} gains one to reach stable $3d^5$; Cr^{3+} is already stable $3d^3$
- 2017 Q9(a) — how does variability of oxidation state differ between transition and p-block? Transition differ by 1 unit; p-block by 2
- 2026 Q21(a) — why is Cu a transition element despite a filled d orbital? Cu^{2+} is $3d^9$ — partially filled in its common oxidation state

Magnetism and colour

- 2020 Q6 — spin-only moment of Co^{2+} , $Z = 27$. $3d^7$, $n = 3$, $\mu = 3.87$ BM
- 2022 Q12 — which of Ti^{4+} , Cr^{3+} , V^{3+} is colourless and why? Ti^{4+} — $3d^0$, no d–d transition possible
- 2020 Q10 — Sc^{3+} or Cr^{3+} : which shows magnetic behaviour? Cr^{3+} ($3d^3$, paramagnetic); Sc^{3+} is $3d^0$, diamagnetic
- 2026 Q21(b) — KMnO_4 or K_2MnO_4 : which is paramagnetic? K_2MnO_4 — Mn is +6, $3d^1$, one unpaired electron

Equations

- 2025 Q4 — balance $\text{Cr}_2\text{O}_7^{2-}$ with Fe^{2+} in acid. $\text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \longrightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$
- 2026 Q25(b) — MnO_4^- with I^- in alkaline medium. $2\text{MnO}_4^- + \text{H}_2\text{O} + \text{I}^- \longrightarrow 2\text{MnO}_2 + 2\text{OH}^- + \text{IO}_3^-$
- 2026 Q25(a) — identify A, B, C: black-brown solid fused with KOH in air $\rightarrow$ dark green $\rightarrow$ electrolytic oxidation $\rightarrow$ dark purple. A = MnO_2 , B = K_2MnO_4 , C = KMnO_4
- 2023 — what happens when KMnO_4 is heated? $2\text{KMnO}_4 \longrightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$ at 513 K

f block

- 2019 Q19 — why is Eu^{2+} a strong reducing agent? Loses one electron to reach the stable +3 lanthanoid state
- 2019 Q19(b) — why is separating a lanthanoid mixture difficult? Similar radii from lanthanoid contraction $\rightarrow$ similar chemical properties
- 2019 Q14(a) — general electronic configuration of lanthanoids. $4f^{1-14} 5d^{0-1} 6s^2$
- 2025 Q24(a) — what is misch metal, and one use? ~95% lanthanoid + 5% Fe with traces; used in Mg alloys, bullets, shells, lighter flints

Built from Sourabh Raina's d and f Block one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 4 (Rationalised 2022–23). First-contact material: every term marked **[exposure]** is explained from scratch before use.

Coordination Compounds

NCERT Class XII Chemistry • Chapter 5 • Coordination Compounds

11 marks, and entirely new ground. This chapter has more vocabulary than any other in the paper, so nothing here assumes you've met a term before — every one tagged [exposure] is defined in plain words, with a worked example, before it gets used again.

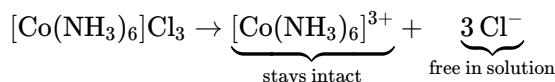
What a coordination compound actually is — *start here* — the whole chapter builds on this one picture

[exposure] A **coordination compound** is a compound in which a central metal atom or ion is surrounded by, and bonded to, a fixed number of molecules or negative ions. Transition metals form these in huge numbers because they have small, highly charged ions and empty d orbitals ready to accept electron pairs. The bond isn't the usual "one electron each" kind — the surrounding species donates *both* electrons. A standard example is $[\text{Co}(\text{NH}_3)_6]^{3+}$: one cobalt ion at the centre, six ammonia molecules attached around it.

They matter well beyond the exam: **chlorophyll** is a coordination compound of magnesium, **haemoglobin** of iron, and **vitamin B₁₂** of cobalt. Industrially they're used in electroplating, dyeing, medicine, metal extraction and catalysis.

The square bracket is the whole notation

Everything **inside** the square bracket travels together as one unit and does *not* break apart in water. Everything **outside** is a separate ion that dissociates freely.



Complex vs double salt — the distinction they examine

	Double salt	Complex
In water	Dissociates completely into all its ions	The bracketed part keeps its identity
Example	Mohr's salt $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6 \text{H}_2\text{O}$	$\text{K}_4[\text{Fe}(\text{CN})_6]$
What you detect	Fe^{2+} gives a positive test	No free Fe^{2+} — the complex ion survives

Also double salts: carnallite $\text{KCl} \cdot \text{MgCl}_2 \cdot 6 \text{H}_2\text{O}$ and potash alum $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24 \text{H}_2\text{O}$. They exist only in the solid state.

Examiner asks: "difference between a complex and a double salt" (2019) — answer with the water behaviour and one example each.

Werner's theory — the 1898 idea that made the whole subject make sense

Alfred Werner explained a puzzle: why does CoCl_3 combine with different numbers of NH_3 to give differently coloured compounds with different behaviour? His answer was that a metal has **two kinds of valence**.

[exposure] **Primary valence** is the metal's ordinary oxidation number — the charge it carries. It is **ionisable**, meaning whatever satisfies it can break away in water, and it is satisfied by **negative ions**. In $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ cobalt is +3, so its primary valence is 3, satisfied by the three chloride ions sitting outside the bracket — which is exactly why those three come free in solution.

[exposure] Secondary valence is the number of species directly bonded to the metal — what we now call the coordination number. It is **non-ionisable**, so these stay attached in water, and it's satisfied by neutral molecules or negative ions that donate a lone pair. It is **fixed for a given metal**. In $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ the secondary valence is 6, satisfied by the six ammonia molecules inside the bracket. Those ligands arrange themselves in a definite shape in space — the **coordination polyhedron**.

How Werner proved it — the silver nitrate experiment

Add excess AgNO_3 to each compound. Only **free** chloride ions precipitate as AgCl , so counting the moles of AgCl tells you how many chlorides are outside the bracket:

Formula	Colour	Moles AgCl	Actual structure
$\text{CoCl}_3 \cdot 6 \text{NH}_3$	Yellow	3	$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
$\text{CoCl}_3 \cdot 5 \text{NH}_3$	Purple	2	$[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
$\text{CoCl}_3 \cdot 4 \text{NH}_3$	Green	1	$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

Notice the coordination number stays **6** throughout — as ammonia leaves the bracket, chloride moves in to take its place.

The vocabulary — nine terms, each with its own example

[exposure] Central atom or ion — the metal at the middle, the one everything else attaches to. It acts as a **Lewis acid**: it's electron-deficient, so it *accepts* electron pairs. In $[\text{Ni}(\text{H}_2\text{O})_4\text{Cl}_2]$ the central ion is Ni^{2+} ; in $[\text{Fe}(\text{CN})_6]^{3-}$ it's Fe^{3+} .

[exposure] Ligand — a molecule or ion attached to the central metal. It must have at least one **lone pair of electrons** to donate, which is what lets it act as a **Lewis base**. Ligands range from simple ions (Cl^- , CN^-) through small molecules (H_2O , NH_3) to whole proteins. The specific atom in the ligand that donates the pair is the **donor atom** — in NH_3 it's the nitrogen.

[exposure] Coordinate bond — the bond formed when *one* species supplies *both* electrons of the shared pair. It exists as a separate name because an ordinary covalent bond takes one electron from each partner. Once formed it's indistinguishable from a normal covalent bond; only its origin differs. Every metal–ligand bond in this chapter is one.

[exposure] Coordination number — the number of donor atoms directly bonded to the metal. Count **donor atoms, not ligands**, and count only sigma bonds. In $[\text{Co}(\text{NH}_3)_6]^{3+}$ six ammonias give six donor atoms, so $\text{CN} = 6$. But in $[\text{Co}(\text{en})_3]^{3+}$ there are only three ligands — each supplies *two* nitrogen donors, so $\text{CN} = 3 \times 2 = 6$ again.

[exposure] Coordination sphere — the central metal plus its ligands, written inside the square bracket together with the overall charge. Whatever sits outside is the **counter ion**, which is the ionisable part. In $\text{K}_4[\text{Fe}(\text{CN})_6]$ the coordination sphere is $[\text{Fe}(\text{CN})_6]^{4-}$ and the counter ions are four K^+ .

[exposure] Coordination polyhedron — the shape the ligands make in space around the metal. It exists as a term because the geometry, not just the count, determines the compound's properties and isomers. Six ligands give an **octahedron** (90° angles), four give either a **tetrahedron** (109.5°) or a **square planar** arrangement (90° , all in one plane).

Oxidation number of the central atom

The charge the metal would carry if every ligand were removed along with its donated electron pair. Write it as a Roman numeral in brackets after the metal's name.

Method: set the metal as x , add each ligand's charge, set the sum equal to the overall charge on the bracket. For $[\text{Cu}(\text{CN})_4]^{3-}$: $x + 4(-1) = -3$, so $x = +1 \rightarrow$ copper(I).

Homoleptic vs heteroleptic

Homoleptic — one type of ligand only, e.g. $[\text{Co}(\text{NH}_3)_6]^{3+}$. **Heteroleptic** — more than one type, e.g. $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.

Types of ligand — denticity, ambidentate, and the chelate effect

[exposure] Denticity is simply how many donor atoms a single ligand uses to grip the metal — from Latin *dens*, tooth. The word exists because a ligand with several donor atoms behaves very differently from one with a single donor: it can clamp onto the metal at multiple points at once. A ligand with one donor site is **unidentate**, two is **didentate**, three or more is **polydentate**.

Type	Donor sites	Examples
Unidentate	1	Cl^- , NH_3 , H_2O , OH^- , CN^-
Didentate	2	ethane-1,2-diamine (en); oxalate $\text{C}_2\text{O}_4^{2-}$ (ox)
Polydentate	3+	EDTA^{4-} — hexadentate

EDTA^{4-} is worth knowing precisely: it binds through **two nitrogen atoms and four oxygen atoms**, six donor sites in total.

[exposure] An ambidentate ligand is a unidentate ligand that has *more than one kind* of donor atom available, and can attach through either one — but only one at a time. It's a distinct category because it creates isomers that differ purely in which atom did the bonding. NO_2^- can bind through nitrogen (called **nitrito-N**) or through oxygen (**nitrito-O**). SCN^- can bind through sulphur (**thiocyanato**) or nitrogen (**isothiocyanato**).

[exposure] A chelate ligand is a didentate or polydentate ligand that grips one metal ion at two or more points, forming a **ring** that includes the metal — the word comes from the Greek for a crab's claw. It matters because chelate complexes are markedly **more stable** than equivalent complexes made from separate unidentate ligands. In $[\text{Pt}(\text{en})\text{Cl}_2]$, the "en" ligand bonds through both its nitrogens, closing a five-membered ring around the platinum.

Why chelates are more stable: replacing six separate unidentate ligands with three didentate ones releases more free particles into solution, so entropy rises; combined with a favourable enthalpy this makes ΔG more negative. More rings means more stability.

IUPAC nomenclature — the rules in order, then worked examples

Naming a complex — the rules

1. **Cation first, then anion** — as in any ionic compound.
2. **Ligands before the metal**, in **alphabetical order** regardless of charge.
3. **Anionic ligands end in -o**: chlorido, bromido, cyanido, hydroxido, sulphato, oxalato.
4. **Neutral ligands** keep special names: H_2O = **aqua**, NH_3 = **ammine** (two m's), CO = **carbonyl**, NO = nitrosyl.
5. **Prefixes** di, tri, tetra for simple ligands; **bis, tris, tetrakis** when the ligand's own name already contains a prefix (then bracket the ligand).
6. Prefixes are **ignored** when alphabetising.

7. **Oxidation number** in Roman numerals, in brackets, right after the metal.
8. If the complex ion is an **anion**, the metal name ends in **-ate** (ferrate, cuprate, argentate, zincate, cobaltate).

Worked — three names built step by step

$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$: alphabetise ammine, aqua, chlorido. Four ammines $\rightarrow$ tetraammine. Charge: $x + 0 + 0 - 1 = +2$, so $x = +3$.

$\rightarrow$ **tetraamminaquachloridocobalt(III) chloride**

$[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$: neutral overall, so $x - 3 = 0$, $x = +3$.

$\rightarrow$ **triamminetrichloridochromium(III)**

$[\text{PtCl}_2(\text{en})_2]^{2+}$: "c" before "e", so chlorido first. "en" already contains "di", so use **bis** and bracket it. $x + 0 - 2 = +2$, $x = +4$.

$\rightarrow$ **dichloridobis(ethane-1,2-diamine)platinum(IV)**

$\text{K}_4[\text{Fe}(\text{CN})_6]$: anionic complex, so iron becomes **ferrate**. $x - 6 = -4$, $x = +2$.

$\rightarrow$ **potassium hexacyanidoferrate(II)**

Writing the formula from a name

1. Metal first, then ligands in **alphabetical order**.
2. Polydentate and abbreviated ligands go in **parentheses**; alphabetise abbreviations by their first letter (en $\rightarrow$ e, ox $\rightarrow$ o).
3. Whole coordination entity in **square brackets**, no spaces inside.
4. Charge as a right superscript outside the bracket, number before sign.

Trap: "ammine" with two m's is NH_3 ; "amine" with one m is an organic amine like ethane-1,2-diamine. NCERT uses both, in these exact senses. Also: current IUPAC is **chlorido**, not chloro.

Isomerism — two families, seven types — the biggest mark-earner in the chapter

[exposure] Isomers are compounds with the same formula but a different arrangement of atoms. In coordination chemistry they split into two families: **stereoisomers**, where the connections are the same but the spatial arrangement differs, and **structural isomers**, where what is bonded to what actually differs. The distinction matters because the two families are detected in completely different ways — stereoisomers by geometry and optical activity, structural ones by simple chemical tests.

Stereoisomerism 1 · Geometrical (cis–trans)

Same ligands, different positions. **Cis** = identical ligands adjacent; **trans** = opposite.

- **Square planar** MX_2L_2 — shows cis and trans. Example: $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$.
- **Octahedral** MX_2L_4 — shows cis and trans. Example: $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$.
- **Octahedral** $[\text{CoCl}_2(\text{en})_2]^+$ — cis and trans.
- **Octahedral** MA_3B_3 — gives **facial (fac)** where the three identical ligands occupy one triangular face, and **meridional (mer)** where they lie around a meridian. Example: $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$.

Trap: tetrahedral complexes do not show geometrical isomerism. In a tetrahedron every position is adjacent to every other — all angles stay 109.5° however you swap ligands, so there's no "opposite" to be

trans to. This is a standing exam question.

Stereoisomerism 2 · Optical

[exposure] A complex is **optically active** when it has no plane of symmetry, so its mirror image cannot be superimposed on it — like your left and right hands. Such a molecule is **chiral**, and the two mirror-image forms are **enantiomers**: one rotates plane-polarised light to the right (**dextro**, **d**), the other equally to the left (**laevo**, **l**). It matters because two compounds identical in every other property can be told apart this way.

Most common in octahedral complexes with didentate ligands:

- $[\text{Cr}(\text{ox})_3]^{3-}$ — type $\text{M}(\text{AA})_3$, optically active.
- $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ — type $\text{M}(\text{AA})_2\text{X}_2$: the **cis** form is optically active, the **trans** form is not (its mirror image is superimposable).

Examiner asks: "draw the geometrical isomer of $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$ which is optically active" — draw the **cis** form. Trans is always the optically inactive one in these.

Structural isomerism — four types

Type	What differs	Example pair
Linkage	Which atom of an ambidentate ligand bonds	$[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]^{2+}$ vs $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$
Ionisation	Ligand and counter ion swap places	$[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ vs $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$
Hydrate	Water inside vs outside the sphere	$[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (violet) vs $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
Coordination	Ligands swap between a cationic and anionic complex	$[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ vs $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$

The chemical test for ionisation isomers (asked directly): add BaCl_2 — the isomer with free sulphate gives a white BaSO_4 precipitate. Add AgNO_3 — the isomer with free chloride gives white AgCl . Whichever ion is *outside* the bracket is the one that reacts.

Valence bond theory — the hybridisation method — work every NCERT example the same way

VBT says the metal makes empty **hybrid orbitals** of equal energy, and each ligand donates a lone pair into one of them.

CN	Hybridisation	Geometry
4	sp^3	Tetrahedral
4	dsp^2	Square planar
6	d^2sp^3	Octahedral (inner orbital)
6	sp^3d^2	Octahedral (outer orbital)

[exposure] A **strong field ligand** pushes hard enough on the metal's d electrons to force them to **pair up**, freeing inner d orbitals for bonding. A **weak field ligand** doesn't, so the electrons stay unpaired. This single distinction decides the hybridisation, the geometry, and the magnetism of every complex below. **Strong:** CN^- , CO , and NH_3 when the metal is highly charged (Co^{3+} , Cr^{3+}). **Weak:** F^- , Cl^- , H_2O .

The method — five steps, every time

1. Find the metal's oxidation number, then its d-electron count.
2. Draw the d orbitals and fill them.
3. Ask: strong ligand (pair up) or weak (leave unpaired)?
4. Count how many orbitals the ligands need, and take that many empty ones — **inner** 3d if pairing freed some, otherwise **outer** 4d.
5. Name the hybridisation, the geometry, and the magnetism.

Complex	Metal	Ligand	Hybrid	Geometry	Magnetism
$[\text{Co}(\text{NH}_3)_6]^{3+}$	$\text{Co}^{3+}, 3d^6$	Strong	$d^2 sp^3$	Octahedral	Diamagnetic
$[\text{CoF}_6]^{3-}$	$\text{Co}^{3+}, 3d^6$	Weak	$sp^3 d^2$	Octahedral	Paramagnetic
$[\text{NiCl}_4]^{2-}$	$\text{Ni}^{2+}, 3d^8$	Weak	sp^3	Tetrahedral	Paramagnetic
$[\text{Ni}(\text{CN})_4]^{2-}$	$\text{Ni}^{2+}, 3d^8$	Strong	dsp^2	Square planar	Diamagnetic
$[\text{Ni}(\text{CO})_4]$	$\text{Ni}, 3d^8 4s^2$	Strong	sp^3	Tetrahedral	Diamagnetic

Worked — $[\text{Ni}(\text{CO})_4]$, the one that catches people out

CO is neutral, so nickel's oxidation state is **0** — configuration $3d^8 4s^2$.

CO is a strong ligand: it forces pairing. The two 4s electrons move into 3d, filling it to $3d^{10}$ and leaving 4s empty.

Four CO ligands need four orbitals: one 4s + three 4p $\rightarrow sp^3$, tetrahedral. All electrons paired $\rightarrow$ **diamagnetic**.

Trap: $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ have the same metal in the same oxidation state but *different* geometry and magnetism — purely because Cl^- is weak and CN^- is strong. This exact pair is a favourite comparison.

[exposure] Inner orbital (or low spin, or spin-paired) complexes use the metal's *inner* 3d orbitals, which only become free when a strong ligand forces pairing — so they have few or no unpaired electrons.

Outer orbital (high spin, spin-free) complexes use the outer 4d orbitals because a weak ligand left the 3d electrons unpaired and in the way. The names are just describing which shell supplied the orbitals:

$[\text{Co}(\text{NH}_3)_6]^{3+}$ is inner/low-spin, $[\text{CoF}_6]^{3-}$ is outer/high-spin.

Magnetic moment

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

Measuring μ tells you n , which tells you whether the ligand was strong or weak — this is how VBT is tested experimentally.

Limitations of VBT — asked as a list

1. Involves many assumptions.
2. Gives no quantitative account of magnetic data.
3. Does not explain the **colour** of complexes.
4. Gives no quantitative account of stability.
5. Cannot predict whether a 4-coordinate complex will be tetrahedral or square planar.
6. Does not distinguish weak from strong ligands — it just assumes the answer.

Crystal field theory — *the model that finally explains colour*

[exposure] Crystal field theory treats the metal–ligand bond as purely **electrostatic attraction** — the positive metal ion pulling on negative ions or on the negative end of polar molecules. It replaced VBT's covalent picture because it can explain what VBT could not: why complexes are coloured, and why some are high spin and others low. Anionic ligands like Cl^- are treated as point negative charges; neutral ones like H_2O as dipoles pointing their negative end at the metal.

[exposure] Crystal field splitting is the central idea. In a free metal ion all five d orbitals have identical energy — they are **degenerate**. If you surrounded the ion with an evenly spread sphere of negative charge, all five would rise in energy equally and stay degenerate. But real ligands arrive from *specific directions*, so some d orbitals get pushed much harder than others. The degeneracy breaks and the five orbitals split into two groups at different energies. That energy gap is what visible light interacts with — which is where colour comes from.

Octahedral field

Ligands approach **along the x, y and z axes**.

- $d_{x^2-y^2}$ and d_{z^2} point **along** the axes → head-on repulsion → energy **raised**. These two are the e_g set.
- d_{xy} , d_{yz} , d_{zx} point **between** the axes → less repulsion → energy **lowered**. These three are the t_{2g} set.

The gap is the **crystal field splitting energy**, Δ_o .

High spin or low spin — the Δ_o vs P contest

[exposure] Pairing energy (P) is the energy cost of forcing a second electron into an orbital that already has one, against their mutual repulsion. It matters because the fourth d electron faces a genuine choice: pay P and pair up in the lower t_{2g} set, or pay Δ_o and go alone into the upper e_g set. Whichever costs less, wins.

- $\Delta_o < P$ (weak field ligand): cheaper to jump up. Fourth electron enters $e_g \rightarrow t_{2g}^3 e_g^1 \rightarrow$ more unpaired electrons → **high spin**.
- $\Delta_o > P$ (strong field ligand): cheaper to pair. Fourth electron pairs in $t_{2g} \rightarrow t_{2g}^4 \rightarrow$ fewer unpaired → **low spin**.

Tetrahedral field

The geometry inverts: now d_{xy} , d_{yz} , d_{zx} lie closer to the incoming ligands and are **raised** (t_2), while $d_{x^2-y^2}$ and d_{z^2} are **lowered** (e).

The splitting is much smaller, because there are only four ligands instead of six and they don't point directly at any orbital:

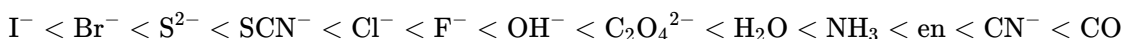
$$\Delta_t = \frac{4}{9} \Delta_o$$

Since Δ_t is always smaller than P , **tetrahedral complexes are essentially always high spin**.

Examiner asks: "how is Δ_o related to Δ_t ?" (2024) — quote $\Delta_t = \frac{4}{9} \Delta_o$ and say why: fewer ligands, none pointing directly at an orbital.

Spectrochemical series

Ligands arranged by the size of Δ they produce, determined experimentally from light absorption. Weak field on the left, strong on the right:



Colour, explained at last

An electron absorbs light of exactly the energy Δ_o and jumps from t_{2g} to e_g — a **d–d transition**. The compound transmits what it didn't absorb, so we see the **complementary** colour. A complex with d^0 or d^{10} has no possible jump and is colourless.

Examiner asks: why does a complex lose its colour when the ligands are removed? Without ligands there is no crystal field, so no splitting, so no d–d transition — it turns colourless.

Stability and applications — the short sections that still carry marks

Stability constant

For $\text{M} + 4 \text{L} \rightleftharpoons \text{ML}_4$, the stability (formation) constant is $K = \frac{[\text{ML}_4]}{[\text{M}][\text{L}]^4}$. Higher K means a more stable complex. Stability rises with the metal's charge, with smaller metal size, and — strongly — with chelation.

Applications worth naming

- **Wilkinson's catalyst**, $[(\text{Ph}_3\text{P})_3\text{RhCl}]$ — catalyses the **hydrogenation of alkenes**.
- **Cisplatin**, $\text{cis-}[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ — anti-cancer drug. Only the *cis* isomer works.
- **EDTA** — estimation of water hardness; treatment of lead poisoning.
- **Extraction of metals** — gold and silver via cyanide complexes.
- **Biological** — chlorophyll (Mg), haemoglobin (Fe), vitamin B₁₂ (Co).
- **Electroplating** and photography (silver thiosulphate complex).

Patterns, collected — the four things you'll be asked to actually do

A · IUPAC name ↔ formula — 2–3 marks

Recognise it: "write the IUPAC name" or "write the formula".

1. Get the oxidation number first — it's needed either way.
2. Naming: ligands alphabetically, anionic ones in -o, prefixes ignored for alphabetising, Roman numeral, -ate if the complex is an anion.
3. Formula: metal first, then ligands alphabetically, abbreviations in parentheses.

B · Hybridisation, geometry, magnetism — 3 marks

Recognise it: an atomic number is given alongside a complex.

1. Oxidation number → d-electron count.
2. Strong or weak ligand? Pair or don't.
3. Count the orbitals needed; inner 3d if free, else outer 4d.
4. State hybridisation, geometry, and paramagnetic/diamagnetic.

C · Identify the isomerism — 2–3 marks

Recognise it: "what type of isomerism is shown by...".

1. Ambidentate ligand (NO_2^- , SCN^-) present → **linkage**.
2. An ion could swap in/out of the bracket → **ionisation**.

3. Water present both inside and out → **hydrate**.
4. Two complex ions in one compound → **coordination**.
5. Otherwise check geometry: square planar or octahedral → **geometrical**; didentate ligands → check **optical** too.

D · Magnetic moment — 2 marks

Recognise it: "calculate the magnetic moment" or "predict the magnetic behaviour".

1. d-count, then apply the ligand's field strength.
2. Count unpaired n .
3. $\mu = \sqrt{n(n+2)}$ BM.

Past year questions · question types — ranked by how often they turn up

From the last 10 years of boards — 25 questions, grouped by pattern.

1 · IUPAC naming / formula writing — 2–3 marks

Recognise it: a complex given with "name this" or a name given with "write the formula".

1. Oxidation number.
2. Alphabetical ligands, correct endings.
3. -ate if anionic.

Trap: "ammine" vs "amine", and forgetting **bis/tris** when the ligand name already has a prefix.

2 · Isomerism identification and drawing — 2–3 marks

Recognise it: "what type of isomerism", or "draw the isomer which is optically active".

1. Scan for ambidentate ligands, swappable counter ions, or two complex ions.
2. If none, it's stereoisomerism — check cis/trans, then optical.
3. Optically active = **cis** with didentate ligands.

3 · Hybridisation and magnetic behaviour — 3 marks

Recognise it: atomic number supplied.

1. Oxidation number → d count.
2. Ligand strength → pairing.
3. Hybridisation → geometry → magnetism.

Trap: in $[\text{Ni}(\text{CO})_4]$ nickel is in oxidation state **zero**, not +2.

4 · Crystal field theory reasoning — 3 marks

Recognise it: Δ_o , Δ_t , high/low spin, or "why is it coloured".

1. Compare Δ_o with P .
2. $\Delta_o < P \rightarrow$ high spin; $\Delta_o > P \rightarrow$ low spin.
3. For colour, name the d–d transition.

5 · Definitions and comparisons — 2 marks

Recognise it: "define", "difference between", "what is denticity/chelate/ambidentate".

1. Give the definition in NCERT's own words.

2. Add one concrete example — it's usually worth half the mark.

6 · Chemical tests and applications — 2–3 marks

Recognise it: "give a chemical test", or "write the formula and use of...".

1. Ionisation isomers → BaCl_2 for sulphate, AgNO_3 for chloride.
2. Named catalysts: Wilkinson's for alkene hydrogenation.

Past year questions · mark slots — *what each type is worth*

Question type	Slot	Time
IUPAC naming / formula	2–3 marker	3 min
Isomerism ID + drawing	2–3 marker	3–4 min
Hybridisation + magnetism	3-marker	4 min
CFT reasoning	3-marker; full splitting diagram is a 5	4–6 min
Definitions / differences	2-marker	2 min
Chemical test / application	2-marker	2 min
Multi-part "answer the following"	5-marker	7 min

The video splits these as short-answer type I (2 marks), type II (3 marks) and long answer (5 marks) — the multi-part sets combining naming, isomerism and CFT are where the 5-markers sit.

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2026 Q1 · 2019 Q4 · 2023 Q2

IUPAC naming — including naming the **linkage isomer** of a red complex, and writing a formula from a given name. Naming appears in essentially every paper.

2017 Q14 · 2022 Q19 · 2026 Q5

"What type of isomerism is shown by..." — run across linkage, ionisation and coordination isomers in the same question format.

2025 Q13 · 2022 Q16

Hybridisation and magnetic behaviour of $[\text{Ni}(\text{CO})_4]$ with $Z = 28$ supplied. The oxidation-state-zero trap is the point of it.

2024 Q24(a) · 2026 Q17

$\Delta_t = \frac{4}{9}\Delta_o$, and the Δ_o vs pairing-energy comparison deciding high vs low spin.

2019 Q25(a) · lecture PYQ

Difference between a complex and a double salt, with examples. Straight recall, always available.

2025 Q18 · 2022 Q23

Geometrical isomerism: which complexes show it, fac vs mer for $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$, and why tetrahedral complexes don't.

2025 Q22 · 2026 Q20

Wilkinson's catalyst formula and use, and why a complex becomes colourless when ligands are absent.

Past year questions · cold practice — *answers only* — *work them before you look*

Naming and formulae

- 2019 — write the formula for pentaamminenitrito-O-cobalt(III). $[\text{Co}(\text{NH}_3)_5(\text{ONO})]^{2+}$
- Lecture PYQ — IUPAC name of $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$. triamminetrichloridochromium(III)
- Lecture PYQ — IUPAC name of $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)]\text{Cl}$. pentaamminecarbonatocobalt(III) chloride
- Lecture PYQ — IUPAC name of $[\text{Zn}(\text{OH})_4]^{2-}$. tetrahydroxidozincate(II) ion

Isomerism

- 2026 Q5 — isomerism in $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$. Linkage — NO_2^- is ambidentate
- 2017 Q14 — isomerism in $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$. Ionisation — sulphate and chloride can swap
- 2017 Q14 — isomerism in $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$. Coordination — ligands exchange between the two spheres
- 2025 Q18 — draw the optically active geometrical isomer of $[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$. The **cis** form; trans is optically inactive
- Lecture PYQ — which shows geometrical isomerism: $[\text{Co}(\text{NH}_3)_6]^{3+}$ or $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$? The second — fac and mer. The first has all identical ligands

Bonding

- 2025 Q13 — hybridisation and magnetic behaviour of $[\text{Ni}(\text{CO})_4]$, $Z = 28$. sp^3 , tetrahedral, diamagnetic — Ni is in oxidation state 0
- Lecture PYQ — compare $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$. sp^3 tetrahedral paramagnetic vs dsp^2 square planar diamagnetic
- Lecture PYQ — why is $[\text{CoF}_6]^{3-}$ paramagnetic but $[\text{Co}(\text{NH}_3)_6]^{3+}$ diamagnetic? F^- weak (no pairing, sp^3d^2); NH_3 strong with Co^{3+} (pairs, d^2sp^3)

CFT and applications

- 2024 Q24(a) — relate Δ_o and Δ_t . $\Delta_t = \frac{4}{9}\Delta_o$ — fewer ligands, none pointing directly at an orbital
- 2026 Q17 — what does $\Delta_o < P$ mean for the complex? Weak field, high spin — the fourth electron goes to e_g rather than pairing
- 2025 Q22 — formula and use of Wilkinson's catalyst. $[(\text{Ph}_3\text{P})_3\text{RhCl}]$; hydrogenation of alkenes
- 2026 Q6 — chemical test distinguishing $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$ from $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$. $\text{AgNO}_3 \rightarrow$ white AgCl for the first; $\text{BaCl}_2 \rightarrow$ white BaSO_4 for the second

Built from Sourabh Raina's Coordination Compounds one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 5 (Rationalised 2022–23). Nomenclature verified against NCERT: **ammine** (two m's) for NH_3 versus **amine** for organic amines, and **chlorido** rather than chloro. First-contact material: every term marked **[exposure]** is explained from scratch before use.

Haloalkanes and Haloarenes

NCERT Class XII Chemistry • Chapter 6 • Haloalkanes and Haloarenes

6 marks, whole chapter 6.1–6.8, and your first organic chapter of the paper. Organic chemistry runs on a small set of ideas repeated everywhere — nucleophile, leaving group, carbocation, steric hindrance. Every one of them is tagged [exposure] and explained from scratch here, because once you have those four, most of this chapter answers itself.

What these compounds are — *start here* — one sentence defines the whole chapter

[exposure] A **haloalkane** (also called an **alkyl halide**, written $R-X$) is what you get when you take a hydrocarbon chain and replace one or more hydrogen atoms with a halogen — fluorine, chlorine, bromine or iodine. That's the entire definition. Ethane CH_3CH_3 with one hydrogen swapped for chlorine becomes chloroethane CH_3CH_2Cl . The defining structural feature: the carbon carrying the halogen is sp^3 **hybridised** — it has four single bonds, no double bond anywhere on it.

[exposure] A **haloarene** (or **aryl halide**, $Ar-X$) is the same swap performed on an *aromatic* ring instead. Benzene with one hydrogen replaced by chlorine is chlorobenzene. The defining feature here: the carbon carrying the halogen is sp^2 **hybridised**, because it is part of the ring's double-bond system. This single difference — sp^3 versus sp^2 — is why haloalkanes and haloarenes behave almost oppositely, and it comes up in the exam again and again.

Throughout the chapter, X means any halogen and R means any alkyl group.

Why the chapter exists at all

Halogen-containing organic compounds are everywhere, natural and synthetic:

- **Chloramphenicol** — a chlorine-containing antibiotic made by soil microorganisms, used against typhoid.
- **Thyroxine** — the iodine-containing hormone your own thyroid produces. Too little iodine in the diet and you get **goitre**.
- **Chloroquine** — synthetic, used against malaria.
- **Halothane** — a synthetic anaesthetic used during surgery.
- **Fully fluorinated compounds** — every hydrogen replaced by fluorine. They dissolve and deliver oxygen to tissue, so they are being explored as **blood substitutes** in surgery.

Examiner asks: one-mark recall of a named compound and its use. Chloramphenicol/typhoid and thyroxine/goitre are the two that appear.

6.1 Classification — *three separate cuts* — count, hybridisation, and where the halogens sit

By number of halogen atoms

Mono, **di** or **poly** halogen compounds, depending on whether they carry one, two, or more halogen atoms. CH_3Cl is monohalo; CH_2Cl_2 is dihalo; $CHCl_3$ is trihalo. The same labelling works on rings: one chlorine on benzene is a monohaloarene, two is a dihaloarene.

By hybridisation of the carbon holding the halogen

This is the classification that actually predicts behaviour, so learn these five names properly.

Compounds with an sp^3 C–X bond:

[exposure] Alkyl halides — the halogen sits on an ordinary sp^3 carbon of a chain. They are sub-classified by *what else* that carbon is attached to: if the carbon carries one other carbon it is **primary (1°)**, two others **secondary (2°)**, three others **tertiary (3°)**. This count matters more than anything else in the chapter — it decides which mechanism the compound follows and how fast. $\text{CH}_3\text{CH}_2\text{Cl}$ is primary; $(\text{CH}_3)_2\text{CHCl}$ is secondary; $(\text{CH}_3)_3\text{CCl}$ is tertiary.

[exposure] An allylic halide has the halogen on an sp^3 carbon that sits *next to* a carbon–carbon double bond. The word exists because that neighbouring double bond changes everything: when the halogen leaves, the positive charge left behind can spread into the double bond by resonance, so allylic halides react far faster than ordinary ones. Example: $\text{CH}_2=\text{CH}-\text{CH}_2-\text{Cl}$ — the chlorine-bearing carbon has four single bonds, the carbon beside it has the double bond.

[exposure] A benzylic halide is the same idea with a benzene ring instead of a double bond: the halogen is on an sp^3 carbon directly attached to an aromatic ring. $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ (benzyl chloride) is the standard one. Benzylic halides are the *most* reactive of all in this respect, because the ring gives the intermediate the largest resonance stabilisation available. These too can be 1° , 2° or 3° by the same counting rule.

Compounds with an sp^2 C–X bond:

[exposure] A vinylic halide has the halogen bonded *directly* to a carbon of a carbon–carbon double bond. Not next to it — on it. $\text{CH}_2=\text{CH}-\text{Cl}$ is vinyl chloride. Because that carbon is sp^2 , the C–X bond is shorter and stronger, and these are very unreactive towards substitution.

[exposure] An aryl halide has the halogen bonded directly to an sp^2 carbon of an aromatic ring — chlorobenzene. Same story as vinylic, for the same reason plus resonance.

Trap: allylic/benzylic (fast) versus vinylic/aryl (slow) is decided by *one* question — is the halogen **on** the unsaturated carbon, or **next to** it? On it $\rightarrow sp^2 \rightarrow$ unreactive. Next to it $\rightarrow sp^3 \rightarrow$ very reactive. This exact discrimination appeared as 2017 Q5(b).

Dihaloalkanes: where the two halogens sit

[exposure] Geminal (gem) dihalides carry both halogens on the *same* carbon — from Latin *geminus*, twin. CH_3CHCl_2 is one. Their common name is built from the alkylidene group: CH_3CHCl_2 is **ethylidene chloride**.

[exposure] Vicinal (vic) dihalides carry the halogens on *adjacent* carbons — from Latin *vicinus*, neighbour. $\text{CH}_2\text{ClCH}_2\text{Cl}$ is one. Their common name uses the alkylene group: **ethylene dichloride**. Vicinal dihalides matter because that's what you get when bromine adds across a double bond.

Examiner asks: "write the structures of all dihalogen derivatives of propane" (NCERT intext 6.3) — there are four: 1,1- and 2,2- (both geminal), 1,2- (vicinal), and 1,3-.

6.2 Nomenclature — IUPAC and common names — both are examinable

IUPAC rules, in the order you apply them

1. Find the **longest carbon chain** containing the principal functional group. Count carbons only — hydrogens don't matter.
2. The halogen is a **substituent**, named as a prefix: **fluoro-**, **chloro-**, **bromo-**, **iodo-**.
3. If there is a **double or triple bond**, it takes **priority** for the lowest number, not the halogen.
4. With no multiple bond, number so the **substituent** gets the lowest locant.

- If numbering from both ends gives the same set of numbers, give the lowest number to whichever substituent comes **first alphabetically**. Bromo beats methyl; chloro beats methyl.
- Write substituents alphabetically in the name.

Structure	IUPAC name	Common name
$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$	1-bromopropane	<i>n</i> -propyl bromide
$(\text{CH}_3)_2\text{CHCl}$	2-chloropropane	isopropyl chloride
$\text{CH}_3\text{CH}_2\text{CHClCH}_3$	2-chlorobutane	<i>sec</i> -butyl chloride
$(\text{CH}_3)_3\text{CBr}$	2-bromo-2-methylpropane	<i>tert</i> -butyl bromide
$(\text{CH}_3)_2\text{CHCH}_2\text{Br}$	1-bromo-2-methylpropane	isobutyl bromide
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$	1-chlorobutane	<i>n</i> -butyl chloride

The straight-chain compound always takes ***n***- in the common name. A three-carbon group halogenated at the middle carbon is **isopropyl**. Four carbons branched at the end is **isobutyl**; branched at the halogen-bearing carbon is *sec*- or *tert*-butyl depending on the count.

Common names you're expected to know cold

Formula	IUPAC	Common
CHCl_3	trichloromethane	chloroform
CHBr_3	tribromomethane	bromoform
CHI_3	triiodomethane	iodoform
CCl_4	tetrachloromethane	carbon tetrachloride
CH_2Cl_2	dichloromethane	methylene chloride

Aromatic compounds

For haloarenes the common name and the IUPAC name are the same word — chlorobenzene is both. Three side-chain names you must recognise:

- Benzyl** — benzene ring plus CH_2 . So $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ is benzyl chloride.
- Benzal** — ring plus CH carrying two halogens. $\text{C}_6\text{H}_5\text{CHCl}_2$ is benzal dichloride.
- Benzo-** — ring plus a carbon with no hydrogen left. $\text{C}_6\text{H}_5\text{CCl}_3$ is benzo-trichloride.

For two substituents on a ring, positions 1,2 are **ortho (o-)**, 1,3 are **meta (m-)**, 1,4 are **para (p-)**. So 1,4-dibromobenzene is *p*-dibromobenzene. Three substituents at 1,3,5 is **sym-** (symmetrical).

Examiner asks: "give the IUPAC name of the following" is a standing 3-marker (2017 Q14, 2024 Q11a). Always check for a double bond before you number.

6.3 Nature of the C–X bond — short section, but it explains every reaction that follows

Halogens are more electronegative than carbon, so the shared electron pair sits closer to the halogen. The result is a **polar bond**: the carbon carries a partial positive charge $\delta+$ and the halogen a partial negative charge $\delta-$.



That $\delta+$ carbon is the target of essentially every reaction in this chapter. Anything electron-rich will go for it. Down the group $F \rightarrow Cl \rightarrow Br \rightarrow I$, the halogen gets bigger, so the **bond length increases** and the **bond enthalpy decreases**. A longer, weaker bond breaks more easily — which is why **R–I is the most reactive alkyl halide and R–F the least** in substitution reactions. Under ordinary conditions R–F barely reacts at all.

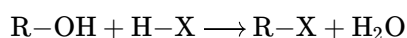
Trap: the C–Cl bond in a **haloalkane is 177 pm**, but in a **haloarene only 169 pm** — shorter, therefore stronger, therefore harder to break. That number is the physical reason haloarenes resist nucleophilic substitution, and it is worth quoting.

6.4 Preparation of haloalkanes — *four routes — from alcohols, from alkanes, from alkenes, by exchange*

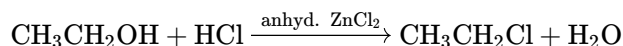
A · From alcohols

The general move: replace the –OH group with a halogen.

(i) With halogen acids (HX)

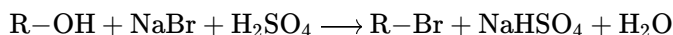


For **chloroalkanes**, use HCl with **anhydrous** $ZnCl_2$ as catalyst:



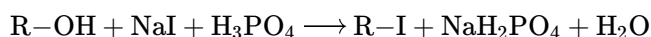
The $ZnCl_2$ works as a **dehydrating agent**: it soaks up the water product. By Le Chatelier's principle, removing a product drives the equilibrium forward, so you get more chloroethane.

For **bromoalkanes**, heat the alcohol with NaBr (or KBr) and concentrated H_2SO_4 :



[exposure] The HBr is generated **in situ** — Latin for "in place", meaning the reagent is made inside the reaction vessel at the moment it's needed rather than being bought and stored. Chemists do this whenever a reagent is nasty or unstable to keep around. HBr is a corrosive acid that gives off heavy fumes, so making it in the flask from $NaBr + H_2SO_4$ is simply safer. You can also just use 48% concentrated HBr directly.

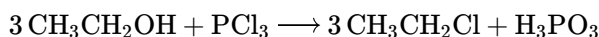
For **iodoalkanes**, use NaI (or KI) with **95% phosphoric acid**, H_3PO_4 — *not* sulphuric acid:



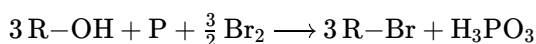
Trap — this is NCERT intext 6.2 and it appears in boards: why not H_2SO_4 with KI? Concentrated H_2SO_4 *would* make HI, but it is also a **strong oxidising agent** and immediately oxidises that HI to I_2 . The HI never survives to reach the alcohol. H_3PO_4 is not an oxidising agent, so it works.

Reactivity orders to memorise — halogen acid: $HI > HBr > HCl$ (because HI has the longest, weakest bond, so it ionises most easily). Alcohol: **tertiary > secondary > primary** (because these reactions go through a carbocation, and a tertiary carbocation is the most stable).

(ii) With phosphorus halides



For bromides and iodides, use **red phosphorus with Br₂ or I₂**. The PBr₃ or PI₃ is generated in situ, again because these are too unstable to store:



(iii) With thionyl chloride — the preferred method

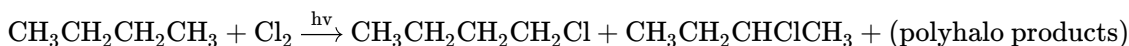


Examiner asks: "why is thionyl chloride preferred for preparing alkyl chlorides from alcohols?" (2024 Q11b) — because both by-products, SO₂ and HCl, are **gases that escape**, leaving pure alkyl chloride behind with no separation step. Say "escapable gases, hence pure product" and you have the mark.

Trap: none of these routes work on phenol. In phenol, the oxygen's lone pair delocalises into the ring, giving the C–O bond **partial double-bond character**. It's too strong to break, so you cannot make an aryl halide from phenol this way. This is a stock 2-marker and it also shows up embedded in "major monohalo product" questions where a molecule has two –OH groups: only the non-phenolic one reacts.

B · From hydrocarbons — free radical halogenation

Treat an alkane with Cl₂ or Br₂ in the presence of UV light or heat, and hydrogens get replaced one at a time:



The problem: you get a **complex mixture** of isomeric mono- and polyhalogenated products whose boiling points are so close they are almost impossible to separate. So this is **not a preferred preparative method** — but it is heavily examined for *predicting* products.

When asked for the **major** free-radical product, use the free-radical stability order: **benzylic ≈ allylic > tertiary > secondary > primary**. Benzylic and allylic radicals win because the ring or double bond stabilises them by resonance.

C · From alkenes

(i) Addition of a halogen acid

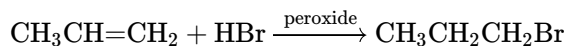
With a **symmetrical** alkene, there's no choice to make:



[exposure] With an **unsymmetrical** alkene the two carbons of the double bond are different, so you need a rule to say which end the halogen goes to. **Markovnikov's rule** says the **negative part of the adding molecule attaches to the carbon that carries fewer hydrogens**. The reason underneath is carbocation stability — the proton adds first, wherever it produces the more stable carbocation, and the halide then goes to that carbon. So propene plus HBr gives 2-bromopropane, not 1-bromopropane.

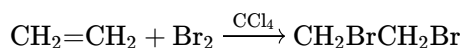


Anti-Markovnikov (peroxide / Kharasch effect): add a peroxide such as benzoyl peroxide and the rule flips — bromine goes to the carbon with *more* hydrogens.



Trap: the peroxide effect works with HBr **only**. It does not happen with HCl or HI. If a question gives you peroxide with HCl, the answer is still Markovnikov.

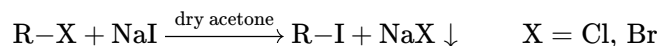
(ii) Addition of a halogen



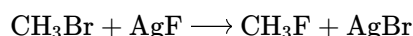
This gives a **vicinal dibromide**. The reddish-brown colour of the bromine solution **discharges** as the reaction proceeds — which is exactly the standard laboratory **test for unsaturation**.

D · Halogen exchange

[exposure] The **Finkelstein reaction** makes alkyl *iodides* by swapping out a chloride or bromide, using NaI in **dry acetone**. It needs a named trick because the swap is an equilibrium that wouldn't otherwise favour the product. The trick: NaCl and NaBr are **insoluble in dry acetone and precipitate out**, while NaI is soluble. Removing a product drags the equilibrium forward (Le Chatelier again).



[exposure] The **Swarts reaction** makes alkyl *fluorides*, which the ordinary routes can't touch. Heat an alkyl chloride or bromide with a **metallic fluoride** — AgF, Hg₂F₂, CoF₂ or SbF₃:

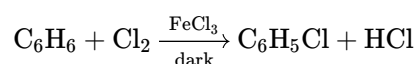


Examiner asks: "what happens when methyl bromide is treated with silver fluoride?" (2025 Q20a) — name it as Swarts and give the equation.

6.5 Preparation of haloarenes — *two routes — and one of them is a named reaction you must know*

A · From arenes, by electrophilic substitution

Treat benzene with Cl₂ or Br₂ in the presence of a **Lewis acid catalyst** — FeCl₃, FeBr₃ or iron filings — in the dark:



Mechanically: the Lewis acid is electron-deficient, so it pulls Cl[−] off Cl₂, leaving Cl⁺. That Cl⁺ is the **electrophile** that attacks the electron-rich ring.

With **toluene**, the —CH₃ group is ortho/para directing, so you get a mixture of *o*- and *p*-chlorotoluene. Conveniently, these two are **easy to separate** because their melting points differ widely.

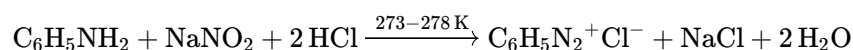
Two limits on this method:

- **Fluoro compounds can't be made this way** — fluorine is too reactive and the reaction is uncontrollable.
- **Iodination is reversible** and gives poor yield. Fix it by adding an **oxidising agent** (HNO₃ or HIO₄), which oxidises the HI by-product to I₂ and so pulls the equilibrium forward.

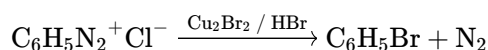
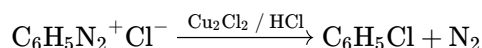
B · From amines — the Sandmeyer reaction

[exposure] A **diazonium salt** is an aromatic compound carrying the group $-\text{N}_2^+$ — a nitrogen pair on its way out. It exists as an intermediate because that group is an *outstanding* leaving group: it departs as harmless N₂ gas, dragging the reaction forward. That makes it the standard way to put almost anything onto a benzene ring. Diazonium salts are only stable cold, which is why the next step specifies 0–5 °C.

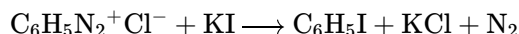
Step 1 — diazotisation. Treat a primary aromatic amine (aniline) with nitrous acid, generated in situ from NaNO₂ + HCl, at **273–278 K (0–5 °C)**:



Step 2 — Sandmeyer. Treat the diazonium salt with a **cuprous halide**:



Trap: for the **iodo** compound you need **no copper salt at all** — simply shaking the diazonium salt with KI does it. This exception is asked directly.



6.6 Physical properties — every point here is a "give the reason" question

Physical state and colour

CH₃Cl, CH₃Br, CH₃CH₂Cl and some chlorofluoromethanes are **gases** at room temperature; higher members are liquids or solids. Pure haloalkanes are **colourless**, but bromo- and iodo- compounds develop colour on standing in light.

Boiling point — four separate rules

1. **RF < RCl < RBr < RI**. Down the group the halogen's mass and size increase, so the **van der Waals forces** between molecules get stronger and more energy is needed to separate them.
2. **A haloalkane always boils higher than the hydrocarbon of comparable mass**. Two reasons together: the C–X bond is polar, so molecules attract each other by **dipole–dipole interaction**; and the heavy halogen raises the molecular mass, strengthening van der Waals forces.
3. **Among isomers, boiling point falls as branching rises**. Branching makes a molecule more compact and spherical, cutting the **surface area of contact** between molecules, which weakens van der Waals forces.

4. **More halogen atoms → higher boiling point**, again through mass.



Worked: Arrange in increasing boiling point: 1-chloropropane, 2-chloropropane, 1-chlorobutane. 1-chlorobutane has four carbons — highest mass, so it boils highest. The other two are isomers, so compare branching: 2-chloropropane is branched, so it boils lowest. Answer: **2-chloropropane < 1-chloropropane < 1-chlorobutane**.

Melting point — the *para* exception

The boiling points of *o*-, *m*- and *p*-dihalobenzenes are nearly identical. But the **para isomer melts much higher** than the other two, because it is more **symmetrical** and so packs more tightly into the crystal lattice. More energy is needed to break a well-packed lattice apart.

Density

Bromo, iodo and polychloro derivatives are **heavier than water**. Density increases with the number of carbon atoms, the number of halogen atoms, and the atomic mass of the halogen.

Solubility — a favourite "give the reason"

Haloalkanes are polar, yet they are **insoluble in water**. Dissolving anything in water requires breaking the strong **hydrogen bonds between water molecules** and also the dipole–dipole attractions between haloalkane molecules. The energy released when new haloalkane–water attractions form is **less** than the energy needed to break the old ones — so it doesn't happen.

They *are* soluble in organic solvents (alcohol, ether, benzene), because there the new attractions formed are about as strong as the ones broken, so the books balance.

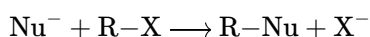
Examiner asks: "why are alkyl halides, though polar, immiscible with water?" (2017 Q5a) — the two energies, in one sentence each. It is a repeat question.

6.7 Reactions of haloalkanes — nucleophilic substitution — the core of the chapter; get the two words first

[exposure] A **nucleophile** is a species that is **electron-rich** — it has a lone pair or a negative charge — and goes looking for a positive centre to donate that pair to. The name means "nucleus-loving". OH^- , CN^- , NH_3 and RO^- are all nucleophiles. In this chapter the positive centre it attacks is always the $\delta+$ carbon of the C–X bond.

[exposure] A **leaving group** is the species that departs, taking the bonding electron pair with it. Here it's the halide ion X^- . A group leaves easily when the resulting ion is stable and the bond was weak — so I^- **is the best leaving group and F^- the worst**, following bond strength. That single fact explains the reactivity order $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$.

Put those together and the reaction reads: a nucleophile attacks the $\delta+$ carbon, the halide leaves, and you have a new compound. The substitution only happens if the incoming nucleophile is **stronger than the halide it displaces**.



The substitution products you must be able to write

Reagent	Product	Class of product
aqueous KOH / NaOH	R–OH	alcohol
NaOR (sodium alkoxide)	R–O–R	ether
KCN	R–CN	alkyl cyanide (nitrile)
AgCN	R–NC	alkyl isocyanide
NH ₃	R–NH ₂	primary amine
KNO ₂	R–O–N=O	alkyl nitrite
AgNO ₂	R–NO ₂	nitroalkane
R'COOAg (silver carboxylate)	R'COOR	ester
LiAlH ₄	R–H	alkane (same carbon count)
R'–Li (or Na)	R–R'	alkane (carbons added)

[exposure] Look at the four pairs above that come in twos. An **ambident nucleophile** is a nucleophile with **two different donor atoms**, either of which can do the attacking — though only one at a time. The category exists because *which* atom attacks decides which product you get, and that in turn depends on whether the reagent is ionic or covalent. The two examples are CN[–] (attacks through C or N) and NO₂[–] (attacks through N or O).

Trap — and one of the most-asked questions in the chapter: KCN is **ionic**, so it dissociates and gives a free CN[–] ion which attacks through its **carbon** → alkyl **cyanide**. AgCN is **covalent**, so it does not dissociate; only the **nitrogen** lone pair is available → alkyl **isocyanide**. The same logic runs the KNO₂ / AgNO₂ pair: ionic gives attack through oxygen (nitrite ester), covalent gives attack through nitrogen (nitroalkane). Remember it as "**silver goes through nitrogen**".

Mechanism 1 · S_N2 — bimolecular

[exposure] S_N2 stands for **Substitution, Nucleophilic, bimolecular**. "Bimolecular" means **two** species are involved in the slow, rate-deciding step, so the rate depends on the concentration of *both*:

$$\text{Rate} = k [\text{R–X}][\text{Nu}^-]$$

Everything happens in **one single step**. The nucleophile attacks the carbon from the side **opposite** the halogen — it has to, because the halogen's lone pairs would repel it if it came from the front. As the new C–Nu bond forms, the old C–X bond breaks, **simultaneously**.

[exposure] Halfway through, the carbon is momentarily bonded to **five** things at once — the three groups it keeps, the incoming nucleophile, and the leaving halogen. That arrangement is the **transition state**. It is not an intermediate: it is a fleeting, very high-energy structure that **cannot be isolated**, existing only at the top of the energy hill. In S_N2 there is **no intermediate at all** — no carbocation, no free radical.

Because the nucleophile comes in from the back and the halogen leaves the front, the other three groups get pushed through, like an umbrella flipping in the wind. The result is **inversion of configuration**, called Walden inversion.

Worked: (–)-2-bromooctane treated with OH[–] gives (+)-octan-2-ol. The OH ends up in the position the Br did not occupy — the configuration is inverted, and the sign of optical rotation flips.

[exposure] Steric hindrance is physical crowding: bulky groups around the reaction centre get in the way of an incoming reagent and slow the reaction down. It matters here because the S_N2 nucleophile must physically reach the carbon. A methyl halide has only three small hydrogens in the way, so it reacts fastest. A tertiary halide has three fat CH_3 groups blocking the approach, so it reacts slowest.

S_N2 reactivity: $\text{CH}_3\text{X} > \text{primary} > \text{secondary} > \text{tertiary}$.

Trap: when both compounds in a pair are primary, compare **how close the branching is to the halogen**. A methyl group nearer the C–X carbon crowds it more and slows S_N2 further. That comparison is NCERT intext 6.7 and appears in boards.

Mechanism 2 · S_N1 — unimolecular

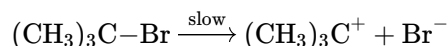
[exposure] S_N1 is Substitution, Nucleophilic, unimolecular. "Unimolecular" means only **one** species appears in the slow step — the alkyl halide. So the rate depends only on *its* concentration; changing the nucleophile's concentration does nothing at all:

$$\text{Rate} = k [\text{R-X}]$$

It runs in **two steps**, and it requires a **polar protic solvent** — polar (it has $\delta + / \delta -$ ends) and protic (it contains an O–H or N–H that can hand out protons). Water, alcohols and acetic acid all qualify.

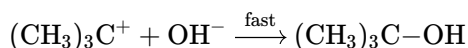
[exposure] A carbocation is a carbon atom bearing a **full positive charge**, left behind when a leaving group departs with both bonding electrons. It has only three bonds, so it is sp^2 hybridised and **flat (planar)**. Being electron-deficient it is very unstable and short-lived — but its stability varies enormously, and that variation drives this whole mechanism. **Tertiary > secondary > primary > methyl**, because the surrounding alkyl groups push electron density in and cushion the positive charge. Benzylic and allylic carbocations beat all of these, because resonance spreads the charge over several atoms.

Step 1 (slow, reversible — the rate-determining step):



Breaking that C–Br bond costs energy. The energy comes from **solvation**: the protic solvent's $\delta +$ hydrogens crowd around the departing Br^- , stabilise it, and release energy in doing so. That is exactly why a polar protic solvent is required.

Step 2 (fast): the nucleophile attacks the carbocation.



S_N1 reactivity: **tertiary > secondary > primary** — the exact opposite of S_N2 , and for a completely different reason (carbocation stability, not crowding).

Trap: for allylic and benzylic halides, **both** mechanisms are fast — S_N1 because their carbocations are resonance-stabilised, S_N2 because the carbon is usually primary and uncrowded. Don't let "it's primary" push you to the wrong answer on an S_N1 question about benzyl chloride: benzyl chloride is the *faster* S_N1 substrate against cyclohexyl chloride, because the benzyl carbocation is resonance-stabilised.

S_N1 vs S_N2 — the comparison table

	<i>S_N1</i>	<i>S_N2</i>
Order of reaction	first order	second order
Rate depends on	[RX] only	[RX] and [Nu]
Steps	two	one
Goes through	carbocation intermediate	transition state, no intermediate
Stereochemistry	racemisation	inversion
Favoured by	3° > 2° > 1°	1° > 2° > 3°
Solvent	polar protic	polar aprotic

Examiner asks: "write two differences between *S_N1* and *S_N2*" (2021 Q21) — pick order + steps, or intermediate + stereochemistry. Two rows, two marks.

Optical isomerism — the stereochemistry vocabulary — six terms, each needed for the *S_N* answers

[exposure] Optical activity is a property some substances have of **rotating the plane of plane-polarised light**. Ordinary light vibrates in every direction; pass it through a polariser and it vibrates in one plane only. Send that through certain liquids or solutions and the plane comes out twisted by some angle. Substances that do this are **optically active**; those that don't are optically inactive. The angle is measured with an instrument called a **polarimeter**.

A substance rotating the plane **clockwise (right)** is **dextrorotatory**, written **d** or **(+)**. One rotating it **anticlockwise (left)** is **laevorotatory**, written **l** or **(-)**.

[exposure] A chiral carbon (also called an asymmetric carbon or stereocentre) is an *sp*³ carbon attached to **four different atoms or groups**. It is marked with an asterisk. It matters because it is the structural cause of optical activity — with four different groups the molecule has no plane of symmetry, and that asymmetry is what twists the light. In CH₃CH(Br)CH₂CH₃ the second carbon holds methyl, ethyl, bromine and hydrogen: four different things, so it is chiral.

[exposure] A chiral molecule has no plane of symmetry, and so is **non-superimposable on its mirror image** — exactly like your left and right hands, which is where the name comes from (Greek *cheir*, hand). Lay one over the other and they never match. A molecule with a chiral carbon is normally chiral, and chiral molecules are optically active.

[exposure] Enantiomers are the two mirror-image forms of a chiral molecule. They are stereoisomers — same connections, different spatial arrangement. They rotate plane-polarised light by the **same angle in opposite directions**: if one is dextro, the other is laevo. Every other physical property — melting point, boiling point, solubility, refractive index — is **identical**.

Examiner asks: "why are the d- and l-isomers of butan-2-ol difficult to separate by fractional distillation?" (2024 Q22) — because fractional distillation separates on boiling point, and enantiomers have **identical** boiling points. One line, full marks.

[exposure] A racemic mixture is a **1:1 (equimolar)** mixture of the two enantiomers. It is **optically inactive**, because the rotation produced by one is exactly cancelled by the other. It's written **dl** or **(±)**, as in (±)-butan-2-ol. The process of converting a single enantiomer into such a mixture is **racemisation**.

[exposure] Three words for what a substitution does to a chiral centre. **Retention** — the new group takes the same spatial position the old one had; the configuration is unchanged. **Inversion** — the new group takes the position *opposite*; the configuration is flipped. **Racemisation** — you get roughly 50% of each, so the product is optically inactive.

Which mechanism gives which:

- $S_N2 \rightarrow$ **inversion**. The nucleophile can only attack from the back, so there is only one possible outcome.
- $S_N1 \rightarrow$ **racemisation**. The carbocation intermediate is **planar**, so the nucleophile can attack from either face with equal ease. Two products form in equal amounts, they are enantiomers, and the mixture is racemic.

Examiner asks: "out of S_N1 and S_N2 , which occurs with inversion and which with racemisation?" (2026 Q23c) — and expect to have to *explain* it: back-side attack for one, planar carbocation for the other.

Elimination and reactions with metals — *the other two things a haloalkane does*

Elimination — dehydrohalogenation

[exposure] **Dehydrohalogenation** means removing H and X from adjacent carbons to make a double bond. It is also called **β -elimination**, because of where the two pieces come from: the carbon bearing the halogen is the **α -carbon**, the one next to it is the **β -carbon**, and the reaction takes the halogen from α and a hydrogen from β . The reagent is **alcoholic KOH** — and the "alcoholic" is the whole point, because aqueous KOH gives substitution instead.



Trap — the single most reliable trick in the chapter: aqueous KOH $\rightarrow$ alcohol (substitution). Alcoholic KOH $\rightarrow$ alkene (elimination). One word in the question decides the whole answer, and it is tested constantly.

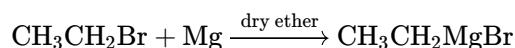
[exposure] When a molecule has **more than one β -hydrogen**, more than one alkene can form, and **Saytzeff's rule** (NCERT spells it **Zaitsev**, "also pronounced as Saytzeff") tells you which one dominates: the **more substituted alkene** — the one whose double-bonded carbons carry the most alkyl groups — is the major product. It exists as a rule because more-substituted alkenes are more stable, and the reaction settles into the more stable product.

Worked: 2-bromopentane with alcoholic KOH. There are β -hydrogens on both sides, so two alkenes are possible. Removing one gives pent-1-ene, the other gives pent-2-ene. Pent-2-ene has an alkyl group on *each* double-bonded carbon, pent-1-ene has only one — so **pent-2-ene is major, 81% against 19%**.

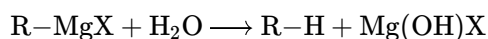
Reaction with metals

[exposure] An **organometallic compound** is any compound containing a direct **carbon–metal bond**. They matter because the metal, being electropositive, pushes electron density onto the carbon, giving that carbon a δ^- charge — the exact reverse of its usual state. A carbon that is normally attacked becomes a carbon that attacks.

[exposure] The **Grignard reagent**, R-Mg-X (alkyl magnesium halide), is the most important organometallic compound in the syllabus. Make it by reacting a haloalkane with **magnesium metal in dry ether**:

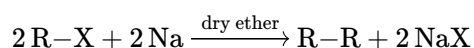


The C–Mg bond is covalent but **highly polar**: carbon δ^- , magnesium δ^+ . That δ^- carbon is so electron-rich that it grabs a proton from **any** source — water, alcohol, an amine — and turns into an alkane:



Examiner asks: "why should Grignard reagents be prepared under anhydrous conditions?" (2024 Q13c) — because they react instantly with any water present to give the alkane, destroying the reagent before it can be used. Repeat question; know it verbatim.

[exposure] The **Wurtz reaction** joins two alkyl halides end to end using **sodium in dry ether**, producing an alkane with **double the carbon count**. It exists as a named reaction because it's the standard way to build a longer chain, and because it's the only route that *doubles* — which is what lets you work backwards from the product in exam questions.



Trap: Wurtz only gives a clean product with a **symmetrical** alkane. So when a question gives you the product and asks for the starting halide, **cut the product exactly in half** and put X on the cut. 2,5-dimethylhexane cuts into two 1-bromo-2-methylpropane molecules. That reverse-engineering step is asked repeatedly (2026 Q10b).

Reactions of haloarenes — *why they behave backwards, and the two named ring reactions*

Why haloarenes resist nucleophilic substitution

This is a guaranteed question. There are four reasons; two or three earn full marks.

1. **Resonance.** The halogen's lone pair delocalises into the ring, so the C–X bond acquires **partial double-bond character**. A partial double bond is far harder to break than a single one.
2. **sp^2 hybridisation.** The carbon holding the halogen is sp^2 , so it has more **s-character** and is more electronegative than an sp^3 carbon. It holds the bonding pair more tightly. The bond is also shorter — 169 pm against 177 pm in a haloalkane.
3. **Instability of the phenyl cation.** An S_N1 route would need a phenyl carbocation, which gets no resonance stabilisation and is therefore not formed.
4. **Electronic repulsion.** The ring's π -electron cloud repels an approaching nucleophile.

The nitro group changes everything

Put a **strongly electron-withdrawing group** such as $-\text{NO}_2$ at the **ortho or para** position and the picture flips. The nitro group pulls electron density away, weakening the C–X bond and activating the ring towards nucleophilic attack. Each extra nitro group makes it easier:

Compound	Conditions needed
chlorobenzene	NaOH, 623 K, 300 atm
<i>p</i> -nitrochlorobenzene	NaOH, 443 K, then acidify
2,4-dinitrochlorobenzene	NaOH, 368 K, then acidify
2,4,6-trinitrochlorobenzene	warm water alone → picric acid

Trap: a nitro group at the **meta** position has **no effect at all**. Resonance only pushes and pulls electron density at the ortho and para positions — meta is never in the delocalisation path. Stating this earns a mark on its own.

Electrophilic substitution — the halogen's split personality

Halogens are **ortho/para directing but deactivating**. Both halves of that need explaining:

- **Ortho/para directing** because the halogen's lone pair delocalises *into* the ring (**+R effect**), and resonance raises the electron density specifically at the ortho and para positions. An incoming electrophile goes where the electrons are.
- **Deactivating** because the halogen is also strongly electronegative and pulls electron density *out* along the sigma bond (**–I effect**). Overall the ring ends up poorer in electrons than plain benzene, so the reaction is slower and needs harsher conditions.

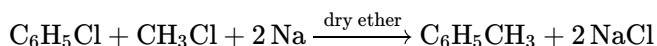
Examiner asks: "although chlorine shows a strong –I effect, why is it ortho/para directing?" (2026 Q25a) — answer with the two effects working in opposite directions: –I decides *how fast*, +R decides *where*.

The four standard electrophilic substitutions on chlorobenzene, all giving ortho + para with **para as the major product** (it's more symmetrical, and less crowded than ortho):

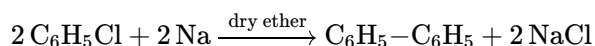
- **Halogenation** — Cl_2 , anhydrous FeCl_3 → 1,2- and 1,4-dichlorobenzene.
- **Nitration** — conc. HNO_3 + conc. H_2SO_4 → 1-chloro-2-nitrobenzene and 1-chloro-4-nitrobenzene.
- **Sulphonation** — conc. H_2SO_4 , heat → 2- and 4-chlorobenzenesulphonic acid.
- **Friedel–Crafts**, anhydrous AlCl_3 — **alkylation** with CH_3Cl gives 1-chloro-2/4-methylbenzene; **acylation** with CH_3COCl gives 2- and 4-chloroacetophenone.

Reactions with metals

[exposure] The **Wurtz–Fittig reaction** is the Wurtz reaction run with a **mixture** of an alkyl halide and an aryl halide, using sodium in dry ether. The alkyl group ends up attached to the ring, giving an **alkylarene**. It's the standard way to put a side chain on benzene: chlorobenzene plus CH_3Cl gives **toluene**.



[exposure] The **Fittig reaction** is the same setup with **aryl halide alone** — so two rings couple to each other, giving **diphenyl (biphenyl)**.



Trap: one reagent tells you which. **Both** halides present → Wurtz–Fittig → alkylarene. **Only** the aryl halide → Fittig → biphenyl. "How do you convert chlorobenzene to biphenyl?" (2025 Q12a) is Fittig, not Wurtz–Fittig.

6.8 Polyhalogen compounds — *pure recall — uses and harmful effects, and it's free marks*

[exposure] A **polyhalogen compound** is simply a carbon compound carrying **more than one halogen atom**. The section exists in the syllabus not for the chemistry but for the applications and the environmental damage — so the marks here come from recall, not reasoning.

Dichloromethane, CH₂Cl₂ (methylene chloride)

Uses: solvent, paint remover, propellant in aerosols, degreasing agent.

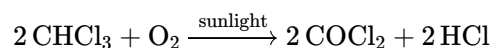
Harm: affects the **central nervous system**. At low levels in air, impairs hearing and vision. At high levels, causes dizziness, nausea, and tingling or numbness in the fingers and toes. Direct contact burns the skin; contact with the eye burns the cornea.

Trichloromethane, CHCl₃ (chloroform)

Uses: solvent for fats, alkaloids and iodine; feedstock for making **Freon R-22**. It was once used as a general anaesthetic in surgery but has been replaced by ether, which is less harmful.

Harm: inhaling it **depresses the central nervous system**. Breathing about 900 ppm briefly causes dizziness, fatigue and headache. Chronic exposure damages the **liver and kidneys**.

Examiner asks: "why is chloroform stored in dark coloured bottles?" (2026 Q24a) — because in **light and air it is slowly oxidised to phosgene**, COCl₂, an extremely poisonous gas. The bottles are dark **and filled to the brim**, so no air is left inside. Repeat question — memorise the equation:



Triiodomethane, CHI₃ (iodoform)

Formerly used as an **antiseptic**, working through the free iodine it liberates. Now replaced by other iodine formulations because of its **objectionable smell**.

Tetrachloromethane, CCl₄ (carbon tetrachloride)

Uses: refrigerant manufacture, aerosol propellant, feedstock for chlorofluorocarbons, solvent. Until the 1960s it was a cleaning fluid — an industrial **degreasing agent** and a household spot remover.

Harm: long exposure causes **liver cancer**. Inhalation causes dizziness, light-headedness, nausea and vomiting, and can permanently damage nerve cells. Severe cases lead to stupor, coma, unconsciousness, irregular heartbeat or death. It also **depletes the ozone layer**.

Freons

Freons are the **chlorofluorocarbons (CFCs)** of methane and ethane. They are extremely stable, unreactive, non-toxic, non-corrosive and easily liquefiable — which is exactly what made them so useful and so damaging. The commonest is **Freon-12**, CCl₂F₂, made from CCl₄ by the **Swarts reaction**.

Uses: aerosol propellant, refrigeration, air conditioning.

Harm: because they are unreactive, they survive the journey up to the **stratosphere** unchanged. There they start a **chain reaction with ozone** and destroy the ozone layer.

DDT

p,p'-Dichlorodiphenyltrichloroethane — two benzene rings each carrying a chlorine at the para position, joined by a carbon bearing a CCl₃ group. Prepared by the **condensation of chloral with**

chlorobenzene in sulphuric acid.

Use: the first modern insecticide — effective against the mosquitoes that spread malaria and the lice that spread typhus.

Harm: many insect species **developed resistance** to it. It is highly toxic to fish. And it is chemically very stable and **fat-soluble**, so animals cannot metabolise it — it accumulates in fatty tissue and builds up along the food chain.

Patterns, collected — *the six things you'll actually be asked to do*

A · Predict the product — 2–3 marks

Recognise it: a reaction arrow with reagents written over it, and "write the major product" or "identify A and B".

1. **Read the reagent first** — it decides everything. Aqueous KOH → alcohol. Alcoholic KOH → alkene. Mg/dry ether → Grignard. Na/dry ether → Wurtz. NaI/acetone → Finkelstein. AgF → Swarts. SOCl₂ → alkyl chloride.
2. If a double bond is being attacked: Markovnikov unless "peroxide" appears, and peroxide only counts with HBr.
3. If an alkene is being formed: apply Saytzeff — more substituted is major.
4. If a ring is being attacked: ortho and para, para major.

B · Which reacts faster, and why — 2–3 marks

Recognise it: a pair of compounds and "which undergoes faster S_N1/S_N2".

1. Identify which mechanism the question names. They run in opposite directions.
2. S_N2 → think **steric hindrance**. Fewer/smaller groups around the carbon wins. 1° > 2° > 3°.
3. S_N1 → think **carbocation stability**. Tertiary, benzylic or allylic wins.
4. If the halogens differ, the **leaving group** decides: I > Br > Cl > F.
5. Always give the *reason* word — "less steric hindrance" or "resonance-stabilised carbocation". The reason carries the mark, not the choice.

C · Give the reason — 1–2 marks

Recognise it: "account for the following" or "why is...".

Almost every one of these reduces to one of five stock answers: **resonance / partial double bond** (haloarene unreactivity, phenol won't react, allyl beats propyl), **sp² vs sp³ and s-character** (dipole moment, bond length), **energy of bonds broken vs formed** (water insolubility), **van der Waals and surface area** (boiling points), **escaping gaseous by-products** (thionyl chloride).

D · Arrange in order — 1–2 marks

Recognise it: three or four compounds, "increasing order of...".

1. **Boiling point** → molecular mass first, branching second.
2. **Reactivity to S_N1** → carbocation stability.
3. **Reactivity to S_N2** → steric hindrance.
4. **Reactivity of RX generally** → C–X bond strength: RI > RBr > RCl > RF.

E · Conversions — 3 marks

Recognise it: "how would you convert X into Y".

1. Count the carbons on both sides. Same count → substitution or addition. Doubled → Wurtz.
2. Alkene → halide: HX, checking Markovnikov versus peroxide.
3. Need an iodide → make the bromide first, then Finkelstein.
4. Halide → alkene: alcoholic KOH, then check Saytzeff.
5. Halide → alkane: Mg/dry ether then water, or LiAlH_4 directly.

F · Work backwards from the product — 2–3 marks

Recognise it: the product is given and the starting material is A, or "identify the hydrocarbon".

1. **Wurtz product** → cut the alkane in half at the middle and add X.
2. **Grignard then water** → the alkane's carbon skeleton is the halide's; put MgX back where the new hydrogen went.
3. **"Only one monochlorination product"** → all the hydrogens must be equivalent → **2,2-dimethylpropane (neopentane)**.
4. **"Three isomeric products"** → pentane. **"Four"** → 2-methylbutane.

Past year questions · question types — ranked by how often they turn up

From the last 10 years of boards — 25 questions, grouped by pattern.

1 · Predict the product / identify A, B, C — 3–5 marks

Recognise it: a reaction scheme, sometimes a chain of three or four arrows.

1. Work one arrow at a time; never skip to the end.
2. Name the reaction under each arrow before you draw anything — Finkelstein, Wurtz, Friedel–Crafts, Sandmeyer.
3. Where two products are possible, say which is major and give the rule by name (Markovnikov, Saytzeff, para-major).

Trap: a molecule with two –OH groups, one on the ring and one on a side chain, treated with SOCl_2 or HCl. Only the **side-chain** –OH reacts; the phenolic one has partial double-bond character and stays put.

2 · S_N1 / S_N2 comparison — 2–3 marks

Recognise it: a pair of halides, or "write the mechanism of the following".

1. For a mechanism question, draw **both steps** of S_N1 and label the slow step as rate-determining, or draw the S_N2 transition state in square brackets.
2. For a comparison, name the deciding factor before you name the winner.

Trap: S_N2 ordering is $1^\circ > 2^\circ > 3^\circ$; S_N1 is the exact reverse. Writing the reactivity order without checking which mechanism the question asked about is the commonest way to lose these marks.

3 · "Give the reason" / "account for the following" — 1–2 marks

Recognise it: a statement of fact, then "why?".

1. Identify which of the five stock explanations applies (see Patterns, C).
2. Write it as cause → effect in one or two sentences. No padding.

4 · IUPAC naming and structure drawing — 1–3 marks

Recognise it: "give the IUPAC name" or "draw the structure of..."

1. Longest chain, then check for a double bond — **it takes numbering priority over the halogen.**
2. Substituents alphabetically in the name; lowest locants where there is a tie.

Trap: numbering from the halogen out of habit when a double bond is present. 4-chlorobut-1-ene, not 1-chlorobut-3-ene.

5 · Conversions — 3 marks

Recognise it: "how can the following conversions be carried out".

1. Count carbons. Choose the route. Write reagents above every arrow — the reagent is what's being marked.

6 · Definitions and stereochemistry — 1–2 marks

Recognise it: "define the following term", or a pair of structures with "which is chiral?".

1. For chirality: find the carbon with **four different** groups. Two identical groups anywhere on it and it is not chiral.
2. For definitions, give the definition and **one example** — the example is usually worth half the mark.

Past year questions · mark slots — *what each type is worth*

Slot	What turns up there
1 mark	Name the type of halide (vinylic / allylic / benzylic). One-line reason questions. Assertion–reason on reactivity orders. (<i>inferred</i>)
2 marks	"Give the reason" pairs. Two differences between S_N1 and S_N2 . Arrange in order of boiling point. One-step "what happens when".
3 marks	IUPAC naming (three compounds). Three-part "give reason for the following". Conversions. Identify A and B in a two-step scheme.
5 marks	Multi-part sets: a reactivity comparison plus a halide-type identification plus a polyhalogen recall, or a product-prediction set combined with definitions.

The video groups these as short answer type I (2 marks, Q1–10), short answer type II (3 marks, Q11–23) and long answer (5 marks, Q24–25). At 6 marks in your blueprint this chapter will most likely appear as a mix of 1- and 2-markers rather than a 5-marker — but the 5-mark sets are still the best revision, because they bundle four separate ideas into one question.

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2021 Q7 · 2024 Q13(b) · 2026 Q24(a)

Which of a pair reacts faster by S_N2 , with reason. Runs across RI vs RCl (leaving group), tertiary vs primary (steric hindrance), and two primaries differing in where the branch sits. The single most repeated question shape in the chapter.

2026 Q13(a) · 2024 Q22(a)

KCN gives cyanide but AgCN gives isocyanide — the ambident nucleophile question. Ionic versus covalent is the whole answer.

2020 Q4 · 2026 Q10(a) · 2021 Q21

S_N1 : write the mechanism, name the most reactive isomer, or give two differences from S_N2 . Carbocation stability is the thread through all three.

2017 Q5(a) · lecture PYQ

Why alkyl halides, though polar, are immiscible with water. Energy of bonds broken versus energy released.

2023 Q8(a) · 2024 Q11(c) · lecture PYQ

Arrange in increasing order of boiling point. Appears in both a chain-length form and a halogen-count form.

2024 Q11(b) · lecture PYQ

Why thionyl chloride is preferred for making alkyl chlorides from alcohols. Escapable gaseous by-products.

2026 Q10(b) · 2020 Q18

Wurtz reaction run backwards — the product is given, find the starting halide. Cut the symmetrical alkane in half.

2024 Q13(c) · lecture PYQ

Why Grignard reagents must be prepared under anhydrous conditions. Water destroys them, giving the alkane.

2025 Q19(b) · 2026 Q25(b) · lecture PYQ

Friedel–Crafts on a haloarene, alkylation or acylation, asking for the **major** product. Always para, always because of symmetry and less crowding.

2025 Q15 · 2026 Q23(b) · 2020 Q18

Multi-step "identify A, B, C, D, E" schemes built from Grignard, Wurtz, alcoholic KOH and HBr addition. Three separate papers, near-identical shape.

Past year questions · cold practice — *answers only* — *work them before you look*

Reactivity and mechanism

- 2021 Q7 — which shows faster S_N2 : *n*-propyl iodide or *n*-propyl chloride? The iodide — I^- is a better leaving group because of its large size and weak C–I bond
- 2021 Q7 — faster S_N2 : *tert*-butyl chloride or ethyl chloride? Ethyl chloride — primary, so far less steric hindrance
- 2026 Q13(b) — why is cyclohexylmethyl chloride more reactive to S_N2 than cyclohexyl chloride? The first is primary, the second secondary — less crowding at a primary carbon
- 2026 Q10(a) — most reactive C_4H_9Br isomer towards S_N1 . 2-bromo-2-methylpropane — it forms the most stable, tertiary carbocation
- 2021 Q21 — faster S_N1 : cyclohexylmethyl chloride or benzyl chloride? Benzyl chloride — the benzyl carbocation is resonance-stabilised by the ring
- 2026 Q24(a) — more reactive to S_N1 : 2-bromo-2-methylbutane or 1-bromopentane? 2-bromo-2-methylbutane — tertiary carbocation beats primary
- 2025 Q20(b) — why is allyl chloride hydrolysed faster than *n*-propyl chloride? The allyl carbocation is resonance-stabilised; the propyl one is not
- 2026 Q23(c) — which of S_N1/S_N2 gives inversion, which gives racemisation? S_N2 inversion (back-side attack); S_N1 racemisation (planar carbocation, attack from either face)

Reason questions

- Lecture PYQ Q1 — why are haloarenes less reactive to nucleophilic substitution? Resonance gives the C–X bond partial double-bond character, and the sp^2 carbon holds the pair more tightly
- 2024 Q16 — why is the dipole moment of chlorobenzene lower than that of cyclohexyl chloride? The sp^2 carbon is more electronegative, so less charge separation; resonance also shortens the C–Cl bond
- 2020 Q9(b) — why does $-\text{NO}_2$ at ortho/para increase haloarene reactivity to nucleophilic substitution? It withdraws electron density, weakening the C–X bond and activating the ring
- 2026 Q25(a) — chlorine shows $-\text{I}$ yet is ortho/para directing. Why? $-\text{I}$ deactivates the whole ring, but $+\text{R}$ resonance raises electron density specifically at ortho and para
- 2024 Q22(c) — why are d- and l-butan-2-ol hard to separate by fractional distillation? Enantiomers have identical boiling points
- 2026 Q24(a) — why is chloroform stored in dark bottles, filled to the brim? Light and air oxidise it to phosgene, COCl_2 , a poisonous gas

Products and conversions

- 2023 Q3 — isopropyl alcohol + PCl_5 , then AgCN . Isopropyl chloride, then isopropyl isocyanide $(\text{CH}_3)_2\text{CHNC}$
- 2023 Q3 — 1-chloropropane + alcoholic KOH , then HBr . Propene, then 2-bromopropane (Markovnikov)
- 2020 Q9(a) — convert but-1-ene to 1-iodobutane. HBr /peroxide $\rightarrow$ 1-bromobutane, then NaI /acetone (Finkelstein)
- 2025 Q12(a) — convert chlorobenzene to biphenyl. Na /dry ether — Fittig reaction
- 2025 Q12(c) — convert 2-bromobutane to but-2-ene. Alcoholic KOH ; but-2-ene is major by Saytzeff
- 2025 Q20(a) — methyl bromide + AgF . Methyl fluoride — Swarts reaction
- 2025 Q20(b) — 2,4,6-trinitrochlorobenzene, hydrolysed. 2,4,6-trinitrophenol, i.e. picric acid
- 2026 Q25(a) — chlorobenzene + conc. H_2SO_4 , heat: major product. 4-chlorobenzenesulphonic acid
- 2026 Q25(b) — bromobenzene + CH_3Cl / anhydrous AlCl_3 : major product. 1-bromo-4-methylbenzene (para major)
- 2025 Q19(a) — a nitro-substituted ethylbenzene heated with Br_2 : major product. Bromination at the benzylic carbon — that free radical is resonance-stabilised
- 2026 Q25(b) — dehydrohalogenation of 1-bromo-1-methylcyclohexane. 1-methylcyclohexene — the more substituted alkene, by Saytzeff

Naming, structure and identification

- 2026 Q2(a) — draw 4-bromo-3-methylpent-2-ene. $\text{CH}_3-\text{CH}=\text{C}(\text{CH}_3)-\text{CHBr}-\text{CH}_3$
- 2024 Q11(a) — IUPAC name of a four-carbon chain with a double bond and Cl at the far end. 4-chlorobut-1-ene — the double bond takes numbering priority
- 2017 Q14 — IUPAC names of $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$, 1,3-dibromobenzene, and a chloropropene. 2-bromobutane; 1,3-dibromobenzene; 3-chloroprop-1-ene
- 2026 Q23(a) — a hydrocarbon C_5H_{12} gives only one monochlorination product. Identify it. 2,2-dimethylpropane — all twelve hydrogens are equivalent

- 2026 Q10(b) — which $\text{C}_4\text{H}_9\text{Br}$ isomer gives 2,5-dimethylhexane with Na/dry ether? 1-bromo-2-methylpropane — cut the product in half
- 2026 Q24(a) — halogen present in $\text{CH}_2=\text{CH}-\text{Cl}$? Vinylic — chlorine sits directly on a doubly-bonded carbon
- 2017 Q5(b) — which of the pair is the allylic halide? The one where Cl is on an sp^3 carbon *next to* the double bond, not on it
- 2026 Q17(b) — which of a pair is chiral? The one whose carbon carries four *different* groups; two identical groups anywhere and it isn't
- 2026 Q24(b) — define ambident nucleophile and racemic mixture. Two donor atoms, attacks through either (CN^- , NO_2^-); and a 1:1 mixture of enantiomers, optically inactive because the rotations cancel

Built from Sourabh Raina's Haloalkanes and Haloarenes one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 6 (Rationalised 2022–23). NCERT spellings used throughout: **Zaitsev** (NCERT notes it is "also pronounced as Saytzeff"), **ambident** nucleophile, **vicinal** and **geminal** dihalides, **laevorotatory**. C–Cl bond lengths (177 pm in haloalkanes, 169 pm in haloarenes) are NCERT's. First-contact material: every term marked **[exposure]** is explained from scratch before use.

Two stretches of both transcripts were lost at automatic-transcription chunk boundaries and were recovered by re-transcribing those windows from the source audio: the S_N2 mechanism and ambident-nucleophile block in the lecture, and PYQ questions 12, 13, 17, 18, 22 and 23. All twenty-five PYQ questions are accounted for here.

Alcohols, Phenols and Ethers

NCERT Class XII Chemistry • Chapter 7 • Alcohols, Phenols and Ethers

6 marks in the board paper (CBSE 2026-27 curriculum). Three families of compounds, all built round one oxygen atom: an alcohol has an -OH on a carbon chain, a phenol has an -OH on a benzene ring, an ether has the oxygen sitting between two carbon groups. Most of the chapter is one idea used again and again — what the oxygen does with its lone pairs, and which of its two bonds (O-H or C-O) breaks. Every new word is tagged [exposure] and explained from scratch. The reactions come in a fixed list; learn the reagent that goes with each and the marks follow.

What the three families are, and where they matter — one oxygen atom, three different places to put it

[exposure] A **hydrocarbon** is a compound of only carbon and hydrogen (methane CH_4 , benzene C_6H_6). Replacing a hydrogen by another atom or group gives a new compound with new properties. The group that carries the property is called the **functional group**: it is the part of the molecule where the chemistry happens. In this chapter the functional group is built on oxygen.

[exposure] The **hydroxyl group** is -OH : an oxygen bonded to a hydrogen, and to the rest of the molecule through the oxygen.

- **Alcohol** — an -OH on a carbon of an **aliphatic** system (a chain or a non-aromatic ring of carbons). Example: methanol CH_3OH , ethanol $\text{C}_2\text{H}_5\text{OH}$.
- **Phenol** — an -OH on a carbon of an **aromatic** system (a benzene ring). Example: phenol itself, $\text{C}_6\text{H}_5\text{OH}$.
- **Ether** — the hydrogen of the -OH is replaced by an alkyl or aryl group, so the oxygen links two carbon groups: R-O-R' . Example: methoxymethane CH_3OCH_3 . You can picture an ether as an alcohol or phenol whose hydroxyl hydrogen has been swapped for a carbon group.

Why the chapter matters outside the exam: alcohols, phenols and ethers are the starting points for **detergents**, **antiseptics** and **fragrances** respectively. Ethanol is the spirit used to polish wood; the sugar in food, the cotton in cloth and the paper in books all contain many -OH groups.

Examiner asks: nothing directly from this opening, but "alcohol or phenol?" depends on whether the carbon holding the -OH is in a ring of benzene type or not, and that decides almost every property later.

Classification — count the -OH groups; then look at the carbon that carries them

[exposure] **Monohydric, dihydric, trihydric, polyhydric** only say how many -OH groups a molecule has: one, two, three, or many. This applies to alcohols and to phenols.

Number of -OH	Alcohol example	Phenol example
Monohydric (1)	ethanol $\text{CH}_3\text{CH}_2\text{OH}$	phenol $\text{C}_6\text{H}_5\text{OH}$
Dihydric (2)	ethane-1,2-diol $\text{HOCH}_2\text{CH}_2\text{OH}$	catechol, resorcinol, hydroquinone
Trihydric (3)	glycerol (propane-1,2,3-triol)	—

Classifying a monohydric alcohol by the carbon that carries -OH

[exposure] **Hybridisation** is the way a carbon atom mixes its orbitals before it bonds. A carbon with four single bonds is sp^3 hybridised; a carbon in a benzene ring or a C=C (three attached groups, one double bond) is sp^2 hybridised. The first rule: ask whether the -OH carbon is sp^3 or sp^2 .

The –OH is on an sp^3 carbon:

- **[exposure] Primary (1°)** alcohol: the –OH carbon is attached to **one** other carbon (or none, as in CH_3OH). **Secondary (2°)**: attached to **two** carbons. **Tertiary (3°)**: attached to **three** carbons. Examples: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (1°), $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ (2°), $(\text{CH}_3)_3\text{C}-\text{OH}$ (3°). This count decides how an alcohol oxidises, dehydrates and reacts with HCl, so you will use it all chapter.
- **[exposure] Allylic alcohol**: the sp^3 carbon bearing –OH is directly next to a $\text{C}=\text{C}$ double bond, as in $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$. **Benzylic alcohol**: that sp^3 carbon is directly next to a benzene ring, as in $\text{C}_6\text{H}_5-\text{CH}_2\text{OH}$. Allylic and benzylic alcohols can each be primary, secondary or tertiary.

The –OH is on an sp^2 carbon: the –OH sits on a carbon of a $\text{C}=\text{C}$, either an ordinary one (a **[exposure] vinylic alcohol**, $\text{CH}_2=\text{CH}-\text{OH}$) or a benzene-ring carbon (a phenol). These are a different class and behave very differently, as the acidity section shows.

Classifying ethers

[exposure] If the two groups on the oxygen are the same the ether is **simple (symmetrical)**: $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$, diethyl ether. If they differ it is **mixed (unsymmetrical)**: $\text{C}_2\text{H}_5\text{OCH}_3$ and $\text{C}_2\text{H}_5\text{OC}_6\text{H}_5$. This matters because the two kinds are made and cleaved differently.

NCERT intext 7.1 and 7.2 ask you to classify six drawn alcohols. NCERT's answers: primary (i), (ii), (iii); secondary (iv), (v); tertiary (vi); allylic (ii) and (vi). The method is always the same: find the –OH carbon, count the carbons attached to it, then check for a neighbouring $\text{C}=\text{C}$.

Examiner asks: "classify as primary, secondary or tertiary", and later "which of these is a tertiary alcohol?" in a reasoning question. Count the carbon neighbours of the –OH carbon.

Nomenclature — *common names are quick; IUPAC names carry the marks*

Alcohols

[exposure] IUPAC is the international system for naming compounds, so every compound has exactly one correct name. For an alcohol: take the longest carbon chain that **contains the carbon bearing –OH**, name the matching alkane, **drop its final "e" and add "ol"**. Number from the end **nearest to the –OH**, and write the position of –OH and of every other group.

- Common name: name the alkyl group, then the word **alcohol**: CH_3OH methyl alcohol; $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ n-propyl alcohol.
- Two or more –OH groups: **keep the "e"** of the alkane and add **diol, triol** with locants: $\text{HOCH}_2\text{CH}_2\text{OH}$ = **ethane-1,2-diol** (ethylene glycol). Glycerol is **propane-1,2,3-triol**.
- Cyclic alcohols: prefix **cyclo**, with the carbon bearing –OH as C-1: cyclohexanol; 2-methylcyclopentanol.

Compound	Common name	IUPAC name
CH_3OH	methyl alcohol	methanol
$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	n-propyl alcohol	propan-1-ol
$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	isopropyl alcohol	propan-2-ol
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$	n-butyl alcohol	butan-1-ol
$\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	sec-butyl alcohol	butan-2-ol
$(\text{CH}_3)_2\text{CHCH}_2\text{OH}$	isobutyl alcohol	2-methylpropan-1-ol
$(\text{CH}_3)_3\text{C}-\text{OH}$	tert-butyl alcohol	2-methylpropan-2-ol
$\text{HOCH}_2\text{CH}_2\text{OH}$	ethylene glycol	ethane-1,2-diol

Worked example · NCERT Example 7.1(i)

Name $\text{CH}_3-\text{CH}(\text{Cl})-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)-\text{CH}_2\text{OH}$.

- Longest chain containing the $-\text{OH}$ carbon: five carbons, so **pentane** → **pentan-1-ol** (the CH_2OH end is C-1, the end nearest $-\text{OH}$).
- Substituents: CH_3 on C-2 and C-3, Cl on C-4.
- Write them alphabetically with the parent: chloro before methyl.

Answer: **4-chloro-2,3-dimethylpentan-1-ol**.

Phenols

The simplest hydroxy derivative of benzene is **phenol**; that is its common name and an accepted IUPAC name. In substituted phenols the ring carbon bearing $-\text{OH}$ is C-1, and the positions are called **ortho** (neighbour, 1,2), **meta** (1,3) and **para** (opposite, 1,4). Methylphenols are **cresols**: o-cresol = 2-methylphenol; m-cresol = 3-methylphenol; p-cresol = 4-methylphenol. The dihydroxybenzenes are **catechol** (benzene-1,2-diol), **resorcinol** (benzene-1,3-diol) and **hydroquinone** or quinol (benzene-1,4-diol). Example: 2,6-dimethylphenol.

Ethers

- Common name:** the two groups in alphabetical order as separate words, then **ether**: $\text{CH}_3\text{OC}_2\text{H}_5$ ethyl methyl ether. Identical groups: **di-**: $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ diethyl ether.
- IUPAC name:** treat the ether as a hydrocarbon carrying an **alkoxy** group $-\text{OR}$ (or aryloxy $-\text{OAr}$). The **larger group** is the parent hydrocarbon; the smaller one becomes "methoxy", "ethoxy" and so on.

Compound	Common name	IUPAC name
CH_3OCH_3	dimethyl ether	methoxymethane
$\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$	diethyl ether	ethoxyethane
$\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$	methyl n-propyl ether	1-methoxypropane
$\text{C}_6\text{H}_5\text{OCH}_3$	methyl phenyl ether (anisole)	methoxybenzene
$\text{C}_6\text{H}_5\text{OC}_2\text{H}_5$	ethyl phenyl ether (phenetole)	ethoxybenzene
$\text{CH}_3-\text{CH}(\text{CH}_3)-\text{O}-\text{CH}_3$	methyl isopropyl ether	2-methoxypropane
$\text{CH}_3\text{OCH}_2\text{CH}_2\text{OCH}_3$	—	1,2-dimethoxyethane

Other NCERT answers worth knowing: $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{O}-\text{CH}_2\text{CH}_3$ is **2-ethoxypropane** (Example 7.1(ii)); the ring compound with OC_2H_5 and NO_2 on neighbouring carbons is **1-ethoxy-2-nitrocyclohexane**; intertext 7.3 answers are 4-chloro-3-ethyl-2-(1-methylethyl)butan-1-ol, 2,5-dimethylhexane-1,3-diol, 3-bromocyclohexanol, hex-1-en-3-ol and 2-bromo-3-methylbut-2-en-1-ol.

Trap: when an alcohol also has a double bond, the $-\text{OH}$ gets the lowest number, even if the double bond then gets a high one. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}=\text{CH}_2$ is **hex-5-en-2-ol** (numbered from the $-\text{OH}$ end), not hex-1-en-5-ol. Both "en" and "ol" appear in the name, with the "e" of "ene" dropped before the vowel of "ol".

Examiner asks: "write the IUPAC name of the following" — a standing 2-marker (for example 3-phenylprop-2-en-1-ol, 2017; 2-methoxy-2-methylpropane, 2019). See the PYQ section.

Structure of the functional groups — *why the bond angles and bond lengths come out as they do*

[exposure] A **sigma (σ) bond** is a single bond formed by head-on overlap of two orbitals. In an alcohol the oxygen of $-\text{OH}$ is sp^3 hybridised and joins the sp^3 carbon through a sigma bond. **[exposure]** A **lone pair** is a pair of valence electrons on an atom that is not used in bonding. Oxygen has two lone pairs, and they are what make alcohols, phenols and ethers react.

NCERT Fig. 7.1 gives the numbers to remember:

Compound	Bond angle	C–O bond length
Methanol CH_3OH	C–O–H = 108.9°	142 pm
Phenol $\text{C}_6\text{H}_5\text{OH}$	C–O–H = 109°	136 pm
Methoxymethane CH_3OCH_3	C–O–C = 111.7°	141 pm

- **Alcohols: angle slightly below the tetrahedral 109.5° (109°28').** The two lone pairs on oxygen repel the bonded pairs and squeeze the C–O–H angle. Lone pair against bond is a stronger repulsion than bond against bond.
- **Phenol: C–O is shorter (136 pm against 142 pm in methanol).** Two reasons: (i) the oxygen's lone pair is shared with the benzene ring (**[exposure] conjugation:** a lone pair next to a ring spreads onto it), which gives the C–O bond **partial double-bond character**; (ii) the carbon is sp^2 hybridised, and an sp^2 carbon holds its bonds a little closer than an sp^3 carbon.
- **Ethers: angle slightly above tetrahedral (111.7°).** The two bulky alkyl groups on oxygen repel each other and push the angle open. The C–O length (141 pm) is about the same as in alcohols.

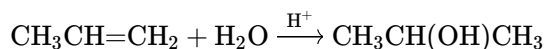
Examiner asks: two reasons that come up directly: "why is the C–O bond in phenol shorter than in methanol?" (2023) and "why is the C–O–C angle in ethers slightly greater than the tetrahedral angle?" (2025). Write the reason, not just the numbers.

Preparing alcohols from alkenes — *two ways to add water across a C=C — one follows Markovnikov, one does the opposite*

[exposure] An **alkene** has a $\text{C}=\text{C}$ double bond. **Hydration** means adding water ($\text{H}-\text{OH}$) across it, so one carbon takes an H and the other takes an OH; the product is an alcohol. **[exposure]** **Markovnikov's rule** says that when a reagent $\text{H}-\text{X}$ adds to an **unsymmetrical** alkene (the two carbons of $\text{C}=\text{C}$ carry different groups), the H goes to the carbon that already has **more** hydrogens and the negative part goes to the carbon with **fewer** hydrogens. You met it in haloalkanes.

1. Acid-catalysed hydration (Markovnikov)

An alkene and water in the presence of acid (H^+) as a **[exposure] catalyst** (a substance that speeds up a reaction and is given back unchanged at the end).



Here OH goes to the carbon with fewer hydrogens, giving **propan-2-ol**.

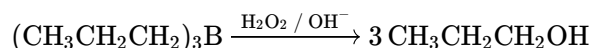
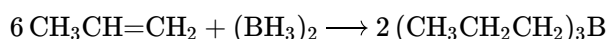
Mechanism, three steps (keep going until the stable product forms):

1. **Protonation of the alkene to a carbocation, by electrophilic attack of H_3O^+** . First the acid makes the hydronium ion: $\text{H}_2\text{O} + \text{H}^+ \longrightarrow \text{H}_3\text{O}^+$. **[exposure]** An **electrophile** is a species that is short of electrons and attacks electron-rich places; here the H^+ attacks the π electrons of the double bond. A **[exposure] carbocation** is a carbon that has lost a bonding pair and carries a positive charge, with only three bonds. $\text{CH}_2=\text{CH}_2 + \text{H}_3\text{O}^+ \longrightarrow \text{CH}_3-\text{CH}_2^+ + \text{H}_2\text{O}$.
2. **Nucleophilic attack of water on the carbocation**. A **[exposure] nucleophile** is an electron-rich species with a lone pair that attacks positive carbon. The oxygen of a second water molecule uses its lone pair to bond to the positive carbon: $\text{CH}_3\text{CH}_2^+ + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{CH}_2-\text{OH}_2^+$. The oxygen now carries a positive charge.
3. **Deprotonation to the alcohol**. A third water molecule takes the H^+ from that oxygen, giving $\text{CH}_3\text{CH}_2\text{OH}$ and regenerating H_3O^+ .

The acid is given back at the end, which is why it counts as a catalyst. A tertiary carbocation is easier to form than a secondary one, so the rule "hydrogen to the carbon with more hydrogens" is really a rule about forming the more stable carbocation.

2. Hydroboration–oxidation (anti-Markovnikov)

[exposure] Diborane is $(\text{BH}_3)_2$, behaving like two molecules of BH_3 . It adds to an alkene to give a **trialkylborane** R_3B (three alkene molecules add to one boron). Treating this with hydrogen peroxide and aqueous sodium hydroxide **oxidises** it to an alcohol.



The boron attaches to the sp^2 carbon that carries **more hydrogens**, so the alcohol looks as if water had added **opposite to Markovnikov's rule**: propene gives **propan-1-ol** here but propan-2-ol by acid hydration. The yield is excellent. The method was reported by H. C. Brown in 1959; he shared the 1979 Nobel Prize in Chemistry for it.

Trap: "propene to propan-1-ol" needs **hydroboration–oxidation** (B_2H_6 , then $\text{H}_2\text{O}_2/\text{OH}^-$); "propene to propan-2-ol" is **acid-catalysed hydration** (H_3O^+). A question that names only the product decides the reagent for you — check which carbon the OH is on.

Examiner asks: the mechanism of hydration of ethene (2017, asked twice), and the product of hydroboration–oxidation of propene (2020). The mechanism is the three steps above; write each step's name, then the species.

Preparing alcohols from carbonyl compounds and Grignard reagents — *reduction goes down from C=O; Grignard builds the carbon chain*

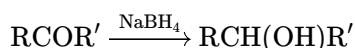
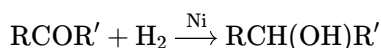
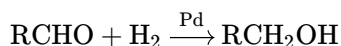
[exposure] A carbonyl compound contains the **carbonyl group** C=O: an **aldehyde** RCHO, a **ketone** RCOR', a **carboxylic acid** RCOOH, or an **ester** RCOOR'. These are the next chapter's topic, so here you only need what they become. **[exposure]** **Reduction** of an organic compound means adding hydrogen to it (or removing oxygen).

1. Reducing aldehydes and ketones

Two kinds of reducing method:

- **Catalytic hydrogenation:** add H₂ over a finely divided metal catalyst (platinum, palladium or nickel).
- **Chemical reducing agents:** sodium borohydride NaBH₄ or lithium aluminium hydride LiAlH₄.

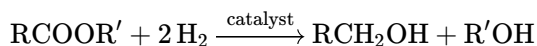
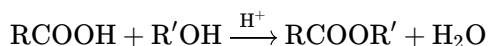
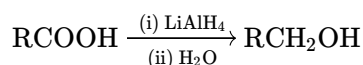
An **aldehyde** gives a **primary** alcohol; a **ketone** gives a **secondary** alcohol.



[exposure] NaBH₄ is a **mild reducing agent**: it reduces aldehydes and ketones but **not** carboxylic acids, esters or amides. LiAlH₄ is a strong reducing agent and reduces all of them. This difference is a popular exam trap.

2. Reducing carboxylic acids and esters

Acids are reduced to primary alcohols, in excellent yield, by LiAlH₄ (then water). LiAlH₄ is **expensive**, so it is used only for special chemicals. To make an alcohol in bulk, first turn the acid into an **ester** (reacting it with an alcohol under acid), then reduce the ester by catalytic hydrogenation. That also hands back the alcohol you used.



Example: ethanoic acid with ethanol gives ethyl ethanoate; hydrogenation over platinum gives **two molecules of ethanol**, one of which was the alcohol you started with.

3. Grignard reagents with aldehydes and ketones

[exposure] A **Grignard reagent** is RMgX (X = Cl, Br, I): an alkyl or aryl group bonded to magnesium. Magnesium is a metal (electropositive), so the carbon on it carries a partial negative charge (δ[−]) and behaves as a **nucleophile**. The C=O carbon carries a partial positive charge (δ⁺) because oxygen is more electronegative and pulls the electrons. So the Grignard carbon attacks the carbonyl carbon.

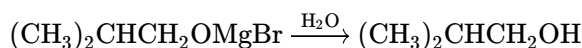
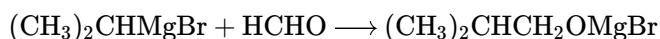
1. **Nucleophilic addition** gives an **[exposure] adduct** (the addition product): the π electrons of $C=O$ move onto oxygen, which takes MgX and becomes $-O-MgX$.
2. **Hydrolysis** (adding water or dilute acid) replaces MgX by H and gives the alcohol, with $Mg(OH)X$ as by-product.

Carbonyl compound	Alcohol class	Equation
Methanal $HCHO$	primary	$HCHO + RMgX \longrightarrow RCH_2OMgX \xrightarrow{H_2O} RCH_2OH$
Any other aldehyde	secondary	$R'CHO + RMgX \longrightarrow R'CH(R)OMgX \xrightarrow{H_2O} R'CH(OH)R$
A ketone	tertiary	$R'COR'' + RMgX \longrightarrow R'R''C(R)OMgX \xrightarrow{H_2O} R'R''C(R)OH$

Remember this table; it is a standing question. It also tells you how to **design** a Grignard route: the R on the Grignard reagent is the group that ends up on the carbon that carries the $-OH$.

Worked example · NCERT intext 7.4 — make 2-methylpropan-1-ol and cyclohexylmethanol from methanal

1. Both alcohols are primary and end in $-CH_2OH$, so the CH_2OH comes from **methanal**.
2. The rest of the molecule is the group that sat on magnesium. For $(CH_3)_2CH-CH_2OH$ the group is isopropyl: Grignard reagent $(CH_3)_2CH-MgBr$.
3. For cyclohexylmethanol it is cyclohexyl: $C_6H_{11}-MgBr$.



Worked example · NCERT Example 7.2 — products of three preparations

1. Catalytic reduction of butanal $CH_3CH_2CH_2CHO$: an aldehyde gives a primary alcohol, **butan-1-ol**.
2. Hydration of propene with dilute H_2SO_4 : Markovnikov, **propan-2-ol**.
3. Propanone CH_3COCH_3 with CH_3MgBr then water: a ketone gives a tertiary alcohol, $(CH_3)_2C(OH)CH_3$ = **2-methylpropan-2-ol**.

Worked example · NCERT intext 7.5 — products with H_3O^+ and with $NaBH_4$

1. Propene with H_3O^+ : acid hydration, **propan-2-ol**.
2. The cyclohexanone that has a CH_2COOCH_3 side chain, with $NaBH_4$: the ketone becomes a secondary alcohol and the **ester stays untouched** (mild reagent).
3. 2-Methylbutanal $CH_3CH_2CH(CH_3)CHO$ with $NaBH_4$: aldehyde to primary alcohol, **2-methylbutan-1-ol**.

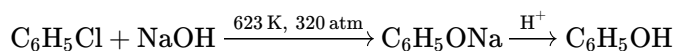
Examiner asks: "write the product of (a ketone or ester compound) with $NaBH_4$ " (2018); "a Grignard reagent with an aldehyde or ketone, then hydrolysis" (2024); "convert propanone to 2-methylpropan-2-ol" and "convert butan-2-one to butan-2-ol". Use the table above and name which reagent leaves the ester alone.

Preparing phenols — four routes — three lab, one industrial

Phenol is also called **carbolic acid**. It was first isolated in the early nineteenth century from **coal tar** (the sticky by-product of heating coal). Today it is made synthetically, and most of the world's phenol comes from cumene.

1. From a haloarene (chlorobenzene)

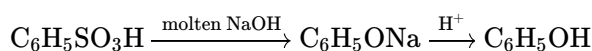
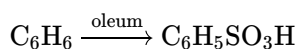
Chlorobenzene is **fused** with sodium hydroxide (heated molten with it) at **623 K and 320 atm** pressure. This gives **sodium phenoxide** $\text{C}_6\text{H}_5\text{ONa}$, a salt; acidifying it releases phenol.



2. From benzenesulphonic acid

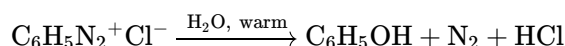
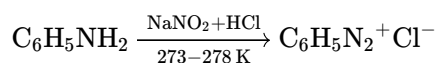
[exposure] Oleum is fuming sulphuric acid, $\text{H}_2\text{S}_2\text{O}_7$ (concentrated H_2SO_4 with dissolved SO_3).

Sulphonation puts an $-\text{SO}_3\text{H}$ group on the ring. Benzene is sulphonated with oleum to give **benzenesulphonic acid**; heated with **molten sodium hydroxide** this becomes sodium phenoxide, and acidification gives phenol.



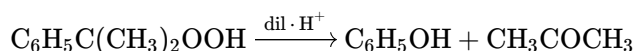
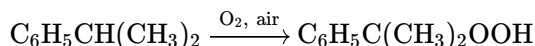
3. From a diazonium salt

[exposure] A **diazonium salt** ArN_2^+X^- is made by treating an **aromatic primary amine** (an $-\text{NH}_2$ directly on a benzene ring, such as aniline) with **nitrous acid**, which is made in the flask from $\text{NaNO}_2 + \text{HCl}$, at **273–278 K**. The $\text{C}-\text{N}_2^+$ bond is weak, so the N_2^+ is easy to displace. **Warming with water** (or treating with dilute acid) hydrolyses it to phenol and releases nitrogen gas.



4. From cumene (the industrial route)

[exposure] **Cumene** is isopropylbenzene, $\text{C}_6\text{H}_5-\text{CH}(\text{CH}_3)_2$: a benzene ring on the middle carbon of propane. It is **oxidised by the oxygen of air** to **cumene hydroperoxide**, in which the benzylic carbon carries an $-\text{O}-\text{O}-\text{H}$ group. Treating that with **dilute acid** splits it into **phenol and acetone**.



This is the preferred commercial method for two reasons: it starts from a cheap hydrocarbon, and the by-product **acetone** is itself an important compound obtained in large quantity. Two useful products from one process.

Examiner asks: "preparation of phenol from cumene" (2025), "convert chlorobenzene to phenol" (2019) and "convert aniline to phenol" (year not stated). Write the reagent and condition over each arrow. The chlorobenzene route needs the high temperature and pressure.

Physical properties: boiling point and solubility — *one explanation — hydrogen bonding through the O–H — covers both*

Alcohols and phenols have two parts: an alkyl or aryl group, and the hydroxyl. The properties come chiefly from the –OH; the carbon group only modifies them.

[exposure] A **hydrogen bond** is an attraction between a hydrogen that is bonded to a very electronegative atom (here oxygen) and a lone pair on a neighbouring oxygen. The oxygen pulls the shared electrons, so the oxygen carries δ^- and the hydrogen carries δ^+ , and the δ^+ hydrogen of one molecule is attracted to the δ^- oxygen of the next. **Intermolecular** means *between* separate molecules; **intramolecular** means *inside* one molecule. It is much stronger than the ordinary **van der Waals forces**, the weak attractions that exist between any two molecules and grow with surface area.

Boiling point

- **Rises with the number of carbons:** a bigger molecule has more surface contact and larger van der Waals forces.
- **Falls with branching:** a branched molecule is more compact, so less surface area and weaker van der Waals forces. Butan-2-ol boils lower than butan-1-ol.
- **Alcohols and phenols boil much higher than hydrocarbons, ethers, haloalkanes and haloarenes of similar molar mass**, because the –OH forms **intermolecular hydrogen bonds** that those others lack. Boiling needs the molecules to be pulled apart, so these bonds must be broken first.

Compound	Type	Molar mass	b.p. (K)
Ethanol C ₂ H ₅ OH	alcohol	46	351
Methoxymethane CH ₃ OCH ₃	ether	46	248
Propane C ₃ H ₈	hydrocarbon	44	231

Same size, three very different boiling points; only the alcohol can hydrogen-bond *to itself*.

Solubility in water

Alcohols and phenols dissolve in water because they form hydrogen bonds with water. The –OH hydrogen is attracted to a water oxygen, and a water hydrogen to the –OH oxygen. The carbon group is **[exposure] hydrophobic** ("water-hating": non-polar, it does not attract water and gets in the way of hydrogen bonding), so **solubility falls as the alkyl or aryl group gets bigger**. **[exposure]** Two liquids are **miscible** if they mix in all proportions; methanol and ethanol are miscible with water in all proportions, while liquids that do not mix are **immiscible**.

Worked example · NCERT Example 7.3 — arrange in increasing boiling point

(a) pentan-1-ol, butan-1-ol, butan-2-ol, ethanol, propan-1-ol, methanol. (b) pentan-1-ol, n-butane, pentanal, ethoxyethane.

1. (a) All are alcohols, so rank by carbon count, then branching: methanol (1 C), ethanol (2), propan-1-ol (3). Butan-2-ol and butan-1-ol both have 4 carbons; the branched butan-2-ol is lower. Then pentan-1-ol (5) is highest.
2. (b) n-Butane is non-polar, only weak van der Waals forces: lowest. Ethoxyethane is an ether, weakly polar. Pentanal is an aldehyde with a strongly polar C=O and stronger dipole–dipole attraction. Pentan-1-ol hydrogen-bonds: highest.

Answer: (a) methanol < ethanol < propan-1-ol < butan-2-ol < butan-1-ol < pentan-1-ol. (b) n-butane < ethoxyethane < pentanal < pentan-1-ol.

Examiner asks: "why does propanol have a higher boiling point than butane?" (2019), the boiling-point arrangement of four compounds (ethoxyethane, butanal, butanol, butane; 2023). NCERT exercise 7.22 asks the same thing of ethanol against methoxymethane. The fixed answer: hydrogen bonding.

Two ways an alcohol reacts — *which bond breaks sorts the whole reaction list*

An alcohol is **versatile**: it can behave as a nucleophile and also, once protonated, as an electrophile.

- **As a nucleophile:** the oxygen has lone pairs, so it is electron-rich. It attacks a positive carbon, and in the process the **O–H bond breaks** (the hydrogen leaves as H^+).
- **As an electrophile** (only after it picks up H^+): the oxygen takes a proton from an acid and becomes a **[exposure] protonated alcohol** ROH_2^+ , also called an **alkyloxonium ion**. The positive oxygen drags electrons away from the carbon, so the carbon now carries $\delta+$ and a nucleophile can attack it; water leaves as the leaving group, so the **C–O bond breaks**. A plain alcohol does not act as an electrophile; only the protonated one does.

On that basis the reactions fall into two lists, and the rest of the chapter follows it:

O–H bond breaks	C–O bond breaks (alcohols only)
Reaction with metals (acidity)	
Reaction with NaOH (phenols only)	
Esterification and acetylation	Reaction with HX and the Lucas test
Reaction with phosphorus halides and $SOCl_2$	
Dehydration to alkenes	
(Oxidation breaks O–H and C–H, a separate case)	

Phenols show C–O cleavage only with zinc dust, because the carbon of a phenol is sp^2 and the C–O bond is strong (partial double-bond character).

Examiner asks: nothing separately, but it organises every answer: before writing a mechanism ask "which bond is breaking?"

Acidity of alcohols and phenols — *proton donors; the weaker the conjugate base's grip on a negative charge, the stronger the acid*

[exposure] A **Brønsted acid** is a species that **donates a proton** (H^+) to a base. The **conjugate base** is what is left after it gives up the proton: ROH gives RO^- . A **stronger acid** gives up its proton more easily, which happens when the conjugate base is **more stable**. Two things make an acid strong: the O–H bond is more polar (so the hydrogen leaves easily), and the negative charge left on oxygen is spread out and stabilised.

Reactions that show the acidity

- **With active metals** (sodium, potassium, aluminium: metals that displace hydrogen): alcohols give **alkoxides**, phenols give **phenoxides**, and hydrogen gas escapes. $2 ROH + 2 Na \longrightarrow 2 RONa + H_2$; $2 C_2H_5OH + 2 Na \longrightarrow 2 C_2H_5ONa + H_2$ (sodium ethoxide); $2 C_6H_5OH + 2 Na \longrightarrow 2 C_6H_5ONa + H_2$.
- **With aqueous NaOH — phenols only:** $C_6H_5OH + NaOH \longrightarrow C_6H_5ONa + H_2O$, an acid-plus-base neutralisation to a salt and water. Alcohols do **not** do this, which is the evidence that **phenols are more acidic than alcohols and water**.

Acidity of alcohols: weak, and the carbon group makes it weaker

The acidity of an alcohol comes from the polar O–H bond: oxygen is electronegative, so oxygen is δ^- and hydrogen δ^+ , and the electrons are held towards oxygen. An alkyl group is an **[exposure] electron-releasing** group (it pushes electron density towards the atom it is attached to; this is the +I effect). It pushes electrons *onto the oxygen*, which lowers the polarity of O–H, so the hydrogen leaves less easily. More alkyl groups, less acidity:

Acid strength, strongest first: methanol > primary > secondary > tertiary alcohols.

A primary alcohol has one pushing alkyl group, a secondary has two and a tertiary three. Alcohols are also **weaker acids than water**: water has no alkyl group to push electrons. Proof: an alkoxide takes a proton from water, $\text{RO}^- + \text{H}_2\text{O} \longrightarrow \text{ROH} + \text{OH}^-$, so water is the better proton donor, and an **alkoxide ion is a stronger base than hydroxide** (sodium ethoxide is a stronger base than sodium hydroxide). Alcohols are also **Brønsted bases** as well as acids: the lone pairs on oxygen let them *accept* a proton.

Acidity of phenol: why it beats alcohols and water

[exposure] Resonance is the delocalisation of electrons over several atoms, so that the true structure is a blend of several drawings (**resonance structures**) that differ only in where the electrons sit. A charge spread over several atoms is more stable than one stuck on a single atom. Phenol is acidic for two reasons.

1. **The –OH carbon is sp^2 , not sp^3 .** An sp^2 orbital has 33% s-character against 25% for sp^3 , and more s-character means the carbon holds its electrons more tightly (**more electronegative**). So in phenol the carbon pulls electron density off the oxygen, the O–H bond becomes more polar, and the proton leaves more easily. In an alcohol the alkyl carbon is less electronegative and also pushes electrons onto oxygen.
2. **Phenoxide ion is stabilised by resonance; alkoxide ion is not.** In RO^- the whole negative charge is stuck on one oxygen atom — concentrated charge, unstable. In $\text{C}_6\text{H}_5\text{O}^-$ the negative charge spreads from oxygen onto the ortho and para carbons of the ring (five resonance structures), so it is spread out and the ion is far more stable. A more stable conjugate base means a stronger acid.

Phenol itself also has resonance structures, but they all have **charge separation** (a plus on oxygen and a minus on a ring carbon), which is destabilising; so the phenoxide ion gains much more from resonance than phenol does. That difference of stability is what drives ionisation. Phenol is about a million times more acidic than ethanol (pK_a 10.0 against 15.9).

Substituted phenols

[exposure] pK_a is a number that measures acid strength: **the smaller the pK_a , the stronger the acid** ($\text{pK}_a = -\log K_a$). **[exposure] An electron-withdrawing group** (such as $-\text{NO}_2$) pulls electron density towards itself; an electron-releasing group (such as $-\text{CH}_3$) pushes it away.

- **Electron-withdrawing groups ($-\text{NO}_2$) increase acidity**, most strongly at **ortho and para**, because the negative charge of the phenoxide ion can be delocalised right onto the nitro group.
- **Electron-releasing groups ($-\text{CH}_3$, alkyl) decrease acidity**: cresols are less acidic than phenol.

Compound	pK_a	Compound	pK_a
o-Nitrophenol	7.2	o-Cresol	10.2
m-Nitrophenol	8.3	m-Cresol	10.1
p-Nitrophenol	7.1	p-Cresol	10.2
Phenol	10.0	Ethanol	15.9

Read the table: p- and o-nitrophenol (7.1, 7.2) are stronger acids than m-nitrophenol (8.3), because the meta nitro cannot take the negative charge by resonance. All three nitrophenols are stronger than phenol (10.0), and the cresols (about 10.2) are weaker.

Worked example · NCERT intext 7.8 — why o- and p-nitrophenol are more acidic than phenol

1. Remove H^+ from o- or p-nitrophenol to get the nitrophenoxide ion. As in phenoxide, the negative charge on oxygen moves onto the ring carbons ortho and para to the oxygen.
2. The nitro group is itself at an ortho or para carbon. Charge arriving there can move on into the $\text{N}=\text{O}$ part of the nitro group, ending up on its oxygen atoms. That is two more resonance structures than plain phenoxide has.
3. The charge is delocalised over the ring **and** onto the nitro group; phenoxide has only the ring. A more stable conjugate base means a stronger acid.

Worked example · NCERT Example 7.4 — increasing acid strength

Propan-1-ol, 2,4,6-trinitrophenol, 3-nitrophenol, 3,5-dinitrophenol, phenol, 4-methylphenol.

1. The alcohol is weakest of all (no resonance in its conjugate base).
2. Among the phenols, the methyl (electron-releasing) is weaker than phenol itself.
3. Then nitrophenols, in order of the number of nitro groups: one, two, three. 2,4,6-Trinitrophenol (picric acid) has three nitro groups, two of them at ortho and para, so it is the strongest.

Answer: propan-1-ol < 4-methylphenol < phenol < 3-nitrophenol < 3,5-dinitrophenol < 2,4,6-trinitrophenol.

Trap: an alcohol is **not** a stronger acid because "alkyl groups are bigger" — the order of alcohol acidity is primary > secondary > tertiary because alkyl groups push electrons *onto* the oxygen, which makes the proton harder to lose. And for phenols the nitro group helps most at ortho and para; at meta it helps less.

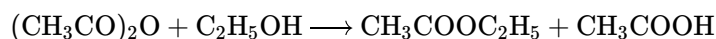
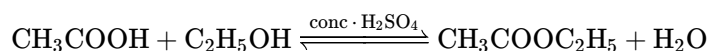
Examiner asks: "why is phenol more acidic than cyclohexanol" (2023) and "why is an alcohol less acidic than phenol" (2019): resonance-stabilised phenoxide against an unstabilised alkoxide; "arrange p-cresol, p-nitrophenol and phenol" (2017); "ortho and para nitrophenols are more acidic than phenol, draw the resonance structures". Always give the conjugate-base argument.

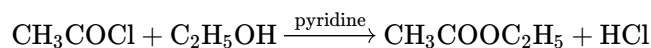
Esterification and acetylation — *the O–H bond breaks; the oxygen attacks an acid derivative*

[exposure] An **ester** is a compound RCOOR' made from a carboxylic acid and an alcohol (or phenol).

Esterification is the making of an ester. The group $\text{RCO}-$ (the acid with its $-\text{OH}$ removed) is called the **[exposure] acyl group**; it combines with the $\text{R}'\text{O}-$ part of the alcohol. Alcohols and phenols give esters with three kinds of acid derivative:

Reacts with	Conditions	By-product
Carboxylic acid RCOOH	a little conc. H_2SO_4 (catalyst, and it removes water)	H_2O
Acid anhydride $(\text{RCO})_2\text{O}$	a little conc. H_2SO_4	RCOOH
Acid chloride RCOCl	in the presence of a base, pyridine	HCl (taken up by the pyridine)

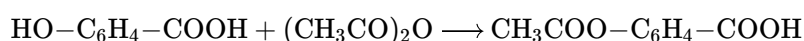




All three give **ethyl ethanoate**. **[exposure]** The reaction with the acid is **reversible** (it runs in both directions until an equilibrium is reached). To make more ester, **remove the water as soon as it forms**: by Le Chatelier's principle, taking away a product pushes the equilibrium to the right. With the acid chloride, the base neutralises the HCl that forms, which also shifts the equilibrium to the right.

Acetylation and aspirin

[exposure] **Acetylation** is putting an **acetyl** group $\text{CH}_3\text{CO}-$ onto the oxygen of an alcohol or phenol. Salicylic acid (2-hydroxybenzoic acid) has a phenolic $-\text{OH}$ and a $-\text{COOH}$. Acetic anhydride acetylates the $-\text{OH}$ and leaves the $-\text{COOH}$ alone: the product is **acetylsalicylic acid, aspirin**, with ethanoic acid as by-product.



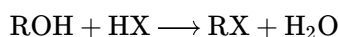
with CH_3COOH as the by-product.

Aspirin is used as a medicine: it is an **analgesic** (pain reliever), **anti-inflammatory** and **antipyretic** (fever reducer).

Examiner asks: acetylation of salicylic acid to give aspirin (2020, "name the product and its use"); why pyridine is used with an acid chloride. Say: it neutralises HCl and shifts the equilibrium forward.

Reactions of alcohols that break the C–O bond: halides and the Lucas test — *protonate the oxygen, let water leave, let a halide in*

1. With hydrogen halides (and the Lucas test)



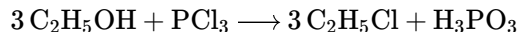
The alcohol is protonated (the oxygen takes H^+), water leaves, and the halide ion attacks the carbon. The three classes of alcohol react at **very different speeds**, and that is how they are told apart. **[exposure]** The **Lucas reagent** is a mixture of **concentrated HCl and anhydrous ZnCl_2** . All the alcohols start by dissolving in it. As the alcohol turns into an alkyl chloride, the chloride is **immiscible** (does not mix) with the solution, so it separates as tiny droplets and the solution goes cloudy. That cloudiness is **[exposure]** **turbidity**.

Alcohol class	What you see with Lucas reagent at room temperature
Tertiary	turbidity immediately : fastest
Secondary	turbidity after about five minutes
Primary	no turbidity at room temperature (it reacts only on heating)

Reactivity: **tertiary > secondary > primary**, because a tertiary alcohol forms its more stable carbocation fastest. A primary alcohol, such as propan-1-ol, gives no cloudiness in the cold; 2-methylpropan-2-ol gives it at once. That pair is a favourite question.

2. With phosphorus trihalides and thionyl chloride

Alcohols react with PCl_3 , PBr_3 and PI_3 to give alkyl halides; the phosphorus ends up as phosphorous acid H_3PO_3 . With **thionyl chloride** SOCl_2 the OH is replaced by Cl, and the by-products are both gases, SO_2 and HCl , which escape and make the chloride easy to isolate.



Worked example · NCERT intext 7.6 — butan-1-ol and 2-methylbutan-2-ol with (a) HCl/ZnCl_2 , (b) HBr , (c) SOCl_2

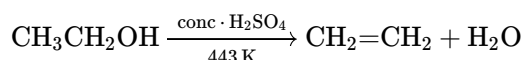
In every case $-\text{OH}$ is replaced by the halogen; the carbon skeleton does not change.

- Butan-1-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$: (a) 1-chlorobutane (slowly, primary); (b) **1-bromobutane**; (c) **1-chlorobutane** + SO_2 + HCl .
- 2-Methylbutan-2-ol, $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)_2\text{OH}$: (a) **2-chloro-2-methylbutane** (immediately, tertiary); (b) **2-bromo-2-methylbutane**; (c) 2-chloro-2-methylbutane.

Examiner asks: the product of butan-2-ol with SOCl_2 (2-chlorobutane, 2017); "distinguish 1-propanol from 2-methyl-2-propanol" (2019): Lucas test; HCl with a compound that has both a phenolic and a benzylic $-\text{OH}$ (2020, see PYQ).

Dehydration of alcohols to alkenes — *remove water from one molecule; the temperature and acid strength depend on the class*

[exposure] Dehydration means removing a molecule of water. Heated with a **[exposure] protic acid** (an acid that gives up H^+ : concentrated H_2SO_4 , or H_3PO_4), or over catalysts such as **anhydrous** ZnCl_2 or **alumina** Al_2O_3 , an alcohol loses OH from one carbon and H from the neighbouring carbon and gives an **alkene**.



The class of the alcohol sets how hard you must push. **Primary** needs strong acid and high temperature; **secondary and tertiary** dehydrate under **milder conditions**:

Alcohol	Conditions	Product
Ethanol (1°)	conc. H_2SO_4 , 443 K	ethene
Propan-2-ol (2°)	85% H_3PO_4 , 440 K	propene
tert-Butyl alcohol (3°)	20% H_3PO_4 , 358 K	2-methylpropene

Relative ease of dehydration: tertiary > secondary > primary. The reason is the same as in the Lucas test: the step that decides the speed forms a carbocation, and tertiary carbocations are the most stable and so form most readily.

Mechanism (ethanol to ethene), three steps

- Formation of protonated alcohol** (fast). The oxygen has lone pairs and takes H^+ from the acid:
 $\text{CH}_3\text{CH}_2\text{OH} + \text{H}^+ \longrightarrow \text{CH}_3\text{CH}_2-\text{OH}_2^+$ (an **ethyloxonium ion**; **[exposure]** an **oxonium ion** is an oxygen carrying a positive charge and three bonds).

- 2. Formation of carbocation (slowest step).** The positive oxygen pulls the bonding electrons towards itself, the C–O bond breaks and water leaves: $\text{CH}_3\text{CH}_2\text{--OH}_2^+ \longrightarrow \text{CH}_3\text{CH}_2^+ + \text{H}_2\text{O}$. Because it is the slowest step of the reaction, it is the **rate-determining step**: the overall speed is the speed of this step.
- 3. Formation of ethene by elimination of a proton.** The carbocation loses H^+ from the neighbouring carbon; the C–H electrons form the C=C: $\text{CH}_3\text{CH}_2^+ \longrightarrow \text{CH}_2=\text{CH}_2 + \text{H}^+$.

The acid used in step 1 is released in step 3, so it behaves as a catalyst. The reaction is an equilibrium, so to push it to the right **ethene is removed as it forms** (Le Chatelier's principle again).

Worked example · NCERT intext 7.7 — major product of acid-catalysed dehydration

- 1-Methylcyclohexanol:** a tertiary alcohol. OH leaves from the ring carbon that carries the methyl, and an H leaves from a neighbouring ring carbon: **1-methylcyclohexene**, the alkene with three alkyl groups on the double bond (the most stable).
- Butan-1-ol:** gives a **mixture of but-1-ene and but-2-ene**, with **but-2-ene as the major product**. It has two alkyl groups on the C=C and is more stable than but-1-ene; NCERT's explanation is that the primary carbocation **rearranges** (a hydride shift) to the more stable secondary carbocation before losing H^+ .

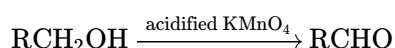
Trap: "which dehydrates faster, tert-butyl alcohol or n-butanol?" The answer is **tert-butyl alcohol**, and the argument must be about the **carbocation** (tertiary is more stable, so the slow step is faster) — not just "tertiary has more carbons". Write the protonation and the carbocation for both and compare.

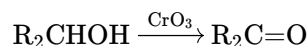
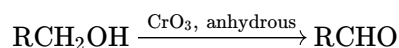
Examiner asks: the mechanism of dehydration of ethanol to ethene at 443 K (2019); "tert-butyl alcohol heated with 20% H_3PO_4 at 358 K" (2020); "which undergoes dehydration faster" (2020). Always give the three steps with their names.

Oxidation of alcohols — *primary, secondary and tertiary alcohols give three different outcomes — choose the reagent*

[exposure] In organic chemistry **oxidation** of an alcohol means forming a C=O bond at the carbon that carried the –OH, which needs **that carbon to have a hydrogen to lose**, together with the hydrogen of the O–H. It is also called **dehydrogenation** (loss of H_2). The oxidising agent decides how far a primary alcohol goes.

Alcohol	Oxidising agent	Product
Primary	Strong: acidified KMnO_4 or $\text{K}_2\text{Cr}_2\text{O}_7$	aldehyde first, which is oxidised further to a carboxylic acid with the same number of carbons
Primary	Mild: CrO_3 in anhydrous medium, or PCC	stops at the aldehyde
Secondary	CrO_3 (chromic anhydride), $\text{K}_2\text{Cr}_2\text{O}_7$, PCC	ketone with the same number of carbons
Tertiary	normal oxidants	no reaction





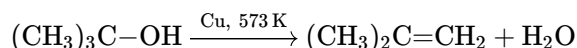
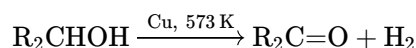
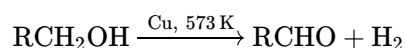
[exposure] PCC is pyridinium chlorochromate, a complex of chromium trioxide with pyridine and HCl. It is the **better reagent** for primary alcohol to aldehyde: good yield of aldehyde and it does not touch a C=C double bond in the molecule. Why do we need a mild reagent? Because an aldehyde is easier to oxidise than the alcohol; a strong oxidant would carry it on to the acid.

Tertiary alcohols resist oxidation

The carbon that carries –OH has no hydrogen, so no C=O can form without breaking a **C–C bond**, which is hard. Under severe conditions (strong oxidant such as KMnO_4 and high temperature) C–C bonds do break, and you get a **mixture of carboxylic acids with fewer carbons**.

Catalytic dehydrogenation

When the **vapour** of a primary or secondary alcohol is passed over **heated copper at 573 K**, hydrogen gas is removed: a primary alcohol gives an **aldehyde**, a secondary alcohol a **ketone**. A tertiary alcohol has no hydrogen on the –OH carbon, so under the same conditions it **dehydrates to an alkene** instead (it loses OH and a β hydrogen).



Why methanol is poisonous

In the body methanol is oxidised first to methanal and then to methanoic acid, which can cause **blindness and death**. Patients are treated with intravenous dilute ethanol: ethanol uses up the same enzyme, giving the kidneys time to excrete the methanol.

Examiner asks: "butan-2-ol with CrO_3 " (2017: butanone); "ethanol to ethanal" (2019: PCC, not KMnO_4); "tert-butyl alcohol over copper at 573 K" (2020: 2-methylpropene); "why are tertiary alcohols resistant to oxidation" (2025). Match the reagent to the class before you write anything.

Reactions of phenol on the ring — –OH makes the ring electron-rich, so it reacts readily and at the ortho and para positions

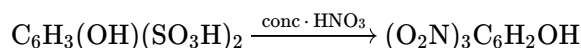
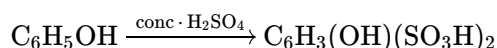
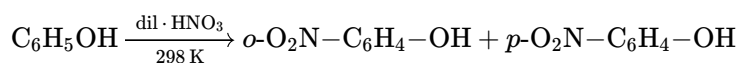
[exposure] Electrophilic aromatic substitution is a reaction in which an electron-poor species (an electrophile) replaces a hydrogen on a benzene ring. A group already on the ring either **activates** it (makes it react faster than benzene) or **deactivates** it, and **directs** the new group to certain positions.

The –OH in phenol **activates** the ring and directs the new group to the **ortho and para positions**. The oxygen has a lone pair that it can push into the ring by **resonance** (the +R effect). The lone pair pushes

electrons onto the carbons ortho and para to the $-\text{OH}$ (two ortho carbons and one para), so these positions are electron-rich and attract an electrophile.

1. Nitration

- **Dilute HNO_3 at low temperature (298 K):** a mixture of **o-nitrophenol** and **p-nitrophenol**.
- **Concentrated HNO_3 :** three nitro groups (two ortho, one para): **2,4,6-trinitrophenol, picric acid**. The yield is poor. The better way: phenol with conc. H_2SO_4 gives phenol-2,4-disulphonic acid, which is then nitrated with conc. HNO_3 ; the sulphonic acid groups are replaced and picric acid forms in good yield. Picric acid is a strong acid because three electron-withdrawing nitro groups spread the negative charge.

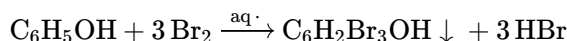
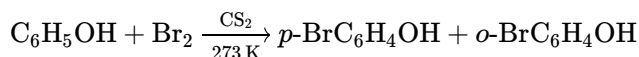


The last product is 2,4,6-trinitrophenol, picric acid.

[exposure] Steam distillation separates liquids by boiling them with steam: a compound that is **steam volatile** (easily carried over as vapour with steam) comes over first. Ordinary ortho and para isomers are hard to separate, but here you can: **o-nitrophenol is steam volatile** because it has an **intramolecular hydrogen bond** (between the $-\text{OH}$ and the neighbouring $-\text{NO}_2$ of the *same* molecule), so its molecules do not stick to each other and vaporise easily. **p-Nitrophenol is less volatile:** its hydrogen bonds are **intermolecular** (the $-\text{OH}$ of one molecule bonds to the $-\text{NO}_2$ of another), so molecules are **associated**, and heating must first break these bonds before they vaporise.

2. Halogenation

- **Br_2 in a solvent of low polarity (CS_2 or CHCl_3) at low temperature (273 K):** **monobromophenols**, p-bromophenol the major product and o-bromophenol the minor one.
- **Bromine water: 2,4,6-tribromophenol, a white precipitate.** Water is very polar, so it polarises Br_2 further and all three positions react.



The precipitate is 2,4,6-tribromophenol.

Benzene needs a Lewis acid (FeBr_3) to polarise Br_2 into Br^+ ; **phenol does not need it**, because the ring is so electron-rich that an approaching Br_2 is polarised by the ring itself: **the highly activating effect of $-\text{OH}$** .

Worked example · NCERT Example 7.5 — nitration of 3-methylphenol

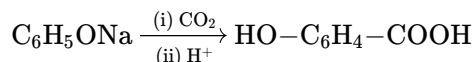
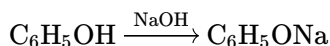
Put $-\text{OH}$ at C-1 and $-\text{CH}_3$ at C-3. The $-\text{OH}$ directs to C-2, C-4 and C-6; the methyl (also ortho/para-directing) directs to C-2, C-4 and C-6 as well, so they agree. C-2 is between the two groups and too crowded for the incoming group, so the nitro group goes to C-4 or C-6.

1. Mononitration: **3-methyl-4-nitrophenol** and **5-methyl-2-nitrophenol**.
2. Dinitration: both C-4 and C-6: **5-methyl-2,4-dinitrophenol**.

3. Mononitration of phenyl methanoate (the ester $\text{C}_6\text{H}_5\text{OCHO}$): the ester oxygen also donates a lone pair, so the group is ortho/para-directing, and the major product is the para nitro compound.

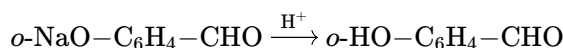
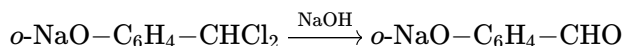
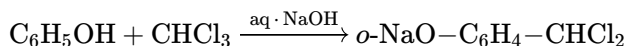
3. Kolbe's reaction

Phenol with NaOH gives **sodium phenoxide**, which is **even more reactive** than phenol towards electrophiles, because the ion carries a full negative charge pushing into the ring. That is enough for it to react with **carbon dioxide**, which is only a **weak electrophile**. CO_2 goes to the ortho position, and acidification gives **2-hydroxybenzoic acid, salicylic acid**.



4. Reimer–Tiemann reaction

Phenol is treated with **chloroform** CHCl_3 in **aqueous** NaOH. A $-\text{CHCl}_2$ group goes onto the ortho position (the phenol is present as the phenoxide), giving an intermediate **substituted benzal chloride**; the alkali hydrolyses this to a $-\text{CHO}$ group, and acidification gives **2-hydroxybenzaldehyde, salicylaldehyde**. So the $-\text{CHO}$ group is introduced at the ortho position.



5. Zinc dust, oxidation, and the iron(III) test

- **With zinc dust**, heated: phenol is reduced to **benzene**, and zinc is oxidised to ZnO .
 $\text{C}_6\text{H}_5\text{OH} + \text{Zn} \longrightarrow \text{C}_6\text{H}_6 + \text{ZnO}$. This is the only C–O cleavage a phenol shows.
- **Oxidation** with chromic acid ($\text{Na}_2\text{Cr}_2\text{O}_7$ with H_2SO_4) gives **benzoquinone**, a **[exposure] conjugated diketone** (a ring with two $\text{C}=\text{O}$ groups and two $\text{C}=\text{C}$ in alternation; $\text{O}=\text{C}_6\text{H}_4=\text{O}$). In air, phenol slowly oxidises to a dark mixture containing quinones.
- **Neutral FeCl_3 test** (asked in board papers): phenol gives a **violet colour** (an iron–phenoxide complex, $[\text{Fe}(\text{OC}_6\text{H}_5)_6]^{3-}$); alcohols give nothing. It is the standard way to tell a phenol from an alcohol. This test is not printed on the NCERT pages checked for these notes; it comes from the PYQ video.

Examiner asks: "write the Reimer–Tiemann and Kolbe reactions" (2016), "phenol to picric acid" (year not stated), "phenol to benzene / toluene" (2019), "why is o-nitrophenol steam volatile" (2019), "phenol with bromine water" (2024), "phenol with bromine in CS_2 " (2024). Name the product and say what it looks like (white precipitate; yellow solid for picric acid).

Methanol and ethanol, the two commercial alcohols — where they come from, the numbers, and why one is a poison

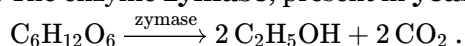
Methanol, CH₃OH

- Called **wood spirit** because it was once made by the **[exposure] destructive distillation of wood** (heating wood strongly in the absence of air so that it breaks down).
- Today it is made by **catalytic hydrogenation of carbon monoxide** at high temperature and pressure over a ZnO—Cr₂O₃ catalyst: $\text{CO} + 2 \text{H}_2 \xrightarrow[573-673 \text{ K, } 200-300 \text{ atm}]{\text{ZnO}-\text{Cr}_2\text{O}_3} \text{CH}_3\text{OH}$. (The numeric conditions are quoted in the lecture; NCERT says only "high pressure and temperature".)
- Colourless liquid, boiling point **337 K**, **highly poisonous**: small quantities cause blindness, large ones death.
- Used as a solvent in paints and varnishes, and chiefly for making methanal (formaldehyde).

Ethanol, C₂H₅OH

[exposure] Fermentation is the breaking down of sugar by the action of yeast, in the **absence of air** (**anaerobic** conditions), giving alcohol and carbon dioxide. An **[exposure] enzyme** is a protein that speeds up a reaction in a living thing; each enzyme works on one kind of reaction.

1. The sugar in molasses, sugarcane or fruits such as grapes is **sucrose**. The enzyme **invertase** hydrolyses it into **glucose and fructose**, both of formula C₆H₁₂O₆ (same formula, different structures).
2. The enzyme **zymase**, present in **yeast**, ferments glucose or fructose:



In wine making, grapes supply both sugar and yeast: as grapes ripen the sugar content rises and yeast grows on the outer skin; when grapes are crushed, the sugar and the yeast enzyme meet and fermentation starts.

- **The action of zymase stops once the alcohol exceeds about 14%.** To get stronger alcohol you must **distil**.
- If **air gets into the fermentation mixture**, its oxygen oxidises ethanol to **ethanoic acid**, which destroys the taste of the drink.
- Ethanol is a colourless liquid, boiling point **351 K**. Uses: a solvent in the paint industry, and to make many carbon compounds. Today much ethanol is made by **hydration of ethene**.

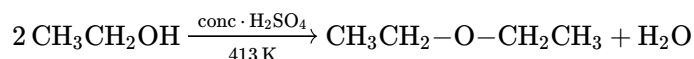
[exposure] Denatured alcohol: commercial ethanol is made unfit to drink by adding **copper sulphate** (to give it a colour) and **pyridine** (a foul-smelling liquid), so that nobody drinks it. This is called **denaturation** of alcohol.

Examiner asks: recall questions: the catalyst for methanol (ZnO—Cr₂O₃), the enzymes invertase and zymase, why fermentation stops at 14%, and what denatured alcohol contains. Not a heavy section, but easy marks.

Preparing ethers — two routes — dehydration of a primary alcohol, and the Williamson synthesis

1. Dehydration of alcohols (bimolecular)

The same acid dehydration can give an alkene or an ether, depending on the conditions. With ethanol and conc. H₂SO₄: at **443 K** the product is **ethene**; at the lower temperature of **413 K**, with excess alcohol, the product is **ethoxyethane**. This is called **bimolecular** dehydration because **two alcohol molecules** combine, losing one water molecule.



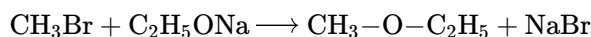
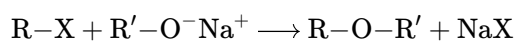
Mechanism (a nucleophilic bimolecular, **SN2**, reaction: one step in which the nucleophile arrives while the leaving group departs):

1. **Protonation:** $\text{CH}_3\text{CH}_2\text{OH} + \text{H}^+ \longrightarrow \text{CH}_3\text{CH}_2\text{—OH}_2^+$.
2. **Attack of a second alcohol molecule** on the protonated alcohol: its oxygen lone pair attacks the carbon (which is $\delta+$ because the oxygen is positive), and water leaves in the same step. This gives a protonated ether: $\text{CH}_3\text{CH}_2\text{—OH}_2^+ + \text{CH}_3\text{CH}_2\text{OH} \longrightarrow \text{CH}_3\text{CH}_2\text{—O}^+(\text{H})\text{—CH}_2\text{CH}_3 + \text{H}_2\text{O}$.
3. **Loss of a proton** from that oxygen gives the ether and returns H^+ : the acid is a catalyst.

Limits. The method works only for ethers with **primary alkyl groups**: the alkyl group must be unhindered (no branching) and the temperature kept low. With secondary and tertiary alcohols the carbon is crowded, one alcohol molecule cannot attack another, and **elimination wins over substitution**, so an alkene forms (tert-butyl alcohol with H_2SO_4 gives 2-methylpropene). The method also gives only **symmetrical** ethers, because it makes the same group on both sides. (NCERT asks why it is not appropriate for ethyl methyl ether: a mixture of alcohols gives a mixture of three ethers, CH_3OCH_3 , $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ and $\text{CH}_3\text{OC}_2\text{H}_5$, not one product.)

2. Williamson synthesis

The important laboratory method, and the only one for **unsymmetrical** ethers (it also gives symmetrical ones). An **alkyl halide** reacts with a **sodium alkoxide**; the alkoxide ion is the nucleophile and displaces the halide by **SN2**.

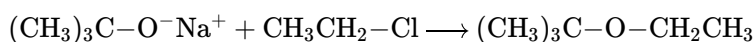


The **alkoxide may be primary, secondary or tertiary**, so an ether with a bulky group can be built, but **the alkyl halide must be primary**. An alkoxide is not only a nucleophile but also a **strong base**. With a secondary or tertiary halide it removes a hydrogen instead (**elimination**) and an alkene forms, with no ether at all.

Worked example · NCERT Example 7.6 — t-butyl ethyl ether

$(\text{CH}_3)_3\text{C—Cl} + \text{C}_2\text{H}_5\text{ONa}$ is "not an appropriate reaction" for making $(\text{CH}_3)_3\text{C—O—C}_2\text{H}_5$. What forms, and what is the right route?

1. The halide is **tertiary**: three methyl groups crowd the carbon and block the nucleophile (steric hindrance). Ethoxide therefore acts as a **base**, removes a hydrogen from a methyl group and the chloride leaves: **2-methylpropene** $(\text{CH}_3)_2\text{C=CH}_2$ is the major product.
2. Swap the roles: bulky group on the alkoxide, primary halide.



plus NaCl .

Worked example · NCERT intext 7.10 and 7.11

1. **Make 2-ethoxy-3-methylpentane from ethanol and 3-methylpentan-2-ol.** The ether has an ethyl on one side and a secondary group on the other. Primary halide, secondary alkoxide: make the alkyl

halide from ethanol, $\text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{HBr}} \text{C}_2\text{H}_5\text{Br}$, and the alkoxide from the secondary alcohol, $\text{R}-\text{OH} \xrightarrow{\text{Na}} \text{R}-\text{ONa}$; then combine: $\text{C}_2\text{H}_5\text{Br} + \text{R}-\text{ONa} \longrightarrow \text{R}-\text{O}-\text{C}_2\text{H}_5 + \text{NaBr}$.

2. **Which pair makes 1-methoxy-4-nitrobenzene: (i) 4-bromonitrobenzene + CH_3ONa , or (ii) sodium 4-nitrophenoxide + CH_3Br ?** Pair (ii). It uses a primary alkyl halide for $\text{S}_\text{N}2$. Pair (i) fails because an **aryl halide does not undergo nucleophilic substitution easily**: the C–Br bond has a partial double-bond character from resonance, and the carbon is sp^2 .

Phenols give ethers by the same method, with phenol as the phenoxide: phenol with NaOH gives sodium phenoxide, then CH_3Br gives **anisole**, $\text{C}_6\text{H}_5\text{OCH}_3$.

Examiner asks: the mechanism of ether formation from ethanol at 413 K (2023); the Williamson route to 2-methyl-2-methoxypropane (the alkyl halide must be CH_3Br , the alkoxide sodium tert-butoxide); "tert-butyl bromide with sodium methoxide gives an alkene, not an ether — give reason" (2019); phenol to anisole.

Physical properties of ethers — *polar bonds, but a boiling point like an alkane's*

The C–O bonds in an ether are polar (oxygen is more electronegative), so the molecule has a net **[exposure] dipole moment** (a measure of how unevenly charge is distributed). But the polarity is weak: the two alkyl groups sit either side of the oxygen and partly cancel it, so the ether is only weakly polar.

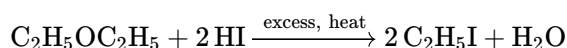
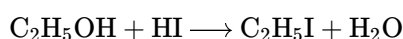
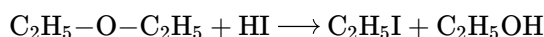
- **Boiling point** is comparable to that of an **alkane** of similar molar mass, and **much lower than an alcohol's**, because ethers cannot hydrogen-bond to each other (no O–H).
- **Solubility in water** is like that of an alcohol of the same molar mass: the ether oxygen can **accept a hydrogen bond from water**, even though the ether cannot hydrogen-bond to itself. An alkane cannot, and stays out of water.

Compound	Class	b.p. (K)	Solubility in 100 mL water
n-Pentane $\text{CH}_3(\text{CH}_2)_3\text{CH}_3$	alkane	309.1	essentially immiscible
Ethoxyethane $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$	ether	307.6	7.5 g
Butan-1-ol $\text{CH}_3(\text{CH}_2)_3\text{OH}$	alcohol	390	9 g

Examiner asks: NCERT exercise 7.22 (why is ethanol's boiling point higher than methoxymethane's) and the solubility comparison above ("can you explain this observation?" — the ether oxygen hydrogen-bonds with water, so ethoxyethane and butan-1-ol are about equally soluble while pentane is not).

Reactions of ethers: cleavage with HI — *ethers are the least reactive class — only a strong acid at high temperature breaks them*

Ethers are the **least reactive** of the functional groups here: no acidic hydrogen, weak polarity. The C–O bond breaks only under **drastic conditions: excess concentrated HI or HBr**, at high temperature. A dialkyl ether gives **two alkyl halide molecules** with excess acid. Order of reactivity of the hydrogen halides: **HI > HBr > HCl**, because HI is the strongest acid.

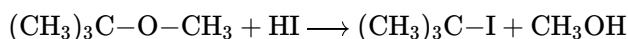


Mechanism for ethers with primary or secondary groups (SN2)

1. **Protonation** of the ether oxygen by HI: $R-O-R' + HI \longrightarrow R-O^+(H)-R' + I^-$ (an oxonium ion). A protonated ether has a much better leaving group than the plain ether.
2. **Iodide attacks** and the alcohol leaves, in one step (SN2). Iodide is a good nucleophile and attacks the **less hindered, smaller** alkyl group (a big alkyl group blocks the approach and repels the iodide). So **the iodide ends up on the smaller group**: $CH_3OC_2H_5 + HI \longrightarrow CH_3I + C_2H_5OH$.
3. With excess HI and heat, the alcohol formed reacts further to the alkyl iodide.

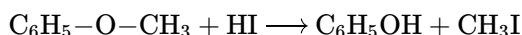
When one group is tertiary: SN1

If one group is **tertiary**, the C–O bond to it breaks first, in a slow step, giving a **stable tertiary carbocation** (the other half leaves as alcohol); iodide then attacks the carbocation quickly. The mechanism is **SN1** (two steps), and the **tertiary alkyl iodide** forms.



Alkyl aryl ethers: phenol and an alkyl halide

In **anisole** $C_6H_5-O-CH_3$ the two C–O bonds differ. The $O-C_6H_5$ bond is **stronger**: the carbon is sp^2 , and the oxygen lone pair is shared with the ring, giving the bond **partial double-bond character**. So after protonation iodide attacks the **methyl** carbon (the $O-CH_3$ bond is weaker) and the products are **phenol and methyl iodide**. The phenol does not react on to give an aryl halide, because the sp^2 carbon cannot undergo the nucleophilic substitution that would be needed.



Worked example · NCERT Example 7.7 — heating each ether with HI

1. $CH_3CH_2CH(CH_3)CH_2-O-CH_2CH_3$ (both groups primary): SN2; iodide takes the **smaller** group, ethyl: $CH_3CH_2I + CH_3CH_2CH(CH_3)CH_2OH$.
2. $CH_3CH_2CH_2-O-C(CH_3)_2CH_2CH_3$ (one tertiary group): SN1; tertiary iodide: $CH_3CH_2C(CH_3)_2I + CH_3CH_2CH_2OH$ (2-iodo-2-methylbutane and propan-1-ol).
3. $C_6H_5-CH_2-O-C_6H_5$ (benzyl phenyl ether): the aryl–O bond is the strong one; iodide takes the benzyl carbon: $C_6H_5CH_2I + C_6H_5OH$.

Worked example · NCERT intext 7.12 — products with HBr and HI

1. $CH_3CH_2CH_2-O-CH_3 + HBr$: both primary, SN2, bromide on the smaller (methyl) group: $CH_3Br + CH_3CH_2CH_2OH$.
2. $(CH_3)_3C-O-C_2H_5 + HI$: one tertiary group, SN1: $(CH_3)_3C-I + C_2H_5OH$.

Trap: do not give the same answer for every ether. Check each side: **primary/secondary on both** → **iodide on the smaller group (SN2)**; **a tertiary group present** → **the tertiary iodide (SN1)**; **aryl on one side** → **phenol + alkyl iodide**, never an aryl iodide.

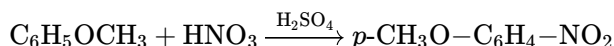
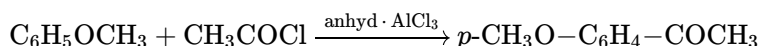
Examiner asks: "how can phenol be prepared from anisole? give reason" (2023); "ether with a tertiary butyl group heated with HI" (2025); "benzyl phenyl ether with HI" (2020). The reason for the aryl case is always the partial double bond.

Reactions of aryl alkyl ethers on the ring — *–OR behaves like –OH: an activator that directs ortho and para*

An alkoxy group –OR on a benzene ring **activates** it and **directs ortho and para**, exactly as the –OH of phenol does, through the same resonance donation from the oxygen lone pair. The para product is the major one (anisole's para isomer is the favoured product: less crowded). Three reactions, all with anisole:

Reaction	Reagent	Major product	Minor product
Bromination	Br ₂ in ethanoic acid, no FeBr ₃ needed	p-bromoanisole (about 90%)	o-bromoanisole
Friedel–Crafts alkylation	CH ₃ Cl, anhydrous AlCl ₃	4-methoxytoluene	2-methoxytoluene
Friedel–Crafts acylation	CH ₃ COCl, anhydrous AlCl ₃	4-methoxyacetophenone	2-methoxyacetophenone
Nitration	conc. H ₂ SO ₄ + conc. HNO ₃	4-nitroanisole	2-nitroanisole

[exposure] A **Friedel–Crafts reaction** puts an **alkyl** group (alkylation) or an **acyl** group RCO– (acylation) on a benzene ring, using an alkyl or acyl halide and a Lewis acid catalyst. A **[exposure] Lewis acid** is an electron-pair acceptor, such as AlCl₃, which is electron-deficient. It pulls the chloride off the reagent: CH₃Cl + AlCl₃ → CH₃⁺ + AlCl₄[–] and CH₃COCl + AlCl₃ → CH₃CO⁺ + AlCl₄[–]. The positive CH₃⁺ or CH₃CO⁺ is the electrophile that the ring attacks. Acetophenone is a benzene ring with a COCH₃ group, which is a ketone.



Each reaction also gives a little of the ortho isomer.

Bromination needs no FeBr₃ for the same reason as in phenol: the methoxy group makes the ring so electron-rich that it polarises Br₂ itself.

Examiner asks: "Friedel–Crafts acylation of anisole" (2016, 3 marks), "nitration of anisole" (2025), "anisole with CH₃Cl and anhydrous AlCl₃" (2024). Name both isomers and say which is major.

Patterns, collected — *the orderings, tests, reagent choices and conversions you will actually be asked to produce*

1. Rank by acidic strength

Worked example

Arrange in increasing acidic strength: p-cresol, p-nitrophenol, phenol (2017).

1. Look at the substituent: –CH₃ releases electrons, so p-cresol is weaker than phenol. –NO₂ withdraws electrons (and delocalises the charge from the para position), so p-nitrophenol is stronger.

Answer: p-cresol < phenol < p-nitrophenol.

- Increasing acidic strength: ethanol, phenol, water — answer only: ethanol < water < phenol
- Increasing acid strength of methanol, propan-2-ol, ethanol, 2-methylpropan-2-ol — answer only: 2-methylpropan-2-ol < propan-2-ol < ethanol < methanol

- Which is more acidic, o-nitrophenol or m-nitrophenol? — answer only: o-nitrophenol (pK_a 7.2 against 8.3); the ortho nitro group takes the charge by resonance

2. Rank by boiling point

Worked example

Increasing boiling point: ethoxyethane, butanal, butan-1-ol, n-butane (2023).

1. n-Butane: non-polar, only weak van der Waals forces: lowest.
2. Ethoxyethane: weakly polar; then butanal: strongly polar $C=O$, dipole–dipole attraction.
3. Butan-1-ol: hydrogen bonds, the strongest: highest.

Answer: n-butane < ethoxyethane < butanal < butan-1-ol.

- Higher boiling point: propan-1-ol or butane — answer only: propan-1-ol (hydrogen bonding)
- Higher boiling point: butan-1-ol or butan-2-ol — answer only: butan-1-ol (less branched, larger surface area)

3. Distinguish by a test

Worked example

Distinguish propan-1-ol from 2-methylpropan-2-ol (2019).

Add Lucas reagent (conc. HCl + anhydrous $ZnCl_2$) to each at room temperature. The tertiary alcohol, 2-methylpropan-2-ol, turns cloudy immediately (alkyl chloride is immiscible); the primary alcohol gives no turbidity at room temperature.

- Distinguish ethanol from dimethyl ether (2019) — answer only: iodoform test, $I_2 + NaOH$; ethanol (a $CH_3CH(OH)-$ group) gives a yellow precipitate of CHI_3 , the ether gives none
- Distinguish phenol from cyclohexanol (2023) — answer only: neutral $FeCl_3$; phenol gives a violet colour, cyclohexanol none
- Tell primary, secondary and tertiary alcohols apart — answer only: Lucas test; turbidity immediate (3°), in about five minutes (2°), none at room temperature (1°)

4. Choose the oxidising or reducing agent

Worked example

Ethanol to ethanal (2019); butan-2-ol with CrO_3 (2017).

Use a **mild** oxidant: PCC (or anhydrous CrO_3) stops a primary alcohol at the aldehyde, $CH_3CH_2OH \xrightarrow{PCC} CH_3CHO$. Strong $KMnO_4$ or $K_2Cr_2O_7$ would give ethanoic acid. Butan-2-ol is secondary, so CrO_3 gives the ketone **butanone**.

- Ethanol to ethanoic acid — answer only: acidified $KMnO_4$ (or acidified $K_2Cr_2O_7$)
- Cyclohexenol (a ring with a $C=C$ and an $-OH$) to the unsaturated ketone — answer only: PCC; it oxidises the alcohol without touching the double bond
- Butan-2-one to butan-2-ol — answer only: $NaBH_4$ or $LiAlH_4$, or H_2/Ni

5. Williamson synthesis: pick the halide and the alkoxide

Worked example

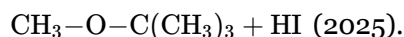
2-Methyl-2-methoxypropane, $(CH_3)_3C-O-CH_3$ (Q10(b), year not stated).

Split at the oxygen. One side is tertiary, the other methyl. The tertiary group must be the alkoxide and the methyl the halide: $(\text{CH}_3)_3\text{C}-\text{ONa} + \text{CH}_3\text{Br} \longrightarrow (\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3 + \text{NaBr}$ (SN2). The other way round would give an alkene.

- 1-Propoxypropane — answer only: $\text{CH}_3\text{CH}_2\text{CH}_2\text{ONa} + \text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
- Ethoxybenzene — answer only: sodium phenoxide + bromoethane ($\text{C}_6\text{H}_5\text{ONa} + \text{C}_2\text{H}_5\text{Br}$); not bromobenzene + sodium ethoxide
- 1-Methoxyethane — answer only: $\text{CH}_3\text{ONa} + \text{CH}_3\text{CH}_2\text{Br}$ (or $\text{CH}_3\text{CH}_2\text{ONa} + \text{CH}_3\text{Br}$)

6. Predict the products of an ether with HI

Worked example



A tertiary group is present: SN1, a tertiary carbocation forms, iodide attacks it. Products: $(\text{CH}_3)_3\text{C}-\text{I} + \text{CH}_3\text{OH}$.

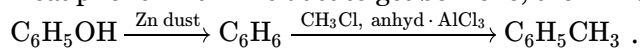
- Anisole with HI — answer only: phenol + CH_3I
- Ethoxyethane with excess HI — answer only: $2 \text{ C}_2\text{H}_5\text{I} + \text{H}_2\text{O}$
- $\text{CH}_3-\text{O}-\text{CH}_2\text{CH}_3 + \text{HI}$ — answer only: $\text{CH}_3\text{I} + \text{CH}_3\text{CH}_2\text{OH}$ (iodide on the smaller group)

7. Convert, using the reagent table

Worked example

Phenol to toluene (2019).

Heat phenol with zinc dust to get benzene; then Friedel–Crafts alkylation:



- Phenol to picric acid — answer only: conc. HNO_3 (poor yield), or conc. H_2SO_4 first and then conc. HNO_3 (good yield): 2,4,6-trinitrophenol
- Phenol to benzoquinone — answer only: $\text{Na}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$ (chromic acid)
- Chlorobenzene to phenol — answer only: fuse with NaOH at 623 K and 320 atm to $\text{C}_6\text{H}_5\text{ONa}$, then acidify
- Aniline to phenol — answer only: $\text{NaNO}_2 + \text{HCl}$ at 273–278 K, then warm with water
- Propanone to 2-methylpropan-2-ol — answer only: CH_3MgBr , then H_3O^+ (a ketone gives a tertiary alcohol)
- Ethanol to propan-2-ol — answer only: PCC to ethanal, then CH_3MgBr and hydrolysis (an aldehyde gives a secondary alcohol)
- Ethylmagnesium chloride to propan-1-ol (NCERT 7.20(iii)) — answer only: react with methanal, then hydrolyse
- Phenol to anisole — answer only: NaOH to sodium phenoxide, then CH_3Br
- Propene to propan-1-ol and to propan-2-ol — answer only: propan-1-ol by hydroboration–oxidation (B_2H_6 , then $\text{H}_2\text{O}_2/\text{OH}^-$); propan-2-ol by acid-catalysed hydration

8. The reasons that keep coming back

- **Alcohol boils higher than hydrocarbon or ether of the same mass:** intermolecular hydrogen bonding through the O–H.

- **Phenol is more acidic than an alcohol:** the phenoxide ion is stabilised by resonance; alkoxide has none (and the sp^2 carbon withdraws electrons).
- **o-Nitrophenol is steam volatile, p-nitrophenol is not:** intramolecular against intermolecular hydrogen bonding.
- **Tertiary alcohols resist oxidation:** no hydrogen on the $-OH$ carbon.
- **tert-Butyl halide with an alkoxide gives an alkene:** steric hindrance, so the alkoxide acts as a base (elimination).
- **Phenol is easily nitrated or brominated:** $-OH$ pushes electrons into the ring at ortho and para.
- **Phenol does not protonate readily:** the oxygen lone pair is delocalised into the ring and is less available.
- **Anisole with HI gives phenol, not an aryl iodide:** the aryl-O bond has partial double-bond character, and an sp^2 carbon will not undergo nucleophilic substitution.
- **The C-O bond in phenol is shorter than in methanol:** partial double-bond character (resonance, sp^2 carbon).

Past year questions · question types — ranked by how often they turn up

One video feeds this section: the "25 PYQ" video for CBSE 2026-27, which covers Q1–Q25, with questions from 2016 to 2025. Question numbers are the video's own. Years are as spoken in the video; where a year is spoken once for a numbered question it is applied to all its parts, and where no year was spoken the question is cited by number only. There is no second PYQ video for this chapter. Counts are by question part: 61 parts in all.

1 · Predict the product / complete the reaction — 2 or 3 marks · about 27 parts

Recognise it: "write the product of ...", "write the structure of the major product", "complete the following", "what happens when ...", "write the equation for ...".

1. Find the functional group that reacts and name the reaction type: reduction, oxidation, hydration, dehydration, ether cleavage, Grignard addition, electrophilic substitution, acetylation.
2. Match the reagent to the class (primary / secondary / tertiary; alkyl or aryl).
3. Write the organic product, plus the small by-product (H_2O , HCl , $NaBr$) if the question wants an equation.
4. For ring substitution give **both** ortho and para products and mark the para as major.

Trap: the reagent *selects* the product. $NaBH_4$ reduces a ketone and leaves an ester; PCC stops at an aldehyde and leaves a $C=C$; HI on an alkyl aryl ether gives phenol and alkyl iodide (never an aryl iodide); anisole needs no $FeBr_3$; copper at 573 K dehydrates a tertiary alcohol instead of dehydrogenating it.

2 · Give a reason / account for — 2 or 3 marks · about 12 parts

Recognise it: "give reason", "why is ...", "explain why ...", "account for ...".

1. Name the structural feature: hydrogen bonding (inter- or intra-molecular), resonance, partial double-bond character, steric hindrance, an electron-releasing or electron-withdrawing group, a missing hydrogen.
2. Say how that feature causes the behaviour, in one sentence, and compare with the other compound.

Trap: the answer is a mechanism-level sentence, not a restatement. "Phenol is more acidic" must go on to "because phenoxide is stabilised by resonance and alkoxide is not". "o-Nitrophenol is steam volatile" must

name **intramolecular** against **intermolecular** hydrogen bonding.

3 · Convert A to B — 2 to 5 marks · about 10 parts

Recognise it: "how will you convert ...", "carry out the following conversion".

1. Say what changes: a functional group (alcohol to aldehyde), a carbon count (ethanol to propan-2-ol: one more carbon), or the class (primary to secondary).
2. Pick the named route: Grignard for adding carbon, a mild oxidant or reductant for changing oxidation level, diazonium for aniline, fusion with NaOH for chlorobenzene.
3. Write each step with its reagent over the arrow.

Trap: a secondary alcohol comes from an aldehyde (plus Grignard) or a ketone (plus a reducing agent); a tertiary alcohol comes from a ketone plus Grignard. Count the carbon neighbours of the product's -OH carbon, then work backwards.

4 · Write the mechanism — 2 to 5 marks · 4 parts

Recognise it: "write the mechanism of ...", "show the curved-arrow notation ...".

1. Name the reaction and write the steps in order: protonation, nucleophilic attack or carbocation formation, loss of a proton.
2. Show the movement of electron pairs from the electron-rich atom to the electron-poor one, and the charges that result. Say which step is slow.
3. End by saying the acid is regained, so it is a catalyst.

Trap: hydration of an alkene (ethene plus water, H^+) is **three steps** and the first makes H_3O^+ ; ether formation at 413 K is an **SN2** step (a second alcohol attacks the protonated one); alkene formation at 443 K goes through a **carbocation**, and that step is the slowest.

5 · Distinguish between two compounds — 2 or 3 marks · 3 parts

Recognise it: "give one chemical test to distinguish between ...".

1. Phenol against alcohol: neutral FeCl_3 , violet colour for phenol.
2. Primary against tertiary alcohol: Lucas reagent, turbidity at once for tertiary.
3. Ethanol against an ether: iodoform test, yellow precipitate for ethanol.

Trap: state the reagent, the observation for **each** compound (one shows, one does not), and not just "gives a colour".

6 · Arrange in order — 1 to 3 marks · 2 parts

Recognise it: "arrange in increasing order of acidic strength" and "arrange in increasing order of boiling point".

1. Acidity: electron-withdrawing group raises it, electron-releasing group lowers it, an alcohol is below any phenol.
2. Boiling point: hydrogen bonding > strong dipole (aldehyde) > weak dipole (ether) > non-polar alkane, and compare carbon counts last.

Trap: read "increasing" and write the order from weakest or lowest to strongest or highest.

7 · Write the IUPAC name — 1 mark · 2 parts

Recognise it: a drawn alcohol or ether, with "write the IUPAC name".

1. Alcohol: longest chain containing the –OH carbon, number from the –OH end, replace "e" by "ol".
2. Ether: the longer chain is the parent; the other side is "...oxy". A phenyl group on a chain is a "phenyl" substituent.

Trap: in 3-phenylprop-2-en-1-ol the –OH gets number 1; the ring is a substituent, so it is named "phenyl" and gets a locant.

8 · Identify A, B, C, D, E — 5 marks · 1 part

Recognise it: "an organic compound A of formula ... reacts with ... to give B ...; identify A to E and write the reactions".

1. Use the molecular formula and the first clue (reaction with sodium gives hydrogen: an alcohol) to fix A.
2. Then follow each reagent: each gives one unique class of product.

Trap: C₂H₆O could be ethanol or methoxymethane, but only the alcohol reacts with sodium. Always use the "reacts with sodium" clue to decide which.

Past year questions · mark slots — what each type is worth

The video states its own marks: Q1–Q10 are 2-mark short answers, Q11–Q23 are 3-mark short answers, and Q24–Q25 are 5-mark long answers. A "(Qn)" with no year was given without one in the video.

Question type	2 marks	3 marks	5 marks
Predict the product / complete	2017 Q1(b), 2018 Q2(a)(b), 2017 Q7(a)(b), 2020 Q9(a)(b), Q10(b)	2016 Q11, 2020 Q12, 2025 Q14, 2020 Q15, 2024 Q20, 2024 Q23	—
Give a reason	2023 Q6(a), 2020 Q8	2023 Q13(a)(c), 2019 Q16, 2019 Q19(b), 2025 Q22	2019 Q25(c)
Convert A to B	2019 Q5(a)(b), Q10(a)	Q17, 2019 Q19(a)	2019 Q25(a)
Mechanism	2017 Q3(b), 2023 Q4	2017 Q18	2019 Q25(b)
Distinguish by a test	2023 Q6(b)	2019 Q21(a)	—
Arrange in order	2017 Q3(a)	2023 Q13(b)	—
IUPAC name	2017 Q1(a)	2019 Q21(b)	—
Identify A–E	—	—	2016 Q24

Past year questions · repeat offenders — appeared more than once — highest probability in the chapter

Q1(b) · 2020 Q12(c) · 2023 Q13(c) · 2025 Q14(c)

Heating an ether with HI. The decision tree: primary and secondary on both sides, iodide goes on the smaller group (S_N2); one tertiary group, tertiary iodide (S_N1); alkyl aryl ether, phenol plus alkyl iodide because the aryl–O bond has partial double-bond character.

2018 Q2(b) · 2017 Q3(b) · 2017 Q18 · 2024 Q20(b)

Acid-catalysed hydration of an alkene, including the three-step mechanism for ethene to ethanol (asked twice in 2017) and the Markovnikov product for styrene and but-1-ene.

2020 Q15(iii) · Q17(ii) · 2024 Q20(a) · 2024 Q23(a)

A Grignard reagent with a ketone or an aldehyde, then hydrolysis: methylmagnesium bromide with propanone (tertiary alcohol), with ethanal (secondary), with cyclohexanone; cyclohexylmagnesium bromide with ethanal.

2019 Q5(b) · 2017 Q7(a) · 2020 Q9(b) · 2020 Q15(i) · 2024 Q23(d) · 2025 Q22(a)

Oxidation and dehydrogenation of alcohols: PCC to an aldehyde, CrO_3 to a ketone, copper at 573 K, and why tertiary alcohols resist oxidation.

2017 Q3(a) · 2019 Q16(c) · 2023 Q6(a)

Why phenol is more acidic than an alcohol, and the order for p-cresol, phenol, p-nitrophenol: resonance-stabilised phenoxide ion against unstabilised alkoxide.

Q10(a) · 2019 Q19(a) · 2023 Q13(c) · 2025 Q14(a)

Preparing phenol: from aniline (diazonium salt, warm water), from chlorobenzene (fuse with NaOH at 623 K, 320 atm), from anisole (cleave with HI) and from cumene (the industrial route, with acetone).

2023 Q4 · 2019 Q25(b)

The two mechanisms of acid dehydration of ethanol: 413 K gives ethoxyethane through $\text{S}_{\text{N}}2$; 443 K gives ethene through a carbocation.

2016 Q11(c) · 2024 Q23(c) · 2025 Q14(b)

Electrophilic substitution of anisole: Friedel–Crafts acylation and alkylation, and nitration; para major, ortho minor.

2016 Q11(a)(b) · 2019 Q25(c) · 2024 Q20(c) · 2024 Q23(b) · Q17(i)

Electrophilic substitution of phenol: Reimer–Tiemann, Kolbe, bromine water (white precipitate of 2,4,6-tribromophenol), bromine in CS_2 (mono), and nitration to picric acid; and why phenol reacts more readily than benzene.

Q10(b) · 2019 Q16(a)

Williamson synthesis: the pair that works (primary halide with a bulky alkoxide) and why a tertiary halide gives an alkene.

2019 Q16(b) · 2019 Q19(b) · 2023 Q13(b)

Hydrogen bonding as the explanation of boiling point and steam volatility: propanol against butane, o- against p-nitrophenol, the four-compound boiling-point order.

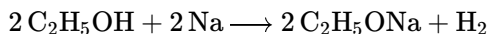
Past year questions · numerical types — *no calculations in this chapter, so these are the reaction and structure types — one worked model each, then cold practice*

A · Identify A to E in a sequence

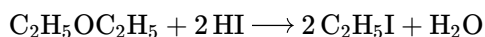
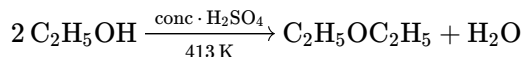
Worked example · 2016 Q24 — compound A, $\text{C}_2\text{H}_6\text{O}$, with sodium gives B and hydrogen; with I_2/NaOH gives C; with conc. H_2SO_4 at 413 K gives D, $\text{C}_4\text{H}_{10}\text{O}$; D with excess HI gives E

1. A reacts with sodium and releases hydrogen, so it has an O–H: an alcohol, not the isomeric ether methoxymethane. $\text{C}_2\text{H}_6\text{O}$ as an alcohol is **ethanol** = A. B = sodium ethoxide.

- The iodoform test (iodine and sodium hydroxide) needs a $\text{CH}_3\text{CH}(\text{OH})-$ group, which ethanol has: **C** = **iodoform** CHI_3 , a yellow precipitate, with HCOONa as by-product.
- Conc. H_2SO_4 at **413 K** is the temperature for the **ether** (443 K would give the alkene): two $\text{C}_2\text{H}_5\text{OH}$ minus H_2O gives $\text{C}_4\text{H}_{10}\text{O}$ = **D** = ethoxyethane.
- Excess HI cleaves both ethyl groups: **E** = **ethyl iodide**.



plus $5 \text{NaI} + 5 \text{H}_2\text{O}$.

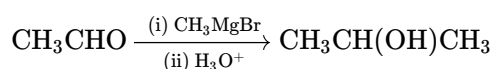
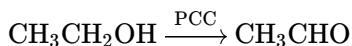


- 2016 Q24, cold — answer only: A ethanol; B sodium ethoxide; C iodoform (plus sodium methanoate); D ethoxyethane; E ethyl iodide

B · Convert A to B

Worked example · 2019 Q25(a)(ii) — ethanol to propan-2-ol

- Propan-2-ol is a **secondary** alcohol with one more carbon than ethanol. A secondary alcohol is made from an aldehyde plus a Grignard reagent, so the route is: ethanol to ethanal, then add one carbon as CH_3- .
- Oxidise ethanol with a **mild** oxidant (PCC or CrO_3) to ethanal.
- Add CH_3MgBr to ethanal, then hydrolyse.



- 2019 Q5(a) — phenol to toluene — answer only: heat with zinc dust to benzene, then CH_3Cl with anhydrous AlCl_3 (Friedel–Crafts)
- 2019 Q5(b) — ethanol to ethanal — answer only: PCC (a mild oxidising agent)
- Q10(a) — aniline to phenol — answer only: NaNO_2/HCl at $0-5^\circ\text{C}$, then warm with water
- Q17(i) — phenol to picric acid — answer only: concentrated HNO_3 , giving 2,4,6-trinitrophenol
- Q17(ii) — propanone to 2-methylpropan-2-ol — answer only: CH_3MgBr then hydrolysis (a ketone gives a tertiary alcohol)
- Q17(iii) — butan-2-one to butan-2-ol — answer only: LiAlH_4 or NaBH_4 , or H_2/Ni
- 2019 Q19(a) — phenol to benzoquinone, and chlorobenzene to phenol — answer only: $\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$; and fuse with NaOH at 623 K and then acidify
- 2019 Q25(a)(i) — phenol to anisole — answer only: NaOH to sodium phenoxide, then CH_3Br (Williamson)

C · Predict the product with a named reagent

Worked example · 2020 Q15 — three reactions

1. A cyclic alcohol with a C=C, treated with **PCC**: PCC oxidises the secondary alcohol to a ketone and leaves the double bond alone: an unsaturated ketone (cyclohexenone).
 2. Salicylic acid with acetic anhydride and acid: the phenolic –OH is acetylated, giving **acetylsalicylic acid, aspirin** (a medicine).
 3. Cyclohexanone with CH_3MgBr then hydrolysis: the methyl carbon attacks the carbonyl carbon and, after hydrolysis, gives the tertiary alcohol **1-methylcyclohexanol**, with $\text{Mg}(\text{OH})\text{Br}$ as by-product.
- 2018 Q2(a) — a ketone with an ester side chain, with NaBH_4 — answer only: the ketone becomes a secondary alcohol; the ester is not reduced
 - 2018 Q2(b) — styrene $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$ with water and H^+ — answer only: $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CH}_3$ (Markovnikov; OH on the carbon carrying the ring)
 - 2017 Q7(a)(b) — butan-2-ol with CrO_3 , and with SOCl_2 — answer only: butanone; 2-chlorobutane
 - 2020 Q9(a) — a compound with a benzylic $-\text{CH}_2\text{OH}$ and a phenolic $-\text{OH}$, heated with HCl — answer only: the benzylic $-\text{OH}$ becomes $-\text{Cl}$; the phenolic $-\text{OH}$ stays (partial double bond in the phenol C–O)
 - 2020 Q9(b) — tert-butyl alcohol heated with copper at 573 K — answer only: 2-methylpropene and water (dehydration)
 - 2020 Q12(a)(b) — propene by hydroboration–oxidation; tert-butyl alcohol with 20% H_3PO_4 at 358 K — answer only: propan-1-ol; 2-methylpropene
 - 2024 Q20 — ethanal with CH_3MgBr then hydrolysis; but-1-ene with dilute H_2SO_4 ; phenol with bromine water — answer only: propan-2-ol; butan-2-ol; 2,4,6-tribromophenol (white precipitate)
 - 2024 Q23 — cyclohexylmagnesium bromide with ethanal; phenol with Br_2/CS_2 ; anisole with $\text{CH}_3\text{Cl}/\text{AlCl}_3$; propan-2-ol over copper at 573 K — answer only: 1-cyclohexylethanol; o- and p-bromophenol (para major); 2- and 4-methoxytoluene (para major); propanone
 - 2016 Q11 — Reimer–Tiemann; Kolbe's reaction; Friedel–Crafts acylation of anisole — answer only: salicylaldehyde; salicylic acid; 4-methoxyacetophenone (major) and 2-methoxyacetophenone
 - 2025 Q14(a)(b) — phenol from cumene; nitration of anisole — answer only: cumene, then O_2 (air), then dilute acid, gives phenol and acetone; 4-nitroanisole (major), 2-nitroanisole

D · Write the mechanism

Worked example · 2017 Q18 — hydration of ethene to ethanol

1. **Protonation of the alkene to a carbocation by electrophilic attack of H_3O^+ .**
 $\text{H}_2\text{O} + \text{H}^+ \longrightarrow \text{H}_3\text{O}^+$. The positive oxygen pulls the bonding electrons, so H_3O^+ gives up H^+ (and becomes H_2O). The H^+ is attacked by the π electrons of $\text{C}=\text{C}$: a C–H bond forms and the other carbon is left with a positive charge: $\text{CH}_2=\text{CH}_2 + \text{H}_3\text{O}^+ \longrightarrow \text{CH}_3-\text{CH}_2^+ + \text{H}_2\text{O}$.
 2. **Nucleophilic attack of water on the carbocation.** A lone pair on a water oxygen bonds to the positive carbon; the oxygen is now positive: $\text{CH}_3\text{CH}_2^+ + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{CH}_2-\text{OH}_2^+$.
 3. **Deprotonation to form the alcohol.** The positive oxygen pulls the O–H electrons, a water molecule takes the H^+ , and $\text{CH}_3\text{CH}_2\text{OH}$ forms; H_3O^+ is regenerated, so it is a catalyst.
- 2017 Q3(b) — ethene with H_3O^+ giving CH_3CH_2^+ and water, curved arrows — answer only: the O–H electrons go to oxygen and release water plus H^+ ; the π electrons attack the H^+ ; a carbocation forms

- 2023 Q4 — ethoxyethane from ethanol at 413 K — answer only: protonation; $\text{S}_{\text{N}}2$ attack of a second ethanol on the protonated alcohol with water leaving; loss of a proton, regenerating H^+
- 2019 Q25(b) — ethene from ethanol at 443 K — answer only: protonated alcohol (fast); carbocation (slow, rate-determining); loss of a β proton gives ethene and returns H^+

E · Arrange in order

Worked example · 2023 Q13(b) — increasing boiling point: ethoxyethane, butanal, butanol, butane

1. n-Butane: non-polar, weak van der Waals forces: lowest.
2. Ethoxyethane: weakly polar, weak dipole interaction.
3. Butanal: strongly polar carbonyl, strong dipole–dipole attraction.
4. Butan-1-ol: hydrogen bonds, the strongest: highest.

Answer: butane < ethoxyethane < butanal < butan-1-ol.

- 2017 Q3(a) — increasing acidic strength: p-cresol, p-nitrophenol, phenol — answer only: p-cresol < phenol < p-nitrophenol

F · Distinguish by a chemical test

Worked example · 2019 Q21(a)(i) — propan-1-ol and 2-methylpropan-2-ol

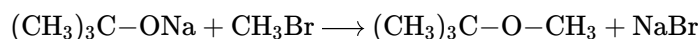
Add Lucas reagent (conc. HCl + anhydrous ZnCl_2) to each. The tertiary alcohol, 2-methylpropan-2-ol, forms the insoluble tertiary chloride at once and the solution turns cloudy. The primary alcohol, propan-1-ol, gives no turbidity at room temperature.

- 2019 Q21(a)(ii) — ethanol and dimethyl ether — answer only: iodoform test ($\text{I}_2 + \text{NaOH}$): ethanol gives a yellow precipitate of CHI_3 ; the ether gives none
- 2023 Q6(b) — phenol and cyclohexanol — answer only: neutral FeCl_3 : phenol gives a violet complex, cyclohexanol does not react

G · Ethers: Williamson pairing and cleavage

Worked example · Q10(b) — 2-methyl-2-methoxypropane by Williamson synthesis

1. The ether is $(\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3$. Split at the oxygen: one side tertiary, the other methyl.
2. The alkyl halide must be **primary** (here methyl), so the tertiary group goes on the alkoxide.



The other pairing, $(\text{CH}_3)_3\text{C}-\text{Br} + \text{CH}_3\text{ONa}$, would give 2-methylpropene by elimination (2019 Q16(a)).

- Q1(b) — $\text{CH}_3\text{CH}_2-\text{O}-\text{CH}(\text{CH}_3)_2$ with HI — answer only: $\text{CH}_3\text{CH}_2\text{I} + (\text{CH}_3)_2\text{CHOH}$ (iodide on the less hindered ethyl)
- 2020 Q12(c) — benzyl phenyl ether with HI (structure as drawn in the video; the spoken products are the benzyl iodide and phenol) — answer only: $\text{C}_6\text{H}_5\text{CH}_2\text{I} + \text{C}_6\text{H}_5\text{OH}$
- 2023 Q13(c) — phenol from anisole — answer only: heat with HI : $\text{CH}_3\text{I} + \text{phenol}$; the aryl–O bond has partial double-bond character so it does not break
- 2025 Q14(c) — methyl tert-butyl ether with HI — answer only: $(\text{CH}_3)_3\text{C}-\text{I} + \text{CH}_3\text{OH}$ ($\text{S}_{\text{N}}1$, tertiary carbocation)

- 2019 Q16(a) — why tert-butyl bromide with sodium methoxide gives an alkene — answer only: three methyl groups block attack (steric hindrance), so methoxide acts as a base and removes a hydrogen: elimination to 2-methylpropene

Aldehydes, Ketones and Carboxylic Acids

NCERT Class XII Chemistry • Chapter 8 • Aldehydes, Ketones and Carboxylic Acids

8 marks in the board paper (CBSE 2026-27 curriculum), the heaviest of the four organic chapters you are adding, and the one with the most reactions. The whole chapter hangs on one idea: the carbon–oxygen double bond, $C=O$. In aldehydes and ketones that carbon is eager to be attacked, and most of the chapter is "what attacks it, and what do you get". In carboxylic acids the same carbon is less eager, and the interest moves to the acidic $O-H$. Every new term is tagged [exposure] and explained from scratch.

The carbonyl family — one functional group, six families — and why they smell nice

[exposure] The **carbonyl group** is a carbon doubly bonded to oxygen, $C=O$. A **functional group** is a small unit of atoms that decides how a whole organic compound behaves, so every compound that carries it reacts in a family way. The carbonyl group is the centre of this chapter, and these are the families it defines:

Family	General formula	Example
Aldehyde	$R-CHO$	CH_3CHO
Ketone	$R-CO-R'$	CH_3COCH_3
Carboxylic acid	$R-COOH$	CH_3COOH
Acid (acyl) halide	$R-COX$	CH_3COCl
Acid anhydride	$R-CO-O-CO-R$	ethanoic anhydride
Ester	$R-COOR'$	ethyl ethanoate
Amide	$R-CONH_2$	ethanamide

The last four are **derivatives** of carboxylic acids: each is an acid with its $-OH$ swapped for something else. An **anhydride** forms when two acid molecules lose one water between them.

Aldehydes, ketones and carboxylic acids are widespread in plants and animals and play a part in life's biochemistry. Many are responsible for fragrance and flavour: **vanillin** (from vanilla beans), **salicylaldehyde** (from meadowsweet) and **cinnamaldehyde** (from cinnamon).

Naming aldehydes — common names come from the acid; IUPAC drops the "e" and adds "-al"

Common names

An aldehyde's common name comes from the common name of the matching carboxylic acid: replace **-ic acid** with **-aldehyde**. $HCOOH$ is formic acid, so $HCHO$ is **formaldehyde**; acetic acid gives **acetaldehyde** (CH_3CHO); propionic acid gives propionaldehyde; butyric acid gives **butyraldehyde**. Substituents are located with Greek letters: the carbon next to $-CHO$ is α , then β , γ , δ . So $CH_3CH(Br)CH_2CHO$ is β -bromobutyraldehyde.

IUPAC names

1. Take the **longest chain that includes the $-CHO$ carbon**. The aldehyde carbon is always **carbon 1**, so it needs no locant.
2. Drop the final "e" of the alkane and add **-al**: CH_3CHO = **ethanal**; CH_3CH_2CHO = **propanal**; $CH_3CH_2CH_2CHO$ = **butanal**. $CH_3CH(Br)CH_2CHO$ = **3-bromobutanal**.

3. **Aldehyde on a ring:** name the ring and add **-carbaldehyde**: cyclopentanecarbaldehyde; benzenecarbaldehyde (= **benzaldehyde**, which IUPAC also accepts).
4. **Aldehyde on a side chain of a ring:** the chain is the parent and the ring is a substituent: $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CHO}$ is 3-phenylpropanal.
5. **Three or more –CHO groups:** they cannot be counted as chain carbons. Number only the backbone and call them **carbaldehydes**: propane-1,2,3-tricarbaldehyde.

Naming ketones, and which group wins — *the ketone is "-one"; as a prefix it becomes "oxo"*

Common names

Name the two groups on the carbonyl carbon alphabetically, then write **ketone**: CH_3COCH_3 is dimethyl ketone (and **acetone**, its historical name); $\text{CH}_3\text{COC}_2\text{H}_5$ is **ethyl methyl ketone**. Substituents are located with α , α' , β , β' . A ketone with a phenyl group is named with an **acyl** prefix and **-phenone**: $\text{C}_6\text{H}_5\text{COCH}_3$ is **acetophenone**, $\text{C}_6\text{H}_5\text{COC}_6\text{H}_5$ is **benzophenone**.

IUPAC names

1. Longest chain containing the carbonyl; number it so the carbonyl gets the **lowest number**.
2. Drop the "e" and add **-one**: CH_3COCH_3 = **propanone** (propan-2-one); pentan-3-one; cyclopentanone.
3. Substituents are prefixes with locants: 2,4-dimethylpentan-3-one.

[exposure] When a compound carries **two functional groups**, only one can be the "suffix" (the ending of the name). The other has to become a **prefix**. The ranking that matters here: **carboxylic acid** > **aldehyde** > **ketone** > **alcohol**. A ketone that loses out becomes the prefix **oxo-**; an alcohol that loses out becomes **hydroxy-**. Example: a chain with a –CHO at one end and a ketone at C3 is **3-oxopentanal**, and a chain with a ketone and an alcohol is **4-hydroxypentan-2-one** (2014 Q1).

Examiner asks: "write the IUPAC name" for a two-group compound (2014 Q1), and "draw the structure of 4-methylpent-3-en-2-one" or "p-methylbenzaldehyde" (2015 Q20). NCERT intext 8.1 asks for structures from names; the pattern is always chain first, then group positions.

Structure of the carbonyl group — *flat, polar, and electron-poor at carbon — the cause of everything that follows*

The carbonyl carbon is sp^2 **hybridised**. It makes three sigma bonds (one to oxygen, two to R or H) lying in **one plane at about 120°** . Its leftover unhybridised p orbital overlaps sideways with a p orbital on oxygen to form the **pi bond**, above and below the plane.

Oxygen is more electronegative than carbon, so the pi electrons sit nearer oxygen and the bond is **polar**: $\text{C}^{\delta+} = \text{O}^{\delta-}$. Resonance puts the same picture another way — one contributor has a positive carbon and a negative oxygen.

[exposure] An **electrophile** is an electron-seeking species and a **nucleophile** is an electron-rich one that attacks a positive centre (you met the nucleophile in Haloalkanes). So the **carbonyl carbon is an electrophilic centre** (a Lewis acid) and the **carbonyl oxygen is a nucleophilic centre** (a Lewis base). Because of the strong dipole, aldehydes and ketones are more polar than ethers.

Preparing aldehydes and ketones — *first the methods that give either; then the aldehyde-only and ketone-only routes*

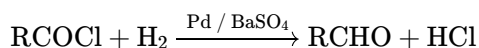
Methods that give aldehydes or ketones

1. **Oxidation of alcohols.** A **primary** alcohol gives an aldehyde, a **secondary** alcohol gives a ketone — with a *mild* oxidant that stops there. **[exposure]** PCC (pyridinium chlorochromate) and anhydrous CrO_3 are the two mild chromium oxidants; PCC is preferred for primary alcohols because it stops at the aldehyde and leaves a $\text{C}=\text{C}$ alone. A strong oxidant would carry an aldehyde on to the acid.
2. **Dehydrogenation of alcohols.** Alcohol vapour passed over copper (or silver) at **573 K** loses hydrogen: primary $\rightarrow$ aldehyde, secondary $\rightarrow$ ketone. It is an industrial method.
3. **Ozonolysis of alkenes.** O_3 and then $\text{Zn} / \text{H}_2\text{O}$ cut the double bond and put $\text{C}=\text{O}$ on each end. Propene gives ethanal and methanal; 2-methylbut-2-ene gives ethanal and propanone (an aldehyde and a ketone).
4. **Hydration of alkynes.** Water adds to a triple bond with HgSO_4 and dilute H_2SO_4 , following Markovnikov. The first product, an enol, flips to the carbonyl form (**tautomerism**). Ethyne gives **ethanal**; every other alkyne gives a **ketone** (propyne gives propanone).

Methods for aldehydes only

Rosenmund reduction — an acid chloride, hydrogenated over **palladium on barium sulphate**.

[exposure] The BaSO_4 **poisons** the palladium (partly deactivates it) on purpose, so it stops at the aldehyde instead of carrying on to the alcohol.



Stephen reaction — a nitrile is reduced to an **imine** $\text{RCH}=\text{NH}$ with stannous chloride and HCl ; hydrolysis gives the aldehyde. **DIBAL-H** (diisobutylaluminium hydride) does the same on nitriles *and* esters, and it does not touch a $\text{C}=\text{C}$.



Oxidising a ring methyl to an aldehyde (toluene $\rightarrow$ benzaldehyde). Strong oxidants overshoot to benzoic acid, so a reagent is needed that stops at an intermediate. Three ways:

- **Etard reaction:** chromyl chloride CrO_2Cl_2 in CS_2 gives a chromium complex; hydrolysis gives benzaldehyde.
 - CrO_3 in **acetic anhydride** (273–283 K) gives **benzylidene diacetate**, which hydrolyses to benzaldehyde.
 - **Side-chain chlorination:** Cl_2 in light (free-radical) puts two chlorines on the methyl to give **benzal chloride** $\text{C}_6\text{H}_5\text{CHCl}_2$; hydrolysis at 373 K gives benzaldehyde. This is the commercial method.
1. **Gattermann–Koch reaction** — benzene with CO and HCl over anhydrous AlCl_3 (and CuCl) gives benzaldehyde. $\text{CO} + \text{HCl}$ gives HCOCl ; AlCl_3 pulls off Cl^- to leave the **formyl cation** HCO^+ , the electrophile that attacks the ring.

Methods for ketones only

1. **From an acid chloride and dialkylcadmium.** A Grignard reagent with CdCl_2 gives R_2Cd ; an acyl chloride then transfers one R to give a ketone. $2 \text{CH}_3\text{COCl} + (\text{CH}_3)_2\text{Cd} \rightarrow 2 \text{CH}_3\text{COCH}_3 + \text{CdCl}_2$.

2. **From a nitrile and a Grignard reagent**, then hydrolysis: $\text{CH}_3\text{CN} + \text{CH}_3\text{MgBr}$ gives an adduct that hydrolyses to **propanone**.
3. **Friedel–Crafts acylation** — benzene with an acid chloride over anhydrous AlCl_3 :
 $\text{C}_6\text{H}_6 + \text{CH}_3\text{COCl} \xrightarrow{\text{AlCl}_3} \text{C}_6\text{H}_5\text{COCH}_3 + \text{HCl}$ (acetophenone). The AlCl_3 is a Lewis acid that pulls off Cl^- and leaves the **acylium ion** CH_3CO^+ , which attacks the ring.

Trap: know which reagent belongs to which product. Choosing between oxidants: **PCC** for primary alcohol $\rightarrow$ aldehyde (and with an allylic double bond present, PCC leaves it alone); **anhydrous** CrO_3 for secondary alcohol $\rightarrow$ ketone; CrO_2Cl_2 **then** H_3O^+ for toluene $\rightarrow$ benzaldehyde; O_3 **then** $\text{Zn}/\text{H}_2\text{O}$ for but-2-ene $\rightarrow$ two ethanal. This is NCERT Example 8.1 and it is asked as "name the reagent".

Examiner asks: the Rosenmund (2019 Q9), Stephen and Etard equations (2015 Q22), Friedel–Crafts acylation (Q10), the dialkylcadmium route (2022 Q28(a)), DIBAL-H (2017 Q2), and conversions that use them (2022 Q24(a)).

Physical properties of aldehydes and ketones — *dipole–dipole forces put them between hydrocarbons and alcohols*

- **State.** Methanal is a gas; ethanal is a volatile liquid; the rest are liquids or solids.
- **Boiling point.** Higher than hydrocarbons and ethers of similar mass, because of **dipole–dipole attraction** between the polar $\text{C}=\text{O}$ groups. Lower than alcohols of similar mass, because they have **no O–H** and so cannot hydrogen-bond to each other. NCERT's set (masses 58–60): **n-butane 273 K < methoxyethane 281 K < propanal 322 K < propanone 329 K < propan-1-ol 370 K**. Between the two isomers, the ketone (two R groups, more surface area) boils slightly higher than the aldehyde.
- **Solubility.** The lower members — methanal, ethanal, propanone — mix with water in all proportions, because the carbonyl oxygen hydrogen-bonds to water. Solubility falls quickly as the alkyl chain grows (the chain is hydrophobic). All are fairly soluble in organic solvents.
- **Odour.** Lower aldehydes smell sharp and pungent; as the molecule grows the smell becomes less pungent and more fragrant, so they are used in perfumes and flavourings.

Examiner asks: "arrange in increasing boiling point" — propane < ethanal < ethanol (2022 Q5(b)); NCERT example 8.2 (an aldehyde, an alcohol, an ether and a hydrocarbon of similar mass: hydrocarbon < ether < aldehyde < alcohol).

Nucleophilic addition — the master reaction — *a two-step attack on the carbonyl carbon, and why aldehydes win over ketones*

[exposure] Nucleophilic addition is the characteristic reaction of the carbonyl group: a nucleophile attacks the electron-poor carbon and something else adds to the oxygen, so the $\text{C}=\text{O}$ becomes a single bond and the carbon turns from sp^2 to sp^3 . It happens in two steps:

1. **Slow step.** The nucleophile, coming at roughly right angles to the flat carbonyl plane, bonds to the carbon. The pi electrons move to oxygen, which becomes negative. This gives a **tetrahedral alkoxide intermediate** (the carbon is now sp^3).
2. **Fast step.** The oxygen picks up H^+ from the medium, giving the neutral addition product.

Aldehydes react faster than ketones — two reasons

- **Steric effect.** A ketone has **two** alkyl groups next to the carbonyl, an aldehyde has one (or none). The crowding blocks the nucleophile's approach.

- **Electronic effect.** Alkyl groups push electrons (+I), which **reduces the positive charge** on the carbonyl carbon. A ketone has two such groups, so its carbon is less electrophilic.

Both factors favour the aldehyde. The same logic orders any set: the **fewer and smaller the groups on the carbonyl, the faster** the reaction.

Worked example · NCERT intext 8.4 — rank ethanal, propanal, propanone, butanone for nucleophilic addition

Count the groups. Ethanal has one methyl; propanal one ethyl; propanone two methyls; butanone a methyl and an ethyl. Most crowded and most electron-pushing is butanone, then propanone, then propanal, then ethanal. Increasing reactivity: **butanone < propanone < propanal < ethanal**.

Aromatic aldehydes are less reactive than aliphatic ones

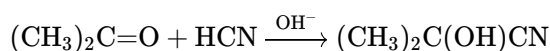
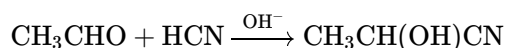
In benzaldehyde the ring pushes electrons into the carbonyl by **resonance** (the $-R$ pull of the carbonyl draws ring electrons towards it), so the carbonyl carbon is **less positive**. Benzaldehyde is therefore less reactive than propanal in nucleophilic addition (2022 Q15(a), 2025 Q1). NCERT example 8.3 and intext 8.4(ii): for aromatic carbonyls the order of reactivity runs from the ketone (acetophenone), through *p*-tolualdehyde (a methyl donates), benzaldehyde, to ***p*-nitrobenzaldehyde** (the nitro group withdraws, so the carbon is most positive and reacts fastest).

Trap: do not answer "benzaldehyde is less reactive because it is bulky". The reason is **resonance with the ring lowering the positive charge** on the carbonyl carbon, and you should say so, with "less electrophilic".

Nucleophilic addition: HCN, bisulphite, Grignard — *three reagents that add across C=O and give three different products*

Hydrogen cyanide → cyanohydrin

[exposure] A **cyanohydrin** carries an $-OH$ and a $-CN$ on the same carbon. Pure HCN is slow, so a trace of **base** is added: it removes H^+ from HCN and gives the cyanide ion CN^- , a much stronger nucleophile. CN^- attacks the carbonyl carbon, and the oxygen then takes H^+ . Cyanohydrins are useful synthetic intermediates.



Sodium hydrogensulphite → addition product

$NaHSO_3$ adds to give a crystalline **hydrogensulphite addition compound**. For most aldehydes the equilibrium lies to the product side; for most ketones it lies to the reactant side, because of steric hindrance. The compound is water-soluble. Warm it with dilute mineral acid or alkali and the carbonyl compound is regenerated — so the reaction is used to **separate and purify aldehydes**: make the adduct, filter away the insoluble impurities, then decompose the adduct (2023 Q33(b)).

Grignard reagent → alcohol

The $R^{\delta-}$ of a Grignard reagent attacks the carbonyl carbon; hydrolysis then gives an alcohol. What you get depends on the carbonyl you started with:

Carbonyl	With RMgX, then H ₂ O
Methanal HCHO	primary alcohol RCH ₂ OH
Any other aldehyde	secondary alcohol RCH(OH)R'
A ketone	tertiary alcohol RR'C(OH)R''

Example (2022 Q8): propanone + CH₃MgI, then water, gives 2-methylpropan-2-ol (tertiary). Ethanal + CH₃MgBr gives propan-2-ol (secondary).

Nucleophilic addition: alcohols — *hemiacetals, acetals and ketals* — and why dry HCl is there

[exposure] An **aldehyde** reacts with **one** equivalent of a monohydric alcohol (one –OH) in the presence of **dry HCl** to give a **hemiacetal**: a carbon holding both an –OH and an –OR. With a second equivalent of alcohol the –OH is replaced by another –OR, giving an **acetal** (a **gem-dialkoxy** compound: two –OR groups on the same carbon).



A **ketone** does not react with a monohydric alcohol, but it does react with **ethylene glycol** (HOCH₂CH₂OH, two –OH groups) under the same conditions, forming a cyclic product, an **ethylene glycol ketal**.

Why dry HCl: it **protonates the carbonyl oxygen**, which makes the carbonyl carbon more positive (more electrophilic), so the weakly nucleophilic alcohol can attack. Acetals and ketals are reversed by **dilute aqueous mineral acid**, which gives back the aldehyde or ketone.

Examiner asks: "draw the structure of the hemiacetal of ethanal" and "cyanohydrin of cyclobutanone" (2020 Q23(b)).

Nucleophilic addition–elimination: ammonia and its derivatives — H₂N–Z *attacks, water leaves, and C=N appears*

Ammonia and its derivatives H₂N–Z (Z may be an alkyl or aryl group, –OH, –NH₂, –NHC₆H₅, –NHCONH₂ ...) add to C=O with acid as a catalyst, and then **lose water** by rapid dehydration of the intermediate. The net change is C=O → C=N–Z. Because the dehydration is fast, the equilibrium is driven to the product.

Reagent H ₂ N–Z	Product of C=O
Ammonia NH ₃	imine C=NH
A primary amine RNH ₂	substituted imine, Schiff's base C=NR
Hydroxylamine NH ₂ OH	oxime C=N–OH
Hydrazine NH ₂ NH ₂	hydrazone C=N–NH ₂
Phenylhydrazine C ₆ H ₅ NHNH ₂	phenylhydrazone
2,4-Dinitrophenylhydrazine (2,4-DNP)	2,4-dinitrophenylhydrazone — a yellow, orange or red solid
Semicarbazide NH ₂ NHCONH ₂	semicarbazone

Aldehyde products take the suffix by name: from ethanal you get acetaldehyde oxime, acetaldehyde hydrazone, and so on; from a ketone, acetone semicarbazone. The derivatives are **crystalline solids with**

sharp melting points, so they are used to **identify** a carbonyl compound — and the **2,4-DNP test** (a yellow/orange/red precipitate) is the standard way to show that a compound is an aldehyde or ketone at all.

Trap: semicarbazide has **two** -NH_2 groups, but only **one** forms the semicarbazone. The lone pair on the nitrogen next to C=O is in resonance with it, so that nitrogen is not nucleophilic enough; the terminal NH_2 , whose lone pair is free, does the attacking (2014 Q13(b)).

Examiner asks: NCERT intext 8.5 (predict the products: cyclopentanone + NH_2OH gives the oxime; cyclohexanone + 2,4-DNP gives the 2,4-dinitrophenylhydrazone; a compound with both C=C and CHO and semicarbazide still keeps the C=C); the semicarbazone of ethanal (2022 Q27(c)).

Reduction of aldehydes and ketones — *to alcohols, or all the way down to hydrocarbons*

To alcohols

NaBH_4 , LiAlH_4 or catalytic hydrogenation (H_2 with Ni, Pt or Pd) reduce an **aldehyde to a primary alcohol** and a **ketone to a secondary alcohol**. $\text{CH}_3\text{CHO} \xrightarrow{\text{H}_2 / \text{Pt}} \text{CH}_3\text{CH}_2\text{OH}$;
 $\text{CH}_3\text{COCH}_3 \xrightarrow{\text{LiAlH}_4} \text{CH}_3\text{CH}(\text{OH})\text{CH}_3$.

To hydrocarbons — *remove the oxygen completely*

The C=O becomes CH_2 in either of two reactions, one in acid and one in base:

- **Clemmensen reduction** — zinc amalgam and concentrated HCl.
 $\text{CH}_3\text{CH}_2\text{CHO} \xrightarrow{\text{Zn-Hg} / \text{conc. HCl}} \text{CH}_3\text{CH}_2\text{CH}_3$. **[exposure] Zinc amalgam** is zinc alloyed with mercury; the metal and acid together make the hydrogen that does the reducing.
- **Wolff–Kishner reduction** — hydrazine first (giving the hydrazone, losing water), then heating with KOH or NaOH in a **high-boiling solvent** such as ethylene glycol; nitrogen gas escapes and hydrogen goes on in its place. $\text{CH}_3\text{COCH}_3 \xrightarrow{\text{NH}_2\text{NH}_2} (\text{CH}_3)_2\text{C=NNH}_2 \xrightarrow{\text{KOH, ethylene glycol, } \Delta} \text{CH}_3\text{CH}_2\text{CH}_3 + \text{N}_2$.

Trap: choose by what else is in the molecule. Clemmensen uses strong acid, Wolff–Kishner strong base — pick the one that the rest of the molecule survives. For a plain "reduce the carbonyl to CH_2 " question, either is accepted; name the one you write.

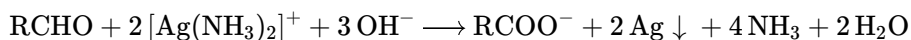
Examiner asks: Wolff–Kishner and Clemmensen as "explain/write the reaction" questions (2022 Q11, Q28(b), 2025 Q6(b)).

Oxidation: the tests that tell aldehydes from ketones — *Tollens', Fehling's, and the haloform reaction*

Aldehydes oxidise easily to carboxylic acids with the *same* number of carbons, even with **mild** oxidants. They can do so because the carbonyl carbon carries a **hydrogen** that can turn into -OH without any bond being broken. **Ketones** have no such hydrogen. They oxidise only under vigorous conditions (a strong oxidant and heat), by breaking a carbon–carbon bond next to the carbonyl; the result is a **mixture of acids with fewer carbons**. This difference is the basis of the distinguishing tests.

Tollens' test

[exposure] Tollens' reagent is **ammoniacal silver nitrate**, $[\text{Ag}(\text{NH}_3)_2]^+$ in alkaline solution, always made fresh. It is a mild oxidant. Warmed with an **aldehyde**, it is reduced to silver metal, which coats the tube as a **bright silver mirror**; the aldehyde becomes a carboxylate. A ketone gives nothing.



Fehling's test

[exposure] Fehling's reagent is two solutions mixed in equal amounts just before use: **Solution A**, aqueous copper(II) sulphate, and **Solution B**, alkaline sodium potassium tartrate (**Rochelle salt**). Warmed with an aldehyde, it gives a **reddish-brown precipitate** of copper(I) oxide Cu_2O . Ketones do not react, and **aromatic aldehydes (benzaldehyde) do not respond** to Fehling's, because it is too weak an oxidant for them.

The haloform (iodoform) reaction

Any aldehyde or ketone with a **methyl group on the carbonyl carbon**, $\text{CH}_3\text{—CO—}$ (a **methyl ketone**), is oxidised by sodium hypohalite NaOX . The methyl group leaves as a **haloform** CHX_3 and the rest becomes a carboxylate with **one carbon fewer**.



For the **iodoform test** you use I_2 with NaOH (which makes NaOI in the flask; " $\text{NaOH} + \text{I}_2$ " in a question means the same). A **yellow precipitate of CHI_3** (iodoform) shows a $\text{CH}_3\text{CO—}$ group, or a $\text{CH}_3\text{CH}(\text{OH})\text{—}$ group, which is first oxidised to $\text{CH}_3\text{CO—}$. The reaction does not touch a $\text{C}=\text{C}$.

Worked example · NCERT Example 8.4 — compound A, $\text{C}_8\text{H}_8\text{O}$

A gives an orange-red precipitate with 2,4-DNP, so it is a carbonyl compound. It gives a yellow precipitate with NaOH / I_2 , so it carries $\text{CH}_3\text{CO—}$. It does **not** reduce Tollens' or Fehling's, so it is a **ketone**. It does not decolourise bromine water or Baeyer's reagent, so the unsaturation in the formula is not an alkene; it is an aromatic ring. The formula fits a ring bearing COCH_3 : **acetophenone**. Drastic oxidation gives benzoic acid $\text{C}_7\text{H}_6\text{O}_2$, and the iodoform reaction gives sodium benzoate and CHI_3 .

Examiner asks: distinguish an aldehyde from a ketone (2020 Q23(a), 2017 Q16(b), 2025 Q7(b)); the same acetophenone puzzle (2025 Q3); the iodoform reaction in conversions (2017 Q18(a), 2023 Q32).

α -Hydrogen reactions: aldol condensation — *why the hydrogens next to $\text{C}=\text{O}$ are acidic, and what that lets two molecules do*

[exposure] The carbon next to the carbonyl is the **α -carbon**, and the hydrogens on it are **α -hydrogens**. They are noticeably **acidic** — a base can remove one. Two reasons: the carbonyl is strongly **electron-withdrawing**, which weakens the C—H bond; and the resulting carbanion is **resonance-stabilised**, because its negative charge can sit on the carbonyl oxygen. (2023 Q34(b))

[exposure] Aldol condensation. An aldehyde or ketone with at least one α -hydrogen, in **dilute alkali**, joins two of its molecules. The base takes an α -hydrogen to make a carbanion; that carbanion attacks the carbonyl carbon of a second molecule; the oxygen then takes H^+ . The product of an **aldehyde** is a **β -hydroxy aldehyde**, an **aldol** (aldehyde + alcohol); that of a **ketone** is a **β -hydroxy ketone**, a **ketol**. On **heating** they lose water to give an **α,β -unsaturated carbonyl compound**, the aldol condensation product.



That is ethanal $\rightarrow$ 3-hydroxybutanal (the aldol) $\rightarrow$ but-2-enal. Propanone gives 4-hydroxy-4-methylpentan-2-one (the ketol) and then 4-methylpent-3-en-2-one on heating.

Cross aldol condensation

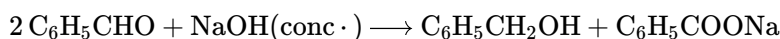
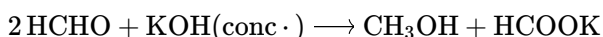
Two *different* carbonyl compounds, both with α -hydrogens, give a mixture of **four** products — two "self" and two "cross". Ethanal + propanal gives: but-2-enal and 2-methylpent-2-enal (self), and 2-methylbut-2-enal and pent-2-enal (cross). If one partner has **no α -hydrogen** (such as benzaldehyde), it can only be the acceptor and a single product forms: benzaldehyde + acetophenone gives **1,3-diphenylprop-2-en-1-one** (benzalacetophenone).

Trap: the condition for aldol is **at least one α -hydrogen**. Benzaldehyde, methanal and $(\text{CH}_3)_3\text{C}-\text{CHO}$ have none and cannot self-aldol; they do the Cannizzaro reaction instead (next section). Do not forget to heat if the question asks for the condensation product (the α,β -unsaturated compound), not just the aldol.

Examiner asks: describe aldol condensation (2015 Q17(b)); ethanal to but-2-enal (2022 Q24(c)); $\text{CH}_3\text{CN} \rightarrow$ ethanal $\rightarrow$ but-2-enal (2017 Q26); the 2025 puzzle where B is made from ethanal and dilute NaOH (2025 Q2); cross-aldol of benzaldehyde with ethanal (2025 Q4(b)).

Cannizzaro reaction; ring substitution; uses — *what an aldehyde with no α -hydrogen does, and where the carbonyl directs a ring*

[exposure] Cannizzaro reaction. An aldehyde with **no α -hydrogen**, heated with **concentrated alkali**, **reacts with itself:** one molecule is **oxidised** to the carboxylate salt, the other is **reduced** to the alcohol. A reaction in which one substance is both oxidised and reduced is a **disproportionation**.



Electrophilic substitution in the ring. In an aromatic aldehyde or ketone the carbonyl pulls electrons out of the ring (the $-R$ effect) and puts a partial positive charge at the ortho and para positions, so it is **deactivating** and **meta-directing**. Benzaldehyde + $\text{HNO}_3 / \text{H}_2\text{SO}_4$ at 273–283 K gives **3-nitrobenzaldehyde** (*m*-nitrobenzaldehyde).

Uses. 40% aqueous methanal is **formalin**, a preservative for biological specimens; methanal is also the monomer for **Bakelite** (phenol–formaldehyde resin) and urea–formaldehyde glues. Ethanal is the starting material for ethanoic acid, ethyl ethanoate, vinyl acetate, polymers and drugs. Benzaldehyde goes into perfumes and dyes; propanone and butanone are industrial solvents; butyraldehyde, vanillin, acetophenone and camphor are used for odour and flavour.

Examiner asks: the Cannizzaro reaction (2022 Q11, 2013 Q19(a), 2025 Q5(a)); the compound that gives sodium formate and methanol (Q29 in the 30+ video — methanal); nitration of benzaldehyde (2013 Q19(b)).

Carboxylic acids: structure and names — *the carboxyl group* $-\text{COOH}$, and a table of names you must know cold

[exposure] A carboxylic acid carries the **carboxyl group** $-\text{COOH}$: a carbonyl and a hydroxyl on the same carbon (the name is "carbonyl" + "hydroxyl"). It is the one family in this chapter that is an **acid**, which is why it turns blue litmus red. If the group sits on an alkyl chain the acid is **aliphatic**; if it sits on a benzene

ring it is **aromatic**. The higher aliphatic acids (C_{12} – C_{18}) are **fatty acids**; palmitic acid $C_{15}H_{31}COOH$ is an example. In natural fats they are present as esters of glycerol.

Common names — where they come from

Carboxylic acids were among the first organic compounds isolated, so most carry a common name from the Latin or Greek for their source: **formic** acid (from *formica*, ant), **acetic** acid (*acetum*, vinegar), **butyric** acid (*butyrum*, butter — it is in rancid butter).

IUPAC names

1. **One** –COOH: replace the final "e" of the alkane with **-oic acid** (the carboxyl carbon is C1): methanoic, ethanoic, propanoic acid.
2. **Two** –COOH **groups**: both carbons are counted in the chain, and the "e" is **kept**: propane-1,3-dioic acid = **propanedioic acid** (no locants needed when they sit at both ends).
3. **Three or more** –COOH **groups**: the carboxyl carbons are **not** counted; they are named as **carboxylic acids** on the backbone: propane-1,2,3-tricarboxylic acid.
4. **On a ring**: ring name + **carboxylic acid**: 2-methylcyclopentanecarboxylic acid. C_6H_5COOH is benzoic acid, also benzenecarboxylic acid.

Structure	Common name	IUPAC name
HCOOH	formic acid	methanoic acid
CH ₃ COOH	acetic acid	ethanoic acid
CH ₃ CH ₂ COOH	propionic acid	propanoic acid
CH ₃ CH ₂ CH ₂ COOH	butyric acid	butanoic acid
(CH ₃) ₂ CHCOOH	isobutyric acid	2-methylpropanoic acid
HOOC–COOH	oxalic acid	ethanedioic acid
HOOC–CH ₂ –COOH	malonic acid	propanedioic acid
HOOC–(CH ₂) ₂ –COOH	succinic acid	butanedioic acid
HOOC–(CH ₂) ₃ –COOH	glutaric acid	pentanedioic acid
HOOC–(CH ₂) ₄ –COOH	adipic acid	hexanedioic acid
C ₆ H ₅ COOH	benzoic acid	benzenecarboxylic acid
C ₆ H ₅ CH ₂ COOH	phenylacetic acid	2-phenylethanoic acid
benzene with two adjacent –COOH	phthalic acid	benzene-1,2-dicarboxylic acid

More examples (NCERT intext 8.6): $C_6H_5CH_2CH_2COOH$ = 3-phenylpropanoic acid; $(CH_3)_2C=CHCOOH$ = 3-methylbut-2-enoic acid; 2,4,6-trinitrobenzoic acid.

Structure of the carboxyl group — why the carbonyl carbon here is less electrophilic than in an aldehyde

The carboxyl carbon is sp^2 hybridised and **trigonal planar** (all bonds in one plane, about 120°). But it is **less electrophilic** than the carbonyl carbon of an aldehyde or ketone. The reason is **resonance**: a lone pair on the –OH oxygen flows towards the carbon, so the carbon receives electron density from the oxygen and its partial positive charge is smaller. A nucleophile therefore has much less to attack.

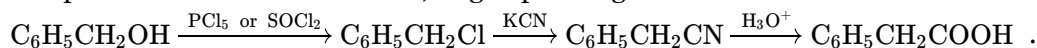
Trap: this is the standard answer to "why do carboxylic acids not give the characteristic reactions of the carbonyl group" (2017 Q6, 2025 Q5(b)(i)). Write: the lone pair on the –OH oxygen is in **resonance** with the C=O, which **reduces the electrophilicity** of the carbon, so the nucleophilic-addition reactions do not occur.

Preparing carboxylic acids — *six routes; the carbon count tells you which one*

1. **Oxidising primary alcohols and aldehydes.** Use a **strong** oxidant: KMnO_4 (acidic, alkaline or neutral), $\text{K}_2\text{Cr}_2\text{O}_7$ (acidic), or CrO_3 in acid, known as the **Jones reagent**. (Anhydrous CrO_3 without the acid is mild.) Aldehydes also oxidise with the *mild* Tollens' and Fehling's reagents.
2. **Oxidising an alkylbenzene.** Vigorous oxidation with hot alkaline KMnO_4 (or chromic acid) turns the **whole side chain, however long, into** –COOH: toluene, ethylbenzene and propylbenzene all give potassium benzoate, and acidification gives **benzoic acid** (2019 Q4).
3. **Hydrolysing nitriles and amides.** RCN with H^+ or OH^- goes to the amide and then to the acid. Mild conditions stop at the amide; complete hydrolysis gives the acid. An amide heated with water gives the acid plus NH_3 .
4. **Grignard reagent and dry ice.** RMgX attacks the carbon of solid CO_2 (dry ice) in dry ether; acidifying the salt gives the acid. This route **adds one carbon**:
$$\text{CH}_3\text{MgBr} + \text{CO}_2 \longrightarrow \text{CH}_3\text{COOMgBr} \xrightarrow{\text{H}_3\text{O}^+} \text{CH}_3\text{COOH} .$$
5. **Hydrolysing acid chlorides and anhydrides.** Water gives the acid; aqueous base is faster and gives the carboxylate, which you acidify.
6. **Hydrolysing esters.** Acid hydrolysis gives acid + alcohol directly. Base hydrolysis is faster and gives the carboxylate salt, which you acidify. $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOC}_2\text{H}_5 \xrightarrow{\text{NaOH}} \text{CH}_3\text{CH}_2\text{CH}_2\text{COONa} + \text{C}_2\text{H}_5\text{OH} .$

Worked example · NCERT Example 8.5 — benzyl alcohol to phenylethanoic acid

The product has one carbon more, so go up through the nitrile.



Worked example · NCERT Example 8.5 — 3-nitrobromobenzene to 3-nitrobenzoic acid

Br must become COOH, one carbon added: make the Grignard reagent (Mg, dry ether), add CO_2 , hydrolyse. The NO_2 survives untouched.

Two more NCERT conversions: butan-1-ol $\rightarrow$ butanoic acid with acidic $\text{K}_2\text{Cr}_2\text{O}_7$; 4-methylacetophenone $\rightarrow$ benzene-1,4-dicarboxylic acid with KMnO_4 / KOH (both the methyl and the ketone are oxidised). Cyclohexene $\rightarrow$ hexanedioic acid with hot KMnO_4 / H_2SO_4 breaks the ring at the double bond.

Examiner asks: ethylbenzene + KMnO_4/KOH then acid (2019 Q4; 2025 Q4(a)); benzene-ring conversions to benzoic acid (NCERT intext 8.7); the Grignard– CO_2 route (2017 Q26).

Physical properties of carboxylic acids — *hydrogen-bonded dimers explain the high boiling points*

- **State and smell.** The first three aliphatic acids are colourless liquids with a sharp smell; the higher ones are waxy solids with almost no smell (low volatility).
- **Boiling point.** Higher than aldehydes, ketones and even alcohols of similar mass. **Reason:** carboxylic acid molecules hydrogen-bond strongly in pairs. The O–H of an acid is **more polar** than that of an alcohol, because the carbonyl group next to it pulls electrons away, so the hydrogen bonds are stronger.

- **[exposure]** A **dimer** is a pair of identical molecules held together; here, two acid molecules joined by two hydrogen bonds. The bonds are so strong that most carboxylic acids stay as dimers **even in the vapour phase** and in **aprotic solvents** (solvents that have no acidic hydrogen to give away).
- **Solubility.** Aliphatic acids up to **four carbons** mix with water (hydrogen bonds to water). Solubility drops as the carbon chain grows. Benzoic acid, the simplest aromatic acid, is insoluble in cold water because of the large hydrophobic ring. Acids dissolve in less polar organic solvents such as benzene, ether, alcohol and chloroform.

Examiner asks: which of ethanoic acid and ethanol has the higher boiling point, and why (2023 Q35(b): ethanoic acid, stronger hydrogen bonding and dimer formation).

Acidity: what carboxylic acids do with metals, alkali and carbonates — *the O–H bond breaks; the test with bicarbonate is the one examiners love*

Carboxylic acids are fairly strong acids because the O–H bond is polar and the **carboxylate ion** RCOO^- left behind is stabilised by resonance (below). They turn blue litmus red and react with:

- **Active metals** (Na, K, Mg): $2 \text{RCOOH} + 2 \text{Na} \longrightarrow 2 \text{RCOONa} + \text{H}_2$. (Alcohols do this too.)
- **Alkalis:** $\text{RCOOH} + \text{NaOH} \longrightarrow \text{RCOONa} + \text{H}_2\text{O}$ (neutralisation). Phenols do this too.
- **Carbonates and bicarbonates:** $\text{RCOOH} + \text{NaHCO}_3 \longrightarrow \text{RCOONa} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$ with a **brisk effervescence** of CO_2 . Alcohols and phenols are too weak to do it.

Trap: the **sodium bicarbonate test** is the standard test to detect a carboxyl group, and the answer to every "distinguish ethanoic acid from ethanol", "benzoic acid from phenol", "benzoic acid from ethyl benzoate" question (2017 Q12(b), 2023 Q34(a), 2025 Q7(b)(i)). Write the observation: brisk effervescence of CO_2 with the acid, nothing with the other.

Putting a number on strength: K_a and pK_a

[exposure] For $\text{RCOOH} + \text{H}_2\text{O} \rightleftharpoons \text{RCOO}^- + \text{H}_3\text{O}^+$ the **acid dissociation constant** is $K_a = \frac{[\text{RCOO}^-][\text{H}_3\text{O}^+]}{[\text{RCOOH}]}$ (the constant concentration of water is folded in), and $pK_a = -\log K_a$. A **larger** K_a means a stronger acid, but the numbers are awkward powers of ten, so the pK_a is used. **The smaller the pK_a , the stronger the acid.** NCERT's bands: below 1, a **strong** acid; 1 to 5, **moderately strong**; 5 to 15, **weak**; above 15, **extremely weak**.

Why carboxylic acids beat alcohols and phenols — *it is all about how stable the left-over negative ion is*

A stronger acid is one whose **conjugate base** (the ion left after H^+ leaves) is more stable, because then the equilibrium lies further towards the ions. Compare the three:

- **Carboxylic acid vs alcohol.** The carboxylate RCOO^- is stabilised by **resonance**: its two structures are **identical**, and the negative charge is spread over **two oxygens**. The alkoxide RO^- of an alcohol has no resonance, and the electron-pushing R group makes its single oxygen even more crowded with charge. So carboxylic acids are much stronger than alcohols. (Ethanoic acid versus ethanol is asked as 2025 Q5(b)(ii).)
- **Carboxylic acid vs phenol.** The phenoxide ion also has resonance, but its charge is shared between **one** oxygen and less electronegative ring carbons. The carboxylate puts the charge on **two electronegative oxygens**, so delocalisation is more effective. So carboxylic acids are stronger than phenols.

The acid itself is also stabilised by resonance, but less so: one of its contributing structures has **charge separation** (a positive and a negative atom), which costs energy.

Ranking: **carboxylic acid > phenol > alcohol**. All are weaker than mineral acids.

What a substituent does to acid strength — *electron-pullers raise it, electron-pushers lower it — four rules*

The same logic: anything that **stabilises the carboxylate** makes a stronger acid.

- **Electron-withdrawing group (EWG)** — NO_2 , CN , halogens, CF_3 — pulls charge towards itself and **stabilises the carboxylate**: stronger acid. An **electron-releasing group (ERG)** — alkyl, OCH_3 — pushes charge onto the carboxylate and **destabilises** it: weaker acid.
- **Number of substituents**: more is stronger. $\text{CH}_3\text{COOH} < \text{ClCH}_2\text{COOH} < \text{Cl}_2\text{CHCOOH} < \text{Cl}_3\text{CCOOH}$.
- **Nature of the substituent**: the more electronegative, the stronger. $\text{CH}_3\text{COOH} < \text{ICH}_2\text{COOH} < \text{BrCH}_2\text{COOH} < \text{ClCH}_2\text{COOH} < \text{FCH}_2\text{COOH}$. Fluorine is the strongest puller.
- **Position**: the **closer** the EWG is to the COOH , the stronger the effect, because its pull falls off with distance. 2-fluorobutanoic acid > 3-fluorobutanoic acid > 4-fluorobutanoic acid.

A surprising one: a **phenyl or vinyl group** attached directly to the carboxyl carbon **increases** acidity relative to a saturated acid. You might expect resonance donation to weaken it, but the carbon the carboxyl group is attached to is sp^2 , which is more electronegative than an sp^3 carbon, and that wins.

Worked example · the benzoic acid series (pK_a from NCERT)

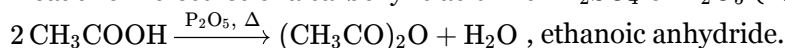
4-Methoxybenzoic acid (pK_a 4.46) < benzoic acid (4.19) < 4-nitrobenzoic acid (3.41) in acid strength. OCH_3 donates electrons and weakens the acid; NO_2 withdraws and strengthens it. A lower pK_a means a stronger acid. 3,4-Dinitrobenzoic acid, with two NO_2 groups, is the strongest of all four (2022 Q21(b)).

Examiner asks: "why is ethanoic acid a stronger acid than ethanol" (2025 Q5(b)(ii)); "4-nitrobenzoic acid is a stronger acid than benzoic acid" (2022 Q15(b)); "arrange CH_3COOH , ClCH_2COOH , FCH_2COOH by acid strength" (2022 Q5(a)); NCERT intext 8.8, which asks which of each pair is stronger.

Carboxylic acid reactions: breaking the C–OH bond — *anhydride, ester, acid chloride, amide — the four derivatives made from the acid*

Anhydride

Heat two molecules of a carboxylic acid with H_2SO_4 or P_2O_5 (P_4O_{10}) and water is removed:



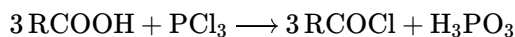
Ester — esterification

[exposure] Esterification is making an ester from a carboxylic acid and an alcohol (or phenol), with an acid catalyst (conc. H_2SO_4 or HCl gas): $\text{RCOOH} + \text{R}'\text{OH} \xrightleftharpoons{\text{H}^+} \text{RCOOR}' + \text{H}_2\text{O}$. It is reversible; esters smell fruity.

Mechanism (all you need is the sequence): (1) H^+ from the catalyst **protonates the carbonyl oxygen**, which makes the carbon more positive. (2) The alcohol's oxygen, with its lone pair, **attacks that carbon** (nucleophilic attack), giving a tetrahedral intermediate. (3) A **proton transfers** between oxygens, so one OH becomes OH_2^+ . (4) **Water leaves**, giving the protonated ester. (5) It **loses** H^+ , giving the ester and handing back the catalyst. The H^+ is used and regained, so it is a catalyst.

Acid chloride

The -OH is replaced by -Cl with PCl_5 , PCl_3 or SOCl_2 (thionyl chloride):



Thionyl chloride is preferred: its by-products, SO_2 and HCl , are gases that leave the mixture, so the acid chloride comes out pure.

Amide

A carboxylic acid with NH_3 gives the ammonium salt; **heating** the salt loses water to give the amide.

$\text{CH}_3\text{COOH} + \text{NH}_3 \longrightarrow \text{CH}_3\text{COO}^-\text{NH}_4^+ \xrightarrow{\Delta} \text{CH}_3\text{CONH}_2 + \text{H}_2\text{O}$. Phthalic acid takes two NH_3 to give ammonium phthalate, which on heating gives **phthalamide**; strong heating drives off a molecule of NH_3 and closes the ring to **phthalimide**.

Examiner asks: CH_3COOH with P_4O_{10} (2025 Q6(a)); phthalamide on strong heating gives phthalimide (2025 Q6(a)); acid + NH_3 then heat gives the amide, which is the middle step of the Hofmann chains you will meet in Amines.

Carboxylic acid reactions: reduction, decarboxylation, hydrocarbon part — LiAlH_4 reaches the acid; soda lime removes CO_2 ; Br₂/red P halogenates the α -carbon

Reduction

LiAlH_4 in ether, or better **diborane** B_2H_6 , reduces a carboxylic acid to a **primary alcohol**. Diborane does not easily reduce esters, nitro or halo groups, so it is selective. **Sodium borohydride** NaBH_4 is too mild and does not reduce a carboxyl group.

Decarboxylation

[exposure] Decarboxylation removes CO_2 from a carboxylic acid, leaving a hydrocarbon with **one carbon less**. The sodium salt of the acid is heated with **soda lime** ($\text{NaOH} + \text{CaO}$ in a **3 : 1** ratio):

$\text{RCOONa} + \text{NaOH} \xrightarrow{\text{CaO}, \Delta} \text{RH} + \text{Na}_2\text{CO}_3$. Examples: sodium ethanoate gives **methane**; sodium benzoate gives **benzene**.

[exposure] Kolbe electrolysis. Electrolysing the **aqueous sodium or potassium salt** of an acid is also a decarboxylation, but the alkyl groups from **two** carboxylate ions join: at the anode

$2 \text{RCOO}^- \longrightarrow \text{R-R} + 2 \text{CO}_2 + 2 \text{e}^-$ and at the cathode $2 \text{H}_2\text{O} + 2 \text{e}^- \longrightarrow \text{H}_2 + 2 \text{OH}^-$. The hydrocarbon has **twice** the carbons of the alkyl group of the acid.

Hell-Volhard-Zelinsky (HVZ) reaction

A carboxylic acid with Cl_2 or Br_2 and a little **red phosphorus**, then water, gets a halogen on the **α -carbon**, giving an **α -halocarboxylic acid**. $\text{CH}_3\text{CH}_2\text{COOH} \xrightarrow[\text{H}_2\text{O}]{\text{Br}_2 / \text{red P}} \text{CH}_3\text{CHBrCOOH}$ (2-bromopropanoic acid).

Aqueous NaOH then replaces the halogen by -OH (ethanoic acid $\rightarrow$ 2-chloroethanoic acid $\rightarrow$ 2-hydroxyethanoic acid, 2017 Q18(b)).

The benzene ring of an aromatic acid

The carboxyl group is **deactivating and meta-directing** (the $-R$ effect pulls ring electrons, leaving the ortho/para positions positive). Benzoic acid + $\text{HNO}_3 / \text{H}_2\text{SO}_4$ gives **3-nitrobenzoic acid**; with $\text{Br}_2 / \text{FeBr}_3$ it gives 3-bromobenzoic acid. It does **not undergo Friedel–Crafts**, for two reasons: the ring is deactivated, and the AlCl_3 catalyst bonds to the electron-rich carboxyl oxygens instead of working as a catalyst (2022 Q14(a), 2023 Q33(a)).

Uses

Methanoic acid goes into rubber, textile, dyeing, leather and electroplating; ethanoic acid is a solvent and the acid of vinegar; hexanedioic acid is a monomer for **nylon-6,6**; esters of benzoic acid are used in perfumery; sodium benzoate is a food preservative; higher fatty acids go into soaps and detergents.

Examiner asks: HVZ (2019 Q7, 2022 Q24(b)); decarboxylation with soda lime (2017 Q12(a), 2022 Q28(c), 2023 Q35, 2025 Q6(b)); the conversion that uses LiAlH_4 vs NaBH_4 .

Patterns, collected — the reagents, tests and orderings you will actually be asked to produce

1. Which reagent for which job

To make	From	Reagent
aldehyde	primary alcohol	PCC (or Cu, 573 K)
ketone	secondary alcohol	anhydrous CrO_3 (or Cu, 573 K)
aldehyde	acid chloride	$\text{H}_2 / \text{Pd}-\text{BaSO}_4$ (Rosenmund)
aldehyde	nitrile	$\text{SnCl}_2 / \text{HCl}$ then H_3O^+ (Stephen), or DIBAL-H
benzaldehyde	toluene	CrO_2Cl_2 (Etard); $\text{CrO}_3 / (\text{CH}_3\text{CO})_2\text{O}$; or $\text{Cl}_2, h\nu$ then hydrolysis
ketone	acid chloride	R_2Cd
ketone	benzene	$\text{RCOCl} / \text{AlCl}_3$ (Friedel–Crafts)
carboxylic acid (+1 C)	alkyl halide	Mg, CO_2 , H_3O^+ — or KCN then hydrolysis
acid (same C)	primary alcohol / aldehyde	KMnO_4 or $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}^+$
hydrocarbon (one C less)	acid salt	soda lime, heat

2. Distinguish by a test

Test	Positive for	Negative for
Tollens' (silver mirror)	aldehyde	ketone
Fehling's (red-brown Cu_2O)	aliphatic aldehyde	ketone, <i>aromatic</i> aldehyde
Iodoform ($\text{NaOH} + \text{I}_2$, yellow CHI_3)	$\text{CH}_3\text{CO}-$ or $\text{CH}_3\text{CH}(\text{OH})-$	others
2,4-DNP (orange ppt)	any aldehyde or ketone	acids, esters, alcohols
NaHCO_3 (effervescence)	carboxylic acid	alcohol, phenol, ester

Worked example

Distinguish propanal from propanone (2020 Q23(a)): warm each with freshly prepared Tollens' reagent. Propanal gives a bright silver mirror ($\text{CH}_3\text{CH}_2\text{CHO} \longrightarrow \text{CH}_3\text{CH}_2\text{COO}^-$, $\text{Ag}^+ \longrightarrow \text{Ag}$); propanone does not react.

- Distinguish ethanol from ethanoic acid — answer only: add NaHCO_3 ; the acid gives brisk effervescence of CO_2 , the alcohol does not
- Distinguish benzaldehyde from acetophenone — answer only: Tollens' gives a silver mirror with benzaldehyde; iodoform test (yellow ppt) is given only by acetophenone

3. Carbon count: how it changes

Worked example

Which reactions change the carbon count? **Gain one:** Grignard + CO_2 ; KCN then hydrolysis; nitrile route.

Lose one: haloform (methyl ketone $\rightarrow$ acid, CHX_3 lost); soda-lime decarboxylation; Hofmann (Chapter 9).

Double: aldol condensation ($2 \text{CH}_3\text{CHO} \longrightarrow \text{C}_4$); Kolbe electrolysis.

- Ethanal to but-2-enal (carbons: $2 \rightarrow 4$) — answer only: aldol condensation, dilute NaOH then heat
- Propanoic acid to 2-bromopropanoic acid — answer only: Br_2 / red P then water (HVZ)

4. Rank by reactivity, acidity or boiling point

Worked example

Increasing acid strength: 4-methoxybenzoic acid, benzoic acid, 4-nitrobenzoic acid, 3,4-dinitrobenzoic acid (2022 Q21(b)). More electron-withdrawing groups give a stronger acid, and an electron donor ($-\text{OCH}_3$) a weaker one: **4-methoxybenzoic < benzoic < 4-nitrobenzoic < 3,4-dinitrobenzoic**.

- Increasing nucleophilic-addition reactivity: ethanal, propanone, propanal, butanone — answer only: butanone < propanone < propanal < ethanal
- Increasing boiling point: propane, ethanal, ethanol — answer only: propane < ethanal < ethanol

5. Solve an "identify A, B, C" puzzle

Worked example · 2023 Q35

A ($\text{C}_4\text{H}_8\text{O}_2$) on acid hydrolysis gives B and C. Oxidising C gives B. The sodium salt of B with soda lime gives methane.

1. Soda lime on a sodium salt removes CO_2 : the salt must be CH_3COONa , so B is **ethanoic acid**.
 2. Oxidising C gives B, and C is a hydrolysis product with two carbons, so C is **ethanol**.
 3. Hydrolysis giving an acid and an alcohol means A is an ester: **ethyl ethanoate** ($\text{CH}_3\text{COOC}_2\text{H}_5$ is $\text{C}_4\text{H}_8\text{O}_2$ ✓).
- $\text{C}_3\text{H}_8\text{O}$ with Cu at 573 K gives B (no Fehling's) which gives a yellow precipitate with I_2/NaOH — answer only: A propan-2-ol; B propanone; the yellow precipitate is CHI_3

Past year questions · question types — ranked by how often they turn up

Two videos feed this section. [30+] is the "30+ PYQ" video for CBSE 2024-25 — Q1–Q35, questions from 2013 to 2023. [2025] is the short "2025 PYQ" video — Q1–Q7, from the 2025 board paper. Question numbers are each video's own numbering, and years come from the video. Where the video gave no year, the question is cited by number only. The CBSE 2026-27 "25 PYQ" video for this chapter was not uploaded when these notes were built; once it is, this section can be refreshed from it. Counts are by question part, from the two videos' lists.

1 · Complete the reaction / write the product / convert — 1 to 5 marks · about 35 parts

Recognise it: "write the product", "complete the reaction", "how will you convert A to B", "give the missing reagent".

1. Name the functional group that changes and the bond that forms or breaks.
2. **Count carbons** in the start and the product: up one → KCN or Grignard + CO₂; down one → haloform or soda lime; doubled → aldol or Kolbe.
3. Write each step with its reagent above the arrow; write the structure of every intermediate.

Trap: the reagent is the mark. Rosenmund needs Pd–BaSO₄ (without the BaSO₄ the aldehyde is reduced on to the alcohol); Wolff–Kishner is hydrazine then KOH/ethylene glycol; and a "benzoic acid from alkylbenzene" question always wants KMnO₄/KOH, heat, then acid, whatever the chain length.

2 · Give a reason / account for — 1 to 5 marks · about 12 parts

Recognise it: "account for ...", "why is X more reactive / a stronger acid / unable to ...".

1. State the structural fact (resonance, +I, –I, steric crowding, α-hydrogen, a lone pair).
2. Say how it causes the behaviour in one sentence.

Trap: one answer covers a lot: **the stability of the conjugate base** (carboxylate) explains acid strength; **steric hindrance plus +I** explains aldehyde vs ketone; **resonance lowering the positive charge** explains benzaldehyde vs propanal and why carboxylic acids do not do carbonyl reactions.

3 · Identify the compounds (A, B, C) — 3 to 5 marks · about 9 parts

Recognise it: a formula plus a list of observations (a reagent that gives or does not give a precipitate), and "identify A, B and C".

1. Turn each clue into a group: positive 2,4-DNP → carbonyl; Tollens'/Fehling's positive → aldehyde, negative → ketone; iodoform positive → CH₃CO–; ozonolysis → cut the double bond and join the two carbonyls back.
2. Check the formula after every guess.
3. Write the equation that links the compounds.

Trap: iodoform needs **a methyl on the carbonyl carbon**, not just any methyl. Ethanal is the *only* aldehyde that gives the iodoform test.

4 · Distinguish by a chemical test — 2 or 3 marks · about 6 parts

Recognise it: "give a simple chemical test to distinguish between ...".

1. Pick the test that separates the pair (the table in the Patterns section): aldehyde vs ketone → Tollens' or Fehling's; acid vs the rest → NaHCO₃; methyl ketone → iodoform.
2. State the observation for *both* compounds, including "no reaction".

Trap: an observation is what you see (silver mirror, brisk effervescence, yellow precipitate), not the equation. Aromatic aldehydes do not respond to Fehling's.

5 · Describe or write a named reaction — 2 to 5 marks · about 8 parts

Recognise it: "describe acetylation / aldol condensation", "write the equation for Stephen / Etard / Cannizzaro / Wolff–Kishner / decarboxylation".

1. One line of what it is, with the reagents.
2. Then a worked equation with an example, and the product named.

Trap: do not mix the three toluene → benzaldehyde methods. CrO_2Cl_2 in CS_2 is Etard; CrO_3 in acetic anhydride gives benzylidene diacetate; Cl_2 in light gives benzal chloride.

6 · Arrange in order; name or draw — 1 to 2 marks · about 9 parts

Recognise it: "arrange in increasing order of acid strength / boiling point / reactivity in nucleophilic addition", "write the IUPAC name", "draw the structure of ...".

1. Orderings: acid strength → more EWG, nearer, stronger; boiling point → hydrogen bonding > dipole–dipole > van der Waals; nucleophilic addition → fewer, smaller groups on the carbonyl, faster.
2. Names: the highest-ranking group is the suffix (acid > aldehyde > ketone > alcohol); the others become prefixes (oxo-, hydroxy-).

Trap: read "increasing" or "decreasing" before you write the order; and in the benzoic-acid series the electron donor (4-methoxy) is the *weakest* acid.

Past year questions · mark slots — what each type is worth

The 30+ video states its own marks: Q1–Q10 are 1-mark, Q11–Q22 are 2-mark, Q23 onward are 3-mark, except that it names Q33 and Q34 (both 2023) as 2-mark. The 2025 video states Q1 as 2 marks and Q4, Q5, Q6 and Q7 as 5 marks each; for its Q2 and Q3 no marks are stated (inferred: 3 each). So this chapter has appeared as a 5-mark long answer in the 2025 paper — the one place in this set where a 5-marker was recorded.

Question type	1 mark	2 marks	3 marks	5 marks
Complete / product / convert	Q2, 2020 Q3, 2019 Q4, 2019 Q7, Q8, 2019 Q9, Q10 [30+]	2017 Q12(a), 2017 Q18, 2013 Q19, 2015 Q22 [30+]	2022 Q24, 2022 Q28, 2020 Q30, 2017 Q26 [30+]	2025 Q4, Q5, Q6, Q7(a) [2025]
Give a reason	2017 Q6 [30+]	2014 Q13, 2022 Q14, 2022 Q15, 2023 Q33, 2023 Q34(b) [30+]; 2025 Q1 [2025]	—	2025 Q5(b) [2025]
Identify A, B, C	—	—	Q25, Q29, 2023 Q32, 2023 Q35 [30+]; 2025 Q2, Q3 [2025, inferred 3]	2025 Q5(a) [2025]
Distinguish by a test	—	2017 Q12(b), 2017 Q16(b), 2023 Q34(a) [30+]	2020 Q23(a) [30+]	2025 Q7(b) [2025]
Named reaction	2019 Q9 [30+]	2022 Q11, 2015 Q17 [30+]	2015 Q22 [30+]	2025 Q6(b) [2025]
Arrange / name / draw	2014 Q1 [30+]	2015 Q20, 2022 Q5, 2022 Q21 [30+]	—	—

Past year questions · repeat offenders — appeared more than once — highest probability in the chapter

2015 Q17(b) · 2022 Q24(c) · 2017 Q26 · 2025 Q2 · 2025 Q4(b)

Aldol condensation — ethanal to but-2-enal (dilute NaOH, then heat), and its cross version with benzaldehyde. The most repeated reaction in the chapter.

2019 Q9 · 2022 Q24(a) · 2023 Q31(b) · 2025 Q7(a)

Rosenmund reduction — an acid chloride with $\text{H}_2/\text{Pd}-\text{BaSO}_4$ gives the aldehyde (benzoyl chloride $\rightarrow$ benzaldehyde).

2022 Q11 · 2013 Q19(a) · Q29 · 2025 Q5(a)

Cannizzaro reaction — no α -hydrogen, concentrated alkali, one molecule oxidised and one reduced (methanal; benzaldehyde).

2020 Q23(a) · 2017 Q16(b) · 2025 Q7(b) · 2025 Q4(a)

Tollens' reagent — the silver-mirror test for an aldehyde against a ketone.

2017 Q12(b) · 2023 Q34(a) · 2025 Q7(b)

The sodium bicarbonate test for a carboxylic acid against an alcohol, phenol or ester.

2014 Q13(a) · 2022 Q15(a) · 2022 Q21(a) · 2022 Q27(a) · 2025 Q1

Reactivity in nucleophilic addition — aldehyde faster than ketone (sterics plus +I); benzaldehyde slower than propanal (resonance). Appears as a reason and as an ordering.

2022 Q5(a) · 2022 Q15(b) · 2022 Q21(b) · 2025 Q5(b)

Acid-strength explanations and orderings — electron-withdrawing groups stabilise the carboxylate; carboxylic acids beat alcohols by resonance.

2019 Q7 · 2022 Q24(b) · 2017 Q18(b)

Hell–Volhard–Zelinsky: $\text{Br}_2/\text{red P}$ (or Cl_2) puts a halogen on the α -carbon of the acid.

2017 Q12(a) · 2022 Q28(c) · 2023 Q35 · 2025 Q6(b)

Decarboxylation with soda lime (sodium salt of the acid $\rightarrow$ hydrocarbon, one carbon fewer; benzoate $\rightarrow$ benzene; ethanoate $\rightarrow$ methane).

2014 Q13(b) · 2022 Q27(c) · 2025 Q7(a)

Semicarbazone — the product, and why only one $-\text{NH}_2$ of semicarbazide reacts.

2014 Q13(a) · 2017 Q16(a) · 2020 Q23(b)

Cyanohydrin formation with HCN (CN^- as the nucleophile).

2022 Q28(a) · 2025 Q6(a) · 2025 Q7(a)

Acid chloride + dimethylcadmium gives a ketone (propanone, acetophenone).

2022 Q11 · 2022 Q28(b) · 2025 Q6(b)

Wolff–Kishner and Clemmensen — carbonyl to CH_2 (propanone $\rightarrow$ propane).

2017 Q18(a) · 2023 Q32 · 2025 Q3 · Q25

The iodoform test and the compounds that give it ($\text{CH}_3\text{CO}-$ or $\text{CH}_3\text{CH}(\text{OH})-$).

2017 Q6 · 2025 Q5(b)

Why carboxylic acids do not give the carbonyl reactions of aldehydes and ketones.

2022 Q14(a) · 2023 Q33(a)

The carboxyl group is deactivating and meta-directing; benzoic acid gives no Friedel–Crafts.

2017 Q2 · 2025 Q2

DIBAL-H reduces a nitrile to an aldehyde and leaves a C=C alone.

2019 Q4 · 2025 Q4(a)

Ethylbenzene with KMnO_4 gives benzoic acid (via potassium benzoate in base).

Past year questions · numerical types — no calculations in this chapter, so these are the reaction and structure types — one worked model each, then cold practice

A · Multi-step conversion

Worked example · 2025 Q4(b) [2025] — benzaldehyde to benzophenone, and to 3-phenylpropan-1-ol

Benzophenone ($\text{C}_6\text{H}_5\text{COC}_6\text{H}_5$): oxidise the aldehyde to benzoic acid ($\text{K}_2\text{Cr}_2\text{O}_7$ in acetic acid), make the acid chloride (SOCl_2 , giving benzoyl chloride), then Friedel–Crafts acylation of benzene with AlCl_3 .

3-Phenylpropan-1-ol ($\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$): add two carbons by a cross aldol with ethanal (dilute NaOH) to $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CH}_2\text{CHO}$; heat to lose water, giving cinnamaldehyde $\text{C}_6\text{H}_5\text{CH}=\text{CHCHO}$; then H_2/Pd reduces *both* the C=C and the $-\text{CHO}$.

- 2017 Q18(a) [30+] — acetophenone to benzoic acid, in two steps — answer only: NaOH / I_2 (iodoform reaction) gives sodium benzoate + CHI_3 ; acidify
- 2017 Q18(b) [30+] — ethanoic acid to 2-hydroxyethanoic acid — answer only: $\text{Cl}_2 / \text{red P}$, then aqueous NaOH
- 2020 Q30 [30+] — benzene to 3-nitroacetophenone — answer only: $\text{CH}_3\text{COCl} / \text{AlCl}_3$ (acylation), then $\text{HNO}_3/\text{H}_2\text{SO}_4$ (the COCH_3 directs meta)
- 2020 Q30 [30+] — benzyl alcohol to phenylethanoic acid — answer only: SOCl_2 , then KCN , then hydrolysis
- 2022 Q24 [30+] — propan-1-ol to 2-bromopropanoic acid — answer only: $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}^+$ to propanoic acid, then $\text{Br}_2 / \text{red P}$

B · Identify the compounds

Worked example · 2023 Q32 [30+] — A is $\text{C}_3\text{H}_8\text{O}$; with Cu at 573 K it gives B; B does not reduce Fehling's but A's product gives a yellow precipitate C with I_2/NaOH

1. Copper at 573 K is dehydrogenation: A is a primary or secondary alcohol, and B is an aldehyde or ketone.
 2. B does not reduce Fehling's, so it is not an aldehyde: B is a **ketone**, and with three carbons it is **propanone**.
 3. A, the alcohol that gives a ketone, is the secondary alcohol **propan-2-ol**.
 4. Propanone has $\text{CH}_3\text{CO}-$, so with I_2/NaOH it gives the yellow precipitate of **iodoform** CHI_3 (C), plus sodium ethanoate.
- 2023 Q35 [30+] — A ($\text{C}_4\text{H}_8\text{O}_2$) hydrolyses to B and C — answer only: A ethyl ethanoate; B ethanoic acid; C ethanol
 - Q25 [30+] — alkene C_5H_{10} whose ozonolysis products are ethanal and propanone — answer only: A 2-methylbut-2-ene; B ethanal; C propanone
 - Q29 [30+] — two moles of A with a strong base give B and C; B with Cu regenerates A; acidifying C gives CH_2O_2 — answer only: A methanal; B methanol; C sodium formate; D formic acid (Cannizzaro)

- 2025 Q3 [2025] — A is C_8H_8O , positive 2,4-DNP, no Tollens', iodoform positive; drastic oxidation gives C — answer only: A acetophenone (1-phenylethanone); with NaOI $\rightarrow$ sodium benzoate + CHI_3 ; C benzoic acid
- 2025 Q2 [2025] — A (C_4H_5N) with DIBAL-H then water gives B, which gives Tollens' but not iodoform, and is also made from ethanal and dilute NaOH on heating — answer only: A but-2-enitrile $CH_3CH=CHCN$; B but-2-enal

C · Give a reason

Worked example · 2022 Q15(a) [30+] / 2025 Q1 [2025] — benzaldehyde is less reactive than propanal in nucleophilic addition

In benzaldehyde the carbonyl is conjugated with the ring, so the ring's π electrons are drawn towards the carbonyl by **resonance**. That **lowers the positive charge** on the carbonyl carbon (it becomes **less electrophilic**), so the nucleophile finds less to attack. Propanal has no resonance, so its carbon is more positive and it reacts faster.

- 2022 Q14(a) [30+] — benzoic acid gives no Friedel–Crafts — answer only: $COOH$ deactivates the ring, and $AlCl_3$ bonds to the carboxyl oxygen
- 2022 Q15(b) [30+] — 4-nitrobenzoic acid is a stronger acid than benzoic acid — answer only: $-NO_2$ is electron-withdrawing and stabilises the conjugate base
- 2017 Q6 [30+] — carboxylic acids do not give the reactions of the carbonyl group — answer only: the lone pair on $-OH$ is in resonance with $C=O$, lowering the carbon's electrophilicity
- 2014 Q13(a) [30+] — CH_3CHO is more reactive than CH_3COCH_3 towards HCN — answer only: two methyls cause more steric hindrance and more +I effect than one
- 2023 Q34(b) [30+] — α -hydrogens of an aldehyde or ketone are acidic — answer only: the carbonyl is electron-withdrawing, and the carbanion is resonance-stabilised

D · Distinguish

Worked example · 2020 Q23(a) [30+] — distinguish propanal from propanone

Warm each with freshly prepared Tollens' reagent (ammoniacal silver nitrate). Propanal is oxidised to propanoate and the silver ion is reduced to a **bright silver mirror**; propanone gives **no reaction**.

- 2017 Q12(b) [30+] — benzoic acid from phenol — answer only: $NaHCO_3$; the acid gives brisk effervescence of CO_2 , phenol does not
- 2017 Q16(b) [30+] — butanal from butan-2-one — answer only: Tollens' reagent; butanal gives a silver mirror
- 2025 Q7(b) [2025] — ethyl benzoate from benzoic acid — answer only: $NaHCO_3$; only the acid gives effervescence

E · Arrange in order

Worked example · 2022 Q21(a) [30+] — nucleophilic-addition reactivity of ethanal, propanone, propanal, butanone

Fewer, smaller groups on the carbonyl means faster. Ethanal has one methyl; propanal one ethyl; propanone two methyls; butanone a methyl and an ethyl. Increasing reactivity: **butanone < propanone < propanal < ethanal**.

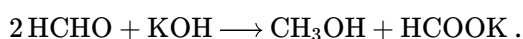
- 2022 Q5(a) [30+] — acid strength of CH_3COOH , ClCH_2COOH , FCH_2COOH — answer only:
 $\text{CH}_3\text{COOH} < \text{ClCH}_2\text{COOH} < \text{FCH}_2\text{COOH}$
- 2022 Q5(b) [30+] — boiling points of CH_3CHO , $\text{CH}_3\text{CH}_2\text{OH}$, $\text{CH}_3\text{CH}_2\text{CH}_3$ — answer only:
 $\text{CH}_3\text{CH}_2\text{CH}_3 < \text{CH}_3\text{CHO} < \text{CH}_3\text{CH}_2\text{OH}$
- 2022 Q21(b) [30+] — acid strength of 4-nitrobenzoic, benzoic, 3,4-dinitrobenzoic and 4-methoxybenzoic acids — answer only: 4-methoxy < benzoic < 4-nitro < 3,4-dinitro

F • Named reactions

Worked example • 2022 Q11 [30+] — Wolff–Kishner reduction and the Cannizzaro reaction

Wolff–Kishner: an aldehyde or ketone is heated with hydrazine to form the hydrazone (water lost), which is then heated with KOH in ethylene glycol; N_2 escapes and the carbonyl becomes CH_2 :

$\text{CH}_3\text{COCH}_3 \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_3$. **Cannizzaro:** an aldehyde with no α -hydrogen is heated with concentrated alkali; one molecule is reduced to the alcohol and one is oxidised to the carboxylate:



- 2015 Q22 [30+] — Stephen reaction — answer only: RCN with SnCl_2/HCl gives the imine, then H_3O^+ gives RCHO
- 2015 Q22 [30+] — Etard reaction — answer only: toluene with CrO_2Cl_2 in CS_2 gives a chromium complex, hydrolysed to benzaldehyde
- 2025 Q5(a) [2025] — toluene with CrO_3 in acetic anhydride (A), hydrolysed (B), then concentrated NaOH (C) — answer only: A benzylidene diacetate; B benzaldehyde; C benzyl alcohol (with sodium benzoate); the video calls the first step Etard, but it is the $\text{CrO}_3/\text{Ac}_2\text{O}$ route
- 2015 Q17 [30+] — acetylation — answer only: introducing $\text{CH}_3\text{CO}-$ into an alcohol, phenol or amine with acetyl chloride or acetic anhydride (aniline gives acetanilide)

G • IUPAC names and structures

Worked example • 2014 Q1 [30+] — name the compound with a ketone at C2 and an alcohol at C4 of a five-carbon chain

The ketone outranks the alcohol, so it is the suffix and gets the lowest number (carbon 2); the alcohol becomes the prefix **hydroxy** at carbon 4. Five carbons, all single bonds: **4-hydroxypentan-2-one**.

- 2015 Q20 [30+] — structure of 4-methylpent-3-en-2-one — answer only: $\text{CH}_3-\text{CO}-\text{CH}=\text{C}(\text{CH}_3)-\text{CH}_3$
- 2015 Q20 [30+] — structure of *p*-methylbenzaldehyde — answer only: benzene ring with $-\text{CHO}$ at C1 and $-\text{CH}_3$ at C4

Amines

NCERT Class XII Chemistry • Chapter 9 • Amines

6 marks in the board paper (CBSE 2026-27 curriculum), and a chapter that is mostly reactions you can predict once you know one fact: the nitrogen has a lone pair of electrons. That lone pair makes amines bases, makes them nucleophiles, and decides what each reagent does to them. Every new word is tagged [exposure] and explained from scratch. The two big payoffs are the basicity order and the diazonium-salt conversions — they carry most of the marks.

What amines are, and where they matter — ammonia with its hydrogens swapped for carbon groups

[exposure] An **amine** is an organic compound you get by replacing one or more hydrogen atoms of **ammonia** NH_3 with alkyl or aryl groups. (An **alkyl** group is a saturated carbon chain such as CH_3- ; an **aryl** group is a benzene-ring group such as C_6H_5- .) Amines matter because the same $-\text{NH}_2$ -type unit sits inside proteins, vitamins, hormones and many drugs. Example: methylamine CH_3NH_2 is ammonia with one hydrogen replaced by CH_3 .

Where they turn up, and what the examiner can ask:

- Biologically active amines: **adrenaline** and **ephedrine** contain a secondary amino group and raise blood pressure.
- **Novocain** — a synthetic amine used as a local anaesthetic in dentistry.
- **Benadryl** — an antihistamine (it treats allergic reactions) with a tertiary amino group.
- **Quaternary ammonium salts** are used as **surfactants** (they lower the surface tension between two liquids that do not mix).
- **Diazonium salts** are intermediates for making a huge range of organic compounds, **dyes** included.

Examiner asks: nothing hard from this section, but the drug names are one-mark recall.

Classification and structure — count the carbon groups on nitrogen; then why the shape is a pyramid

[exposure] Amines are sorted by **how many hydrogens of ammonia were replaced**. This count decides almost every reaction later.

Type	Replaced	General formula	Example
Primary (1°)	one H	RNH_2	CH_3NH_2
Secondary (2°)	two H	R_2NH	$(\text{CH}_3)_2\text{NH}$
Tertiary (3°)	three H	R_3N	$(\text{CH}_3)_3\text{N}$

[exposure] A **quaternary ammonium salt** has *four* carbon groups on nitrogen, replacing all four hydrogens of the ammonium ion NH_4^+ : $\text{R}_4\text{N}^+\text{X}^-$. The nitrogen carries a positive charge and has no lone pair left to give away, which is why such salts are not bases.

[exposure] **Simple versus mixed.** A secondary or tertiary amine is **simple** if all its carbon groups are the same (diethylamine, trimethylamine, diphenylamine) and **mixed** if they differ (ethylmethylamine, ethylphenylamine).

Structure — why the shape is a pyramid

The nitrogen of an amine is sp^3 **hybridised**: one s and three p orbitals blend into four equal hybrid orbitals. Three of them make sigma bonds (to H or R); the fourth holds the **lone pair**. A lone pair repels the bonded pairs more than they repel each other, so the shape is a **pyramid** (the same as ammonia) and the bond angle is squeezed below the tetrahedral 109.5° . In trimethylamine it is about **108°** .

Examiner asks: classify given amines as primary, secondary or tertiary — count the groups on nitrogen (NCERT intext 9.1).

Nomenclature — common names for the pattern; IUPAC "alkanamine" for the marks

Common names

Name the alkyl group(s), then write **amine**. Same groups repeated: **di-**, **tri-**. Different groups: **alphabetical order**. CH_3NH_2 = methylamine; $(\text{CH}_3)_2\text{NH}$ = dimethylamine; $(\text{CH}_3)_3\text{N}$ = trimethylamine; $\text{C}_2\text{H}_5\text{NHCH}_3$ = ethylmethylamine.

IUPAC names

1. **Primary:** drop the final "e" of the alkane and add **amine**: CH_3NH_2 = **methanamine**; $\text{CH}_3\text{CH}_2\text{NH}_2$ = **ethanamine**; $\text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_3$ = **propan-2-amine** (number the carbon that carries $-\text{NH}_2$).
2. **Secondary and tertiary:** the **longest chain** gives the parent name. The smaller group(s) on nitrogen become substituents written with the locant **N**: $\text{CH}_3\text{NHC}_2\text{H}_5$ = **N-methylethanamine**.
3. Two identical small groups: **N,N-** prefix. $(\text{CH}_3)_2\text{NC}_2\text{H}_5$ = **N,N-dimethylethanamine**. Different small groups: alphabetical, each with its own N: **N-ethyl-N-methyl...**
4. **Two $-\text{NH}_2$ groups:** keep the "e" and add **diamine**: $\text{H}_2\text{N}-\text{CH}_2\text{CH}_2-\text{NH}_2$ = **ethane-1,2-diamine**.

Aromatic amines

$\text{C}_6\text{H}_5\text{NH}_2$ is **aniline** (IUPAC also accepts **benzenamine**). A methyl on the ring gives **toluidine** (ortho, meta, para). With IUPAC numbering the $-\text{NH}_2$ carbon is 1: so 2-bromoaniline, and 2,4,6-tribromoaniline. Groups on nitrogen use N: $\text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$ = **N,N-dimethylaniline**.

Trap: in a secondary or tertiary amine the *longest* chain is the parent and every other carbon group is an **N-** substituent. The smaller group goes in front with N, not the bigger one. A tertiary amine with two ethyls and one methyl on N is **N-ethyl-N-methylethanamine**, not "triethyl...".

Examiner asks: write the IUPAC name of a given amine — a standing 1-marker (2017, 2016, 2022) and a 2-marker (2023). See the PYQ section.

Isomers of an amine formula — how $\text{C}_4\text{H}_{11}\text{N}$ gives eight isomers, and the four kinds of isomerism they show

Draw isomers by first fixing the *type* of amine, then moving the groups. For $\text{C}_4\text{H}_{11}\text{N}$ (NCERT intext 9.2) there are **eight**:

Type	Isomers (IUPAC)
Primary (4)	butan-1-amine · butan-2-amine · 2-methylpropan-1-amine · 2-methylpropan-2-amine
Secondary (3)	N-methylpropan-1-amine · N-ethylethanamine · N-methylpropan-2-amine
Tertiary (1)	N,N-dimethylethanamine

[exposure] Isomers are different compounds with the same molecular formula. Amines show four kinds:

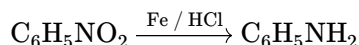
- **Chain isomerism** — the carbon chain differs; the -NH_2 position is the same (butan-1-amine and 2-methylpropan-1-amine).
- **Position isomerism** — the position of the functional group differs (butan-1-amine and butan-2-amine).
- **Metamerism** — same functional group, but the alkyl groups on either side of it differ. It happens in secondary amines (N-methylpropan-1-amine, N-ethylethanamine, N-methylpropan-2-amine).
- **Functional isomerism** — the amine class itself differs: primary, secondary and tertiary amines of one formula are functional isomers of each other.

Examiner asks: $\text{C}_3\text{H}_9\text{N}$ — four isomers: propan-1-amine, propan-2-amine, N-methylethanamine, N,N-dimethylmethanamine. Which of them release nitrogen with nitrous acid? Only the *primary* ones (NCERT intext 9.8); see the nitrous-acid section.

Preparing amines — *six routes — and a one-line reason to prefer each*

1 · Reduction of nitro compounds

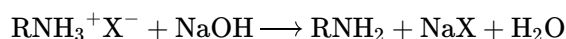
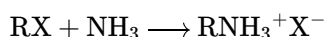
[exposure] A **nitro compound** carries the group -NO_2 . Reducing it (adding hydrogen, removing oxygen) gives the amine. Two ways: **catalytic hydrogenation** (hydrogen gas with a finely divided metal catalyst — Ni, Pd or Pt) or **a metal in acid** (Sn/HCl, Fe/HCl).



Why Fe + HCl is preferred: the FeCl_2 it makes hydrolyses and gives back HCl, so only a *small* amount of acid is needed to start the reaction and it keeps itself going.

2 · Ammonolysis of alkyl halides

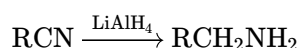
[exposure] **Ammonolysis** means breaking a bond using ammonia. When an alkyl (or benzyl) halide is heated with ethanolic ammonia in a sealed tube at 373 K, the lone pair on NH_3 attacks the carbon and kicks out the halide: nucleophilic substitution. The product is a substituted ammonium salt; a base (NaOH) then frees the amine.



The drawback: the new amine is itself a nucleophile, so it attacks another RX, then another — you get a **mixture of 1°, 2°, 3° amines and a quaternary salt**. To make mainly the *primary* amine, use a **large excess** of NH_3 . Reactivity of the halide: $\text{RI} > \text{RBr} > \text{RCl}$.

3 · Reduction of nitriles

[exposure] A **nitrile** is $\text{R-C}\equiv\text{N}$ (a cyanide group on carbon). Reducing it with LiAlH_4 , sodium–mercury alloy in ethanol, or H_2/Ni gives a primary amine **with one carbon more** than the halide you started from — it is the route for **climbing up** an amine series.



4 · Reduction of amides

An **amide** is RCONH_2 . LiAlH_4 reduces it to the primary amine RCH_2NH_2 (same carbon count).



5 · Gabriel phthalimide synthesis

[exposure] Phthalimide is a ring imide whose N–H is acidic. The **Gabriel synthesis** uses it as a clean source of a single $-\text{NH}_2$ group, so you avoid the mixture problem of ammonolysis. Steps: (1) phthalimide + ethanolic KOH gives the **potassium salt** (nitrogen now carries a negative charge); (2) the salt is heated with an **alkyl halide** — the nitrogen attacks the carbon, giving N-alkylphthalimide; (3) **alkaline hydrolysis** with aqueous NaOH releases RNH_2 and sodium phthalate. It gives **primary aliphatic** amines only.

Trap: Gabriel **cannot make aromatic primary amines** (aniline). An aryl halide does not undergo nucleophilic substitution with the phthalimide anion, because resonance gives the C–X bond partial double-bond character and it will not break. This is a standing 1-mark "give reason" (2017, 2020, 2025).

6 · Hofmann bromamide degradation

[exposure] Hofmann bromamide degradation converts an amide into a primary amine that has **one carbon less**. The amide is treated with Br_2 and aqueous or ethanolic NaOH; the R group migrates from the carbonyl carbon to nitrogen, and the carbonyl carbon is lost as sodium carbonate. It is the route for **stepping down** an amine series.



Example: CH_3CONH_2 (ethanamide, 2 C) gives CH_3NH_2 (methanamine, 1 C).

Examiner asks: the Hofmann equation (2019) and "amide that gives propanamine by Hofmann" (butanamide — NCERT Example 9.3). Nitrile reduction (2015). The Gabriel equation. All are 1- to 3-mark questions.

Conversions: build and trim carbon chains — *three moves cover almost every "convert A to B" question*

Before you pick reagents, **count the carbons** in the start and the product.

- **Same number of carbons:** reduce the nitro compound or amide, or use the Gabriel route.
- **One more carbon in the product:** go via the nitrile. $\text{R-X} \xrightarrow{\text{KCN}} \text{R-CN} \xrightarrow{\text{LiAlH}_4} \text{R-CH}_2\text{-NH}_2$.
- **One fewer carbon in the product:** go via the amide and Hofmann.



Worked from NCERT (Examples 9.2, 9.3, 9.5 and intext 9.3):

Worked example · $\text{CH}_3\text{CH}_2\text{Cl}$ to $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$

Two carbons → three carbons, so go up through the nitrile:

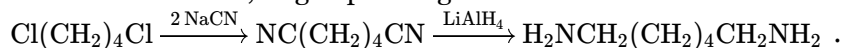
$\text{CH}_3\text{CH}_2\text{Cl} \xrightarrow{\text{KCN}} \text{CH}_3\text{CH}_2\text{CN} \xrightarrow{\text{LiAlH}_4} \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$. The cyanide ion replaces chlorine (nucleophilic substitution) and then LiAlH_4 reduces the $-\text{CN}$ to $-\text{CH}_2\text{NH}_2$.

Worked example · benzene to N, N-dimethylaniline

$\text{C}_6\text{H}_6 \xrightarrow{\text{HNO}_3 / \text{H}_2\text{SO}_4} \text{C}_6\text{H}_5\text{NO}_2 \xrightarrow{\text{Fe} / \text{HCl}} \text{C}_6\text{H}_5\text{NH}_2 \xrightarrow{2 \text{CH}_3\text{Br}} \text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$. Nitrate, reduce, then alkylate twice; each methyl replaces one N–H.

Worked example · $\text{Cl}-(\text{CH}_2)_4-\text{Cl}$ to hexane-1,6-diamine

Four carbons → six, so go up through nitriles on both ends:

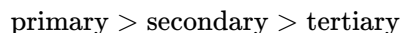


Physical properties — *state, solubility, boiling point — three tables of "why"*

- **State and smell.** Lower aliphatic amines are gases with a fishy smell; primary amines with three or more carbons are liquids; higher ones are solids. Aniline and other arylamines are colourless when pure but turn coloured on standing, because air oxidises them.
- **Solubility.** Lower amines dissolve in water because they form **hydrogen bonds** with it. As the hydrophobic (water-repelling) alkyl part grows, solubility drops, so higher amines are insoluble. Amines dissolve in organic solvents (alcohol, ether, benzene). An alcohol is **more** soluble in water than the amine of similar size, because O is more electronegative than N, so the O–H bond is more polar and bonds more strongly to water.

Boiling point

Primary and secondary amines hydrogen-bond to *each other* (N of one molecule to H of the next). A **primary** amine has two N–H bonds available, a **secondary** has one, a **tertiary** has none. So, for isomeric amines:



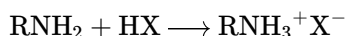
And amines boil **lower than alcohols** of similar mass, because the O–H bond is more polar than N–H. NCERT's numbers (all about 72–74 g/mol): *n*-butanol 390.3 K; *n*-butylamine 350.8 K; diethylamine 329.3 K; ethyldimethylamine 310.5 K; and the hydrocarbon isopentane 300.8 K (no hydrogen bonding at all).

Trap: the boiling-point and solubility questions are decided by the **number of N–H bonds and their polarity**, not by the molar mass. Primary amine, 2 N–H → highest. Tertiary amine, no N–H → lowest. Alcohol O–H beats any N–H. Aniline is the least soluble in water because of its large hydrophobic ring.

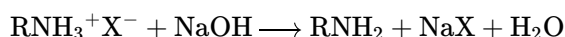
Examiner asks: arrange in increasing boiling point or solubility (2020 Q18; 2025 Q1, Q4); amine versus alcohol — which boils higher, and why (2020 Q3).

Basicity: the idea — *K_b, pK_b, and one rule — the easier the lone pair is handed over, the stronger the base*

[exposure] A **base** here is something that accepts a proton H^+ . The nitrogen of an amine has an **unshared lone pair**, which it can donate to H^+ , so amines are **Lewis bases** (electron-pair donors). They turn acids into salts:



and the salts give the amine back when treated with a base:



Amine salts dissolve in water but not in ether. That is how an amine is **separated from non-basic organic compounds**: turn it into a water-soluble salt, wash away the rest, then add base to get it back.

[exposure] K_b and pK_b put a number on strength. In water an amine does

$\text{RNH}_2 + \text{H}_2\text{O} \rightleftharpoons \text{RNH}_3^+ + \text{OH}^-$. Since water is a pure liquid its concentration is constant and folds into the constant, giving the **base dissociation constant**:

$$K_b = \frac{[\text{RNH}_3^+][\text{OH}^-]}{[\text{RNH}_2]}, \quad pK_b = -\log K_b$$

Larger $K_b \Leftrightarrow$ smaller $pK_b \Leftrightarrow$ stronger base. The pK_b scale runs *opposite* to basic strength — that reversal is where most marks are lost.

Amine (aqueous)	pK _b	Amine (aqueous)	pK _b
methanamine	3.38	ethanamine	3.29
N-methylmethanamine	3.27	N-ethylethanamine	3.00
N,N-dimethylmethanamine	4.22	N,N-diethylethanamine	3.25
ammonia	4.75	phenylmethanamine	4.70
benzenamine (aniline)	9.38	N-methylaniline / N,N-dimethylaniline	9.30 / 8.92

Basicity: alkylamines, ammonia and aniline — three comparisons, each with a different reason

Alkylamine versus ammonia — the alkylamine wins

[exposure] The **+I effect** (positive inductive effect) is an alkyl group's habit of pushing electron density towards the atom it is attached to. On nitrogen it does two things: it makes the lone pair **more available** to take a proton, and it **stabilises the substituted ammonium ion** that results by spreading the positive charge. Both push the same way, so **alkylamines are stronger bases than ammonia** (pK_b of ammonia 4.75; aliphatic amines 3 to 4.22).

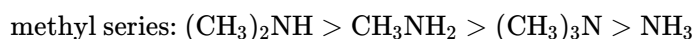
Among alkylamines — gas phase versus water

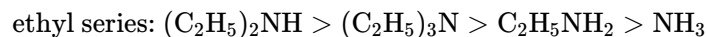
With no solvent (gas phase) only the +I effect acts, so more alkyl groups means more basic: **tertiary > secondary > primary > NH₃**.

In **water** the order is irregular, because three factors compete:

1. **+I effect** — favours tertiary > secondary > primary.
2. **[exposure] Solvation effect.** The protonated amine ion is stabilised when water molecules hydrogen-bond to its N–H hydrogens, releasing hydration energy. A primary ion (RNH_3^+) has **three** N–H to bond with water, a secondary two, a tertiary one. So solvation favours primary > secondary > tertiary — the *reverse* of +I.
3. **[exposure] Steric hindrance.** Bulky groups crowd the nitrogen and block both the proton and the water. A small group like CH_3 blocks hydrogen bonding very little; an ethyl or larger group blocks it more. So changing methyl to ethyl changes the order.

The result, from NCERT:





In both series the **secondary amine is strongest**, because it has a good share of both +I and solvation. In the methyl series methyl is too small to cause steric hindrance, so solvation beats +I and primary > tertiary. In the ethyl series the bigger ethyls hinder solvation, so +I wins and tertiary > primary.

Trap: when a question gives no solvent, **assume aqueous**. And do not use one rule for both series — the methyl and ethyl orders differ (above). Memorise the two lines; they are the single most repeated ordering question (2019 Q11 gas phase; 2020 Q19(b); Q26(a)).

Arylamine versus ammonia — aniline is weaker

In aniline the nitrogen is attached straight to the benzene ring, so its lone pair is **delocalised into the ring** by resonance (the +R effect). Delocalised electrons are **not available** for protonation. And once aniline *does* take a proton, the anilinium ion has only **two** resonance structures against **five** for ammonia, so the ion is less stabilised. Result: aniline (pK_b 9.38) is a much weaker base than ammonia (4.75).

Substituents on aniline

[exposure] An **electron-releasing group (ERG)** such as $-\text{CH}_3$ or $-\text{OCH}_3$ raises the electron density on nitrogen and **increases** basic strength. An **electron-withdrawing group (EWG)** such as $-\text{NO}_2$, $-\text{CN}$, $-\text{COOH}$ or a halogen pulls electron density away and **decreases** it.

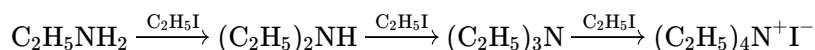
Examiner asks: "arrange in order of basic strength" (NCERT intext 9.4, example 9.4; 2019 Q11; 2025 Q4(a)); "why is methylamine more basic than aniline / than ammonia" (2020 Q4, 2013 Q13(b)); "why $(\text{CH}_3)_2\text{NH}$ is more basic than $(\text{CH}_3)_3\text{N}$ in water" (2020 Q19(b); Q26(a)); "why amides are less basic than amines" — the nitrogen lone pair in an amide is delocalised onto the C=O oxygen (2025 Q1(b), Q16(b)).

Reactions of amines: salts, alkylation, acylation — nitrogen as a nucleophile

An amine can act as a **nucleophile** (an electron-pair donor that attacks a positive centre) because of the lone pair; and the number of N–H bonds decides which reactions are possible.

Alkylation

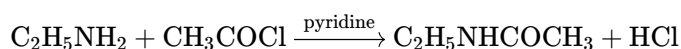
An amine reacts with an alkyl halide, replacing an N–H hydrogen with the alkyl group, so the amine moves up one class each time. It stops only when the nitrogen has four groups.

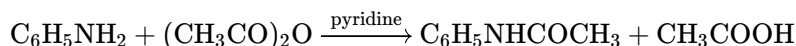


(N-ethylethanamine, N,N-diethylethanamine, then tetraethylammonium iodide.) Aniline does the same with CH_3I : N-methylaniline, N,N-dimethylaniline, and finally N,N,N-trimethylanilinium iodide.

Acylation

[exposure] **Acylation** is replacing an N–H hydrogen of a **primary or secondary** amine by an **acyl group** $\text{R}-\text{CO}-$. The reagent is an acid chloride, an anhydride or an ester; the product is an **amide**. A **tertiary amine cannot** be acylated because it has no N–H to replace. Use a base **stronger than the amine**, such as **pyridine**: it mops up the HCl formed, shifting the equilibrium to the product side.





The second is aniline with ethanoic anhydride, giving **N-phenylethanamide, acetanilide**. With **benzoyl chloride** $\text{C}_6\text{H}_5\text{COCl}$ the same reaction is called **benzoylation**:

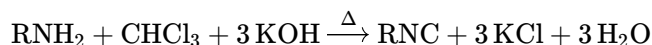
$\text{CH}_3\text{NH}_2 + \text{C}_6\text{H}_5\text{COCl} \longrightarrow \text{CH}_3\text{NHCOC}_6\text{H}_5 + \text{HCl}$ (N-methylbenzamide). Aniline plus benzoyl chloride gives N-phenylbenzamide.

Trap: the role of pyridine is a standing 1-mark question. Answer: it **neutralises the HCl** produced, which shifts the equilibrium forward so the reaction goes to completion.

Carbylamine test, nitrous acid and Hinsberg — *three ways to tell the classes apart*

Carbylamine reaction (the test for primary amines)

[exposure] An **isocyanide** (carbylamine) is $\text{R}-\text{N}\equiv\text{C}$, a nitrogen–carbon triple unit attached through nitrogen. Heating a **primary** amine (aliphatic or aromatic) with **chloroform and ethanolic KOH** gives an isocyanide with a **foul smell**. Secondary and tertiary amines do not give it, so it is the test for a primary amine.



Aniline gives phenyl isocyanide $\text{C}_6\text{H}_5\text{NC}$ the same way.

Reaction with nitrous acid

[exposure] **Nitrous acid** HNO_2 is unstable, so it is made **in situ** ("in place", inside the flask) from sodium nitrite and dilute hydrochloric acid. The three classes behave differently:

- **Primary aliphatic** amine $\rightarrow$ an aliphatic diazonium salt, which is **unstable** (no resonance) and breaks down to **nitrogen gas and an alcohol**: $\text{RNH}_2 + \text{HNO}_2 \longrightarrow \text{ROH} + \text{N}_2 + \text{H}_2\text{O}$. The nitrogen released is measured to estimate amino acids and proteins.
- **Primary aromatic** amine at **273–278 K** $\rightarrow$ a **diazonium salt**, which is stable at that temperature: $\text{C}_6\text{H}_5\text{NH}_2 + \text{NaNO}_2 + 2 \text{HCl} \longrightarrow \text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{NaCl} + 2 \text{H}_2\text{O}$.
- **Secondary and tertiary** amines react "in a different manner" (no nitrogen gas as with primary). The lecture stops at that statement.

The key exam use: **only primary aliphatic amines give off nitrogen gas** with nitrous acid — the NCERT $\text{C}_3\text{H}_9\text{N}$ question (intext 9.8) turns on it: propan-1-amine and propan-2-amine give N_2 ; the secondary and tertiary isomers do not.

Hinsberg's reagent

[exposure] **Hinsberg's reagent** is **benzenesulphonyl chloride**, $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$. It distinguishes the three amine classes and also separates their mixtures. (Nowadays *p*-toluenesulphonyl chloride replaces it.)

Amine	Reaction	Product	In aqueous alkali
Primary RNH ₂	reacts, loses HCl	C ₆ H ₅ SO ₂ NHR (N-alkylbenzenesulphonamide) has an N–H left	soluble (acidic N–H)
Secondary R ₂ NH	reacts, loses HCl	C ₆ H ₅ SO ₂ NR ₂ (N,N-dialkylbenzenesulphonamide), no N–H	insoluble
Tertiary R ₃ N	no reaction	—	—

Why the primary sulphonamide dissolves: the strongly electron-withdrawing –SO₂– group makes the remaining N–H hydrogen **acidic**, so alkali removes it and the compound becomes a water-soluble salt. The secondary product has no N–H, so it cannot.

Trap: the answer for "distinguish primary, secondary, tertiary" needs **both** halves — reaction or not, *and* solubility of the product in alkali. For primary versus secondary alone, the **carbylamine test** is the quicker answer (2020 Q14; but N-methylaniline, being secondary, does not give it, while aniline does).

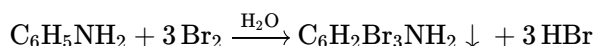
Examiner asks: the carbylamine equation (2025 Q2(b), 2022 Q28, 2014 Q25); distinguish primary from secondary (2020 Q14); distinguish secondary from tertiary with Hinsberg (2013 Q30(a)); "what is Hinsberg's reagent" (Q26(b)).

Ring reactions of aniline — –NH₂ *is a powerful activator; that is both a gift and a nuisance*

[exposure] Electrophilic substitution is the reaction where an electron-seeking group (an **electrophile**) replaces a hydrogen on a benzene ring. A group already on the ring either **activates** it (more reactive than benzene) or **deactivates** it, and it steers the incoming group to a position: **ortho** (next to it), **para** (opposite), or **meta**. In aniline, –NH₂ donates its lone pair into the ring, so the ortho and para positions become rich in electrons. –NH₂ is a **powerful activating, ortho/para-directing group**, so aniline reacts faster than benzene (2013 Q16(a)).

Bromination — too much of a good thing

Aniline reacts with bromine water at room temperature to give a **white precipitate of 2,4,6-tribromoaniline** (three bromines at once).



Controlling the activation — protect the amino group

To get just one substituent, **acetylate** the –NH₂ first (acetic anhydride, pyridine) to acetanilide C₆H₅NHCOCH₃. The nitrogen's lone pair is now shared with the C=O oxygen by resonance, so it is *less* available to the ring, and the activation is weaker. Carry out the substitution, then **hydrolyse** the amide back to the amine.

Worked example · aniline to 4-bromoaniline

Acetylate: C₆H₅NH₂ $\xrightarrow{(\text{CH}_3\text{CO})_2\text{O}}$ C₆H₅NHCOCH₃. Brominate in ethanoic acid: the bromine goes mostly **para** (the symmetric position, the major product). Hydrolyse with acid or base: –NHCOCH₃ $\longrightarrow$ –NH₂. Result: **4-bromoaniline**.

Nitration — two problems

Direct nitration with $\text{HNO}_3 + \text{H}_2\text{SO}_4$ goes wrong in two ways. Nitric acid is a strong oxidant, so aniline is oxidised to **tarry products**. And in the strongly acidic mixture aniline picks up H^+ to become the **anilinium ion** $\text{C}_6\text{H}_5\text{NH}_3^+$, which is **meta-directing** (the positive nitrogen withdraws electrons and deactivates the ortho/para positions). NCERT's yield at 288 K: **47% meta**, 51% para, 2% ortho. So aniline gives meta-nitroaniline in good yield even though $-\text{NH}_2$ is itself o,p-directing (2020 Q19(a)).

The fix is the same: **acetylate first**, nitrate (para-nitroacetanilide is the main product), then hydrolyse to **p-nitroaniline**.

Sulphonation

Aniline with concentrated H_2SO_4 gives anilinium hydrogensulphate, $\text{C}_6\text{H}_5\text{NH}_3^+\text{HSO}_4^-$. Heated at **453–473 K** it rearranges to **p-aminobenzenesulphonic acid, sulphanilic acid**. In water it exists as a **zwitterion**, $^-\text{O}_3\text{S}-\text{C}_6\text{H}_4-\text{NH}_3^+$ (acidic $-\text{SO}_3\text{H}$ gives up H^+ ; basic $-\text{NH}_2$ takes it): 2023 Q32(a).

No Friedel–Crafts

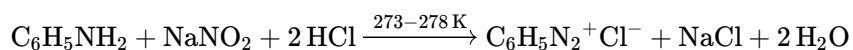
[exposure] Friedel–Crafts reactions (alkylation and acylation of a benzene ring) use AlCl_3 , a Lewis acid, as the catalyst. Aniline does **not** undergo them: its nitrogen lone pair attacks the electron-deficient AlCl_3 and forms a **salt**. The nitrogen is now positively charged, so it strongly withdraws electrons and **deactivates** the ring; no electrophile can attack. This is a 1- or 2-mark question that recurs (2014 Q13, 2017, 2020 Q29(a)).

Diazonium salts: what they are and how they are made — ArN_2^+X^- — one of the most useful functional groups in the course

[exposure] A **diazonium salt** has the general formula ArN_2^+X^- : an aryl group bonded to the $-\text{N}_2^+$ **diazonium group**, with an anion X^- (Cl^- , Br^- , HSO_4^- , BF_4^-). They exist because N_2 is such a good **leaving group** (a group that departs easily, here as nitrogen gas) that the ring can then be given almost any other group. Name them as the parent hydrocarbon + **diazonium** + the anion: $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$ is **benzenediazonium chloride**; $\text{C}_6\text{H}_5\text{N}_2^+\text{HSO}_4^-$ is benzenediazonium hydrogensulphate; $\text{C}_6\text{H}_5\text{N}_2^+\text{BF}_4^-$ is benzenediazonium fluoroborate.

Preparation — diazotisation

Dissolve aniline in HCl and add NaNO_2 at **273–278 K** (0–5 °C). Nitrous acid forms in the flask and converts the primary aromatic amine into the diazonium salt. This is called **diazotisation**.



The salt is not stored; it is **used straight away**.

Stability

An **aromatic** diazonium salt is far more stable than an aliphatic one. In the arenediazonium ion the positive charge is **spread over the benzene ring by resonance**; in an alkyl diazonium ion nothing can spread it, so it loses N_2 at once to give a carbocation. Even so the aromatic one is stable only for a short time, in solution, at **273–278 K** (2018 Q17(b)).

Physical properties

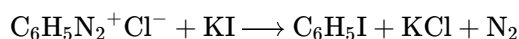
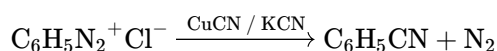
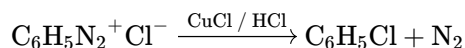
Benzenediazonium chloride is a colourless crystalline solid, soluble in water, stable in the cold, and it reacts with warm water; dry it decomposes easily. The **fluoroborate is insoluble in water and stable at room temperature**, which is why it is used for fluorine and nitro replacements.

Diazonium reactions: displacing nitrogen — swap N₂ for Cl, Br, CN, I, F, H, OH or NO₂ — one reagent per swap

The group –N₂⁺ leaves as nitrogen gas, and the group you want to put on takes its place. The table is the whole lesson:

You want	Reagent	Name
–Cl	CuCl / HCl (or Cu / HCl)	Sandmeyer (or Gattermann)
–Br	CuBr / HBr (or Cu / HBr)	Sandmeyer (or Gattermann)
–CN	CuCN / KCN	Sandmeyer
–I	KI	(no copper needed)
–F	HB ₄ F then heat	via the fluoroborate
–H	H ₃ PO ₂ (or CH ₃ CH ₂ OH)	reduction
–OH	warm water (up to 283 K)	hydrolysis → phenol
–NO ₂	HB ₄ F, then NaNO ₂ / Cu, heat	via the fluoroborate

[exposure] The **Sandmeyer reaction** introduces Cl[–], Br[–] or CN[–] using a **copper(I)** salt. The **Gattermann reaction** does the chlorine/bromine version with **copper powder and the halogen acid** instead. The Sandmeyer gives the better yield (NCERT). Both are named reactions you can be asked by name.



Fluorine: $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{HB}_4\text{F} \longrightarrow \text{C}_6\text{H}_5\text{N}_2^+\text{BF}_4^- + \text{HCl}$; the fluoroborate precipitates and, when heated, gives $\text{C}_6\text{H}_5\text{F} + \text{BF}_3 + \text{N}_2$.

Hydrogen: $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{H}_3\text{PO}_2 + \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_6 + \text{N}_2 + \text{H}_3\text{PO}_3 + \text{HCl}$. The H₃PO₂ is the reducing agent and is itself oxidised to H₃PO₃ . Ethanol does the same job and is oxidised to ethanal CH₃CHO .

Hydroxyl: warm the diazonium solution above the preparation temperature (up to 283 K) and it is hydrolysed: $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^- + \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_5\text{OH} + \text{N}_2 + \text{HCl}$. This is why diazotisation is kept at 273–278 K – a little warmer and phenol forms (2013 Q1: "benzenediazonium chloride + warm water").

Nitro: $\text{C}_6\text{H}_5\text{N}_2^+\text{BF}_4^- + \text{NaNO}_2 \xrightarrow{\text{Cu}, \Delta} \text{C}_6\text{H}_5\text{NO}_2 + \text{N}_2 + \text{NaBF}_4$.

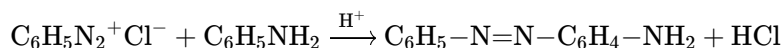
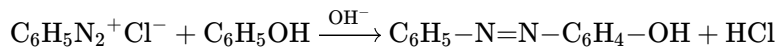
Trap: I and F are the awkward ones. Iodine and fluorine cannot be put on the ring by direct halogenation (fluorine is too violent, iodine too unreactive), and –CN cannot be put on by substituting chlorobenzene.

Diazonium salts are the way to make aryl fluorides, aryl iodides and aryl cyanides. Say so when asked why the diazonium route is used (Q on the importance of diazonium salts).

Diazonium reactions: coupling — the –N=N– link keeps its nitrogen — and makes dyes

[exposure] A **coupling reaction** joins the diazonium ion to another electron-rich benzene ring *without losing the nitrogen*. The positive diazonium end is the electrophile; the partner (phenol or aniline) has a ring

made electron-rich by its $-\text{OH}$ or $-\text{NH}_2$, and the ring is attacked at the **para** position. The product is an **azo compound**, two aromatic rings joined by $-\text{N}=\text{N}-$. That link carries an extended system of alternating double bonds (a **conjugated system**), which absorbs visible light, so these compounds are **coloured and used as dyes**.



- With **phenol** (in a basic solution, where the phenol is present as the more reactive phenoxide): **p-hydroxyazobenzene**, an **orange dye**.
- With **aniline** (acidic medium): **p-aminoazobenzene**, a **yellow dye**.

Coupling is itself an **electrophilic substitution** on the second ring.

Examiner asks: the coupling of benzenediazonium chloride with aniline (2025 Q3(b)); the displacement reactions (2019 Q24; 2020 Q10; 2014 Q15; 2013 Q1); "diazotisation with an equation" (2018 Q17(a)); "why aromatic diazonium salts are more stable than aliphatic" (2018 Q17(b)).

Diazonium conversions — *three NCERT multi-step problems, each built from the table above*

Worked example · NCERT Example 9.5 — 4-nitrotoluene to 2-bromobenzoic acid

1. Brominate: the CH_3 is ortho/para-directing and the para place is taken by NO_2 , so Br goes **ortho to the methyl**.
2. Remove the NO_2 : reduce it to NH_2 (Sn/HCl), diazotise (NaNO_2/HCl , 273–278 K), and replace the diazonium group by hydrogen (H_3PO_2).
3. Oxidise the CH_3 to COOH (KMnO_4 , base).

The diazonium route is how an NH_2 (and so an NO_2) is removed.

Worked example · NCERT intext 9.9 — 3-methylaniline to 3-nitrotoluene

Diazotise ($\text{NaNO}_2 / \text{HCl}$, 273–278 K); make the fluoroborate (HBF_4); then $\text{NaNO}_2 / \text{Cu}$, heat — the diazonium group is replaced by NO_2 .

Worked example · NCERT intext 9.9 — aniline to 1,3,5-tribromobenzene

Bromine water first: $-\text{NH}_2$ directs three Br to **2,4,6**, giving 2,4,6-tribromoaniline. Then diazotise and replace the diazonium group by hydrogen (H_3PO_2). With NH_2 gone, the three bromines are at 1,3,5.

Patterns, collected — *the orderings, tests and conversions you will actually be asked to produce*

1. Rank by basic strength

Worked example

Arrange in decreasing basic strength: $\text{C}_6\text{H}_5\text{NH}_2$, $\text{C}_2\text{H}_5\text{NH}_2$, $(\text{C}_2\text{H}_5)_2\text{NH}$, NH_3 (NCERT Example 9.4).

1. No solvent named, so use the aqueous order. The ethyl series: $(\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3$.
2. Aniline's lone pair is delocalised into the ring, so aniline goes last.

Answer: $(\text{C}_2\text{H}_5)_2\text{NH} > \text{C}_2\text{H}_5\text{NH}_2 > \text{NH}_3 > \text{C}_6\text{H}_5\text{NH}_2$. For pK_b , reverse the order.

- Increasing basic strength: CH_3NH_2 , $(\text{CH}_3)_3\text{N}$, $(\text{CH}_3)_2\text{NH}$ in water — answer only:
 $(\text{CH}_3)_3\text{N} < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH}$
- Gas-phase order of primary, secondary, tertiary — answer only: tertiary > secondary > primary
 (increasing: $1^\circ < 2^\circ < 3^\circ$)
- Increasing pK_b : $(\text{C}_2\text{H}_5)_2\text{NH}$, NH_3 , $\text{C}_2\text{H}_5\text{NH}_2$, $\text{C}_6\text{H}_5\text{NH}_2$ — answer only:
 $(\text{C}_2\text{H}_5)_2\text{NH} < \text{C}_2\text{H}_5\text{NH}_2 < \text{NH}_3 < \text{C}_6\text{H}_5\text{NH}_2$

2. Rank by boiling point or solubility

Worked example

Increasing boiling point: a secondary amine, a primary amine and ethanol (2025 Q1(a)).

Count and compare the hydrogen bonds: secondary has one N–H, primary has two, ethanol has an O–H that is more polar than any N–H. Order: secondary < primary < ethanol.

- Increasing solubility in water: aniline, a secondary amine, a primary amine — answer only: aniline < secondary < primary
- Higher boiling point: propan-1-amine or N,N-dimethylmethanamine — answer only: propan-1-amine (primary, hydrogen bonds)

3. Distinguish by a test

Worked example

Distinguish ethylamine from diethylamine (2020 Q14).

Heat each with chloroform and ethanolic KOH. Ethylamine (primary) gives the foul-smelling ethyl isocyanide; diethylamine (secondary) does not.

- Distinguish a secondary from a tertiary amine — answer only: Hinsberg's reagent; the secondary gives an alkali-insoluble sulphonamide, the tertiary does not react
- Distinguish aniline from N-methylaniline — answer only: carbylamine test; aniline (primary) gives it, N-methylaniline (secondary) does not

4. Convert, tracking the carbon count

Worked example

Ethanamide to methanamine (2022 Q21): one carbon fewer, so Hofmann: $\text{CH}_3\text{CONH}_2 \xrightarrow{\text{Br}_2 / \text{NaOH}} \text{CH}_3\text{NH}_2$.

- Bromoethane to propan-1-amine — answer only: KCN to give $\text{CH}_3\text{CH}_2\text{CN}$, then LiAlH_4 (or H_2/Ni) to give $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
- Benzene to aniline — answer only: nitration ($\text{HNO}_3/\text{H}_2\text{SO}_4$), then Fe/HCl

5. Pick the diazonium reagent

Worked example

Aniline to chlorobenzene (2025 Q5(b)): diazotise with $\text{NaNO}_2 / \text{HCl}$ at 273–278 K to $\text{C}_6\text{H}_5\text{N}_2^+ \text{Cl}^-$, then CuCl / HCl (Sandmeyer) to $\text{C}_6\text{H}_5\text{Cl}$.

- Benzenediazonium chloride to iodobenzene — answer only: KI
- Benzenediazonium chloride to phenol — answer only: warm water
- Benzenediazonium chloride to benzene — answer only: H_3PO_2 (or ethanol)

6. Count the isomers, then test them with nitrous acid

Worked example

$\text{C}_3\text{H}_9\text{N}$: write the isomers and say which release nitrogen with nitrous acid (NCERT intext 9.8).

Primary: propan-1-amine, propan-2-amine. Secondary: N-methylethanamine. Tertiary: N,N-dimethylmethanamine. Only the primary aliphatic amines give nitrogen gas: propan-1-amine and propan-2-amine.

Past year questions · question types — ranked by how often they turn up

Two videos feed this section. [30+] is the "30+ PYQ" video for CBSE 2024-25 — Q1–Q33, questions from 2013 to 2023. [2025] is the short "2025 PYQ" video — Q1–Q5, from the 2025 board paper. Question numbers are each video's own numbering, and years come from the video. Where the video gave no year, the question is cited by number only. The CBSE 2026-27 "25 PYQ" video for this chapter was not uploaded when these notes were built; once it is, this section can be refreshed from it. Counts are by question part, from the two videos' lists.

1 · Convert, complete, or identify A, B, C — 1 to 3 marks · about 22 parts

Recognise it: "how will you convert ...", "complete the reaction", "identify A, B and C", "give the equation for ...", "write the structure of the product".

1. Name the type of change: nitro $\rightarrow$ amine; nitrile $\rightarrow$ amine; amide $\rightarrow$ amine; amine $\rightarrow$ diazonium salt.
2. **Count carbons** in the start and the product: more carbons $\rightarrow$ nitrile route; fewer $\rightarrow$ Hofmann.
3. Write each step with its reagent over the arrow; for A/B/C questions, write the formula and name of each letter.

Trap: the reagents decide the marks. Nitro $\rightarrow$ amine is Fe/HCl or Sn/HCl (not LiAlH_4); nitrile $\rightarrow$ amine is LiAlH_4 or H_2/Ni ; Hofmann needs $\text{Br}_2 + \text{NaOH}$; diazotisation is $\text{NaNO}_2 + \text{HCl}$ at **0–5 °C** (273–278 K) — at room temperature you get phenol.

2 · Give a reason / account for — 1 to 3 marks · about 14 parts

Recognise it: "give reason ...", "account for ...", "why is X more basic / weaker / unable to ...".

1. State the structural fact: a lone pair on N; resonance; N–H count; an electron-pulling or -pushing group.
2. Say how it causes the behaviour in one sentence.

Trap: the same three answers cover most of these: **aniline's lone pair is delocalised into the ring** (weak base; no Friedel–Crafts; ring is activated); **aryl halides do not do nucleophilic substitution** (no Gabriel for aromatic amines); and **the solvation effect** (secondary > tertiary in water).

3 · Arrange in order — 1 to 3 marks · about 10 parts

Recognise it: "arrange in increasing/decreasing order of basic strength / pK_b / boiling point / solubility in water", "in the gaseous phase".

1. Decide which property it is and what controls it: basicity $\rightarrow$ lone-pair availability; boiling point and solubility $\rightarrow$ hydrogen bonding.
2. Basicity: no solvent named $\rightarrow$ aqueous. Use the two NCERT orders for methyl and ethyl series; aniline is last; pK_b runs the opposite way.
3. Boiling point: primary > secondary > tertiary, and alcohol above all of them. Solubility: more N–H and a smaller ring/chain $\rightarrow$ more soluble.

Trap: read "increasing" or "decreasing", and read pK_b versus basic strength — a smaller pK_b means a *stronger* base. Gas phase is the only case where tertiary > secondary > primary.

4 · Distinguish between two compounds — 2 or 3 marks · about 5 parts

Recognise it: "distinguish between ..." or "what is Hinsberg's reagent".

1. Primary versus secondary or tertiary: carbylamine test (chloroform + ethanolic KOH, foul smell, primary only).
2. Secondary versus tertiary: Hinsberg's reagent $C_6H_5SO_2Cl$ — secondary gives a sulphonamide that is insoluble in alkali; tertiary does not react.

Trap: aniline is a primary amine and does give the carbylamine test; N-methylaniline is secondary and does not. Do not treat "aromatic" as "different".

5 · Write the IUPAC name — 1 or 2 marks · 4 parts

Recognise it: a drawn amine, with "write the IUPAC name".

1. Find the longest carbon chain on nitrogen; that is the parent (...anamine). For a ring on N it is aniline.
2. Every other group on nitrogen becomes an N-substituent, alphabetical order.

Trap: in 2,4,6-tribromoaniline the $-NH_2$ carbon is number 1; do not number from a bromine.

Past year questions · mark slots — *what each type is worth*

The 30+ video states its own marks: Q1–Q12 are 1-mark, Q13–Q20 are 2-mark, Q21–Q30 are 3-mark, and the three 2023 questions Q31–Q33 are stated as 2-mark each. The 2025 video states Q1, Q2 and Q4 as 3-mark and Q3 as 2-mark, and does not state the marks for Q5 (inferred: 3). Amines does not appear as a 5-mark long answer in either video.

Question type	1 mark	2 marks	3 marks
Convert / complete / identify	2019 Q9, 2020 Q10, 2014 Q12, 2013 Q1, 2015 Q7, Q5 [30+]	2014 Q15, 2018 Q20, 2023 Q31(a)(b) [30+]; 2025 Q3 [2025]	2022 Q21, 2015 Q23, 2019 Q24, 2014 Q25, 2022 Q28 [30+]; 2025 Q2, Q5 [2025]
Give a reason	2020 Q3, Q4, Q8 [30+]	2014 Q13, 2013 Q16, 2018 Q17(b), 2020 Q19, 2023 Q32(b) [30+]	Q29, 2013 Q30(b) [30+]; 2025 Q1(b) [2025]
Arrange in order	2019 Q11 [30+]	2020 Q18 [30+]	Q26(a), Q27(b) [30+]; 2025 Q1(a), Q4 [2025]
Distinguish / reagent	—	2020 Q14 [30+]	2013 Q30(a), Q26(b) [30+]
IUPAC name	2017 Q2, 2016 Q6 [30+]	2023 Q33(a) [30+]	2022 Q22 [30+]

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2019 Q9 · 2022 Q21(b) · 2015 Q23 · 2014 Q15 · 2014 Q25

Hofmann bromamide degradation — the equation, ethanamide to methanamine, benzamide to aniline. The single most repeated reaction.

2014 Q13(a) · 2017 · 2020 Q29(a)

Why aniline does not undergo Friedel–Crafts: it forms a salt with AlCl_3 , the nitrogen becomes positive and deactivates the ring.

2017 · 2020 Q29(b) · 2025 Q1(b)

Why aromatic primary amines cannot be made by Gabriel's synthesis: aryl halides do not undergo nucleophilic substitution (partial double-bond character).

2020 Q14 · 2022 Q28 · 2014 Q25 · 2025 Q2(b)

Carbylamine test / phenyl isocyanide from aniline — foul smell, primary amines only.

2015 Q7 · 2022 Q21(c) · 2023 Q31(b) · 2014 Q25 · 2025 Q3(a)

Nitrile reduction to an amine, often as the second step after KCN/NaCN — the "go up one carbon" route.

2013 Q1 · 2014 Q15 · 2025 Q5(c)

Benzenediazonium chloride warmed with water gives phenol (and why diazotisation must stay at 0–5 °C).

2020 Q10 · 2019 Q24 · 2025 Q5(b)

Diazonium chloride with CuCN (benzonitrile → benzoic acid on hydrolysis), KI (iodobenzene), CuCl/HCl (chlorobenzene).

2019 Q24 · 2023 Q31(a)

Diazonium chloride with ethanol (or hypophosphorous acid) gives benzene; the ethanol is oxidised to ethanal.

2020 Q5 · 2014 Q15 · 2022 Q21(a) · 2022 Q28 · 2023 Q33(b)

Nitrobenzene reduced to aniline with Fe/HCl or Sn/HCl — the first step in a long list of sequences.

2014 Q12 · 2018 Q20(a) · 2025 Q2(c) · 2023 Q32(b) · 2025 Q5(a)

Acetylation of aniline with acetic anhydride and pyridine (acetanilide), and its use to protect the $-\text{NH}_2$ in nitration and bromination.

2020 Q19(b) · Q26(a) · 2019 Q11 · Q27(b) · 2025 Q4(a)

Basic-strength and pK_b orderings: secondary > tertiary in water (solvation + +I), tertiary highest in the gas phase, aniline last.

2020 Q3 · Q8 · 2020 Q18(b) · 2025 Q1(a) · 2025 Q4(b)

Boiling-point order: primary > secondary > tertiary, and an alcohol above an amine.

2013 Q16(b) · 2025 Q1(b)

Why amides are weaker bases than amines: the nitrogen lone pair is delocalised onto the carbonyl oxygen by resonance.

2025 Q2(a) · 2023 Q32(a)

Sulphanilic acid — how to make it from aniline, and its zwitterion.

2013 Q30(a) · Q26(b) · 2018 Q20(b)

Hinsberg's reagent — secondary versus tertiary; the sulphonamide of a secondary amine.

2017 Q2 · 2016 Q6 · 2022 Q22 · 2023 Q33(a)

An IUPAC name for a tertiary amine, a tribromoaniline, or a mixed N-substituted aniline.

Past year questions · numerical types — no calculations in this chapter, so these are the reaction and structure types — one worked model each, then cold practice

A · Identify A, B, C in a sequence

Worked example · 2015 Q23 [30+] — A is $C_7H_6O_2$; A + aqueous NH_3 (heat) gives B; B with Br_2 and alcoholic KOH gives C, C_6H_7N

1. $Br_2 + KOH$ is the Hofmann reagent, so B is an **amide** and C is the **amine** with one carbon fewer.
 C_6H_7N = aniline = **C**.
2. An aromatic amide with seven carbons is benzamide $C_6H_5CONH_2$ = **B**.
3. A acid + NH_3 , heat, gives the amide, so A is benzoic acid C_6H_5COOH ($C_7H_6O_2$ ✓).



- 2014 Q15 [30+] — nitrobenzene with Sn/HCl , then $NaNO_2/HCl$ at 273 K, then warm water — answer only: A aniline; B benzenediazonium chloride; C phenol
- 2014 Q25 [30+] — CH_3CH_2Br with KCN , then $LiAlH_4$, then HNO_2 at 0 °C — answer only: CH_3CH_2CN ; $CH_3CH_2CH_2NH_2$; propan-1-ol
- 2022 Q28 [30+] — A with iron scrap and HCl gives C_6H_7N ; with $CHCl_3$ and alcoholic KOH gives a foul smell — answer only: A nitrobenzene; B aniline; C phenyl isocyanide
- 2025 Q3(a) [2025] — chloroethane + $NaCN$ to A, then H_2/Ni to B — answer only: A CH_3CH_2CN ; B propan-1-amine

B · Replace the diazonium group

Worked example · 2019 Q24 [30+] — benzenediazonium chloride with $CuCN$, with ethanol, and with KI
 $CuCN$: CN takes the place of $N_2 \rightarrow$ **benzonitrile** C_6H_5CN (Sandmeyer). **Ethanol** is a mild reducing agent: hydrogen takes the place of $N_2 \rightarrow$ **benzene**; the ethanol is oxidised to ethanal. KI : iodide takes the place of $N_2 \rightarrow$ **iodobenzene**. Each time nitrogen gas leaves.

- 2020 Q10 [30+] — benzenediazonium chloride + $CuCN$ gives A; A hydrolysed gives B — answer only: A benzonitrile; B benzoic acid
- 2013 Q1 [30+] — benzenediazonium chloride warmed with water — answer only: phenol (+ $N_2 + HCl$)
- 2023 Q31(a) [30+] — benzenediazonium chloride with ethanol — answer only: benzene; ethanol is oxidised to ethanal
- 2025 Q5(b) [2025] — chlorobenzene from aniline — answer only: $NaNO_2/HCl$ at 0–5 °C, then $CuCl/HCl$

C · Substitute on aniline's ring (protect, react, deprotect)

Worked example · 2025 Q5(a) [2025] — *p*-nitroaniline from aniline

Acetylate: $C_6H_5NH_2 \xrightarrow{(CH_3CO)_2O} C_6H_5NHCOCH_3$. Nitrate with HNO_3/H_2SO_4 : the para product is the major one, *p*-nitroacetanilide. Hydrolyse with acid or base to get back $-NH_2$: ***p*-nitroaniline**. The acetyl group stops oxidation and stops the anilinium ion forming, so the nitro group goes para, not meta.

- 2025 Q2(a) [2025] — sulphanilic acid from aniline — answer only: aniline + conc. $H_2SO_4 \rightarrow$ anilinium hydrogensulphate; heat at 453–473 K
- 2025 Q2(c) / 2014 Q12 [2025, 30+] — acetanilide from aniline — answer only: $(CH_3CO)_2O$ with pyridine $\rightarrow$ N-phenylethanamide

- 2023 Q33(b) [30+] — nitrobenzene to 2,4,6-tribromoaniline — answer only: Sn/HCl to aniline, then Br_2 water

D · Arrange in order

Worked example · 2020 Q18 [30+] — increasing solubility in water: aniline, a secondary amine, a primary amine

Solubility rests on hydrogen bonding with water. The primary amine has two N–H, the secondary one, and aniline has a large hydrophobic ring. Order: **aniline < secondary < primary**. (Boiling point, same question: tertiary < secondary < primary.)

- 2019 Q11 [30+] — basic strength in the gas phase, increasing — answer only: primary < secondary < tertiary
- Q26(a) [30+] — CH_3NH_2 , $(\text{CH}_3)_3\text{N}$, $(\text{CH}_3)_2\text{NH}$ in water, increasing — answer only:
 $(\text{CH}_3)_3\text{N} < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH}$
- Q27(b) [30+] — increasing pK_b : $(\text{C}_2\text{H}_5)_2\text{NH}$, $\text{C}_2\text{H}_5\text{NH}_2$, NH_3 , $\text{C}_6\text{H}_5\text{NH}_2$ — answer only:
 $(\text{C}_2\text{H}_5)_2\text{NH} < \text{C}_2\text{H}_5\text{NH}_2 < \text{NH}_3 < \text{C}_6\text{H}_5\text{NH}_2$
- 2025 Q4(a) [2025] — decreasing pK_b of four amines — answer only: aniline first (highest pK_b), then the methyl-substituted arylamine, then the primary, then the secondary aliphatic amine (the four structures are shown on screen, not spoken)

E · Give a reason

Worked example · 2020 Q19(b) [30+] — $(\text{CH}_3)_2\text{NH}$ is more basic than $(\text{CH}_3)_3\text{N}$ in aqueous solution

Three factors decide basicity in water. **+I effect**: three methyls push more electron density onto nitrogen, favouring the tertiary. **Solvation**: the secondary cation has two N–H to hydrogen-bond with water, the tertiary only one, so the secondary cation is better stabilised. **Steric hindrance**: methyl is small, so it does not block the solvation. Solvation outweighs +I, so the **secondary amine is the stronger base**.

- 2014 Q13(a) / 2020 Q29(a) [30+] — aniline gives no Friedel–Crafts reaction — answer only: forms a salt with AlCl_3 ; nitrogen turns positive and deactivates the ring
- 2018 Q17(b) [30+] — why aromatic diazonium salts are more stable than aliphatic — answer only: resonance spreads the positive charge over the benzene ring
- 2013 Q16(a) [30+] — electrophilic substitution in aromatic amines is easier than in benzene — answer only: the nitrogen lone pair is delocalised into the ring (+R), raising its electron density
- 2025 Q1(b)(ii) [2025] — amides are less basic than amines — answer only: the nitrogen lone pair is in resonance with the C=O group, so it is not available to take a proton

F · Distinguish

Worked example · 2013 Q30(a) [30+] — distinguish a secondary from a tertiary amine

Add benzenesulphonyl chloride (Hinsberg's reagent). The **secondary** amine reacts and loses HCl , forming an N,N-dialkylbenzenesulphonamide that has no N–H, so it is **insoluble in alkali**. The **tertiary** amine has no hydrogen on nitrogen, so there is no reaction.

- 2020 Q14 [30+] — ethylamine versus diethylamine — answer only: carbylamine test; ethylamine gives the foul-smelling isocyanide, diethylamine does not
- 2020 Q14 [30+] — aniline versus N-methylaniline — answer only: carbylamine test; aniline (primary) gives it, N-methylaniline does not

G · IUPAC names

Worked example · 2022 Q22 [30+] — name $(\text{CH}_3\text{CH}_2)_2\text{N}-\text{CH}_3$

Longest chain on nitrogen: two carbons (either ethyl) → **ethanamine**. The other ethyl and the methyl are N-substituents, alphabetical: **N-ethyl-N-methylethanamine**.

- 2017 Q2 [30+] — $(\text{CH}_3)_2\text{N}-\text{CH}_2\text{CH}_3$ — answer only: N,N-dimethylethanamine
- 2016 Q6 [30+] — aniline with three bromines at the 2, 4, 6 positions — answer only: 2,4,6-tribromoaniline
- 2023 Q33(a) [30+] — a tertiary amine with ethyl, methyl and phenyl on nitrogen — answer only: N-ethyl-N-methylaniline

Biomolecules

NCERT Class XII Chemistry • Chapter 10 • Biomolecules

7 marks in the board paper (CBSE 2026-27 curriculum). This chapter is a tour of the molecules that living things are built from: sugars, proteins, enzymes, vitamins, DNA and RNA, hormones. It reads like biology, but the exam asks it as chemistry — a structure, a reagent, a reason. Every new word is tagged [exposure] and explained from scratch. Glucose is the heart of the chapter; get it and half the marks follow.

What a biomolecule is — *three definitions to hold on to before anything else*

[exposure] Biochemistry is the branch of science that studies the chemistry going on inside living things — what they are made of and which reactions run in them. It exists because a living body is just atoms and molecules following chemical rules; to understand digestion or heredity you have to understand the molecules. This chapter is a slice of it.

[exposure] A biomolecule is any complex organic molecule that builds up a living organism and is needed for its growth and upkeep. The four big families are **carbohydrates, proteins, lipids and nucleic acids**. This chapter covers three of them in full (carbohydrates, proteins, nucleic acids), then enzymes, vitamins and hormones.

[exposure] A polymer is a very long molecule made by joining many small identical or similar units, the **monomers**, end to end. Starch is a polymer of glucose; a protein is a polymer of amino acids; DNA is a polymer of nucleotides. You will see the same pattern three times, so keep it in mind: *small unit → joined by one specific bond → long chain*.

Examiner asks: nothing directly from this section, but it names the three monomer–polymer pairs the whole chapter is built on.

10.1 Carbohydrates — what counts as one — *the formula is a trap; the real definition is about groups*

[exposure] Carbohydrates are a huge group of naturally occurring organic compounds, made mostly by plants. Cane sugar, glucose and starch are all carbohydrates. They matter because they are the main fuel and the main structural material of living things. The name means "hydrate of carbon", because most of them fit the general formula $C_x(H_2O)_y$ — carbon with water attached.

Glucose $C_6H_{12}O_6$ fits it as $C_6(H_2O)_6$. But the formula cannot be the definition, in both directions:

- Acetic acid CH_3COOH fits it as $C_2(H_2O)_2$ and is **not** a carbohydrate.
- Rhamnose $C_6H_{12}O_5$ is a carbohydrate and does **not** fit it.

So the real definition is by structure: a carbohydrate is an **optically active polyhydroxy aldehyde or ketone**, or a compound that gives such units on hydrolysis. Three words to unpack:

[exposure] Optically active means the compound rotates the plane of plane-polarised light. A beam of light that vibrates in only one plane is plane-polarised; passing it through some compounds turns that plane to the right or left. It happens when a molecule has a carbon with four different groups on it (you met this as chirality in Haloalkanes). Why it matters here: it is the property that lets chemists tell sugars apart and track their structure.

[exposure] Polyhydroxy means more than one $-OH$ group. An **aldehyde** is the group $-CHO$ and a **ketone** is $>C=O$ inside a chain (the next chapter treats both properly). So a carbohydrate is a chain carrying several $-OH$ groups and one $-CHO$ or $>C=O$.

[exposure] Hydrolysis means breaking a bond by adding water, usually with a little acid or an enzyme to speed it up. The big carbohydrates are made by joining small ones and losing water, so hydrolysis runs it backwards and gives the small units back. Example: sucrose plus water, with a little acid, gives glucose plus fructose (below).

Carbohydrates that taste sweet are called **sugars**: the table sugar at home is **sucrose**; the sugar in milk is **lactose**. Carbohydrates are also called **saccharides**, from the Greek word for sugar.

Trap: " $C_x(H_2O)_y$ means carbohydrate" is false in both directions. Cite acetic acid (fits, not a carbohydrate) and rhamnose (does not fit, is one). Then give the structural definition.

Classifying carbohydrates — *by what hydrolysis gives, and by whether they reduce Tollens' and Fehling's*

1. By behaviour on hydrolysis

[exposure] A **monosaccharide** is a carbohydrate that cannot be hydrolysed any further into simpler polyhydroxy aldehyde or ketone units. It is the single building block. About 20 are known in nature; glucose, fructose and ribose are the ones you need.

[exposure] An **oligosaccharide** gives 2 to 10 monosaccharide units on hydrolysis ("oligo" = few). They are named by count: **di-**, **tri-**, **tetra-saccharide** for 2, 3, 4 units. The common ones are disaccharides. Example: **sucrose** hydrolyses to one glucose plus one fructose (two different units); **maltose** hydrolyses to two glucose (two same units).

[exposure] A **polysaccharide** gives a large number of monosaccharide units on hydrolysis. Starch, cellulose, glycogen and gums are the standard examples. They are not sweet, so they are also called **non-sugars**.

2. By reducing action

[exposure] **Tollens' reagent** and **Fehling's solution** are two standard lab tests for an aldehyde: an aldehyde reduces them (and is itself oxidised), and you see a silver mirror or a red-brown precipitate. A carbohydrate that reduces either is a **reducing sugar**; one that does not is **non-reducing**. The test works only when a *free* aldehyde or ketone group is available to react.

- All **monosaccharides** are reducing sugars, aldose or ketose, because the group is free.
- A **disaccharide** is reducing if at least one of its two aldehyde/ketone groups is still free: **maltose**, **lactose**.
- It is **non-reducing** if both groups are tied up in the bond joining the two units: **sucrose**.

Examiner asks: "name the disaccharide that gives two glucose on hydrolysis" (2020 — maltose); "products of hydrolysis of sucrose" (2019 — glucose plus fructose); "which is a disaccharide, which is a monosaccharide" from a list of starch, maltose, glucose, fructose, cellulose (2015 — maltose; fructose); "why sucrose is non-reducing" (2010 — both reducing groups are used up in the bond).

Monosaccharides — the naming grid — *carbon count gives the stem; aldehyde or ketone gives the prefix*

[exposure] An **aldose** is a monosaccharide that carries an aldehyde group; a **ketose** carries a ketone group. The carbon count goes into the name too, so a six-carbon aldose is an **aldohexose** and a six-carbon ketose is a **ketohexose**. This two-part name tells you in one word what the molecule is built from.

Carbons	General term	Aldose	Ketose
3	triose	aldotriose	ketotriose
4	tetrose	aldotetrose	ketotetrose
5	pentose	aldopentose	ketopentose
6	hexose	aldohexose	ketoheptose
7	heptose	aldoheptose	ketoheptose

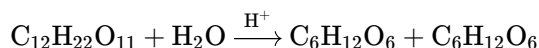
Glucose is an aldohexose. Fructose is a ketohexose. Ribose, which turns up in RNA, is an aldopentose.

Glucose — where it comes from — *two preparations, both ending in glucose*

Glucose occurs both free (sweet fruit, honey, ripe grapes) and combined (inside sucrose, starch, cellulose). It is made in the lab or industry in two ways.

From sucrose (cane sugar)

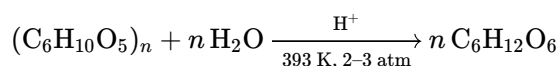
Boil sucrose with dilute HCl or H₂SO₄ in alcoholic solution. It hydrolyses to **equal amounts of glucose and fructose**. Both have the same molecular formula C₆H₁₂O₆.



Glucose and fructose are two different compounds with the same formula.

From starch (the industrial route)

Boil starch (or cellulose) with dilute H₂SO₄ at 393 K under 2–3 atm pressure. Every glucose unit in the chain is released.



Examiner asks: the conditions 393 K and 2–3 atm, and the fact that sucrose gives glucose plus fructose in equal amounts.

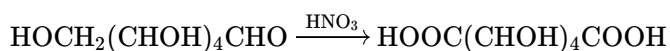
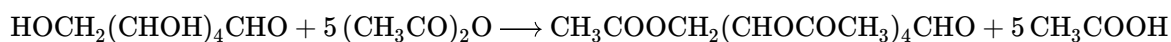
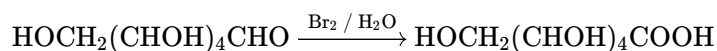
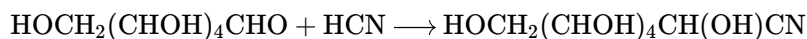
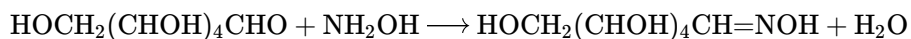
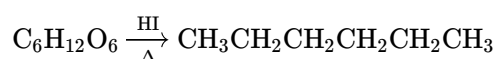
Glucose — proving its open-chain structure — *six reagents, six facts; this table is the single most repeated question in the chapter*

[exposure] The **open-chain structure** of glucose is the straight-chain form: six carbons in a row, an aldehyde group –CHO at one end, a –CH₂OH at the other, and four –CHOH groups in between. Chemists did not see it; they inferred it by reacting glucose with chosen reagents and reading what came out. Each reaction proves one feature, and each one is a standalone exam question.



Reagent	What happens	What it proves
Molecular formula	$C_6H_{12}O_6$	six carbons
HI, prolonged heating	gives <i>n</i> -hexane $CH_3(CH_2)_4CH_3$	all six carbons are in a straight chain
NH_2OH (hydroxylamine)	gives an oxime , $CH=NOH$	a carbonyl group is present
HCN	gives a cyanohydrin	a carbonyl group is present
Br_2 water (mild oxidant)	gives gluconic acid , $-CHO \longrightarrow -COOH$	the carbonyl is an aldehyde , not a ketone
$(CH_3CO)_2O$ (acetic anhydride)	gives glucose pentaacetate	five $-OH$ groups , each on a different carbon
HNO_3 (strong oxidant)	gives saccharic acid , a dicarboxylic acid	a primary alcohol $-CH_2OH$ is present

The reactions, written out:



[exposure] An **oxime** is what an aldehyde or ketone becomes after it condenses with hydroxylamine NH_2OH : the $C=O$ turns into $C=N-OH$ and one water leaves. It exists as a classic way to test for a carbonyl group. Example: glucose gives glucose oxime, $CH=NOH$ at the top of the chain.

[exposure] A **cyanohydrin** forms when HCN adds across a carbonyl: the carbon gains both a $-CN$ and an $-OH$. The CN^- ion is a nucleophile (an electron-rich attacker, as in Haloalkanes), it attacks the electron-poor carbonyl carbon, and the oxygen then picks up H^+ from the solution. Same job as the oxime: it confirms a carbonyl.

[exposure] **Acetylation** means attaching an acetyl group CH_3CO- to an $-OH$, turning each one into an ester. Acetic anhydride does it. Glucose takes *five* acetyls, so it has five $-OH$ groups. Because the product is a stable compound, no two of those $-OH$ groups sit on the same carbon.

Trap: do not mix up which reagent proves which feature. Carbonyl is proved by *oxime or cyanohydrin* (they work on aldehydes *and* ketones). It is **bromine water** that narrows it to an aldehyde. And HNO_3 gives a *dicarboxylic acid* (both ends become $-COOH$), which is why NCERT concludes a **primary alcohol** is present.

Examiner asks: this table is the most repeated question in the chapter. Forms it takes: "write the product of glucose with HI" (2014); "with conc. HNO_3 " (2018); "show the open chain has a straight chain, five $-\text{OH}$, an aldehyde" (2019); "glucose with HCN , Br_2 , acetic anhydride" (2019); "heating glucose with hydroxylamine — which group is confirmed" (2023, answer: carbonyl). See the PYQ section.

Glucose — D, L and the (+) sign — *two labels that students merge into one; they mean different things*

[exposure] The **Fischer projection** is a way of drawing a 3-D molecule flat. The carbon chain runs vertically with the most oxidised carbon ($-\text{CHO}$) at the top. Each carbon sits where a horizontal line meets the vertical one. Horizontal bonds point *toward* you; vertical bonds point *away*. It exists because chemists needed one fixed convention to compare the 3-D arrangement of many sugars on paper. Fischer used it to fix the positions of glucose's $-\text{OH}$ groups after studying many of its properties.

In glucose's Fischer projection, going down from the aldehyde carbon, the $-\text{OH}$ groups sit **right, left, right, right** (on carbons 2, 3, 4, 5).

[exposure] An **asymmetric (chiral) carbon** is one attached to four different atoms or groups. A molecule with one comes in two mirror-image forms called **enantiomers**, and it is optically active. Glyceraldehyde ($\text{HOCH}_2-\text{CH}(\text{OH})-\text{CHO}$) has exactly one, so it exists as two enantiomers. This is why it is chosen as the reference for the D/L system.

[exposure] **D and L** label the *spatial arrangement* of a compound relative to glyceraldehyde. Write the structure with the most oxidised carbon on top. Look at the **lowest asymmetric carbon** (the one furthest from the $-\text{CHO}$). If its $-\text{OH}$ is on the **right**, the compound is **D**; on the **left**, **L**. Glucose's lowest asymmetric carbon (C_5) has $-\text{OH}$ on the right, so glucose is **D-glucose**. Only that one carbon is used for the label.

[exposure] The **(+) or (-)** sign is a separate fact: it says which way the compound rotates plane-polarised light. (+) = dextrorotatory = to the right; (-) = laevorotatory = to the left. It has to be measured; you cannot read it off a drawing. Glucose rotates right, so it is **D-(+)-glucose**. The older lowercase d and l mean the same as (+) and (-).

Trap: D has no relation to optical rotation. D tells you the arrangement; (+) tells you the rotation direction. D-fructose is D but laevorotatory, written D-(-)-fructose. That one pair is the standard example to quote.

Quick relatives of glucose that follow from the same drawing: if the top $-\text{CHO}$ becomes $-\text{COOH}$ it is **gluconic acid**; if both ends are $-\text{COOH}$ it is **saccharic acid**.

Glucose — the ring form — *three facts the straight chain cannot explain; one idea that explains all of them*

The open-chain structure explains almost everything above, but three observations break it:

1. Glucose has an aldehyde group, yet it **does not give Schiff's test** (the pink-colour test for aldehydes) and it **does not form the hydrogensulphite addition product** with NaHSO_3 , which aldehydes normally do.
2. The **pentaacetate of glucose does not react with hydroxylamine**, though a free aldehyde would. So in glucose the $-\text{CHO}$ is not free.
3. Glucose forms **two crystalline forms**: the **α -form** (melting point 419 K, crystallised from concentrated solution at 303 K) and the **β -form** (melting point 423 K, crystallised from hot saturated solution at 371 K).

[exposure] A **hemiacetal** is what forms when an --OH group adds to an aldehyde: the old carbonyl carbon ends up holding an --H , an --OH and an --OR . When the --OH and the aldehyde are on the *same molecule*, the addition closes a ring. This is the fix for all three facts: in glucose the --OH on **carbon 5** adds to the carbon-1 aldehyde, and a **six-membered ring** (five carbons and one oxygen) closes. The --CHO is now used up inside the ring, which is why Schiff's test, NaHSO_3 and hydroxylamine all fail.

The two crystal forms then have an easy explanation: closing the ring turns carbon 1 into a new asymmetric carbon, and the new --OH on it can sit on either side. The two cyclic forms stay in **equilibrium with the open chain** in solution.

[exposure] The **anomeric carbon** is carbon 1, the former aldehyde carbon, now the carbon that holds the new --OH . The two ring forms differ *only* in the position of that one --OH : **on the right** in the Fischer drawing = $\alpha\text{-D-(+)-glucose}$; **on the left** = $\beta\text{-D-(+)-glucose}$. Two isomers that differ only at the anomeric carbon are called **anomers**.

[exposure] A **pyranose** is any sugar ring with six atoms (five carbons and one oxygen), named after **pyran**, the ring compound with five carbons and one oxygen. So the glucose rings are **$\alpha\text{-D-glucopyranose}$** and **$\beta\text{-D-glucopyranose}$** . The **Haworth structure** is the standard flat drawing of this ring, with the ring edge-on, and each group pointing either up or down.

Reading a Haworth drawing off the Fischer one

The rule that does every sugar: **on the right in Fischer $\rightarrow$ down in Haworth; on the left $\rightarrow$ up**. The terminal CH_2OH on carbon 5 points up in every D-sugar.

Carbon	$\alpha\text{-D-glucopyranose}$	$\beta\text{-D-glucopyranose}$
C1 (anomeric)	--OH down	--OH up
C2	--OH down	--OH down
C3	--OH up	--OH up
C4	--OH down	--OH down
C5	$\text{--CH}_2\text{OH}$ up	$\text{--CH}_2\text{OH}$ up

The only line that changes between α and β is C1. That one change is the whole difference between the anomers.

Examiner asks: "enumerate the reactions of glucose that its open structure cannot explain" (2014); "why does glucose not give the 2,4-DNP test" (2019); "why the pentaacetate of glucose does not react with hydroxylamine" (2023); "define anomers" (2014). All four answers are the same idea — the --CHO is not free because it is inside the ring.

Fructose — same formula as glucose, a ketone instead of an aldehyde, and a five-membered ring

Fructose is the ketohexose that you get along with glucose from hydrolysing sucrose. It is found in fruit, honey and vegetables, and it is used as a sweetener. Its molecular formula is the same as glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, but the reactions show a **ketone group at carbon 2**, on the same six-carbon straight chain. It is a D-sugar (the lowest asymmetric carbon has --OH on the right) and it rotates light to the left, so its name is **D-(–)-fructose**.

[exposure] A **furanose** is a sugar ring with **five** atoms (four carbons and one oxygen), named after **furan**, the five-membered ring with four carbons and one oxygen. Fructose closes its ring when the --OH on **carbon**

5 adds to the **carbon-2 ketone**. That ring is five-membered, unlike glucose's six, which is why the name is **fructofuranose**, not "-pyranose". Again there are two anomers, α and β , differing only at the new anomeric carbon (C2): **α -D-(-)-fructofuranose** and **β -D-(-)-fructofuranose**.

	Glucose	Fructose
Type	aldohexose	ketohehexose
Ring formed by	C5 –OH + C1 aldehyde	C5 –OH + C2 ketone
Ring size	6 (pyranose)	5 (furanose)
Anomeric carbon	C1	C2
Rotation	(+) dextro	(-) laevo

Examiner asks: ring size and the names pyranose / furanose; the (–) sign on fructose.

Disaccharides — two sugars joined through oxygen — sucrose, maltose, lactose, and which of them reduce

[exposure] A **glycosidic linkage** is the C–O–C bridge that joins two monosaccharides. It forms when an –OH on one sugar and an –OH on the other lose one water between them, leaving a single oxygen as the link. NCERT calls this an "oxide linkage". This is the one bond that holds every disaccharide and polysaccharide together, and it is the carbohydrate equivalent of the peptide bond you will meet in proteins. A disaccharide hydrolysed with dilute acid or an enzyme gives back two monosaccharide molecules.

Disaccharide	Made of	Linkage	Reducing?
Sucrose	α -D-glucose + β -D-fructose	C1 of glucose – C2 of fructose	No
Maltose	two α -D-glucose	C1 of one – C4 of the other	Yes
Lactose	β -D-galactose + β -D-glucose	C1 of galactose – C4 of glucose	Yes

Sucrose

The common table sugar. On hydrolysis it gives an **equimolar mixture** (equal numbers of molecules) of D-(+)-glucose and D-(-)-fructose. In sucrose the link is made *through the aldehyde carbon of glucose (C1) and the ketone carbon of fructose (C2)*. Both reducing groups are used up in that one bond, so neither is free, and sucrose is a **non-reducing sugar**: it does not reduce Tollens' or Fehling's.

[exposure] **Invert sugar** is the glucose–fructose mixture you get by hydrolysing sucrose. Sucrose itself is dextrorotatory (+66.5°). But the hydrolysis gives glucose (+52.5°) and fructose (–92.4°), and the fructose rotates further left than glucose rotates right, so the **mixture is laevorotatory**. The sign of the rotation *inverts* from (+) to (–) during hydrolysis, which is where the name comes from.

Maltose

Two α -D-glucose units joined C1 to C4. The second glucose still has its C1 hemiacetal, which in solution opens to give a free aldehyde, so maltose **reduces**.

Lactose (milk sugar)

β -D-galactose plus β -D-glucose, joined C1 of galactose to C4 of glucose. The glucose unit still has a free C1, so lactose **reduces**. Galactose differs from glucose only at C4, where the –OH is on the other side. On hydrolysis, lactose gives β -D-glucose and β -D-galactose.

Trap: the reason for "sucrose is non-reducing" must say *both reducing groups, the aldehyde of glucose and the ketone of fructose, are involved in the glycosidic bond*. Saying "it has no aldehyde group" scores nothing; it does, until it is used in the bond.

Examiner asks: products of hydrolysis of lactose (2019); "why is sucrose non-reducing" (2010); name the linkage between two monosaccharides (2018, 2020 — glycosidic); "invert sugar" (see the PYQ section).

Polysaccharides — starch, cellulose, glycogen — the same monomer, three different jobs

Polysaccharides are thousands of monosaccharide units joined by glycosidic linkages. They do two jobs: **food storage** and **structure**.

Starch — the plant's stored food

The main storage polysaccharide of plants and our main dietary carbohydrate (cereals, roots, tubers). It is a polymer of α -glucose and has *two* components:

	Amylose	Amylopectin
Share of starch	15–20%	80–85%
Shape	long, unbranched chain, 200–1000 α -D-(+)-glucose units	branched chain of α -D-glucose
Linkage	C1–C4 glycosidic	C1–C4 along the chain; C1–C6 at the branches
In water	soluble	insoluble

Cellulose — the plant's structure

A straight-chain polymer of **β -D-glucose**, joined C1 to C4. It is the most abundant organic substance in the plant kingdom and the main constituent of plant cell walls. We use it as wood and as cotton.

Glycogen — the animal's stored food

The carbohydrate stored in the animal body, also called **animal starch**. Its structure is like amylopectin but **more highly branched**. It is found in the liver, muscles and brain; when the body needs glucose, enzymes break glycogen down. It is also found in yeast and fungi.

Trap: starch and cellulose are *both* glucose polymers. The structural difference they want is one line: **starch is α -D-glucose, cellulose is β -D-glucose**; starch has an unbranched *and* a branched part, cellulose is purely linear.

Why carbohydrates matter

They are the major part of our food; honey has long been used as an instant energy source by "vaid" in the Ayurvedic system; starch and glycogen are the storage molecules; cellulose builds plant cell walls (NCERT adds bacteria) and gives us wood and cotton fibre. D-ribose and 2-deoxy-D-ribose are the two aldopentoses in nucleic acids, and carbohydrates occur in the body combined with proteins and lipids.

Examiner asks: a carbohydrate found in liver, muscle and brain (2019 — glycogen); the structural difference between starch and cellulose (2016, 2023); the two components of starch (2013); amylose versus amylopectin (2019).

10.2 Amino acids — *the monomer of every protein — what one is, how they are classified, and why they behave like salts*

[exposure] **Proteins** are the most abundant biomolecules in living systems; they form the basis of structure and function and are needed for growth and upkeep. Chief sources: milk, cheese, pulses, peanuts, fish, meat. The word comes from the Greek for "of primary importance". Every protein is a polymer of **α -amino acids**.

[exposure] An **amino acid** is a compound that carries both an amino group —NH_2 (basic) and a carboxyl group —COOH (acidic). In an **α -amino acid** both sit on the same carbon, the α -carbon, along with a hydrogen and a side chain R: $\text{H}_2\text{N—CHR—COOH}$. Only α -amino acids come out of the hydrolysis of proteins. Different R groups make different amino acids. Example: R = H gives **glycine**; R = CH_3 gives **alanine**.

About 20 amino acids make up all plant and animal proteins. Each has a **trivial name** that records a property or the source — glycine is named from the Greek for "sweet" (it tastes sweet) and tyrosine from the Greek for "cheese" (it was first isolated from cheese). Each has a three-letter code and a one-letter code: glycine Gly / G, alanine Ala / A.

Classification 1: acidic, basic, neutral

Count the groups. **Equal** numbers of —NH_2 and —COOH → **neutral** (glycine, alanine). **More** —COOH → **acidic** (extra —COOH in the side chain: aspartic acid, glutamic acid). **More** —NH_2 → **basic** (extra —NH_2 in the side chain: lysine, arginine).

Classification 2: essential and non-essential

[exposure] An **essential amino acid** is one the body cannot make, so it has to come from food. A **non-essential** one the body can make itself. Of the 20, **10 are essential** and 10 are not. Essential: valine, leucine, isoleucine, arginine, lysine, threonine, methionine, phenylalanine, tryptophan, histidine. Non-essential: glycine, alanine, glutamic acid, aspartic acid, glutamine, asparagine, serine, cysteine, tyrosine, proline.

Trap: for a two-example answer, safe picks are **valine and leucine** (essential) and **alanine and glutamic acid** (non-essential). Arginine and lysine are on *both* the "basic" list and the "essential" list, so do not use them as a clean example of either.

Properties

Amino acids are colourless crystalline solids, soluble in water, with high melting points. They behave like **salts** rather than like a simple amine or acid. The reason is the two groups in one molecule:

[exposure] A **zwitterion** (zwitter ion) is a molecule that carries a positive and a negative charge at the same time and is neutral overall. In water, the —COOH gives up H^+ to become —COO^- while the —NH_2 picks it up to become —NH_3^+ . Both charges sit on the same molecule: $\text{H}_3\text{N}^+\text{—CHR—COO}^-$. That ionic character is why amino acids behave like salts, dissolve in water and melt at high temperature.

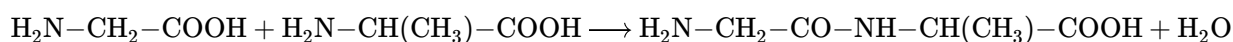
[exposure] **Amphoteric** means able to react with both acids and bases. In zwitterion form, an amino acid has an acidic group *and* a basic group, so it reacts with either. That is the whole answer to "why are amino acids amphoteric".

Optical activity. Every α -amino acid except **glycine** is optically active, because the α -carbon is asymmetric. Glycine's α -carbon carries two hydrogens, so it is not. Amino acids exist as D and L forms; the natural ones are **L**, written with the —NH_2 on the **left**. (For sugars you look at the —OH ; for amino acids you look at the —NH_2 .)

Examiner asks: define an amino acid (2014); what is an essential amino acid with examples (2014, 2018, 2025); why amino acids are amphoteric (2018, 2023, 2025); why they behave like salts (2023).

Peptides and the peptide bond — *how amino acids join; when a chain stops being a peptide and becomes a protein*

[exposure] A **peptide bond** (peptide linkage) is the amide bond —CO—NH— that joins two amino acids. It forms when the —COOH of one amino acid reacts with the —NH_2 of the next and a molecule of water leaves. It exists because that is the only way nature strings amino acids into a chain; it is the protein version of the glycosidic linkage.



That is glycine's carboxyl joining alanine's amino group, giving the dipeptide **glycylalanine**.

Name by count: 2 amino acids $\rightarrow$ **dipeptide**; 3 $\rightarrow$ **tripeptide**; 4, 5, 6 $\rightarrow$ **tetra-, penta-, hexapeptide**; **more than 10** $\rightarrow$ **polypeptide**. Each amino acid inside the chain is called a **residue**, and the chain loses one water per bond: **n amino acids make n – 1 peptide bonds**.

A polypeptide with **more than 100 amino acid residues and a molecular mass above 10,000 u** is called a protein. The line is not sharp: a shorter chain still counts as a protein if it folds into a well-defined protein shape, and the standard example is **insulin, with only 51 amino acids**.

Trap: glycosidic versus peptide linkage is a standing 2019 question. One-liner: **glycosidic = C–O–C between two monosaccharides (carbohydrates); peptide = –CO–NH– amide between two amino acids, with loss of water (proteins)**.

Protein structure and shape — *fibrous or globular; then four levels, each one built on the last*

By overall shape

[exposure] **Fibrous proteins** are thread-like: many polypeptide chains lie side by side, held together by **hydrogen bonds and disulphide bonds** (S–S bridges between sulphur-containing amino acids). They are **insoluble in water**. Examples: **keratin** (hair, wool, silk) and **myosin** (muscle).

[exposure] **Globular proteins** are chains coiled up into a ball. They are generally **soluble in water**, which is why they can travel in blood plasma. Examples: **insulin and albumins**.

The four levels

1. **Primary** — the *sequence* of amino acids in each chain. Change even one amino acid in the sequence and you have a different protein. **Secondary** — the regular shape the backbone folds into, held by **hydrogen bonds** between the C=O of one peptide bond and the N–H of another. Two forms:
 - **α -helix**: one chain twists into a right-handed screw; each N–H bonds to the C=O one turn along. The hydrogen bonds are **within one chain** (intramolecular).
 - **β -pleated sheet**: several chains stretched out and laid side by side like folds in a curtain; the hydrogen bonds are **between different chains** (intermolecular).
2. **Tertiary** — the overall folding of the whole chain, giving the fibrous or globular shape. Held by hydrogen bonds, disulphide linkages, van der Waals forces and electrostatic attraction.

3. **Quaternary** — only in proteins with **two or more chains** (sub-units): the arrangement of the sub-units relative to each other. Haemoglobin is the example: four sub-units.

Trap: the α -helix versus β -sheet question is always about the *location* of the hydrogen bonds: **intramolecular (one chain) versus intermolecular (different chains)**. State both halves.

Examiner asks: an example each of fibrous and globular protein (2016); difference between α -helix and β -pleated sheet (2018, 2023); define primary structure (2024); two differences between globular and fibrous (2013).

Denaturation — *what boiling an egg does to a protein*

[exposure] A **native protein** is a protein as it occurs in a living system, with its own unique 3-D structure and its biological activity (an enzyme, for instance). **Denaturation** is what happens when a native protein is pushed out of that shape by a *physical* change (heat) or a *chemical* change (a change in pH). Hydrogen bonds break, globules unfold, helices uncoil, and the protein **loses its biological activity**. What stays: the **primary structure**, because it is held by peptide bonds, not hydrogen bonds. What is destroyed: the **secondary and tertiary** structure.

Two standard examples: **coagulation of egg white on boiling**, and **curdling of milk** (bacteria make lactic acid, the pH drops, the milk protein coagulates).

Examiner asks: define native and denatured protein with an example (2018, 2025); what changes in egg protein on boiling (2013 — it denatures, hydrogen bonds break, activity lost).

10.3 Enzymes — *biological catalysts — very specific, needed only in tiny amounts*

[exposure] **Enzymes** are biological catalysts: substances that speed up the chemical reactions of living things without being used up. They exist because the body has to run huge numbers of reactions under mild conditions (body temperature, neutral pH), and they only work fast enough with a catalyst. Almost every enzyme is a **globular protein**. Example: the enzyme **maltase** speeds the hydrolysis of maltose into glucose.

- **Specific** — one enzyme for one reaction and one substrate. The **substrate** is the molecule the enzyme acts on.
- **Needed only in small amounts** for the reaction to proceed.
- **Naming:** usually after the compound it works on (maltose $\rightarrow$ *maltase*), sometimes after the reaction (an enzyme catalysing oxidation of one substrate with simultaneous reduction of another is an **oxidoreductase**). The name **ends in "-ase"**.

How they work: like any catalyst, an enzyme lowers the **activation energy** (the energy barrier a reaction has to cross; you met it in Chemical Kinetics). NCERT's number: acid hydrolysis of sucrose needs 6.22 kJ mol^{-1} ; with the enzyme sucrase it needs only 2.15 kJ mol^{-1} .

Examiner asks: define enzymes (2014 — biological catalysts, almost all globular proteins).

10.4 Vitamins — *definition, the two classes, and the source–disease table that is a pure recall mark*

[exposure] **Vitamins** are organic compounds needed in the diet in *small* amounts to perform specific biological functions for normal growth and health. Their deficiency causes specific diseases. Most cannot be made in the human body (so food has to supply them); plants make almost all of them, so vitamins count as "essential food factors". They are named A, B, C, D, E, K, with sub-groups such as B₁, B₂, B₆, B₁₂.

Two classes, by solubility

- **Fat-soluble** — dissolve in fat and oil, not in water: **A, D, E, K**. They are stored in the liver and in adipose (fat-storing) tissue.
- **Water-soluble** — **B group and C**. They must be taken **regularly** because they are readily excreted in urine and cannot be stored — **except vitamin B₁₂**, which is stored in the liver.

Vitamin	Source	Deficiency disease
A (retinol)	fish liver oil, carrot, butter, milk	xerophthalmia (hardening of the cornea); night blindness
B ₁ (thiamine)	yeast, milk, green vegetables, cereals	beri-beri (loss of appetite, retarded growth)
B ₂ (riboflavin)	milk, egg white, liver, kidney	cheilosis (fissures at corners of mouth and lips), digestive disorders, burning skin
B ₆ (pyridoxine)	yeast, milk, egg yolk, cereals, gram	convulsions
B ₁₂	meat, fish, egg, curd	pernicious anaemia (red blood cells deficient in haemoglobin)
C (ascorbic acid)	citrus fruits, amla, green leafy vegetables	scurvy (bleeding gums)
D	sunlight, fish, egg yolk	rickets (bone deformity in children); osteomalacia (soft bones, joint pain in adults)
E	vegetable oils (wheat germ, sunflower)	increased fragility of red blood cells; muscular weakness
K	green leafy vegetables	increased blood-clotting time

Vitamin D is the one that the body can make itself, when skin is exposed to sunlight; every other vitamin has to come from the diet.

Trap: "why can vitamin C not be stored in the body" (2016) — **it is water-soluble and is excreted in urine**. And the 1-mark favourites: bleeding gums → C; pernicious anaemia → B₁₂; bone deformity in children → D; blood clotting → K.

Examiner asks: define vitamin and classify it by solubility; diseases for A, E, C, D, B₁₂; which vitamin the body can make.

10.5 Nucleic acids — what they are made of — **sugar + phosphate + a nitrogen base, then how those units chain together**

[exposure] Heredity is the passing of inherited characters from one generation to the next. It is carried by **chromosomes**, thread-like structures in the cell nucleus made of protein plus a second biomolecule: the **nucleic acids**. They come in two types, **DNA** (deoxyribonucleic acid) and **RNA** (ribonucleic acid). Because each is a long chain polymer of **nucleotides**, nucleic acids are also called **polynucleotides**.

Complete hydrolysis gives three things

1. A **pentose sugar** (five-carbon, in ring form): **β-D-2-deoxyribose** in DNA, **β-D-ribose** in RNA. They differ only at carbon 2: ribose has an –OH there; 2-deoxyribose has just –H (no oxygen).
2. **Phosphoric acid** — the same in both.
3. A **nitrogen base**.

[exposure] The **nitrogen bases** are nitrogen-containing ring compounds ("heterocyclic": a ring containing something other than carbon). Two families: **purines** (double ring): **adenine (A)** and **guanine (G)**; and **pyrimidines** (single ring): **cytosine (C)**, **thymine (T)**, **uracil (U)**. DNA has A, G, C and T; RNA has A, G, C and U. Thymine is uracil with an extra $-\text{CH}_3$. So thymine is found only in DNA, and uracil only in RNA.

Nucleoside and nucleotide

[exposure] A **nucleoside** is a **base joined to the 1' carbon of the sugar**. The carbons of the sugar are written 1', 2', 3', ... (read "one prime") so they cannot be mixed up with the carbons in the base. A **nucleotide** is a nucleoside with **phosphoric acid attached at the 5' position** of the sugar. So: **nucleoside = sugar + base; nucleotide = sugar + base + phosphate**.

[exposure] A **phosphodiester linkage** is how nucleotides chain up. The phosphate on the **5' carbon** of one nucleotide bonds to the $-\text{OH}$ on the **3' carbon** of the next, so the phosphate group is joined to two sugars (an ester at each end, hence "di-ester"). Repeat and you get a polynucleotide whose backbone alternates sugar–phosphate–sugar–phosphate, with the bases hanging off the sides. One end of the chain has a free phosphate on the 5' carbon (the **5' end**); the other has a free $-\text{OH}$ on a 3' carbon (the **3' end**).

Trap: "base + sugar" is the nucleoside; adding the phosphate makes the nucleotide. Questions on nucleoside versus nucleotide (2016, 2019) are lost on one missing word: **phosphate**.

Examiner asks: the unit formed by attaching a base to the 1' position of a sugar (2020 — nucleoside); the type of linkage in nucleic acids (2016 — phosphodiester); nucleoside versus nucleotide (2016, 2019); define nucleotide (2025).

Nucleic acids — structure and functions — primary, the DNA double helix, base pairing, three kinds of RNA, three jobs of DNA

The **primary structure** of a nucleic acid is the *sequence of nucleotides* along the chain — which bases come in which order.

DNA secondary structure — the double helix. Watson and Crick showed that DNA is two polynucleotide strands wound around each other in a helix, running in **opposite directions** (one $5' \rightarrow 3'$, the other $3' \rightarrow 5'$), and held together by **hydrogen bonds between pairs of bases**. The pairing is fixed:

- **Adenine pairs with thymine** — 2 hydrogen bonds ($\text{A}=\text{T}$).
- **Guanine pairs with cytosine** — 3 hydrogen bonds ($\text{G}\equiv\text{C}$).

That is why the two strands are **complementary**, not identical: wherever one has A, the other has T, and G faces C. It also gives the double helix its stability.

RNA secondary structure. RNA is normally a **single-stranded** helix; it sometimes folds back on itself. There are three types, each with a different job: **messenger RNA (m-RNA)** carries the message from DNA; **ribosomal RNA (r-RNA)** sits in ribosomes and helps in protein synthesis; **transfer RNA (t-RNA)** brings amino acids for protein synthesis.

	DNA	RNA
Strands	double helix	single strand
Sugar	deoxyribose	ribose
Bases	A, G, C, T	A, G, C, U
Main job	heredity — carries genetic information to the next generation	protein synthesis

Biological functions of nucleic acids

1. **Heredity** — DNA is the chemical basis of heredity and the store of genetic information.
2. **Replication** — during cell division the DNA makes a copy of itself, and identical DNA goes to both daughter cells.
3. **Protein synthesis** — the various RNAs make the proteins, but the *message* for each protein is in the DNA, passed on through m-RNA.

Examiner asks: "two strands in DNA are not identical but complementary" (2019); structural and functional differences between DNA and RNA (2013, 2023); one function of nucleic acids (2013).

10.6 Hormones — *chemical messengers — three chemical types and what each does*

[exposure] Hormones are molecules that act as **intercellular messengers**: they carry a signal from one group of cells or tissue to another. They are made by **endocrine (ductless) glands** and poured directly into the blood, which carries them to receptors at the site of action. They exist so the body can coordinate itself over distance. Compare with vitamins: hormones are **made in the body** and are messengers; vitamins (except D) have to come from food and are needed in small amounts for growth.

Chemical type	Examples	Role
Steroids (four-ring skeleton)	oestrogens, androgens	sexual development
Polypeptides (chains of amino acids)	insulin, endorphins	insulin lowers blood glucose; endorphins reduce pain
Amino-acid derivatives	epinephrine, norepinephrine	mediate the stress response

What they do

- **Glucose balance** — **insulin** lowers blood glucose (it turns glucose into glycogen); **glucagon** raises it (glycogen back to glucose). The two together hold the level steady.
- **Stress response** — epinephrine and norepinephrine.
- **Growth and development** — growth hormones (height, muscle mass); sex hormones (testosterone, oestradiol) for secondary sexual characters.
- **Thyroid** — the thyroid gland makes **thyroxine**, an iodinated derivative of tyrosine, which controls metabolism. Too little gives hypothyroidism; too much gives hyperthyroidism. A diet low in iodine causes enlargement of the thyroid (**goitre**); iodised salt prevents it.
- **Adrenal cortex** — **glucocorticoids** control carbohydrate metabolism and inflammation; **mineralocorticoids** control the kidney's salt and water balance. If the adrenal cortex fails the result is **Addison's disease**.
- **Progesterone** — prepares the uterus for implantation of the fertilised egg.

Examiner asks: how vitamins and hormones differ in source and function (2013 — hormones made by endocrine glands and released into blood, they transfer information between cells; vitamins come from the diet, in minute quantity, for growth and upkeep).

Patterns, collected — *the conversions, counts and calls you will actually be asked to make*

1. Reagent → product for glucose

Worked example

Glucose is heated with HI for a long time. Name the product and say what it proves.

Straight-chain six-carbon alkane: $\text{C}_6\text{H}_{12}\text{O}_6 \xrightarrow[\Delta]{\text{HI}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3 = n\text{-hexane}$. All six carbons lie in a straight chain, with no branch.

- Glucose + Br_2 water: product and conclusion — answer only: gluconic acid; the carbonyl is an aldehyde
- Glucose + acetic anhydride: product and conclusion — answer only: glucose pentaacetate; five $-\text{OH}$ groups
- Glucose + conc. HNO_3 : product and conclusion — answer only: saccharic acid (dicarboxylic acid); a primary $-\text{OH}$ is present

2. "Why does glucose not give ... but the open chain says it should?"

Worked example

Why does glucose not give Schiff's test even though it has an aldehyde group?

In solution the $-\text{OH}$ on C_5 adds to the C_1 aldehyde (intramolecular hemiacetal formation) and a six-membered ring forms. The $-\text{CHO}$ is no longer free, so no test that needs a free $-\text{CHO}$ can succeed. The same answer covers NaHSO_3 , 2,4-DNP, and hydroxylamine on the pentaacetate.

- Why does glucose pentaacetate not react with hydroxylamine? — answer only: the ring locks the $-\text{CHO}$; no free $-\text{CHO}$

3. Reading D or L from a drawing

Worked example

A Fischer drawing of a sugar has $-\text{OH}$ on the *right* at carbons 2, 4, 5 and on the *left* at carbon 3. D or L?

1. Put the most oxidised carbon ($-\text{CHO}$) on top — it already is.
 2. Find the lowest asymmetric carbon: C_5 (C_6 is CH_2OH , not asymmetric).
 3. Its $-\text{OH}$ is on the right $\rightarrow$ **D**. The other carbons are ignored. (This is D-glucose.)
- The same sugar with $-\text{OH}$ on the left at C_5 — answer only: L
 - Amino acid with $-\text{NH}_2$ on the left of the α -carbon — answer only: L-amino acid (the natural form)

4. Haworth drawing from the Fischer

Worked example

D-glucose, Fischer: $-\text{OH}$ right at C_2 , left at C_3 , right at C_4 , right at C_5 . Draw β -D-glucopyranose.

Right $\rightarrow$ down, left $\rightarrow$ up: C_2 down, C_3 up, C_4 down. $-\text{CH}_2\text{OH}$ on C_5 is up. For β the anomeric $-\text{OH}$ on C_1 is **up** (for α it would be down).

- Where is the C_1 $-\text{OH}$ in α -D-glucopyranose? — answer only: down

5. Sign flip on hydrolysing sucrose

Worked example

Show that hydrolysed sucrose is laevorotatory, using the rotations $+52.5^\circ$ (glucose) and -92.4° (fructose).

Hydrolysis gives glucose and fructose in *equal* molar amounts, so each contributes equally. Mixture rotation $\approx \frac{(+52.5) + (-92.4)}{2} = -19.95^\circ$ — negative. Sucrose started at $+66.5^\circ$, so the sign has *inverted*, hence "invert sugar". (The -19.95° is the average built from the two NCERT values; use it to justify the sign, not as a quoted constant.)

- Which monosaccharide, glucose or fructose, makes the mixture laevorotatory, and why? — answer only: fructose; its -92.4° outweighs glucose's $+52.5^\circ$

6. Is this disaccharide reducing?

Worked example

Maltose and sucrose: which reduces Fehling's?

Check whether either aldehyde/ketone carbon is free. Maltose: the second glucose's C1 is not in the bond (the bond is C1 of the first unit to C4 of the second), so it can open to a free aldehyde → reducing. Sucrose: the bond is C1 of glucose to C2 of fructose, which uses *both* reducing carbons → non-reducing.

- Lactose — reducing? — answer only: yes (C1 of the glucose unit is free)

7. Counting peptides

Worked example

How many peptide bonds and how many water molecules are lost when 5 amino acids join in a chain?

Each bond forms one H_2O and joining n units takes $n - 1$ bonds: $5 - 1 = 4$ peptide bonds and **4** water molecules (a pentapeptide).

- A tripeptide — how many peptide bonds? — answer only: 2
- Which structure level is lost on denaturation: 1° , 2° , 3° ? — answer only: 2° and 3° (1° stays)

8. Pairing the bases

Worked example

One DNA strand reads $5'-\text{A}-\text{T}-\text{G}-\text{C}-3'$. Write the complementary strand and count the hydrogen bonds.

$\text{A} \rightarrow \text{T}$, $\text{T} \rightarrow \text{A}$, $\text{G} \rightarrow \text{C}$, $\text{C} \rightarrow \text{G}$, giving $3'-\text{T}-\text{A}-\text{C}-\text{G}-5'$. Hydrogen bonds: A–T has 2, T–A has 2, G–C has 3, C–G has 3 → **10**.

- A strand $5'-\text{G}-\text{G}-\text{A}-3'$: complementary strand and hydrogen bonds — answer only: $3'-\text{C}-\text{C}-\text{T}-5'$; $3 + 3 + 2 = 8$

Past year questions · question types — ranked by how often they turn up

Two videos feed this section. [30+] is the "30+ PYQ" video for CBSE 2024-25 — Q1–Q32, questions from 2013 to 2023. [2025] is the short "2025 PYQ" video — Q1–Q6, from the 2025 board paper. Question numbers below are the video's own numbering (the year comes from the video too). Where the lecture video solved a past question but gave only the year, it is cited as "year · one-shot". The CBSE 2026-27 "25 PYQ" video for this chapter was not uploaded when these notes were built; once it is, this section can be refreshed from it. Counts are by question part, from the two videos' lists.

1 · Glucose: reagent → product, and what it proves — 1, 2 or 3 marks · about 10 parts

Recognise it: "glucose is treated with ...", "write reactions to show that glucose contains ...", or "why does glucose not give ... test".

1. Name the feature the reagent tests.
2. Write the one-line reaction: $\text{HI} \rightarrow n\text{-hexane}$; Br_2 water → gluconic acid; $(\text{CH}_3\text{CO})_2\text{O} \rightarrow$ pentaacetate; $\text{HNO}_3 \rightarrow$ saccharic acid; $\text{NH}_2\text{OH} \rightarrow$ oxime; $\text{HCN} \rightarrow$ cyanohydrin.
3. Finish with the conclusion sentence — "this shows ...". The conclusion is usually the mark.

4. If it says "why does glucose *not* give ...", the answer is always the ring: the –CHO is tied up in the hemiacetal, so it is not free.

Trap: HI needs *prolonged* heating to reach *n*-hexane. Br₂ water proves an **aldehyde** (it does not touch a ketone); oxime and cyanohydrin only prove a **carbonyl**. Mixing those two conclusions loses the mark.

2 · Differentiate / write the difference — 1 to 3 marks · about 9 parts

Recognise it: "differentiate between ...", "write one structural difference ...", "how are X and Y different in source and function".

1. Give each side one line, *on the same property*.
2. Use the pair you were given: starch/cellulose → α- versus β-glucose; amylose/amylopectin → unbranched versus branched, soluble versus insoluble; DNA/RNA → double versus single strand, deoxyribose versus ribose, T versus U; nucleoside/nucleotide → missing versus present phosphate; fibrous/globular → insoluble versus soluble; α-helix/β-sheet → intra- versus intermolecular hydrogen bonding.
3. For 3 marks give three contrasts (monomer, shape, solubility, linkage, function).

Trap: two separate definitions score less than one contrast. For α-helix versus β-sheet the answer is *where the hydrogen bonds are* — and you must say both.

3 · Define the term — 1 or 2 marks · about 8 parts

Recognise it: "define ...", "what is meant by ...". The terms that have appeared: amino acid, anomers, enzymes, oligosaccharides, invert sugar, denatured protein, native protein, nucleotide, essential amino acid.

1. One-line definition built on the key word.
2. Add the example — it is often a separate mark. Denatured protein → boiled egg white; essential amino acid → valine, leucine; invert sugar → hydrolysed sucrose; oligosaccharide → sucrose, lactose.

Trap: an "define with an example" question (2018 Q20) gives nothing for a bare definition. And "invert sugar" must say the *sign of rotation changes from (+) to (–)*.

4 · Name / identify — 1 mark · about 8 parts

Recognise it: "name the ...", "which of the following is a disaccharide", "write the product of hydrolysis of ...".

1. Answer in one line, no working.
2. Learn the short list: two glucose → **maltose**; sucrose → glucose + fructose; **lactose** → β-D-glucose + β-D-galactose; linkage between monosaccharides → **glycosidic**; linkage in nucleic acids → **phosphodiester**; the two parts of starch → **amylose, amylopectin**.

Trap: the "which is a disaccharide" list always offers starch (polysaccharide), glucose and fructose (monosaccharides), and a decoy like amylose or cellulose. Only maltose, sucrose or lactose is the answer.

5 · Give a reason — 2 or 3 marks · about 6 parts

Recognise it: "why are amino acids amphoteric", "why do amino acids behave like salts", "what changes in egg protein on boiling", "the two DNA strands are complementary, not identical", "water-soluble vitamins must be taken regularly".

1. State the structural fact (acidic and basic group in one molecule; hydrogen bonds break; A pairs with T and G with C; water-soluble so excreted in urine).
2. Say how the fact causes the behaviour.

Trap: vitamin B₁₂ is the one water-soluble vitamin that is stored (liver). Put "except B₁₂" in the answer when the question is about storage.

6 · Vitamin recall — 1 or 3 marks · about 5 parts

Recognise it: "the vitamin whose deficiency causes ...", "classify vitamins and name the one for blood clotting", "the only vitamin made in the body".

1. Classify by solubility: fat-soluble A, D, E, K; water-soluble B group and C.
2. Match the disease from the table in the lecture section: bleeding gums → C; bone deformity → D; night blindness → A; muscular weakness → E; clotting → K.

Trap: blood coagulation is vitamin **K**, and it is also the vitamin whose lack *increases* clotting time. Vitamin D is the one made in the body (sunlight).

Past year questions · mark slots — *what each type is worth*

The 30+ video groups its questions by mark value: Q1–Q18 are 1-mark, Q19–Q23 are 2-mark, Q24–Q32 are 3-mark. Where it names a different value for one question, that is shown. Biomolecules does not appear as a 5-mark long answer in either video.

Question type	1 mark	2 marks	3 marks
Glucose reagent / evidence	2014 Q2, 2018 Q4 [30+]	2023 Q31 [30+]; 2025 Q3, Q5 [2025]	2019 Q25, 2019 Q30 [30+]
Differentiate	2016 Q9, 2016 Q11, 2013 Q15 [30+]	2013 Q21, 2018 Q23(b) [30+]	2013 Q26(a), 2019 Q29 [30+]; 2025 Q6(a) [2025]
Define	2014 Q6, Q13, Q17 [30+]	Q19, 2018 Q20 [30+]; 2025 Q2 [2025]	2014 Q27 [30+]; 2025 Q4 [2025]
Name / identify	2020 Q3, 2016 Q5, 2019 Q8, 2013 Q12, 2015 Q14, 2020 Q16 [30+]	2025 Q1 [2025]	2025 Q6(b) [2025] (inside a 3-mark question)
Give a reason	—	2014 Q22, 2018 Q23(a) [30+]	2013 Q26(b), 2019 Q28, 2023 Q32 [30+]
Vitamin recall	2015 Q1, 2013 Q7, 2013 Q10 [30+]	—	2015 Q24 [30+]; 2025 Q6(c) [2025]

The video gives 2018 Q4 and the other items in the 1-mark block as 1-mark questions; Q15 (vitamins versus hormones) sits in that block too, though its answer is two paragraphs long (inferred: it would carry more marks in a real paper). The 2025 video states Q1 as a 2-mark question, but does not state a mark value for Q2 (inferred: 2).

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2019 Q25 · 2019 Q30 · 2025 Q3(b) · 2025 Q5(a) · 2021 · one-shot

Glucose with bromine water → gluconic acid, to prove an aldehyde. The single most repeated reaction in the chapter.

2019 Q25 · 2019 Q30 · 2025 Q5(b) · 2014 Q2 · 2021 · one-shot

Acetic anhydride → glucose pentaacetate (five –OH), and HI prolonged heating → *n*-hexane (straight chain).

2019 Q30 · 2025 Q3(a) · 2021 · one-shot

Glucose with HCN → cyanohydrin. The CN^- attacks the carbonyl carbon.

2014 Q22 · 2019 Q28(a) · 2023 Q32(a)

Why glucose does not behave like an aldehyde — no Schiff's test, no NaHSO_3 adduct, no 2,4-DNP, and the pentaacetate does not react with hydroxylamine. One answer: the ring has locked the –CHO.

2014 Q27 · 2018 Q20 · 2025 Q2 · 2025 Q4(c)

Essential amino acid — definition and examples (valine, leucine). Four appearances in the two videos.

2018 Q23(a) · 2025 Q2 · 2023 · one-shot

Why amino acids are amphoteric — an acidic –COOH and a basic –NH₂ in the same molecule.

2018 Q20 · 2025 Q4(a) · 2025 Q6(a) · 2013 Q26(b)

Native versus denatured protein; boiling an egg. Hydrogen bonds break, biological activity is lost, the primary structure survives.

2019 Q8 · 2025 Q1 · 2019 · one-shot

Hydrolysis products of lactose (β-D-glucose + β-D-galactose) and of sucrose (D-glucose + D-fructose).

2015 Q14 · 2020 Q3 · 2025 Q6(b) · 2015 · one-shot

Pick the disaccharide from a list (maltose, lactose; never starch, glucose, fructose, amylose).

2015 Q24 · 2025 Q6(c)

The vitamin for blood coagulation — K.

2016 Q11 · 2025 Q4(b) · 2019 · one-shot · 2020 · one-shot

Nucleoside versus nucleotide; the unit formed when a base attaches to the 1' position of a sugar. The phosphate is the whole difference.

2020 Q16 · 2019 Q29(b) · 2018 · one-shot · 2019 · one-shot

Glycosidic linkage, and glycosidic versus peptide linkage.

2016 Q9 · 2013 Q12 · 2019 Q29(a) · 2023 · one-shot

Starch versus cellulose, and amylose versus amylopectin.

2013 Q21 · 2023 · one-shot

DNA versus RNA (strands, sugar; and function — heredity versus protein synthesis).

2018 Q23(b) · 2023 · one-shot

α-helix versus β-pleated sheet.

2023 Q32(c) · 2016 · one-shot

Why water-soluble vitamins (and vitamin C) must be taken regularly — excreted in urine, not stored (B₁₂ excepted).

Past year questions · numerical types — no calculations in this chapter, so these are the reaction and structure types — one worked model each, then cold practice

A · Glucose with a reagent

Worked example · 2019 Q30 [30+] — glucose with HCN, bromine water and acetic anhydride

HCN: CN^- is a nucleophile; it attacks the carbonyl carbon and the oxygen takes H^+ :

$\text{HOCH}_2(\text{CHOH})_4\text{CHO} + \text{HCN} \longrightarrow \text{HOCH}_2(\text{CHOH})_4\text{CH}(\text{OH})\text{CN}$. Product: **glucose cyanohydrin**. It confirms a carbonyl group.

Br_2 **water**: mild oxidant, takes the aldehyde to an acid:

$\text{HOCH}_2(\text{CHOH})_4\text{CHO} \xrightarrow{\text{Br}_2 / \text{H}_2\text{O}} \text{HOCH}_2(\text{CHOH})_4\text{COOH}$. Product: **gluconic acid**. It confirms an aldehyde.

Acetic anhydride: acetylates every $-\text{OH}$: product **glucose pentaacetate**. Five acetyl groups attach, so there are five $-\text{OH}$ groups.

- 2014 Q2 [30+] — product of glucose with HI — answer only: *n*-hexane on prolonged heating; six carbons in a straight chain
- 2018 Q4 [30+] — D-glucose with concentrated HNO_3 — answer only: saccharic acid (D-saccharic acid); aldehyde and primary $-\text{OH}$ both oxidised to $-\text{COOH}$
- 2023 Q31 [30+] — glucose heated with hydroxylamine, and the group confirmed — answer only: glucose oxime, $\text{CH}=\text{NOH}$; a carbonyl group
- 2025 Q5 [2025] — how to show the aldehyde group and the five $-\text{OH}$ groups — answer only: bromine water gives gluconic acid; acetic anhydride gives the pentaacetate

B · Why glucose does not behave like an aldehyde

Worked example · 2019 Q28(a) [30+] — why glucose does not give the 2,4-DNP test

The 2,4-DNP test needs a free $-\text{CHO}$ or $>\text{C}=\text{O}$. In solution the C_5 $-\text{OH}$ adds to the C_1 aldehyde and a six-membered hemiacetal ring forms. The aldehyde is no longer free, so there is no reaction. (Same answer for Schiff's test and for NaHSO_3 .)

- 2023 Q32(a) [30+] — why the pentaacetate of glucose does not react with hydroxylamine — answer only: the $-\text{CHO}$ is part of the ring, not free
- 2014 Q22 [30+] — reactions the open chain cannot explain — answer only: no NaHSO_3 adduct; no Schiff's or 2,4-DNP test; pentaacetate does not react with NH_2OH ; two crystalline forms (α , β)

C · Hydrolysis products

Worked example · 2025 Q1 [2025] — hydrolysis products of sucrose and of lactose

Both are disaccharides, so each gives two monosaccharides in equal numbers. Sucrose gives **D-(+)-glucose and D-(-)-fructose** (invert sugar). Lactose gives **β -D-galactose and β -D-glucose**.

- 2019 Q8 [30+] — hydrolysis product of lactose — answer only: β -D-glucose and β -D-galactose
- 2020 Q3 [30+] — disaccharide that gives two glucose — answer only: maltose
- 2025 Q6(b) [2025] — which of glucose, lactose, amylose, fructose is a disaccharide — answer only: lactose

D · Differentiate a pair

Worked example · 2013 Q21 [30+] — structural and functional difference between RNA and DNA

Structure: DNA is a double helix (two strands); RNA is a single strand. DNA's sugar is deoxyribose; RNA's is ribose. (Bases: DNA has thymine, RNA has uracil.) **Function:** DNA carries genetic information from one generation to the next (heredity); RNA helps in protein synthesis.

- 2016 Q9 [30+] — structural difference between starch and cellulose — answer only: starch is α -D-glucose (linear and branched), cellulose is β -D-glucose (linear)
- 2018 Q23(b) [30+] — α -helix versus β -pleated sheet — answer only: intramolecular versus intermolecular hydrogen bonding
- 2019 Q29(a) [30+] — amylose versus amylopectin — answer only: unbranched and water-soluble (15–20%) versus branched and insoluble (80–85%)
- 2025 Q6(a) [2025] — native versus denatured protein — answer only: native keeps its 3-D structure and activity; denatured has lost its hydrogen bonding and activity after a physical or chemical change

E · Vitamin matching

Worked example · 2013 Q10 [30+] — deficiency diseases of vitamins A and E

Vitamin A → **night blindness** (and xerophthalmia). Vitamin E → **muscular weakness** (and fragile red blood cells).

- 2015 Q1 [30+] — vitamin whose lack causes bleeding gums — answer only: vitamin C
- 2013 Q7 [30+] — the vitamin the body can make, and its deficiency diseases — answer only: vitamin D; rickets in children, osteomalacia in adults
- 2015 Q24 [30+] / 2025 Q6(c) [2025] — vitamin for blood coagulation — answer only: vitamin K; vitamins are classified into fat-soluble (A, D, E, K) and water-soluble (B group, C)

Every Chemistry Formula

Class XII CBSE • Chemistry • Chapters 1–6

Every formula the half-yearly can ask for, each with what its symbols mean, the unit it comes out in, and the one situation that tells you to reach for it. The cue is on the outside and the formula is hidden — so read down the page and say each one before you open it.

- Formulas: 50
- Recall rules: 18
- Must be instant: 40
- Paper: 70 marks

How to use this

● means you should be able to say it before the question finishes loading. ○ means a few seconds of thought is fine — these are the ones you derive or reconstruct rather than recall.

Work down a chapter with everything closed. Say the formula out loud, *then* tap to reveal. A formula you can only recognise is not a formula you have.

Units are not decoration. Most lost marks in Solutions and Electrochemistry are unit slips — molality is per **kilogram of solvent**, molarity per **litre of solution**, and κ in S cm^{-1} with c in mol L^{-1} is what makes the 1000 appear in molar conductivity.

Recognise index

Every cue on one screen. Read the cue, say the formula, tap to check.

1 • Solutions — 15 marks

- c1.1 Concentration by mass, volume, or parts per million
- c1.2 Mole fraction of a component
- c1.3 Molarity, from mass or from moles
- c1.4 Molality, and why it is the one used for colligative properties
- c1.5 A gas dissolving in a liquid under pressure
- c1.6 Vapour pressure of a mixture of two volatile liquids
- c1.7 Vapour pressure when the solute is non-volatile
- c1.8 Relative lowering of vapour pressure
- c1.9 Boiling point raised by a dissolved solute
- c1.10 Freezing point lowered by a dissolved solute
- c1.11 Osmotic pressure across a semipermeable membrane
- c1.12 Molar mass from any one colligative property
- c1.13 The measured effect is bigger or smaller than predicted
- c1.14 Degree of dissociation or association from i

2 • Electrochemistry — 14 marks

- c2.1 Standard cell potential from two electrode potentials

- c2.2 Cell potential at non-standard concentrations
- c2.3 Nernst for a single electrode
- c2.4 Free energy change of a cell reaction
- c2.5 Equilibrium constant from cell potential
- c2.6 Conductance, resistance and the cell constant
- c2.7 Conductivity from measured resistance
- c2.8 Molar conductivity from conductivity
- c2.9 Limiting molar conductivity of an electrolyte from its ions
- c2.10 Degree of dissociation of a weak electrolyte
- c2.11 Dissociation constant of a weak acid from conductivity
- c2.12 Mass deposited during electrolysis
- c2.13 Moles of electrons passed

3 · Chemical Kinetics — 13 marks

- c3.1 Rate of a reaction from a concentration change
- c3.2 Rate written for a balanced equation with coefficients
- c3.3 Rate law and the order of a reaction
- c3.4 Units of the rate constant for any order
- c3.5 Zero order — concentration against time
- c3.6 Zero order half-life
- c3.7 First order rate constant from concentrations
- c3.8 First order half-life
- c3.9 First order reaction in the gas phase, measured by total pressure
- c3.10 Rate constant against temperature
- c3.11 Activation energy from a straight-line plot
- c3.12 Activation energy from two temperatures

4 · d and f Block — 11 marks

- c4.1 Magnetic moment from unpaired electrons, and back again
- c4.2 General electronic configuration of a block

5 · Coordination Compounds — 11 marks

- c5.1 Oxidation number of the metal inside a complex
- c5.2 Coordination number when a ligand grips more than once
- c5.3 Tetrahedral splitting compared with octahedral
- c5.4 Crystal field stabilisation energy of an octahedral complex
- c5.5 High spin or low spin — deciding which
- c5.6 Number of geometrical isomers of a complex

6 · Haloalkanes and Haloarenes — 6 marks

- c6.1 Rate law that identifies the substitution mechanism
- c6.2 Degree of unsaturation from a molecular formula

Constants — worth the same marks as the formulas

- κ Every constant this paper can hand you, with its unit

CH 1 Solutions — 14 formulas · 15 marks

• c1.1 Concentration by mass, by volume, or in parts per million

$$\% w/w = \frac{w_B}{w_A + w_B} \times 100 \quad \% v/v = \frac{V_B}{V_{\text{soln}}} \times 100$$

$$\% w/v = \frac{w_B}{V_{\text{soln in mL}}} \times 100 \quad \text{ppm} = \frac{w_B}{w_{\text{soln}}} \times 10^6$$

Symbol	Meaning	Unit
w_B	mass of solute	g
w_A	mass of solvent	g
V_{soln}	volume of solution	mL or L
all four	the concentration itself	dimensionless

Use it when:

The question gives masses or volumes and no moles anywhere. ppm is for trace quantities — pollutants, dissolved gases, hardness of water.

Trap:

$\% w/w$ divides by the mass of the **solution**, not the solvent. Adding 10 g of solute to 90 g of water gives 10%, not 11.1%.

• c1.2 Mole fraction of a component

$$x_A = \frac{n_A}{n_A + n_B} \quad x_A + x_B = 1$$

Symbol	Meaning	Unit
x_A	mole fraction of the solvent	dimensionless
n_A	moles of solvent, = w_A/M_A	mol
n_B	moles of solute	mol

Use it when:

Raoult's law or Henry's law is involved — both are written in mole fractions, never in molarity.

Trap:

For water as solvent, $M_A = 18$, so n_A is large and x_B is tiny. Do not round x_A to 1 before subtracting — you will get $x_B = 0$.

• **c1.3 Molarity, from moles or straight from mass**

$$M = \frac{n_B}{V_{\text{soln in L}}} = \frac{w_B \times 1000}{M_B \times V_{\text{soln in mL}}}$$

Symbol	Meaning	Unit
M	molarity	mol L^{-1}
n_B	moles of solute	mol
M_B	molar mass of solute	g mol^{-1}
V	volume of solution	L (or mL)

Use it when:

Volume of solution is given or asked. Also the concentration used in osmotic pressure, $\pi = CRT$.

Trap:

Molarity **changes with temperature** because volume does. Molality does not. If a question mentions heating or cooling and then asks which concentration term is unaffected, the answer is molality.

• **c1.4 Molality — and the unit that makes it the colligative one**

$$m = \frac{n_B}{w_A \text{ in kg}} = \frac{w_B \times 1000}{M_B \times w_A \text{ in g}}$$

Symbol	Meaning	Unit
m	molality	mol kg^{-1}
w_A	mass of solvent only	kg (or g)
M_B	molar mass of solute	g mol^{-1}

Use it when:

Anything colligative: ΔT_b , ΔT_f . Both are written in molality precisely because it is temperature-independent.

Trap:

Per kg of **solvent**, not of solution. The 1000 in the second form converts grams of solvent to kilograms — it is not the same 1000 as in molarity, where it converts mL of solution to L.

• **c1.5 A gas dissolving in a liquid under pressure**

$$p = K_H x$$

Symbol	Meaning	Unit
p	partial pressure of the gas above the solution	bar (or kbar)
x	mole fraction of the gas in solution	dimensionless
K_H	Henry's law constant	bar (or kbar)

Use it when:

A gas is being dissolved — soda bottles, deep-sea diving and the bends, oxygen at altitude, aquatic life in warm water.

Trap:

K_H goes **up** with temperature, so solubility goes **down** — which is why warm water holds less dissolved oxygen. A higher K_H means a *less* soluble gas.

• c1.6 Vapour pressure of a mixture of two volatile liquids

$$p_{\text{total}} = p_A^\circ x_A + p_B^\circ x_B = p_B^\circ + (p_A^\circ - p_B^\circ) x_A$$

Symbol	Meaning	Unit
p_A°	vapour pressure of pure A	bar or mm Hg
x_A	mole fraction of A in the liquid	dimensionless
y_A	mole fraction of A in the vapour , $= p_A/p_{\text{total}}$	dimensionless

Use it when:

Both components evaporate. The second form is the equation of a straight line in x_A — that is the plot they ask you to sketch.

Trap:

Vapour composition y_A is not the same as liquid composition x_A . The vapour is always richer in the more volatile component.

• c1.7 Vapour pressure when the solute is non-volatile

$$p_{\text{soln}} = x_A p_A^\circ$$

Symbol	Meaning	Unit
p_{soln}	vapour pressure of the solution	bar or mm Hg
x_A	mole fraction of the solvent	dimensionless
p_A°	vapour pressure of pure solvent	bar or mm Hg

Use it when:

A solid, non-volatile solute (sugar, urea, a salt) is dissolved. Only the solvent contributes vapour.

Trap:

It is x of the **solvent**, not the solute. Using x_B here is the single commonest slip in this chapter.

• **c1.8 Relative lowering of vapour pressure**

$$\frac{p_A^\circ - p_{\text{soln}}}{p_A^\circ} = x_B = \frac{n_B}{n_A + n_B} \approx \frac{w_B M_A}{M_B w_A}$$

Symbol	Meaning	Unit
x_B	mole fraction of solute	dimensionless
M_A	molar mass of solvent	g mol^{-1}
M_B	molar mass of solute — usually what is asked	g mol^{-1}

Use it when:

Two vapour pressures are given (pure solvent and solution) and a molar mass is wanted.

Trap:

The final approximation assumes $n_B \ll n_A$ — valid for a dilute solution only. If the question says concentrated, use the exact form.

• **c1.9 Boiling point raised by a dissolved solute**

$$\Delta T_b = K_b m \quad \Delta T_b = T_b - T_b^\circ$$

Symbol	Meaning	Unit
ΔT_b	elevation of boiling point	K
K_b	molal elevation constant (ebullioscopic constant)	K kg mol^{-1}
m	molality	mol kg^{-1}

Use it when:

A boiling point shifts. For water $K_b = 0.52 \text{ K kg mol}^{-1}$.

Trap:

K_b is a property of the **solvent** alone — it never depends on which solute you dissolved.

• **c1.10 Freezing point lowered by a dissolved solute**

$$\Delta T_f = K_f m \quad \Delta T_f = T_f^\circ - T_f$$

Symbol	Meaning	Unit
ΔT_f	depression of freezing point	K
K_f	molal depression constant (cryoscopic constant)	K kg mol ⁻¹
m	molality	mol kg ⁻¹

Use it when:

A freezing point shifts — antifreeze in a radiator, salt on an icy road. For water $K_f = 1.86 \text{ K kg mol}^{-1}$.

Trap:

Note the order of subtraction is reversed from ΔT_b : freezing point goes **down**, so it is pure minus solution. Both ΔT values come out positive.

• **c1.11 Osmotic pressure across a semipermeable membrane**

$$\pi = CRT = \frac{n_B}{V}RT \quad \pi V = n_B RT$$

Symbol	Meaning	Unit
π	osmotic pressure	bar or atm
C	molarity	mol L ⁻¹
R	gas constant	0.0821 L atm K ⁻¹ mol ⁻¹
T	absolute temperature	K

Use it when:

A membrane, a molar mass of a **protein or polymer**, or the word isotonic/hypertonic appears. This is the method of choice for large molar masses because π is measurably large even when ΔT_f would be tiny.

Trap:

Uses molarity, not molality — the only colligative property that does. Match the R value to the pressure unit you want.

• **c1.12 Molar mass from any one colligative property**

$$M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A} \quad M_B = \frac{K_f w_B \times 1000}{\Delta T_f w_A} \quad M_B = \frac{w_B RT}{\pi V}$$

Symbol	Meaning	Unit
M_B	molar mass of the solute	g mol ⁻¹
w_B	mass of solute	g
w_A	mass of solvent	g
V	volume of solution	L

Use it when:

"Calculate the molar mass of the solute" — the standard 3-marker. All three are just C1.9–C1.11 rearranged; do not memorise them separately.

Trap:

The 1000 is there because w_A is in grams while molality needs kilograms. Drop it and your answer is out by exactly a factor of 1000.

• **c1.13 When the measured effect is bigger or smaller than predicted**

$$i = \frac{\text{observed colligative property}}{\text{calculated colligative property}} = \frac{\text{normal molar mass}}{\text{abnormal molar mass}}$$

$$\Delta T_b = iK_b m \quad \Delta T_f = iK_f m \quad \pi = iCRT$$

Symbol	Meaning	Unit
i	van't Hoff factor	dimensionless
$i > 1$	solute dissociates — e.g. NaCl, $i \rightarrow 2$	dimensionless
$i < 1$	solute associates — e.g. benzoic acid in benzene, $i \rightarrow 0.5$	dimensionless

Use it when:

The solute is ionic, or the question uses the word "abnormal", or the observed ΔT_f does not match the calculated one.

Trap:

Molar mass and i move **opposite** ways. Dissociation ($i > 1$) makes the measured molar mass come out *lower* than the true value.

○ **c1.14 Degree of dissociation or association from i**

$$\alpha_{\text{dissociation}} = \frac{i - 1}{n - 1} \quad \alpha_{\text{association}} = \frac{1 - i}{1 - \frac{1}{n}}$$

Symbol	Meaning	Unit
α	degree of dissociation or association	dimensionless
n	number of particles one formula unit gives (or combines into)	dimensionless
i	van't Hoff factor	dimensionless

Use it when:

i has been found and the percentage dissociation is asked. For K_2SO_4 , $n = 3$; for a dimerising acid, $n = 2$.

Trap:

Read n off the actual dissociation equation, not the number of atoms. $\text{K}_2\text{SO}_4 \rightarrow 2\text{K}^+ + \text{SO}_4^{2-}$ gives three particles, so $n = 3$.

CH 2 Electrochemistry — 13 formulas · 14 marks

• c2.1 Standard cell potential from two electrode potentials

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} = E_{\text{right}}^{\circ} - E_{\text{left}}^{\circ}$$

Symbol	Meaning	Unit
E_{cell}°	standard cell potential	V
E°	standard reduction potential of each electrode	V

Use it when:

Two half-cells and their E° values are given. A positive E_{cell}° means the reaction is spontaneous as written.
Daniell cell: 1.1 V.

Trap:

Both values in the table are **reduction** potentials — subtract, never add, and never flip the sign of the anode value and then also subtract.

• c2.2 Cell potential when concentrations are not 1 M

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303 RT}{nF} \log Q \xrightarrow{298 \text{ K}} E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$$

Symbol	Meaning	Unit
n	moles of electrons in the balanced cell reaction	dimensionless
Q	reaction quotient, products over reactants	dimensionless
F	Faraday constant	96487 C mol ⁻¹
R	gas constant	8.314 J K ⁻¹ mol ⁻¹

Use it when:

Any concentration is given that is not 1 M, or the question says "calculate the emf of the cell".

Trap:

NCERT uses **0.059**, not 0.0591. Pure solids and liquids never appear in Q . Get n from the balanced equation, not from the charge on one ion.

○ c2.3 Nernst equation for a single electrode

$$E_{(\text{M}^{n+}/\text{M})} = E_{(\text{M}^{n+}/\text{M})}^{\circ} - \frac{0.059}{n} \log \frac{1}{[\text{M}^{n+}]}$$

Symbol	Meaning	Unit
$[M^{n+}]$	concentration of the metal ion	mol L^{-1}
n	charge on the metal ion	dimensionless
E	electrode potential	V

Use it when:

Only one half-cell is being asked about, or a concentration cell is given where both electrodes are the same metal at different concentrations.

Trap:

The solid metal is taken as unit activity, so it is the 1 in the numerator. For a concentration cell $E_{\text{cell}}^{\circ} = 0$ and the whole emf comes from the log term.

• **c2.4 Free energy change of a cell reaction**

$$\Delta_r G = -nFE_{\text{cell}} \quad \Delta_r G^{\circ} = -nFE_{\text{cell}}^{\circ}$$

Symbol	Meaning	Unit
$\Delta_r G$	Gibbs energy change of the cell reaction	J mol^{-1}
n	moles of electrons transferred	dimensionless
F	Faraday constant	96487 C mol^{-1}

Use it when:

The question asks for ΔG , for maximum work obtainable, or whether a reaction is feasible.

Trap:

Answer comes out in joules, not kilojoules — divide by 1000 if the options are in kJ. The minus sign is part of the formula: a positive emf gives a negative ΔG .

○ **c2.5 Equilibrium constant from cell potential**

$$\Delta_r G^{\circ} = -RT \ln K_c \quad \Rightarrow \quad E_{\text{cell}}^{\circ} = \frac{0.059}{n} \log K_c$$

Symbol	Meaning	Unit
K_c	equilibrium constant of the cell reaction	dimensionless
E_{cell}°	standard cell potential	V
n	moles of electrons	dimensionless

Use it when:

K_c is asked from E° , or vice versa. It is just the Nernst equation with $E_{\text{cell}} = 0$ and $Q = K_c$ — at equilibrium the cell is dead.

Trap:

Note this is +, not -: setting $E_{\text{cell}} = 0$ moves the log term across the equals sign.

- **c2.6 Conductance, resistance and the cell constant**

$$G = \frac{1}{R} \quad G^* = \frac{l}{A} \quad R = \rho \frac{l}{A}$$

Symbol	Meaning	Unit
G	conductance	S (siemens)
R	resistance	Ω
G^*	cell constant	cm^{-1} or m^{-1}
ρ	resistivity	$\Omega \text{ cm}$

Use it when:

A conductivity cell is described by its electrode area and separation, or the cell constant is given directly.

Trap:

$1 \text{ S} = 1 \Omega^{-1}$, also written mho. The cell constant has units of **inverse** length because it is length $\div$ area.

- **c2.7 Conductivity from a measured resistance**

$$\kappa = \frac{1}{\rho} = G \times G^* = \frac{1}{R} \times \frac{l}{A}$$

Symbol	Meaning	Unit
κ	conductivity (specific conductance)	S cm^{-1} or S m^{-1}
G^*	cell constant	cm^{-1}
R	measured resistance	Ω

Use it when:

The question gives a resistance from a conductivity cell. κ is the conductance of a 1 cm cube of solution.

Trap:

κ **decreases** on dilution (fewer ions per unit volume) while molar conductivity **increases**. Being asked to explain that opposite behaviour is a standing 2-marker.

- **c2.8 Molar conductivity from conductivity**

$$\Lambda_m = \frac{\kappa}{c} = \frac{\kappa \times 1000}{c \text{ in mol L}^{-1}}$$

Symbol	Meaning	Unit
Λ_m	molar conductivity	$\text{S cm}^2 \text{ mol}^{-1}$
κ	conductivity	S cm^{-1}
c	concentration	mol L^{-1}

Use it when:

Converting a measured κ into the quantity you actually compare between electrolytes.

Trap:

The 1000 exists only because κ is per **cm³** while c is per **litre**. In SI (κ in S m^{-1} , c in mol m^{-3}) there is no 1000 and the unit is $\text{S m}^2 \text{ mol}^{-1}$.

• c2.9 Limiting molar conductivity built from its ions

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$$

Symbol	Meaning	Unit
Λ_m°	limiting molar conductivity (infinite dilution)	$\text{S cm}^2 \text{ mol}^{-1}$
$\lambda_\pm^\circ$	limiting molar conductivity of each ion	$\text{S cm}^2 \text{ mol}^{-1}$
$\nu_\pm$	how many of that ion one formula unit gives	dimensionless

Use it when:

Λ_m° of a **weak** electrolyte is wanted — it cannot be measured by extrapolation, so it is assembled from strong electrolytes. This is **Kohlrausch's law of independent migration of ions**.

Trap:

Multiply by ν . For CaCl_2 , $\Lambda_m^\circ = \lambda_{\text{Ca}^{2+}}^\circ + 2\lambda_{\text{Cl}^-}^\circ$ — the 2 is easy to drop.

• c2.10 Degree of dissociation of a weak electrolyte

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

Symbol	Meaning	Unit
α	degree of dissociation	dimensionless
Λ_m	molar conductivity at concentration c	$\text{S cm}^2 \text{ mol}^{-1}$
Λ_m°	limiting molar conductivity	$\text{S cm}^2 \text{ mol}^{-1}$

Use it when:

A weak acid or base is given with both conductivity values.

Trap:

Only meaningful for a **weak** electrolyte. A strong electrolyte is fully dissociated; its Λ_m rises with dilution because ion–ion interference falls, not because more of it dissociates.

○ **c2.11 Dissociation constant of a weak acid from conductivity**

$$K_a = \frac{c\alpha^2}{1-\alpha} = \frac{c\Lambda_m^2}{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)}$$

Symbol	Meaning	Unit
K_a	dissociation constant	mol L^{-1}
c	concentration of the acid	mol L^{-1}
α	degree of dissociation	dimensionless

Use it when:

Both Λ_m and Λ_m° are given for a weak acid and K_a or $\text{p}K_a$ is asked.

Trap:

Do not approximate $1 - \alpha \approx 1$ unless α is genuinely small (below about 0.05).

● **c2.12 Mass deposited or liberated during electrolysis**

$$Q = It \quad w = ZIt \quad Z = \frac{M}{nF}$$

Symbol	Meaning	Unit
Q	charge passed	C
I	current	A
t	time	s
w	mass deposited	g
Z	electrochemical equivalent	g C^{-1}

Use it when:

A current runs for a time and a mass is asked. Faraday's first law of electrolysis.

Trap:

Time must be in **seconds**. A question quoting minutes or hours is testing exactly this.

● **c2.13 Moles of electrons passed, and how much substance that gives**

$$n_{e^-} = \frac{Q}{F} = \frac{It}{96487} \quad \text{moles of product} = \frac{n_{e^-}}{n}$$

Symbol	Meaning	Unit
n_{e^-}	moles of electrons	mol
F	charge on one mole of electrons	96487 C mol^{-1}
n	electrons needed per ion, from the half-equation	dimensionless

Use it when:

Several cells are in series, or a gas volume at STP is asked. Faraday's second law: the same charge deposits amounts in the ratio of their equivalent masses.

Trap:

Get n from the half-reaction. $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$ needs 2 F per mole; Al^{3+} needs 3 F.

CH 3 Chemical Kinetics — 12 formulas · 13 marks

• c3.1 Rate of a reaction from a concentration change

$$r_{\text{av}} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t} \quad r_{\text{inst}} = -\frac{d[R]}{dt}$$

Symbol	Meaning	Unit
$[R]$	concentration of a reactant	mol L^{-1}
t	time	s
r	rate of reaction	$\text{mol L}^{-1} \text{ s}^{-1}$

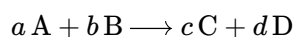
Use it when:

A table of concentration against time is given. Instantaneous rate is the slope of the tangent to that curve.

Trap:

The minus sign exists so the rate comes out positive — reactant concentration is falling. Never report a negative rate.

• c3.2 Rate written for an equation that has coefficients



$$r = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = +\frac{1}{c} \frac{d[C]}{dt} = +\frac{1}{d} \frac{d[D]}{dt}$$

Symbol	Meaning	Unit
a, b, c, d	stoichiometric coefficients	dimensionless
r	the one unique rate of reaction	$\text{mol L}^{-1} \text{ s}^{-1}$

Use it when:

The question gives the rate of disappearance of one species and asks for another. $2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$ makes NO_2 four times as fast as it makes O_2 .

Trap:

You **divide** by the coefficient. Multiplying is the reflex error.

• **c3.3 Rate law, and the order of a reaction**

$$r = k [A]^x [B]^y \quad \text{order} = x + y$$

Symbol	Meaning	Unit
k	rate constant	depends on order — see C3.4
x, y	order with respect to A and B	dimensionless
order	overall order	dimensionless

Use it when:

A table of initial rates against initial concentrations is given. Double one concentration, see what the rate does: unchanged $\rightarrow$ order 0, doubles $\rightarrow$ 1, quadruples $\rightarrow$ 2.

Trap:

Order is found by **experiment** and can be zero or fractional. Molecularity comes from the mechanism, is a whole number, and is never zero. A **pseudo first order reaction** is second order on paper but first order in practice because one reactant is in huge excess.

• **c3.4 Units of the rate constant for any order**

$$[k] = \text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$$

Symbol	Meaning	Unit
$n = 0$	zero order	$\text{mol L}^{-1} \text{s}^{-1}$
$n = 1$	first order	s^{-1}
$n = 2$	second order	$\text{L mol}^{-1} \text{s}^{-1}$
n	overall order	dimensionless

Use it when:

The order is asked and only the units of k are given — this runs backwards perfectly, and it is a favourite one-marker.

Trap:

First order k has no concentration unit at all, which is why a first-order half-life does not depend on concentration.

• **c3.5 Zero order — concentration against time**

$$[R] = -kt + [R]_0 \quad k = \frac{[R]_0 - [R]}{t}$$

Symbol	Meaning	Unit
$[R]_0$	initial concentration	mol L^{-1}
$[R]$	concentration at time t	mol L^{-1}
k	zero order rate constant	$\text{mol L}^{-1} \text{s}^{-1}$

Use it when:

The plot of $[R]$ against t is a straight line of slope $-k$. Real examples: decomposition of NH_3 on hot platinum, of HI on gold.

Trap:

Zero order means the rate is independent of concentration — usually because a catalyst surface is saturated, not because nothing is happening.

• **c3.6 Zero order half-life**

$$t_{1/2} = \frac{[R]_0}{2k}$$

Symbol	Meaning	Unit
$t_{1/2}$	half-life	s
$[R]_0$	initial concentration	mol L^{-1}
k	rate constant	$\text{mol L}^{-1} \text{s}^{-1}$

Use it when:

A zero-order reaction's half-life is asked, or the question tests that $t_{1/2} \propto [R]_0$ here.

Trap:

This one **does** depend on starting concentration — the exact opposite of first order. Getting the two the wrong way round is the standing trap of this chapter.

• **c3.7 First order rate constant from concentrations**

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]} \quad \log[R] = -\frac{k}{2.303}t + \log[R]_0$$

Symbol	Meaning	Unit
k	first order rate constant	s^{-1}
$[R]_0$	initial concentration	mol L^{-1}
$[R]$	concentration at time t	mol L^{-1}

Use it when:

The commonest numerical in the chapter. The second form says a plot of $\log[R]$ against t is a straight line of slope $-k/2.303$.

Trap:

2.303 converts $\ln$ to $\log_{10}$. If you use $\ln$, drop the 2.303 — using both is a factor-of-2.3 error.

• c3.8 First order half-life

$$t_{1/2} = \frac{0.693}{k}$$

Symbol	Meaning	Unit
$t_{1/2}$	half-life	s
k	first order rate constant	s^{-1}
0.693	$\ln 2$	dimensionless

Use it when:

Any half-life question on a first-order reaction, including all radioactive decay.

Trap:

Independent of starting concentration. Being asked to prove that is a standing 2-marker — see the derivations page.

○ c3.9 First order gas-phase reaction followed by total pressure

$$k = \frac{2.303}{t} \log \frac{p_i}{2p_i - p_t}$$

Symbol	Meaning	Unit
p_i	initial pressure of the reactant	bar or atm
p_t	total pressure at time t	bar or atm
k	rate constant	s^{-1}

Use it when:

A gas decomposes into two gas molecules and the data is a table of total pressure against time — e.g. azoisopropane, or N_2O_5 decomposition.

Trap:

The $2p_i - p_t$ form holds only when one mole of gas gives two moles of gas. For a different stoichiometry, rebuild the pressure table from scratch.

- **c3.10 Rate constant against temperature**

$$k = A e^{-E_a/RT}$$

Symbol	Meaning	Unit
A	frequency (pre-exponential) factor	same as k
E_a	activation energy	J mol^{-1}
R	gas constant	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
T	absolute temperature	K

Use it when:

Temperature enters the question at all. The Arrhenius equation.

Trap:

$e^{-E_a/RT}$ is the **fraction of molecules with energy at least E_a** — quoting that meaning is often the mark.

- **c3.11 Activation energy from a straight-line plot**

$$\log k = \log A - \frac{E_a}{2.303 RT}$$

Symbol	Meaning	Unit
slope	$-E_a/2.303R$	K
intercept	$\log A$	dimensionless
x -axis	$1/T$	K^{-1}

Use it when:

A graph of $\log k$ against $1/T$ is given, or asked to be sketched.

Trap:

The slope is **negative** and E_a is positive — remember to take the magnitude when you convert slope to E_a .

- **c3.12 Activation energy from two temperatures**

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$$

Symbol	Meaning	Unit
k_1, k_2	rate constants at T_1 and T_2	same units
E_a	activation energy	J mol^{-1}
T_1, T_2	absolute temperatures	K

Use it when:

Two rate constants at two temperatures are given. Also handles "the rate doubles when temperature rises by 10 K" — set $k_2/k_1 = 2$.

Trap:

Inside the bracket it is $1/T_1 - 1/T_2$ — the **smaller** temperature first. Reversing it flips the sign of E_a .

ch 4 d and f Block Elements — 2 formulas + rules · 11 marks

This chapter is mostly trends and reasons rather than calculation. Two things are computable; everything else is recall, collected in the rules table below.

• c4.1 Magnetic moment from unpaired electrons — and back again

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

Symbol	Meaning	Unit
μ	spin-only magnetic moment	BM (Bohr magneton)
n	number of unpaired electrons	dimensionless
$n = 1, 2, 3$	gives $\mu = 1.73, 2.83, 3.87$	BM
$n = 4, 5$	gives $\mu = 4.90, 5.92$	BM

Use it when:

Either direction: count unpaired electrons from the configuration and get μ , or read μ off the question and work back to n . Same formula serves Chapter 5.

Trap:

n is unpaired electrons in the **ion**, not the atom. For a transition metal ion, remove the $4s$ electrons first, then the $3d$.

○ c4.2 General electronic configuration of each block

d-block: $(n-1)d^{1-10} ns^{1-2}$ lanthanoid: $4f^{1-14} 5d^{0-1} 6s^2$ actinoid: $5f^{1-14} 6d^{0-1} 7s^2$

Symbol	Meaning	Unit
n	period number	dimensionless
Cr	exception: $3d^5 4s^1$, half-filled is stabler	dimensionless
Cu	exception: $3d^{10} 4s^1$, fully-filled is stabler	dimensionless

Use it when:

Writing a configuration, or explaining why Zn, Cd and Hg are not really transition metals — their d subshell is full in both atom and ion.

Trap:

Electrons fill $4s$ before $3d$ but are **removed from $4s$ first**. Fe^{2+} is $3d^6$, not $3d^4 4s^2$.

Rules, not formulas — recall these as facts

Ask	Answer
Highest oxidation state in 3d series	+7 in Mn (as KMnO_4); Cr reaches +6
Most stable oxidation state across the series	+2 for most, becoming more stable to the right
Why coloured	$d-d$ transition of an unpaired d electron; Sc^{3+} (d^0) and Zn^{2+} (d^{10}) are colourless
Lanthanoid contraction	Steady size decrease across 4f, from poor shielding by f electrons
Its main consequence	Zr and Hf (also Nb/Ta) have almost identical radii and are hard to separate
Why alloys form readily	Similar atomic radii let one metal substitute for another in the lattice
Why good catalysts	Variable oxidation states and the ability to adsorb reactants on the surface
Most common lanthanoid oxidation state	+3; actinoids show far more variability

CH 5 Coordination Compounds — 6 formulas · 11 marks**• c5.1 Oxidation number of the metal inside a complex**

$$x + \sum (\text{ligand charges}) = \text{charge on the coordination sphere}$$

Symbol	Meaning	Unit
x	oxidation number of the central metal	dimensionless
neutral	NH_3 , H_2O , CO , en — contribute 0	dimensionless
anionic	Cl^- , CN^- , OH^- contribute -1 ; $\text{C}_2\text{O}_4^{2-}$ contributes -2	dimensionless

Use it when:

Always — it is the first step of every naming, VBT and CFT question. In $\text{K}_4[\text{Fe}(\text{CN})_6]$: $x + 6(-1) = -4$, so $x = +2$.

Trap:

In $[\text{Ni}(\text{CO})_4]$ the ligands are neutral and the sphere is neutral, so nickel is in oxidation state **zero**. This exact complex is the standing trap.

• c5.2 Coordination number when a ligand grips more than once

$$\text{CN} = \sum_{\text{ligands}} (\text{denticity}) = \text{number of donor atoms bonded to the metal}$$

Symbol	Meaning	Unit
CN	coordination number	dimensionless
en	ethane-1,2-diamine — didentate, counts 2	dimensionless
C ₂ O ₄ ²⁻	oxalate — didentate, counts 2	dimensionless
EDTA	hexadentate, counts 6	dimensionless

Use it when:

Working out geometry. [Co(en)₃]³⁺ has three ligands but a coordination number of **6**.

Trap:

Count **donor atoms**, not ligands. This is the difference between getting octahedral and getting the wrong geometry entirely.

• c5.3 Tetrahedral splitting compared with octahedral

$$\Delta_t = \frac{4}{9} \Delta_o$$

Symbol	Meaning	Unit
Δ_o	octahedral crystal field splitting energy	J mol ⁻¹ (or cm ⁻¹)
Δ_t	tetrahedral splitting energy	J mol ⁻¹ (or cm ⁻¹)

Use it when:

Asked to relate the two, or to explain why tetrahedral complexes are essentially always high spin.

Trap:

Because Δ_t is small, it is almost never bigger than the pairing energy — so **no low-spin tetrahedral complexes** in this syllabus. Also note the splitting is *inverted*: *e* below *t₂*.

○ c5.4 Crystal field stabilisation energy of an octahedral complex

$$\text{CFSE} = (-0.4x + 0.6y) \Delta_o \quad \text{for } t_{2g}^x e_g^y$$

Symbol	Meaning	Unit
<i>x</i>	electrons in the lower <i>t_{2g}</i> set	dimensionless
<i>y</i>	electrons in the upper <i>e_g</i> set	dimensionless
CFSE	crystal field stabilisation energy	J mol ⁻¹ (or units of Δ_o)

Use it when:

Comparing the stability of two configurations. Each t_{2g} electron is stabilised by $0.4\Delta_o$; each e_g electron is destabilised by $0.6\Delta_o$.

Trap:

The barycentre must balance: $3 \times 0.4 = 2 \times 0.6$. If your two numbers do not satisfy that, you have the sets the wrong way round.

• **c5.5 High spin or low spin — deciding which**

$$\Delta_o > P \Rightarrow \text{low spin (pair up)} \quad \Delta_o < P \Rightarrow \text{high spin (stay unpaired)}$$

Symbol	Meaning	Unit
Δ_o	crystal field splitting energy	J mol^{-1}
P	pairing energy	J mol^{-1}
strong field	CN^- , CO , NH_3 — large Δ_o , low spin	—
weak field	I^- , Br^- , Cl^- , F^- , H_2O — small Δ_o , high spin	—

Use it when:

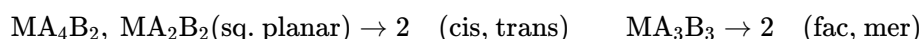
Predicting magnetic behaviour or hybridisation. Only matters for d^4 to d^7 ; below and above that there is no choice to make.

Trap:

Spectrochemical series, weakest to strongest:



○ **c5.6 How many geometrical isomers a complex has**



Symbol	Meaning	Unit
cis	identical ligands adjacent (90°)	—
trans	identical ligands opposite (180°)	—
fac	three identical ligands on one triangular face	—
mer	three identical ligands around a meridian	—

Use it when:

"What type of isomerism is shown by..." — check the formula shape first, then the geometry.

Trap:

Tetrahedral complexes show no geometrical isomerism — every position is adjacent to every other, so there is no "trans". Only the **cis** form of $[\text{M}(\text{en})_2\text{X}_2]$ is optically active.

сн 6 Haloalkanes and Haloarenes — 2 formulas + orders · 6 marks

An organic chapter, so almost everything is a reactivity order rather than an equation. The two computable things come first; the orders that actually earn the marks are collected below them.

• c6.1 The rate law that identifies the substitution mechanism



Symbol	Meaning	Unit
S_N2	second order, one step, transition state, inversion	k in $\text{L mol}^{-1} \text{s}^{-1}$
S_N1	first order, two steps, carbocation, racemisation	k in s^{-1}
$[\text{RX}]$	concentration of the alkyl halide	mol L^{-1}

Use it when:

Asked for two differences between the mechanisms, or told how the rate responds to changing the nucleophile's concentration.

Trap:

The reactivity orders are **opposite**: S_N2 runs $1^\circ > 2^\circ > 3^\circ$ (steric hindrance), S_N1 runs $3^\circ > 2^\circ > 1^\circ$ (carbocation stability). Check which mechanism the question named before you answer.

○ c6.2 Degree of unsaturation from a molecular formula

$$\text{DoU} = \frac{2C + 2 + N - H - X}{2}$$

Symbol	Meaning	Unit
C, H, N	counts of carbon, hydrogen, nitrogen	dimensionless
X	count of halogen atoms	dimensionless
DoU	rings plus π bonds	dimensionless

Use it when:

A molecular formula is given and you must decide how many isomers or what skeleton is possible. Oxygen is ignored — it does not change the count.

Trap:

Halogen counts like hydrogen (subtract it). A benzene ring alone gives DoU = 4: three π bonds plus one ring.

The orders that earn the marks

Comparison	Order, and the reason
Reactivity of RX to substitution	$\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$ — C–X bond weakens down the group
S_N1 rate	$3^\circ > 2^\circ > 1^\circ$ — carbocation stability; benzylic and allylic beat all three
S_N2 rate	$\text{CH}_3\text{X} > 1^\circ > 2^\circ > 3^\circ$ — steric hindrance
Alcohol + HX	$3^\circ > 2^\circ > 1^\circ$, and $\text{HI} > \text{HBr} > \text{HCl}$
Boiling point of isomers	Falls as branching rises — less surface contact, weaker van der Waals
Melting point of dihalobenzenes	para > ortho $\approx$ meta — para packs better in the crystal lattice
Free radical stability	benzylic $\approx$ allylic > $3^\circ > 2^\circ > 1^\circ$ — resonance beats hyperconjugation
Aqueous vs alcoholic KOH	Aqueous $\rightarrow$ substitution (alcohol); alcoholic $\rightarrow$ elimination (alkene)
Markovnikov vs peroxide	Peroxide effect works with HBr only
Saytzeff (NCERT: Zaitsev)	The more substituted alkene is the major elimination product

const Constants and conversions — *know the units too*

• **κ** Every constant this paper can hand you, with its unit

Symbol	Meaning	Value and unit
R	gas constant, energy form	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
R	gas constant, pressure–volume form	$0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$
F	Faraday constant	96487 C mol^{-1}
K_b	ebullioscopic constant, water	$0.52 \text{ K kg mol}^{-1}$
K_f	cryoscopic constant, water	$1.86 \text{ K kg mol}^{-1}$
N_A	Avogadro constant	$6.022 \times 10^{23} \text{ mol}^{-1}$
2.303	converts $\ln$ to $\log_{10}$	dimensionless
0.693	$\ln 2$, for first-order half-life	dimensionless
0.059	$2.303RT/F$ at 298 K, for Nernst	V
T	0°C in kelvin	273.15 K
STP	molar volume of an ideal gas	22.4 L mol^{-1}

Trap:

Match R to the units you want out. Using 8.314 with pressures in atm, or 0.0821 when the answer should be in joules, is a whole-question error and not a rounding one.

Built from Sourabh Raina's one-shot and PYQ videos for Chapters 1–6, and cross-checked against NCERT Class XII Chemistry (Rationalised 2022–23). NCERT conventions used throughout: **0.059** rather than 0.0591 in the Nernst equation, $F = 96487 \text{ C mol}^{-1}$, molar conductivity in $\text{S cm}^2 \text{ mol}^{-1}$, and **Zaitsev** (which NCERT notes is "also pronounced as Saytzeff").

The six chapter pages carry the teaching and the past-year questions; this page is only the recall layer. Derivations for the formulas that have them live on the companion page, **Chemistry, Derived**.

Chemistry, Derived

Class XII CBSE • Chemistry • Chapters 1–3

Twelve derivations, one algebraic move per line, each with the reason it is allowed. Every one ends in a formula that is already on Every Chemistry Formula — the point of this page is that you can rebuild them when memory fails, and that you can write the proof out when the paper asks for it.

- Derivations: 12
- Figures: 12
- Chapters: 3
- Marks covered: 42

How to use this

Chapters 4, 5 and 6 have no derivations worth the name — their marks come from recall and reasoning, and those live on the formula sheet. Everything derivable in this paper is on this page.

Read the **setup** first and draw the figure yourself before you look at the steps. A derivation you can only follow is not one you can write.

The *italic reason* beside a step is what turns it from an assertion into a proof. In the exam, those reasons are usually where the marks are.

CH 3 Chemical Kinetics — 5 derivations • 13 marks

D1 Integrated rate law for a zero order reaction — 2–3 marks

A reaction $R \longrightarrow P$ whose rate does not depend on the concentration of R at all. Start from the definition of rate, put the zero-order rate law into it, and integrate from the start of the reaction to time t . Write $[R]_0$ for the concentration at $t = 0$ and $[R]$ for the concentration at time t .

Figure. Graph of concentration of R against time for a zero order reaction: a straight line starting at the initial concentration on the vertical axis and falling with constant slope minus k , meeting the time axis when all of R is used up.

Zero order is the only order whose concentration–time plot is a straight line. Its slope is $-k$ and its intercept is $[R]_0$, which is how both are read off experimental data.

1. Rate = $-\frac{d[R]}{dt}$ — (definition; minus sign because $[R]$ falls)
2. Rate = $k[R]^0 = k$ — (zero order: anything to the power 0 is 1)
3. $-\frac{d[R]}{dt} = k$
4. $d[R] = -k dt$ — (separating the variables)
5. $\int_{[R]_0}^{[R]} d[R] = -k \int_0^t dt$ — (integrating between the two states)
6. $[R] - [R]_0 = -k(t - 0)$
7. $[R] - [R]_0 = -kt$
8. $[R] = -kt + [R]_0$ — (the straight line in the figure, $y = mx + c$)
9. $kt = [R]_0 - [R]$ — (rearranging for k)
10. $k = \frac{[R]_0 - [R]}{t}$

Result: $[R] = -kt + [R]_0 \quad k = \frac{[R]_0 - [R]}{t}$

Units check: k comes out as concentration $\div$ time, $\text{mol L}^{-1} \text{s}^{-1}$ — which is exactly what C3.4 predicts for $n = 0$.

D2 Half-life of a zero order reaction — 2 marks

Half-life $t_{1/2}$ is the time at which exactly half the reactant is left. Take the integrated law from D1 and substitute that one condition.

Shared setup with D1. Every half-life derivation is the integrated rate law with $[R] = [R]_0/2$ and $t = t_{1/2}$ substituted. If you can do D1 and D3, you can do D2 and D4 without learning anything new.

Figure. Zero order concentration against time, with the half-life marked: the concentration reaches half its initial value at a time equal to the initial concentration divided by twice k , and reaches zero at twice that time.

Because the line is straight, the reaction finishes at exactly $2t_{1/2}$ — a zero order reaction really does run out, which no first order reaction ever does.

1. $k = \frac{[R]_0 - [R]}{t}$ — (from D1)
2. At $t = t_{1/2}$, $[R] = \frac{[R]_0}{2}$ — (definition of half-life)
3. $k = \frac{[R]_0 - \frac{[R]_0}{2}}{t_{1/2}}$ — (substituting both)
4. $k = \frac{\frac{[R]_0}{2}}{t_{1/2}}$
5. $k t_{1/2} = \frac{[R]_0}{2}$
6. $t_{1/2} = \frac{[R]_0}{2k}$

Result: $t_{1/2} = \frac{[R]_0}{2k}$ — directly proportional to the initial concentration

D3 Integrated rate law for a first order reaction — 3 marks

Same reaction $R \rightarrow P$, but now the rate is proportional to $[R]$. The integration produces a natural logarithm, which is then converted to base 10 because every exam table and graph uses $\log$.

Figure. Graph of $\log$ of concentration of R against time for a first order reaction: a straight line with intercept $\log$ of the initial concentration and slope minus k over 2.303.

A first order reaction is straight only after you take the logarithm. That is the practical test: if $\log[R]$ against t is a line, the reaction is first order and k is $2.303 \times$ the magnitude of its slope.

1. Rate $= -\frac{d[R]}{dt}$ — (definition)
2. Rate $= k[R]^1 = k[R]$ — (first order rate law)
3. $-\frac{d[R]}{dt} = k[R]$
4. $\frac{d[R]}{[R]} = -k dt$ — (separating the variables)
5. $\int_{[R]_0}^{[R]} \frac{d[R]}{[R]} = -k \int_0^t dt$
6. $\ln[R] - \ln[R]_0 = -kt$ — (integral of $1/x$ is $\ln x$)

$$7. \ln \frac{[R]}{[R]_0} = -kt \text{ — (law of logarithms)}$$

$$8. \ln \frac{[R]_0}{[R]} = kt \text{ — (inverting the fraction flips the sign)}$$

$$9. 2.303 \log \frac{[R]_0}{[R]} = kt \text{ — } (\ln x = 2.303 \log_{10} x)$$

$$10. k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$$

$$11. \text{Rearranged for a graph: } \log[R] = -\frac{k}{2.303}t + \log[R]_0 \text{ — (the straight line drawn above)}$$

Result: $k = \frac{2.303}{t} \log \frac{[R]_0}{[R]}$

Units check: the log is dimensionless, so k has units of 1/time, s_1^- — matching C3.4 for $n = 1$. That is also why the next derivation comes out independent of concentration.

D4 First order half-life, and why it ignores concentration — 2–3 marks

Substitute the half-life condition into D3. Watch what happens to $[R]_0$ — the whole point of this derivation is that it cancels.

Figure. First order decay curve: concentration falls from its initial value to one half after one half-life, to one quarter after two half-lives and one eighth after three, with the three time intervals all equal.

Each successive halving takes the same time, no matter how little is left. That is the visual statement of the algebraic cancellation below, and it is why radioactive decay has a fixed half-life.

$$1. k = \frac{2.303}{t} \log \frac{[R]_0}{[R]} \text{ — (from D3)}$$

$$2. \text{At } t = t_{1/2}, [R] = \frac{[R]_0}{2} \text{ — (definition of half-life)}$$

$$3. k = \frac{2.303}{t_{1/2}} \log \frac{[R]_0}{[R]_0/2}$$

$$4. \frac{[R]_0}{[R]_0/2} = 2 \text{ — } ([R]_0 \text{ cancels — this is the whole result})$$

$$5. k = \frac{2.303}{t_{1/2}} \log 2$$

$$6. \log 2 = 0.301$$

$$7. k = \frac{2.303 \times 0.301}{t_{1/2}} = \frac{0.693}{t_{1/2}}$$

$$8. t_{1/2} = \frac{0.693}{k}$$

Result: $t_{1/2} = \frac{0.693}{k}$ — independent of the initial concentration

Step 4 is the answer to "show that the half-life of a first order reaction is independent of initial concentration". Do not skip it; it is the mark.

D5 Activation energy from rate constants at two temperatures — 3 marks

Start from the Arrhenius equation, take logarithms to make it linear, write it once at each of two temperatures T_1 and T_2 , and subtract. The frequency factor A is unknown and must disappear — that is what the subtraction is for.

Figure. Arrhenius plot: $\log k$ against $1/T$ is a descending straight line whose intercept is $\log A$ and whose slope is minus the activation energy divided by $2.303 R$.

$T_2 > T_1$, so $1/T_2$ sits to the left of $1/T_1$ and the higher temperature gives the higher $\log k$. Reading the axis backwards is the commonest error on this plot.

1. $k = A e^{-E_a/RT}$ — (Arrhenius equation)

2. $\ln k = \ln A - \frac{E_a}{RT}$ — (taking natural logs of both sides)

3. $\log k = \log A - \frac{E_a}{2.303RT}$ — (converting to base 10)

4. At T_1 : $\log k_1 = \log A - \frac{E_a}{2.303RT_1}$

5. At T_2 : $\log k_2 = \log A - \frac{E_a}{2.303RT_2}$

6. Subtracting: $\log k_2 - \log k_1 = -\frac{E_a}{2.303RT_2} + \frac{E_a}{2.303RT_1}$ — (log A cancels — it is the same constant at both temperatures)

7. $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$

8. $\frac{1}{T_1} - \frac{1}{T_2} = \frac{T_2 - T_1}{T_1 T_2}$ — (common denominator)

9. $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$

Result: $\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_1 T_2} \right]$

The bracket is $\frac{1}{T_1} - \frac{1}{T_2}$, smaller temperature first. If you get a negative E_a , you reversed it at step 7.

CH 2 Electrochemistry — 3 derivations · 14 marks

D6 The Nernst equation — 3 marks

A galvanic cell running the reaction $a A + b B \longrightarrow c C + d D$ with n electrons transferred. Two facts are combined: the thermodynamic relation between Gibbs energy and the reaction quotient, and the electrochemical relation between Gibbs energy and cell potential.

Figure. A galvanic cell: a zinc electrode in zinc sulphate solution on the left as the anode where oxidation occurs, a copper electrode in copper sulphate solution on the right as the cathode where reduction occurs, joined by a salt bridge that dips into both solutions and by an external circuit with a voltmeter through which electrons flow from left to right.

The Daniell cell, $E_{\text{cell}}^\circ = 1.1 \text{ V}$. The Nernst equation says what happens to that reading when the two solutions are not at 1 M — and the sign of the correction follows from Le Chatelier, not from memory.

1. $\Delta_r G = \Delta_r G^\circ + RT \ln Q$ — (thermodynamics: Gibbs energy away from standard state)

2. $\Delta_r G = -nFE_{\text{cell}}$ — (electrical work done by the cell)

3. $\Delta_r G^\circ = -nFE_{\text{cell}}^\circ$ — (same relation at standard state)

4. $-nFE_{\text{cell}} = -nFE_{\text{cell}}^\circ + RT \ln Q$ — (substituting 2 and 3 into 1)

5. Divide throughout by $-nF$:

6. $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \ln Q$

7. $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{2.303RT}{nF} \log Q$ — (converting to base 10)

8. At 298 K: $\frac{2.303 \times 8.314 \times 298}{96487} = 0.059$ — (substituting R , T and F)

9. $E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{0.059}{n} \log Q$

Result: $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303RT}{nF} \log Q \xrightarrow{298 \text{ K}} E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$

Step 8 is worth writing out in full — being asked where the 0.059 comes from is a common follow-up, and "2.303RT/F at 298 K" is the whole answer.

D7 Equilibrium constant from standard cell potential — 2–3 marks

Take the Nernst equation and ask what it says at the one moment the cell stops working. A cell at equilibrium has no driving force, so it reads zero volts — and at that moment the reaction quotient has become the equilibrium constant.

Shared setup with D6. This is the Nernst equation with one substitution. Do not learn it as a separate formula.

Figure. Graph of cell potential against log Q: a descending straight line of slope minus 0.059 over n, with intercept the standard cell potential at log Q equals zero, crossing the horizontal axis at log of the equilibrium constant where the cell is dead.

The cell potential falls linearly as products build up. Where the line crosses zero, Q has reached K_c — the battery is flat. Reading the intercept and the zero-crossing off this one line gives both results.

1. $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log Q$ — (from D6)
2. At equilibrium the cell delivers no work, so $E_{\text{cell}} = 0$ — (a dead cell)
3. At equilibrium $Q = K_c$ — (definition of the equilibrium constant)
4. $0 = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log K_c$
5. $E_{\text{cell}}^{\circ} = \frac{0.059}{n} \log K_c$ — (moving the log term across — this is why the sign is now positive)
6. $\log K_c = \frac{n E_{\text{cell}}^{\circ}}{0.059}$ — (rearranged for K_c)
7. Equivalently, from $\Delta_r G^{\circ} = -nF E_{\text{cell}}^{\circ}$ and $\Delta_r G^{\circ} = -RT \ln K_c$:
8. $nF E_{\text{cell}}^{\circ} = RT \ln K_c$ — (equating the two expressions for $\Delta_r G^{\circ}$)

Result: $E_{\text{cell}}^{\circ} = \frac{0.059}{n} \log K_c \quad \log K_c = \frac{n E_{\text{cell}}^{\circ}}{0.059}$

D8 Molar conductivity from conductivity — where the 1000 comes from — 2 marks

A conductivity cell of electrode area A and separation l . Conductivity κ is defined for a 1 cm cube of solution; molar conductivity Λ_m is defined for however much solution contains exactly one mole. The 1000 is nothing but a cm^3 -to-litre conversion, and this derivation is really a units argument.

Figure. A conductivity cell: two parallel electrodes of area A separated by a distance l , with the cell constant defined as l divided by A , beside a unit cube of solution of side one centimetre whose conductance is the conductivity.

Two different reference volumes: κ is per cubic centimetre, Λ_m is per mole. Converting between them is the entire content of the formula.

1. $\Lambda_m = \kappa \times V$ — (V = volume of solution containing exactly 1 mole of electrolyte)
2. Let the concentration be c mol per litre — (the unit every exam question uses)
3. c moles occupy 1 L, so 1 mole occupies $\frac{1}{c}$ L
4. 1 L = 1000 cm^3 — (the conversion — this is the 1000)

$$5. 1 \text{ mole occupies } \frac{1000}{c} \text{ cm}^3$$

$$6. \Lambda_m = \kappa \times \frac{1000}{c}$$

$$7. \text{Units: } \text{S cm}^{-1} \times \text{cm}^3 \text{ mol}^{-1} = \text{S cm}^2 \text{ mol}^{-1} \text{ — (confirming the standard unit)}$$

$$\text{Result: } \Lambda_m = \frac{\kappa \times 1000}{c} \text{ in S cm}^2 \text{ mol}^{-1}$$

In strict SI (κ in S m^{-1} , c in mol m^{-3}) step 4 disappears and $\Lambda_m = \kappa/c$ in $\text{S m}^2 \text{ mol}^{-1}$. The 1000 is a unit artefact, not physics.

CH 1 Solutions — 4 derivations · 15 marks

D9 Relative lowering of vapour pressure equals the solute's mole fraction — 3 marks

A non-volatile solute B dissolved in a volatile solvent A. Only A can evaporate, so the vapour above the solution is pure A — but there is less of it, because solute particles occupy part of the surface. Start from Raoult's law and rearrange.

Figure. Two beakers side by side: pure solvent on the left with many molecules escaping from its surface, and a solution on the right whose surface is partly occupied by solute particles so fewer solvent molecules escape, giving a lower vapour pressure.

Solute particles (filled circles) take up surface sites, so fewer solvent molecules can leave. The fraction of the surface still available to solvent is x_A — which is exactly what Raoult's law asserts.

$$1. p_A = x_A p_A^\circ \text{ — (Raoult's law; the solute is non-volatile so } p_{\text{soln}} = p_A)$$

$$2. p_A^\circ - p_A = p_A^\circ - x_A p_A^\circ \text{ — (subtracting both sides from } p_A^\circ)$$

$$3. p_A^\circ - p_A = p_A^\circ (1 - x_A)$$

$$4. x_A + x_B = 1, \text{ so } 1 - x_A = x_B \text{ — (mole fractions sum to one)}$$

$$5. p_A^\circ - p_A = p_A^\circ x_B$$

$$6. \frac{p_A^\circ - p_A}{p_A^\circ} = x_B \text{ — (dividing by } p_A^\circ \text{ — the left side is the relative lowering)}$$

$$7. x_B = \frac{n_B}{n_A + n_B} \text{ — (definition of mole fraction)}$$

$$8. \text{For a dilute solution } n_B \ll n_A, \text{ so } n_A + n_B \approx n_A$$

$$9. \frac{p_A^\circ - p_A}{p_A^\circ} \approx \frac{n_B}{n_A} = \frac{w_B/M_B}{w_A/M_A}$$

$$10. \frac{p_A^\circ - p_A}{p_A^\circ} = \frac{w_B M_A}{M_B w_A} \text{ — (rearranged to give molar mass)}$$

$$\text{Result: } \frac{p_A^\circ - p_A}{p_A^\circ} = x_B \quad M_B = \frac{w_B M_A p_A^\circ}{w_A (p_A^\circ - p_A)}$$

Steps 1–6 are exact. Step 8 is the only approximation, and it is only valid for a dilute solution — say so if you use it.

D10 Elevation of boiling point is proportional to molality — 3 marks

A liquid boils when its vapour pressure equals the external pressure. A non-volatile solute lowers the vapour pressure at every temperature (D9), so the solution's curve sits below the solvent's — and has to be taken to a higher temperature before it reaches the 1 atm line.

Figure. Vapour pressure against temperature for pure solvent and for solution: the solution curve lies below the solvent curve everywhere, so it meets the one atmosphere line at a higher temperature, and the gap

between the two crossing points is the elevation of boiling point.

The dashed curve is the solution. It never reaches 1 atm at the solvent's boiling point, so the temperature must be raised by ΔT_b — and for a dilute solution the two curves are near enough parallel that the shift is proportional to how much vapour pressure was lost.

1. For a dilute solution, the lowering of vapour pressure is proportional to the elevation of boiling point:

$$\Delta T_b \propto (p_A^\circ - p_A) \text{ — (the two curves are almost straight and almost parallel over this small range)}$$

$$2. \frac{p_A^\circ - p_A}{p_A^\circ} = x_B \text{ — (from D9)}$$

3. So $\Delta T_b \propto x_B$ — (p_A° is a constant for a given solvent)

$$4. x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A} \text{ — (dilute solution)}$$

$$5. \Delta T_b \propto \frac{n_B}{n_A}$$

$$6. n_A = \frac{w_A}{M_A}, \text{ so } \frac{n_B}{n_A} = \frac{n_B M_A}{w_A}$$

$$7. M_A \text{ is a constant for a given solvent, so } \Delta T_b \propto \frac{n_B}{w_A}$$

$$8. \frac{n_B}{w_A \text{ in kg}} = m, \text{ the molality — (definition of molality)}$$

$$9. \Delta T_b \propto m$$

10. $\Delta T_b = K_b m$ — (introducing the proportionality constant)

$$11. m = \frac{w_B \times 1000}{M_B w_A} \text{ — (molality written from masses, } w_A \text{ in grams)}$$

$$12. \Delta T_b = \frac{K_b w_B \times 1000}{M_B w_A}$$

$$13. M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A}$$

$$\textbf{Result: } \Delta T_b = K_b m \quad M_B = \frac{K_b w_B \times 1000}{\Delta T_b w_A}$$

K_b is the elevation for a 1 molal solution, and depends only on the solvent — for water $0.52 \text{ K kg mol}^{-1}$.

D11 Depression of freezing point is proportional to molality — 3 marks

A solution freezes when the vapour pressure of the liquid equals that of the pure solid solvent. Adding solute lowers the liquid's vapour pressure, so the two curves now meet at a *lower* temperature. Structurally identical to D10 — only the meeting point changes.

Shared setup with D10. Both start from "vapour pressure is lowered by x_B " and end in "proportional to molality". If you can write one, change the meeting condition and you have written the other.

Figure. Vapour pressure against temperature near the freezing point: the solid solvent line crosses the pure solvent curve at the normal freezing point on the right, and crosses the lower solution curve at a lower temperature on the left, the gap between the two crossings being the depression of freezing point.

Freezing happens where the liquid curve crosses the solid curve. Lowering the liquid curve moves that crossing to the left, and the shift is ΔT_f . Note the solid curve is unchanged — solute does not dissolve in the solid solvent.

1. A solution freezes when the vapour pressure of the liquid equals that of the pure solid solvent — (definition of freezing point)

2. For a dilute solution, $\Delta T_f \propto (p_A^\circ - p_A)$ — (same near-linear argument as D10)

$$3. \frac{p_A^\circ - p_A}{p_A^\circ} = x_B \text{ — (from D9)}$$

4. $\Delta T_f \propto x_B$
5. $x_B \approx \frac{n_B}{n_A}$ — (dilute solution)
6. $\Delta T_f \propto \frac{n_B}{w_A}$ — (M_A constant for a given solvent)
7. $\Delta T_f \propto m$ — (definition of molality)
8. $\Delta T_f = K_f m$
9. $m = \frac{w_B \times 1000}{M_B w_A}$
10. $M_B = \frac{K_f w_B \times 1000}{\Delta T_f w_A}$

Result: $\Delta T_f = K_f m$ $M_B = \frac{K_f w_B \times 1000}{\Delta T_f w_A}$

K_f for water is $1.86 \text{ K kg mol}^{-1}$ — larger than K_b , which is why freezing-point depression is the more sensitive method for finding a molar mass.

►12 Degree of dissociation from the van't Hoff factor — 2–3 marks

One mole of an electrolyte A_n that breaks into n particles, with a fraction α actually dissociating. Colligative properties count *particles*, so build the particle total after dissociation and compare it with the total before.

Figure. Particle count before and after dissociation: one mole of solute starts as one particle, and after a fraction α dissociates into n particles the total becomes one minus α plus $n\alpha$, whose ratio to the original is the van't Hoff factor.

Only the fraction α splits; the rest stays whole. Counting both contributions is the entire derivation — i is just the ratio of the two particle totals.

1. $A_n \longrightarrow n A$, starting from 1 mole with a fraction α dissociating
2. Moles of A_n left undissociated = $1 - \alpha$
3. Moles of A produced = $n\alpha$ — (each dissociated mole gives n particles)
4. Total moles of particles after = $(1 - \alpha) + n\alpha$
5. $i = \frac{\text{total particles after}}{\text{total particles before}} = \frac{1 - \alpha + n\alpha}{1}$ — (definition of the van't Hoff factor)
6. $i = 1 - \alpha + n\alpha$
7. $i - 1 = \alpha(n - 1)$ — (collecting the α terms)
8. $\alpha = \frac{i - 1}{n - 1}$
9. For **association**, n molecules combine into 1, so the total after is $1 - \alpha + \frac{\alpha}{n}$
10. $i = 1 - \alpha + \frac{\alpha}{n} = 1 - \alpha \left(1 - \frac{1}{n}\right)$
11. $\alpha = \frac{1 - i}{1 - \frac{1}{n}}$

Result: $\alpha_{\text{dissoc}} = \frac{i - 1}{n - 1}$ $\alpha_{\text{assoc}} = \frac{1 - i}{1 - \frac{1}{n}}$

Dissociation gives $i > 1$ and association $i < 1$; both formulas return a positive α because the numerator flips with them.

Built from Sourabh Raina's one-shot and PYQ videos for Chapters 1–3, and cross-checked against NCERT Class XII Chemistry (Rationalised 2022–23) for constants and conventions — $F = 96487 \text{ C mol}^{-1}$, $R =$

$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $K_b = 0.52$ and $K_f = 1.86 \text{ K kg mol}^{-1}$ for water, and **0.059** rather than 0.0591 in the Nernst equation.

Chapters 4, 5 and 6 are not represented here because they contain no derivations — their marks come from trends, reasons and reactivity orders, all of which are on the companion page, **Every Chemistry Formula**.